

A TREATISE ON THE PRINCIPLES OF FINANCE



*Being a Plain Account of Money, Capital, Risk and the
Management of Funds*

M. MEHU, G. MOMO

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M . M E H U , G . M O M O

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First edition

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*To the one I would avenge from death, if given the
chance; the one with whom I truly shared love; my
first pet, my Cutush and Lokkhi. Pufi*

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TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches A Treatise on the Principles of Finance one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

PREFACE

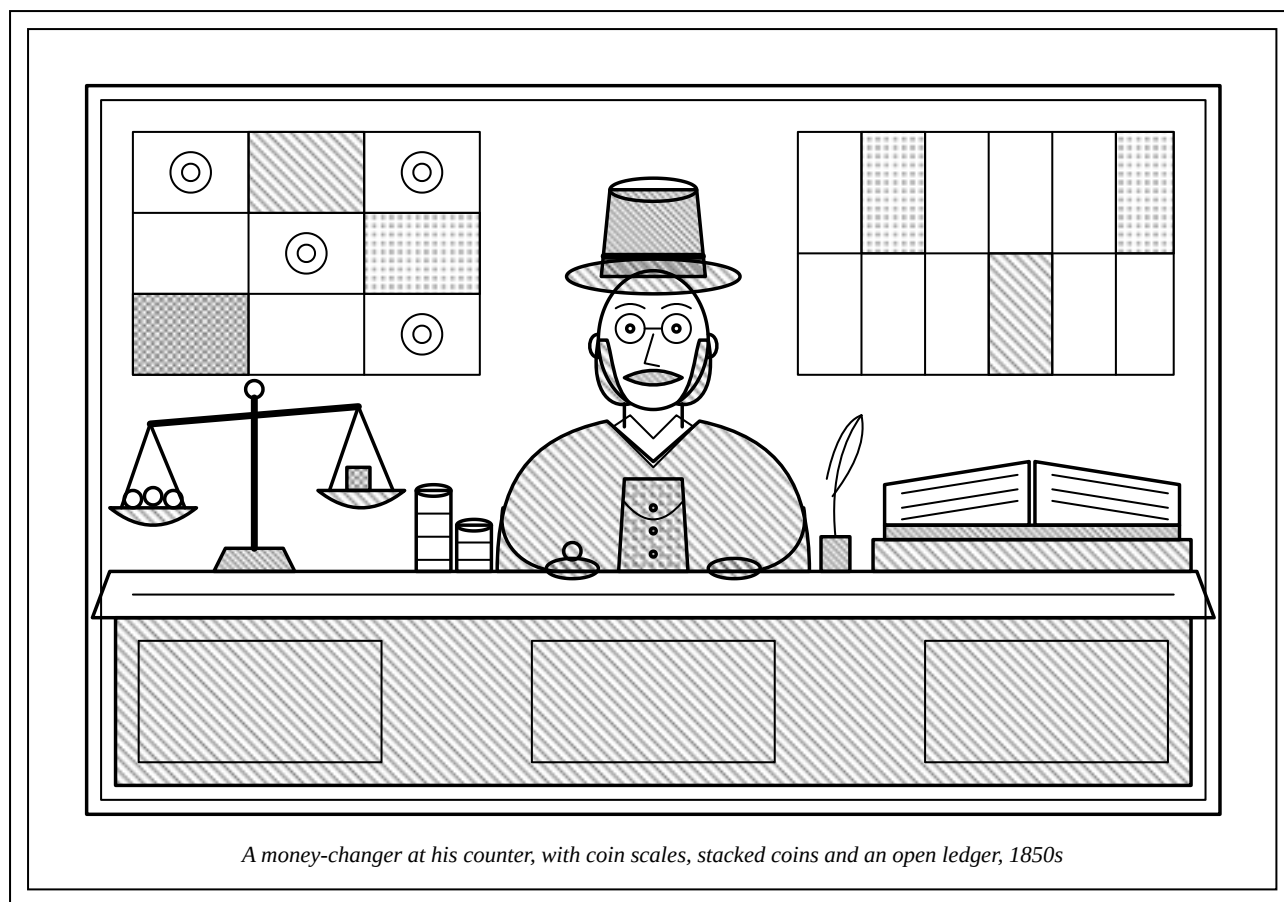
This treatise was written for the reader who has never studied finance and wishes to begin properly. We have tried to say each thing once, plainly, and in its right place.

The book moves from the nature and aims of finance, through the value of money over time and the sources of capital, to the weighing of risk and the choice of investments. A reader who takes the chapters in order will find that each rests upon the one before.

Where a word has a special meaning, it is explained at its first use and gathered into the glossary at the end. The questions at the close of each chapter are answered at the back of the book; we advise the reader to attempt them first.

If this book spares a single student an hour of needless confusion, its labour is repaid.

M. Mehu, G. Momo



CHAPTER I

Money, Finance and the Firm

Where does a business find its money, and what does it do with it?



This chapter introduces finance: what it is, where it sits among the main kinds of work in a business, and how money moves between owners, lenders and the business. A family bookshop in Dhaka supplies the figures, and a banking house in Florence supplies the history.

A stall owner has a full table of goods but too little money to buy more.

Her choices raise three questions, and finance exists to answer them.

You need nothing beforehand except your own experience of spending and borrowing.

Each idea is built in the same order, and small signs help you follow it.

The chapter takes three steps, from the idea of finance to the flow of funds.

Every household already practises finance on a small scale.

GO SLOWLY HERE.

Finance. The work of finding money and deciding how to use it, so a business or person can reach chosen aims.

Finance is finding money and deciding how to use it.

Finance has two halves, and the same two halves appear in every business.

Picture a woman who sells vegetables at a market stall. On Monday evening her table still holds tomatoes, onions and gourds, and at the prices on her sign they are worth Tk 5,000.¹ Yet at five o'clock on Tuesday morning, at the wholesale market, her purse holds only Tk 1,200, and the onions she needs cost Tk 2,000. The stall is full, and she still cannot buy what she needs. Her trouble is older than any bank: the goods a person owns and the money a person holds are not the same thing.

She has choices. She may buy fewer onions. She may ask a cousin for a loan until Friday. She may sell the tomatoes cheaply this evening to receive money at once. Each choice raises the same questions. Where will the money come from? What will it be spent on? What will she owe, and to whom, afterwards? These questions are the beginning of finance. This chapter turns them into plain ideas and gives each idea a name.

You will need: nothing beyond your own experience of spending, saving and borrowing. Every other word is explained when it first appears.

This book builds each idea in the same order: a plain statement, an example, a rule, and then the limits of the rule. A short note in the side column sums up each paragraph in one line. When a new term appears, it is printed in bold, defined at once and repeated in the side column; a fuller meaning is in the glossary. Questions marked Pause let you test yourself before reading on, and their answers are given at the back. This first chapter goes slowly on purpose, because every term in it is new.

The chapter takes three steps. Section 1.1 explains what finance is. Section 1.2 shows where finance sits among the kinds of work that a business performs. Section 1.3 follows money from the people who supply it to the things the business buys with it. A running case, the bookshop Meghna and Sons in Dhaka, appears from section 1.2 onward.

* * *

1.1 WHAT FINANCE IS

Begin with your own week. Money reaches you from somewhere: an allowance, a wage, a gift from a relative. Money leaves you for a purpose: bus fares, food, a phone, fees. Sometimes you keep part of it for later. Sometimes you borrow from a friend and promise to repay. In each case, someone decides where money should come from and where it should go. That someone is practising finance, on a small scale.

Finance is the work of finding money and deciding how to use it, so that a business or a person can reach chosen aims. The stall owner who asks her cousin for Tk 800 and decides to spend it on onions is doing finance twice: once to find the money, and once to choose its use.

Words first, then numbers. The first half asks: where can money be found? The second half asks: what is the best use of it? Suppose the Tk 800 buys onions that she sells for Tk 1,000. She finds Tk 800 and uses Tk 800. After she repays her cousin, she keeps Tk 200. The example is small on purpose, because the same two halves govern a bookshop that handles millions of taka.

¹ Tk is the written short form of taka, the money of Bangladesh. The book writes amounts as Tk 5,000, with a comma after each group of three digits.

Finance has limits worth knowing. It is not only for banks or very large businesses; the stall owner uses it every day. It is also not the same as having money: a business with no money may need finance more than a rich one. And finance is a method for choosing, not a promise that every choice will be right, because a loan used badly must still be repaid.

The viewpoint changes with the person. The owner wants the business to grow. A lender wants to be repaid on the agreed day. An employee wants wages paid on time. A customer wants the shop to be open next month. All four depend on how well the money has been found and used.

Funds are money that a business has available to spend, save or invest. To invest means to use money now in the hope of earning more later. Funds may be notes in a purse or a balance held at a bank. On a morning when the stall owner's purse holds Tk 1,200, her funds are Tk 1,200. The tomatoes on her table are not funds, because she cannot spend a tomato at the wholesale market until it has been sold.

Cash flow is money moving into or out of a business during a period. A period is any stretch of time, such as a day, a month or a year. On Monday customers pay the stall owner Tk 3,000, which is a cash flow into the business, and she pays Tk 2,000 to the wholesaler, which is a cash flow out. Funds are measured at one moment; cash flow is measured over a stretch of time.

The table below compares the two ideas. Read each row from left to right: the first column names the idea, the second says how it is measured, and the third gives an example from the stall.

FUNDS AND CASH FLOW COMPARED

IDEA	HOW IT IS MEASURED	EXAMPLE FROM THE STALL
Funds	At one moment	Tk 1,200 in her purse at five in the morning
Cash flow	Over a period	Tk 3,000 received and Tk 2,000 paid during Monday

A sale and a cash flow are different events. Suppose a customer takes goods worth Tk 5,000 on 28 June and promises to pay on 28 July. The sale happens in June, but the cash flow into the business happens in July. In the business sense of the word, the profit from that sale is counted in June, while the money reaches the business in July. A business can therefore show a good profit and still lack money to pay wages on Friday. This is why finance watches cash flow closely.

PAUSE AND CHECK 1.P1 A tailor accepts an order worth Tk 3,000 today. The customer promises to pay when the suit is ready next week. Is there a cash flow into the business today? Give a reason.

Wealth is the total value of what a person or business owns, after taking away what is owed. Suppose the stall owner owns a table, scales and Tk 1,200 in cash, worth Tk 4,000 together, and owes her cousin Tk 800. Her wealth is Tk 4,000 less Tk 800, which is Tk 3,200. Wealth can be large while funds are small: her table is part of her wealth, but it is not funds.

Before moving on. Finance is the work of finding money and deciding how to use it. Funds are money available at one moment, cash flow is money moving over a period, and wealth is what is owned after taking away what is owed. *You should now be able to say in one sentence what finance is, and why owning goods is not the same as having funds.*

Finance is not only for banks, not the same as having money, and not a promise of good choices.

Each person connected to a business sees finance from a different side.

GO SLOWLY HERE.

Funds. Money that a business has available to spend, save or invest.

Funds are money that is available to use.

Cash flow. Money moving into or out of a business during a period.

Cash flow is money in motion over a period of time.

The table sets the two ideas side by side.

Profit. Word in use. In daily life, the money you gain from a sale. In business, what a sale earns after the costs of making it, counted when the sale is made, even if the customer pays later.

A sale and its cash flow can fall in different months.

Wealth. The total value of what a person or business owns, after taking away what is owed.

Wealth is what is owned after taking away what is owed.

The section ends with a short recap.

* * *

1.2 FINANCE AMONG THE BUSINESS FUNCTIONS

A bookshop needs more than one kind of work, and someone must look after each.

Business function. One main kind of work in a business, such as selling, making or managing money.

A business function is one main kind of work in a business.

Account. Word in use. In daily life, a record of money in a bank. In accounting, a named record of one kind of item, such as Cash or Sales.

The running case shows two functions shared between two brothers.

Managers decide; finance supplies the money to carry out each decision and the facts to judge it.

Finance depends on the other functions as much as they depend on it.

Finance is close to accounting and to economics, but its main question differs.

Think of a bookshop on a busy street. Someone chooses and buys the books. Someone serves the readers. Someone records what was sold. Someone makes sure the shop can pay its suppliers. These are different kinds of work, and no one person can do all of them well at all times. Where does the work of finance sit among them?

A **business function** is one main kind of work in a business, such as selling, making or managing money. In a bookshop, buying books from publishers is one function, selling to readers is another, and looking after money is a third. Larger businesses give each function its own team; a small business may give several functions to one person.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhaka, founded on 12 January 1998 by Abdur Rahim Chowdhury. Since 2019 it has been run by his sons Imran and Tanvir. Imran is responsible for the shop and buying. Tanvir is responsible for accounts and money.

Managers decide what a business will do. They choose goods, prices, staff and suppliers. Every such decision has a money side. Suppose a bookshop wishes to add a shelf of stationery. Someone must find the money for the new goods and the shelf. Someone must judge whether the new goods will bring in enough to repay that money. Finance serves the other functions in this way, by supplying funds and by judging the money side of their decisions. It informs the decision; it does not replace it.

The dependence runs both ways. A business with perfect money records but no customers will fail, because finance cannot create sales. In a larger business, the person who leads this function is called the financial manager; Chapter 2, The Financial Manager's Work, describes the role. In the running case, one brother holds that duty beside his work on accounts.

Two nearby subjects are often confused with finance. Accounting is the work of recording and reporting what has happened to the money of a business. Economics studies how people and societies choose when money and resources are limited. Finance draws on both, but its main question looks forward: what should we do with money, and where do we find it? The table below sets the three side by side, using a tailor's shop. Read each row from left to right: the subject, the question it asks, and an example.

FINANCE, ACCOUNTING AND ECONOMICS COMPARED

SUBJECT	MAIN QUESTION	EXAMPLE AT A TAILOR'S SHOP
Accounting	What has happened to our money?	Recording that Tk 6,000 was received this month
Finance	What should we do with money, and where do we find it?	Deciding whether to borrow to buy a second sewing machine
Economics	How do people and societies choose when resources are limited?	Why the price of cloth rises when cloth is scarce

The three overlap. The tailor uses accounting records to see how much money was received, uses economics to understand why cloth costs more, and uses finance to decide about the machine. A common confusion is to think that finance is only keeping records. Records belong to accounting; finance uses them to make decisions about the future.

Before moving on. A business function is one main kind of work in a business. Finance is one function: it supplies funds to the others and judges the money side of their decisions. It differs from accounting, which records what has happened, and from economics, which studies choice in society.

* * *

1.3 WHERE FUNDS COME FROM AND WHERE THEY GO

Walk into a bookshop and look along the shelves. Every book there was bought with money before it was sold. Somebody supplied that money; the shop did not create it. In any business, then, two questions come before all others: who supplied the money, and what was it used for?

Case file, illustrative. At 1 January 2024, Meghna and Sons held stock of Tk 1,850,000, a cash and bank balance of Tk 240,000 and fittings of Tk 310,000, which together come to Tk 2,400,000. The funds came from a bank loan of Tk 600,000 at 10% a year and from the family's own funds of Tk 1,800,000. Sales in 2023 were Tk 7,200,000.

Two words in the case file need care. Stock here means the books held for sale. Fittings means the shelves and counters fixed in the shop. Both belong to the shop, and both were bought with money.

A **source of funds** is a place a business gets money from, such as its owners or a lender. Meghna and Sons has two: the family and a bank. Each supplied part of the Tk 2,400,000.

A **use of funds** is a purpose a business spends money on, such as stock or equipment. The shop used funds for books to sell and for shelves and counters. Keeping money at the bank is also a use, because the business chose to hold it for later.

An **asset** is something of value that a business owns and uses to earn money. The shop's books, its shelves and counters, and its cash and bank balance are assets. Each asset is the result of a use of funds.

Now the rule that joins the two halves. On any one date, the total from all sources of funds equals the total of the assets that those funds bought or kept. For Meghna and Sons, add the sources: Tk 600,000 plus Tk 1,800,000 is Tk 2,400,000. Add the assets: Tk 1,850,000 plus Tk 240,000 plus Tk 310,000 is also Tk 2,400,000. The two totals agree.

Two confusions are common. First, a source does not choose the use. The bank lent Tk 600,000 without saying whether it should buy books or shelves, because money from any source is the same money once it is in the business. Second, the rule describes one date. The totals will change when the business borrows more, repays a loan or earns a profit, and they will agree again on the new date.

The three subjects overlap, so a real decision often uses all three.

The section ends with a short recap.

Every book on a shelf was bought with someone's money, so whose money was it?

The case file gives the figures that the rest of the section sorts.

Two words in the case have special meanings in the shop.

GO SLOWLY HERE.

Source of funds. A place a business gets money from, such as its owners or a lender.

A source of funds is where the money came from.

Use of funds. A purpose a business spends money on, such as stock or equipment.

A use of funds is what the money was spent on.

Asset. Something of value that a business owns and uses to earn money.

An asset is something the business owns that helps it earn money.

The rule: on any one date, the money obtained equals the assets held.

A source does not choose its use, and the rule holds for one date only.

Equity. Money put in by the owners, who share the business profit or loss.

Capital. Word in use. In daily life, the main city of a country, or a large letter. In finance, money put into a business to start or run it.

Equity is the owners' money, and the owners share profit or loss.

Debt. Money borrowed that must be repaid, with an extra charge for using it.

Debt is borrowed money that must be repaid with an extra charge.

The table compares equity and debt on four points.

Equity is money put in by the owners, who share the business profit or loss. The family's Tk 1,800,000 is equity. Money from owners is often called capital. Owners are not promised a fixed payment: if the shop earns a profit they share it, and if it makes a loss they bear it.

Debt is money borrowed that must be repaid, with an extra charge for using it. The bank loan of Tk 600,000 is debt, and its charge is 10% a year.² The lender is promised repayment on agreed dates, whether the shop earns a profit or not.

The table below compares equity and debt. Read down a column to see one kind of source; read across a row to compare the two on one point.

EQUITY AND DEBT COMPARED

POINT	EQUITY	DEBT
Who supplies it	The owners	A lender
Repayment	No fixed repayment; owners share profit or loss	Must be repaid on agreed dates
Extra charge	None promised; a share of any profit	Agreed in advance: 10% a year
At Meghna and Sons	Tk 1,800,000 from the family	Tk 600,000 from the bank

PAUSE AND CHECK 1.P2 A new bakery begins with Tk 300,000 of the owner's savings and a Tk 100,000 loan from a bank. Which amount is equity and which is debt? Who must be repaid on agreed dates?

Liability. An amount a business owes to someone else.

A liability is any amount the business owes.

Financial market. A place or network where buyers and sellers trade money, loans and ownership parts of businesses.

A financial market is where money, loans and ownership parts are traded.

The figure shows the shop's flows of money in one picture.

A **liability** is an amount a business owes to someone else. The bank loan of Tk 600,000 is a liability. Debt is one kind of liability; an unpaid bill to a supplier is another. Now the wealth of the family in the shop can be found: assets of Tk 2,400,000 less the liability of Tk 600,000 leave Tk 1,800,000, which equals the family's equity.

A **financial market** is a place or network where buyers and sellers trade money, loans and ownership parts of businesses. A bank branch where one borrower meets one lender is a very small example. Large markets let a business find many sources at once. Chapter 13, Valuing Bonds, and Chapter 14, Valuing Common and Preferred Stock, show how such trades set prices.

The figure below shows the same flows as a picture. Read it from left to right. A solid arrow shows money going into the shop from its owners or its lender. A dashed arrow shows what goes back to them. The arrow on the right shows the shop spending funds on assets. Each arrow carries a label in plain words, and the amounts match the case file.

² A charge of 10% a year means Tk 10 for each Tk 100 borrowed, for each year the money is used. Later chapters show how the charge is calculated.

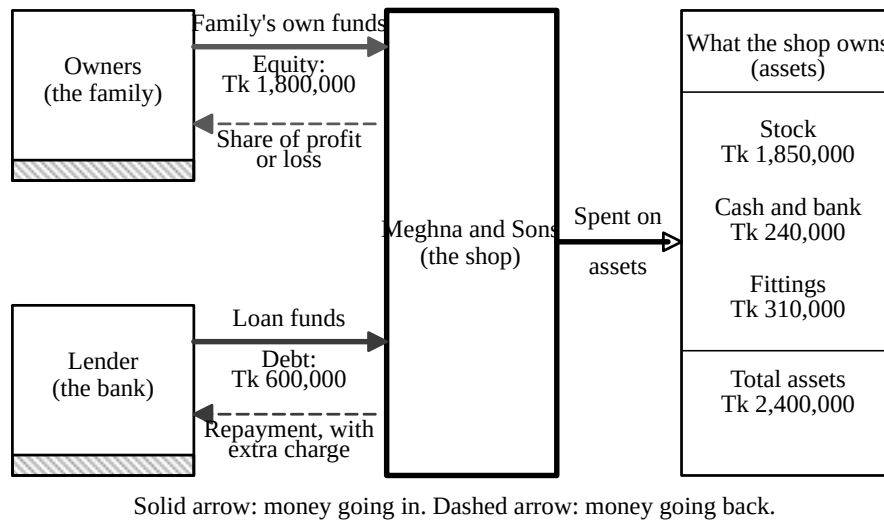


FIG. 1.1. Funds flowing between the owners, the lender, the shop and its assets, Meghna and Sons, 1 January 2024

Before moving on. A source of funds is where money comes from, a use of funds is what it buys or keeps, and assets are the things the business owns. Equity is the owners' money and debt is borrowed money, and on any one date the sources equal the assets. *You should now be able to say which part of a business's funds is equity, which is debt, and what each one bought.*

* * *

WORKED CASE: SOURCES AND USES AT MEGHNA AND SONS

Case file, illustrative. *Situation.* At 1 January 2024, Meghna and Sons records stock of Tk 1,850,000, a cash and bank balance of Tk 240,000 and fittings of Tk 310,000. It owes a bank loan of Tk 600,000 at 10% a year, and the family's own funds are Tk 1,800,000. Sales in 2023 were Tk 7,200,000.

Question. First, which figures are sources of funds and which are uses? Second, do the totals agree? Third, what is the family's wealth in the shop, and does the sales figure belong in the sorting?

Reasoning. Begin by sorting. The bank loan and the family's own funds answer the question "where did the money come from?", so they are sources: debt of Tk 600,000 and equity of Tk 1,800,000. Stock, the cash and bank balance and fittings answer the question "what did the money buy or keep?", so they are uses, and each one is an asset.

Next, compare the totals. Sources: Tk 600,000 plus Tk 1,800,000 is Tk 2,400,000. Assets: Tk 1,850,000 plus Tk 240,000 plus Tk 310,000 is Tk 2,400,000. Then find wealth. Assets of Tk 2,400,000 less the liability of Tk 600,000 leave Tk 1,800,000.

Last, consider sales. The figure of Tk 7,200,000 describes the whole of 2023, which is a period. The sorting describes one date, 1 January 2024. Sales therefore do not belong in the list of sources or uses. They belong with the ideas of period and cash flow, and sales are not the same as cash received if customers pay later.

The section ends with a short recap.

The worked case sorts each figure into its place.

Three tasks are set: sort, compare, and find wealth.

The first step asks of each figure: did the money come from here, or go to here?

The second step adds each side and compares the totals.

The sales figure describes a year, while the sorting describes one date.

The conclusion states what the sorting shows.

The check confirms that nothing is missing or counted twice.

Conclusion. Two sources supplied the shop: debt of Tk 600,000, which is one quarter of the total, and equity of Tk 1,800,000, which is three quarters. Three assets hold the funds: stock, the cash and bank balance, and fittings. The family's wealth in the shop is Tk 1,800,000.

Check. The totals agree at Tk 2,400,000, and the wealth equals the equity. If the totals had differed, a figure would be missing or counted twice. Note also that the cash and bank balance sits among the assets and not among the sources: it is money held, not money obtained.

* * *

SUMMARY

The chapter in brief.

Finance is the work of finding money and deciding how to use it. Funds are money available at one moment, cash flow is money moving over a period, and wealth is what is owned after taking away what is owed. Finance is one business function, beside selling, buying and managing people, and it differs from accounting and economics. A business obtains funds from sources, equity from owners and debt from lenders, and uses them on assets. On any one date, the sources equal the assets.

THINK IT THROUGH

1.T1 A cinema owner takes Tk 40,000 in ticket sales, owes Tk 25,000 for film hire next month and pays a Tk 10,000 loan instalment. Classify each of the three movements, and say which are cash flows today.

1.T2 What is the net cash flow for the cinema owner today? Which of the three movements is a liability, and what will it become next month?

WHERE PEOPLE GO WRONG

Mistake: finance belongs only to banks and large businesses.

Some students believe that finance belongs only to banks and large businesses. In fact, every stall owner who finds money and decides how to use it practises finance. Size changes the amounts, not the questions.

Mistake: goods on the shelves count as funds.

Some students count the books on the shelves as funds. Books are assets; they become funds only when they are sold and paid for. Meghna and Sons holds stock of Tk 1,850,000 but a cash and bank balance of only Tk 240,000.

Mistake: profit and cash flow are the same thing.

A sale made in June and paid for in July creates profit in June, but its cash flow arrives in July. The two measure different things, and a business needs to watch both.

Mistake: equity is free money, or debt is always bad.

Owners supply equity, but they expect a share of any profit, so equity is not free. Debt carries an agreed charge and must be repaid, but it lets a business obtain funds that its owners cannot supply. Neither is good or bad in itself; each comes with terms.

IN HISTORY

A Florentine banking house moved money between cities without moving coin.

In 1397, in Florence, a trading city in what is now Italy, a merchant named Giovanni di Bicci de' Medici founded the Medici Bank. Before founding it he had worked in banking in Rome. His bank grew to have branches in other cities, among them Rome, Bruges and London, and it lasted until 1494.

A merchant of that time faced a problem: carrying coin from one city to another was slow and dangerous. The bank offered a solution, the bill of exchange. A merchant paid money to the bank in one city and received a written order. The order told a branch or agent in another city to pay an agreed sum there. No coin travelled; only the paper did. The bank earned its charge from the difference in the rates at which currencies were exchanged.

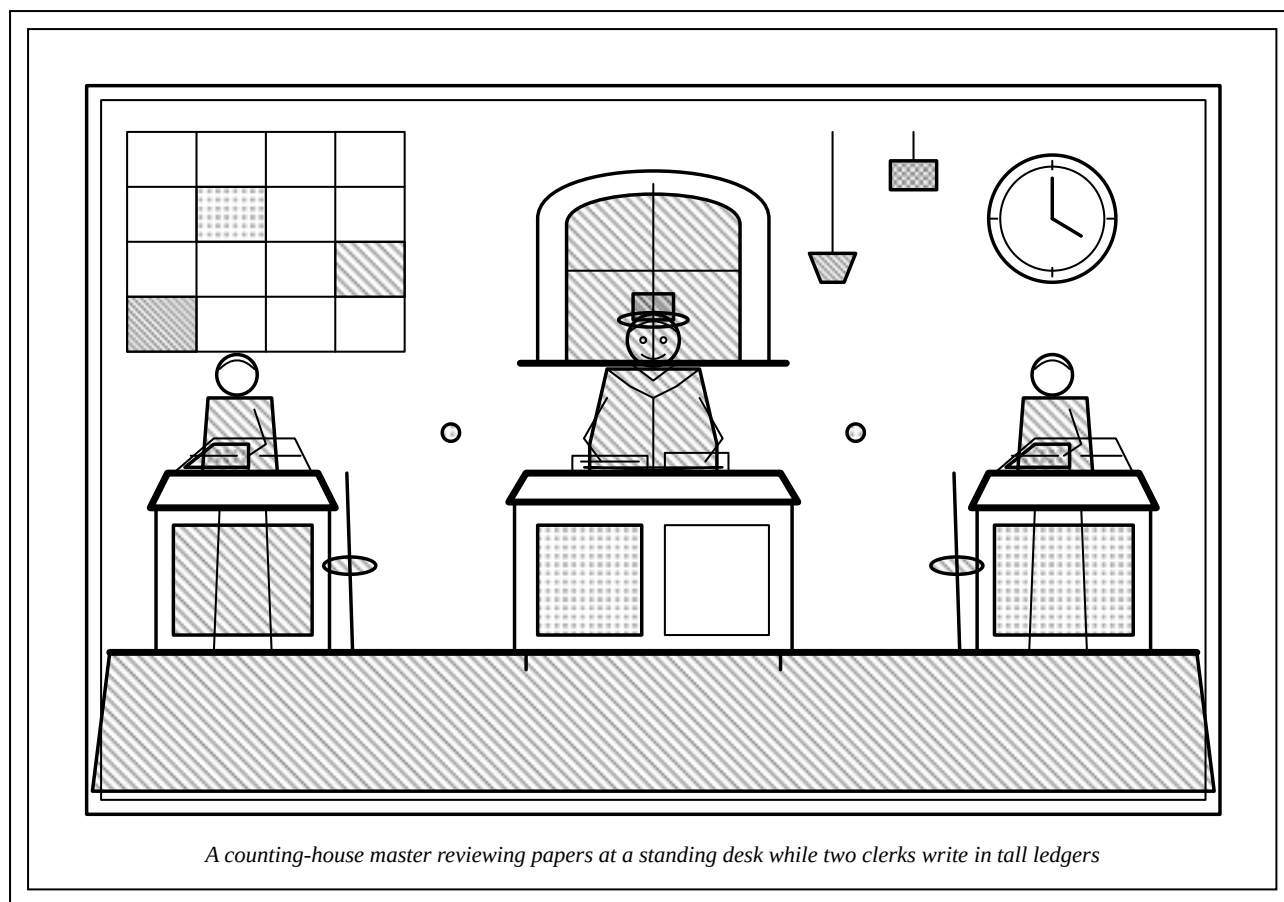
The bill of exchange solved the problem of carrying coin from city to city.

Seen with this chapter's words, the bank received money from depositors, which was a source of funds, and used it to make loans and to finance trade, which were uses. It also moved funds between cities, so that merchants did not have to. The historian Raymond de Roover told the story in *The Rise and Decline of the Medici Bank, 1397-1494* (1963). The bank ended in 1494, when the Medici were expelled from Florence.

In this chapter's words, the bank gathered sources of funds and put them to use.

REVIEW QUESTIONS

- 1.1 A stall owner has Tk 1,200 in her purse and a table full of vegetables. In your own words, explain what finance means for her. Name one thing she must find and one thing she must decide.
- 1.2 A tailor records the money received each month and also decides whether to borrow to buy a second sewing machine. Which task is accounting and which is finance? Explain the difference.
- 1.3 A new bakery begins with the owner's savings and a loan. Name its two sources of funds, say which is equity and which is debt, and give two uses of the funds.
- 1.4 A customer takes goods on 28 June and pays on 28 July. Explain why the sale and the cash flow fall in different months.
- 1.5 On Monday customers pay the stall owner Tk 3,000, and she pays the wholesaler Tk 2,000. State the cash flow in, the cash flow out and the net movement.
- 1.6 The same stall owner now takes a loan of Tk 800 from her cousin. Is this a source or a use of funds? Is it equity or debt? What does she owe afterwards?
- 1.7 A school canteen is run by a committee, using money from a school fund. Name one source of funds and one use of funds for the canteen.
- 1.8 A small clinic is funded by its owners' savings and a bank loan. Name its sources of funds and two assets it might hold. Where does finance sit among the clinic's business functions, and what does it do for the others?



CHAPTER II

The Financial Manager's Work

*Who decides where the money goes, and how do they
decide well?*



*This chapter introduces the financial manager and the three
decisions this person makes about money: what to buy, how to pay
for it, and how much profit to pay out. It ends by showing how
these decisions affect the daily success of a business. A bookshop
in Dhaka supplies the figures, and a London bank supplies a
warning from history.*

One sum of money has three fair claims on it, and only one can win.

Every business meets this puzzle on a larger scale, and someone must answer it.

You need the idea of sources and uses of funds from Chapter 1.

The chapter has three sections, a worked case and a short history.

Recall: business function, Chapter 1, section 1.2, "Finance Among the Business Functions".

A small shop shows that a role can exist before it has a title.

Tanvir holds the money function, and a name exists for the person who does so.

Financial manager. The person who plans, raises and controls the money of a business.

The financial manager is whoever plans, raises and controls the money.

Each of the three verbs describes a separate part of the duty.

Financial planning. Deciding in advance how much money a business will need and when.

Financial planning decides in advance how much money is needed and when.

Suppose you hold Tk 100,000 that you do not need to spend this month. Three people ask for it. A friend wants to start a juice stall and needs equipment. Your parents would like a share, because you could not have saved it without them. A bank is waiting for a loan to be repaid, and every month of waiting adds a charge. Each request is reasonable, and the money cannot go to all three. Who should receive it?

A business meets the same puzzle in larger form. It earns a profit, and its owners, its lenders and its own plans for growth all ask for the money. Somebody must hear the claims, weigh them and choose. That person does not guess. The person follows a method, and the method has three parts. This chapter names the person, names the three parts, and shows how the choices appear in the daily life of a shop.

You will need: Chapter 1, section 1.3, "Where Funds Come From and Where They Go": sources and uses of funds, equity and debt. If those words feel uncertain, read that section again before starting.

Section 2.1 describes the person who does finance in a business. Section 2.2 explains the three decisions this person makes, and it goes slowly, because the three decisions sound alike. Section 2.3 shows how finance affects the daily success of a business. A worked case then follows one day of decisions at Meghna and Sons, and a history shows what can happen when control fails.

* * *

2.1 THE FINANCIAL MANAGER'S ROLE

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhaka. Since 2019 it has been run by Imran, who is responsible for the shop and buying, and by Tanvir, who is responsible for accounts and money.

In Chapter 1 we called finance one business function. Someone must perform it, even in a small shop. In the case file above, Tanvir does so, beside his work on accounts. The person who holds this duty has a name.

A **financial manager** is the person who plans, raises and controls the money of a business. In a small shop this may be one brother. In a large business it may be a team led by a senior person with a grander title, such as chief financial officer.¹ The title varies from business to business. The duties are what count.

Look closely at the three verbs. To plan is to decide in advance what money will be needed and when. To raise is to find the money, from owners or from lenders. To control is to check that money is used as intended, and to correct matters when it is not. A tailor who foresees a large order, borrows to buy cloth and later compares what she spent with what she expected performs all three duties, though nobody has given her a title.

Financial planning means deciding in advance how much money a business will need and when. Suppose the tailor expects to pay Tk 30,000 for cloth on 1 March and Tk 8,000 for rent on 5 March. By 20 February she knows she needs Tk 38,000 in the first week of March, and she has time to arrange it. Without a plan she learns of the shortage on the day the supplier asks for payment. A plan also gives a standard to check against later.

¹ Chief financial officer is the title of the most senior person in charge of finance in a large business. This book uses "financial manager" for the role in every size of business.

Financial control means checking that money is spent and recorded as planned, and acting when it is not. Suppose the tailor planned Tk 30,000 for cloth but the bill is Tk 36,000. Control notices the difference of Tk 6,000 and asks why. Perhaps cloth became more expensive, or perhaps she bought cloth she does not need. Then comes the act: change a price, change a supplier or change the plan. Control that finds a difference and does nothing has no value.

The viewpoint changes with the person. Owners rely on the financial manager to use their money wisely and to report on it honestly. A lender relies on the manager to repay on the agreed day. Employees rely on the manager to make sure money is available for wages. The financial manager therefore answers to several groups at once, which is part of what makes the work demanding.

The role has limits. The financial manager is not the accountant of Chapter 1, section 1.2. The accountant records what has happened, while the financial manager uses those records to decide what should happen next. The financial manager also does not choose which books the shop stocks; Imran does. The role supplies money and judges its use, but it does not run the other functions. In a very small business one person may hold both roles, though the two kinds of work remain different.

Before moving on. A financial manager is the person who plans, raises and controls the money of a business, whatever the title. Planning decides in advance how much money is needed and when. Control checks what happened against the plan and acts on the difference. *You should now be able to say what each of the three verbs means and give a small example of each.*

* * *

2.2 THE THREE CORE DECISIONS

Begin at home. A household has saved Tk 60,000 and has some spare income each month. It faces three questions. First, what should we buy with our money? Perhaps a fridge. Second, how should we pay for it: from our savings, or by borrowing from a relative? Third, how much of our spare income should we give to the grandparents, and how much should we keep? These questions sound alike, because each one concerns the same money. Yet each asks something different, and a household that mixes them up chooses badly.

A business asks the same three questions, and each question has a name. We take them one at a time, because the names are similar and the questions are not.

The first is the investment decision. The **investment decision** is choosing which long-lasting assets or projects a business should buy. A project is a planned piece of work that needs money, such as opening a second branch. A bakery that must choose between a new oven and a delivery van makes an investment decision, because either would serve the bakery for many years.

Case file, illustrative. On 10 February 2024 the owners of Meghna and Sons decided to open a stationery corner costing Tk 350,000. This is an investment decision: the corner is a long-lasting asset, bought to earn money over many years.

Financial control. Checking that money is spent and recorded as planned, and acting when it is not.

Financial control compares what happened with the plan and acts on any difference.

The owner, the lender and the employee each depend on the financial manager in a different way.

Recall: finance beside accounting, Chapter 1, section 1.2, "Finance Among the Business Functions".

The role has limits: it is not accounting, and it does not replace the other managers.

The section ends with a short recap.

GO SLOWLY HERE.

Even a household faces three separate questions about the same money.

A business asks the same three questions, and each has a name.

Investment decision. Choosing which long-lasting assets or projects a business should buy.

The investment decision asks what to buy.

The case file shows the investment decision at Meghna and Sons.

Stock. Word in use. In daily life, a supply of something kept for later use. In a shop, the goods held for sale. In finance, stock can also mean ownership parts of a company, taught in Chapter 14, "Valuing Common and Preferred Stock".

The rule is that an investment commits money for a long time, so it is hard to reverse.

The first decision is complete: it asks what to buy.

Financing decision. Choosing how to raise the money for purchases, from owners or lenders.

The financing decision asks how to pay.

Recall: sources of funds, equity and debt, Chapter 1, section 1.3, "Where Funds Come From and Where They Go".

The case file shows how the corner was paid for.

A preview: the kept profit has a name, which this section gives a little further on.

The two decisions use the same money but ask different questions.

Dividend decision. Choosing how much profit to pay out to owners and how much to keep.

The dividend decision asks how much profit goes to the owners.

Dividend. A share of profit paid out to the owners of a business.

Retained earnings. Profit kept in the business instead of being paid out to owners.

The part paid out is the dividend; the part kept is retained earnings.

Profit. Word in use. In daily life, any money gained. In business, what is left of sales after all costs, and tax, are taken away.

The case file records the dividend decision for 2023.

The rule is that an investment decision commits money to assets for a long time, so it is hard to undo. A corner, once built, cannot be returned to the supplier. There are two limits to the meaning. Here, investment means spending the business's own funds on its own assets, not saving in a bank. And a shorter-lived use of funds, such as the textbook stock that the shop buys before each school term, is treated in Chapter 15, "Working Capital and the Cash Cycle". The method for judging whether a project is worth buying is taught in Chapter 7, "Capital Budgeting: Decisions and Cash Flows".

So far, one idea only: the investment decision asks what to buy.

The second is the financing decision. The **financing decision** is choosing how to raise the money for purchases, from owners or lenders. Recall that equity is money from owners and debt is borrowed money. A bakery that buys an oven may pay from its owners' funds, take a bank loan, or use both. The purchase is the same in each case, but the way of paying is different.

Case file, illustrative. The corner of Tk 350,000 is funded by Tk 150,000 of profit the business had kept and a new bank loan of Tk 200,000 at 10% a year. That choice is the financing decision. The yearly charge on the new loan, called interest, is Tk 20,000. After the new borrowing, the bank loan stands at Tk 800,000.

The kept profit has a name that this section gives a little further on. For now, read it as owners' funds, which are a source of funds, while the loan is debt.

PAUSE AND CHECK 2.P1 A bakery buys an oven for Tk 200,000. It pays Tk 80,000 from the owners' funds and borrows Tk 120,000. Which part of this is the investment decision, and which part is the financing decision?

Notice how the two decisions differ. Investment asks what to buy; financing asks how to pay. A loan with a low charge does not make a poor purchase wise, and a good purchase can still fail if it is paid for in a way the business cannot repay. A common slip is to answer "should we buy?" with "the loan is cheap". That answers the second question, not the first.

The third is the dividend decision. The **dividend decision** is choosing how much profit to pay out to owners and how much to keep. A shopkeeper who earns a profit of Tk 90,000 may take Tk 30,000 home and leave Tk 60,000 in the business. Neither choice is wrong in itself. A payout rewards the owners now. Keeping profit leaves more to fund growth later, without asking a lender.

The part of profit paid out is called a **dividend**, and the part kept is called **retained earnings**. In the shopkeeper's example, the dividend is Tk 30,000 and the retained earnings are Tk 60,000. A business owned by one family may call its payout a withdrawal. This book uses the word dividend for any share of profit paid to owners.

Case file, illustrative. Profit after tax for 2023 was Tk 480,000.² The family withdrawal, which is the dividend, was Tk 120,000, so the retained earnings were Tk 360,000.

² Tax is money that a business pays to the government on its profit. Profit after tax is the profit left once that payment has been made.

Retained earnings connect the third decision to the second. Profit that stays in the business still belongs to the owners, so it is a source of funds from the owners, which Chapter 1 called equity. It is also the easiest source to use, because nobody new must be asked. This is why the decisions cannot be fully separated: the more profit is paid out, the less is left to fund purchases, and the more the business may need to borrow. At Meghna and Sons, Tk 150,000 of the Tk 360,000 kept went into the corner. This is the kept profit that the financing paragraph above described.

PAUSE AND CHECK 2.P2 A tailor earns a profit of Tk 90,000 and pays Tk 25,000 of it to her family, who own the shop. How much is the dividend, and how much are the retained earnings?

The table below sets the three decisions side by side. Read each row from left to right: the decision, the question it asks, and the choice made at Meghna and Sons.

THE THREE CORE DECISIONS COMPARED

DECISION	QUESTION IT ASKS	AT MEGHNA AND SONS
Investment	Which long-lasting assets should we buy?	A stationery corner costing Tk 350,000
Financing	How should we raise the money?	Tk 150,000 of kept profit and a Tk 200,000 bank loan
Dividend	How much profit do we pay out, and how much do we keep?	Tk 120,000 paid out; Tk 360,000 kept

The figure below draws the same decisions as three arrows around a box of funds. Read it arrow by arrow. The arrow into the box shows money arriving from owners and lenders. The arrow to the right shows money leaving to buy assets. The dashed arrow downward shows money leaving as dividends. Each arrow carries the name of its decision in plain words.

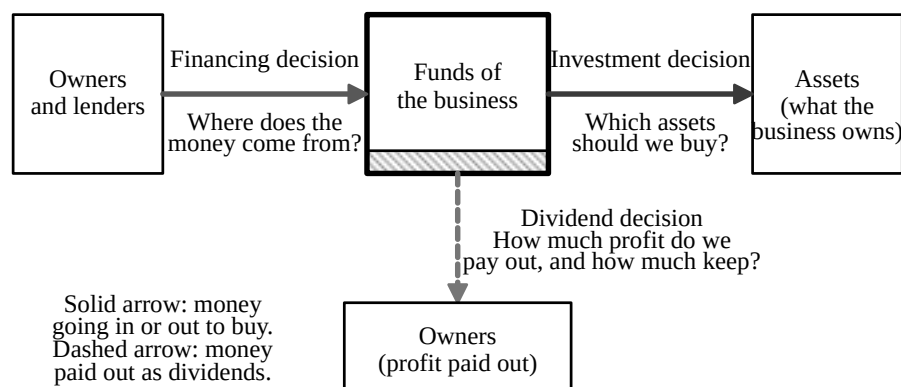


FIG. 2.1. The three decisions as three arrows around the funds of a business

Before moving on. The investment decision asks what to buy, the financing decision asks how to pay, and the dividend decision asks how much profit to pay out and how much to keep. Retained earnings, the profit kept, can fund purchases. You should now be able to say

Retained earnings link the dividend decision to the financing decision.

The table compares the three decisions side by side.

The figure shows the three decisions as three arrows around the funds of a business.

The section ends with a short recap.

which of the three questions a given choice answers, and to give one example of each.

* * *

2.3 HOW FINANCE SHAPES BUSINESS SUCCESS

Choices matter because of what they do to a business every day.

Operating efficiency. How little money and effort a business uses to produce each unit of sales.

Operating efficiency measures how little is used for each unit of sales.

Liquidity. How easily a business can pay its bills on time from cash it has or can quickly raise.

Liquidity is the ability to pay bills on time.

The case file shows why stock is not the same as cash.

Profitability. How much profit a business earns compared with the money invested or the sales made.

Profitability compares profit with money invested or with sales.

The case file gives the profit and sales of 2023.

The three qualities can pull in different directions.

The section ends with a short recap.

Everything so far has described choices. Choices matter because of their effect on a business each day. A shop that buys what it cannot sell, or sells below what the goods cost, or cannot pay a supplier on time, will struggle however loyal its customers are. Three qualities show whether finance is serving the business well.

Operating efficiency means how little money and effort a business uses to produce each unit of sales. Suppose a tailor spends Tk 70 on cloth, thread and labour for every Tk 100 she sells, while a rival spends Tk 80 for the same Tk 100. The first tailor is more efficient. Size does not decide this: a small shop can be more efficient than a large one.

Liquidity means how easily a business can pay its bills on time from cash it has or can quickly raise. Weak liquidity disrupts daily work: a supplier refuses a delivery, wages are late, a chance to buy cheaply passes.

Case file, illustrative. At 1 January 2024, Meghna and Sons held stock of Tk 1,850,000 and a cash and bank balance of Tk 240,000. Books on the shelves cannot pay a supplier until they are sold, but the balance at the bank can.

Profitability means how much profit a business earns compared with the money invested or the sales made. A profit of Tk 20,000 on Tk 200,000 invested is 10% of the money invested. Compare ratios, not amounts, because a large profit can be a poor result on a very large investment.

Case file, illustrative. Profit after tax for 2023 was Tk 480,000, and sales in 2023 were Tk 7,200,000. The profit is about Tk 6.67 for each Tk 100 of sales.

The three qualities can pull in different directions. Buying more stock may raise sales and profit, yet it spends cash and so lowers liquidity. The financial manager's work is to keep all three in view, and each of the three decisions of section 2.2 affects at least one of them.

Before moving on. Operating efficiency measures how little is used for each unit of sales. Liquidity measures the ability to pay bills on time. Profitability compares profit with the money invested or with sales. *You should now be able to say which of the three a business lacks if it earns a profit but cannot pay its wages on time.*

* * *

WORKED CASE: THREE DECISIONS AT MEGHNA AND SONS

The worked case sets out the situation of 10 February 2024.

Case file, illustrative. *Situation.* On 10 February 2024 the owners of Meghna and Sons review the year 2023. Profit after tax was Tk 480,000, and the family withdrew Tk 120,000. The owners decide to open a stationery corner costing Tk 350,000. They fund it with Tk 150,000 of retained earnings and a new bank loan of Tk 200,000 at 10% a year. The other Tk 210,000 of retained earnings buys textbook stock before the new term.

Question. Which choice is the investment decision, which the financing decision and which the dividend decision? How much profit was kept? What does the new loan cost each year, and how large is the bank loan afterwards?

Reasoning. Begin with the profit. The family withdrew Tk 120,000, which is the dividend, so the retained earnings are Tk 480,000 less Tk 120,000, which is Tk 360,000. The choice of how much to pay out and how much to keep is the dividend decision.

Next, the corner. Opening it commits Tk 350,000 to a long-lasting asset, so this is the investment decision. Paying for it is a separate choice. The owners supply Tk 150,000 of retained earnings and a lender supplies Tk 200,000, and Tk 150,000 plus Tk 200,000 is Tk 350,000, which matches the cost. Choosing this mixture of owners and lender is the financing decision.

Then count what remains. Tk 360,000 less Tk 150,000 leaves Tk 210,000 of retained earnings, which buys textbook stock. The bank loan was Tk 600,000, and the new loan adds Tk 200,000, so it becomes Tk 800,000. The yearly charge on the new loan is 10% of Tk 200,000, which is 10 divided by 100 times Tk 200,000, or Tk 20,000.

Conclusion. The dividend decision kept Tk 360,000. The investment decision chose the corner at Tk 350,000. The financing decision paid for it with Tk 150,000 of retained earnings and a Tk 200,000 loan. The loan costs Tk 20,000 a year, and the bank loan after the new borrowing is Tk 800,000.

Check. The Tk 150,000 for the corner and the Tk 210,000 for stock add to Tk 360,000, so all retained earnings are used and none is counted twice. The two sources of the corner add to its cost. The loan grew by one third of its old size, from Tk 600,000 to Tk 800,000, which is a sensible rise for a purchase partly paid by borrowing.

* * *

SUMMARY

The financial manager plans, raises and controls the money of a business, and this duty exists in every business whatever the title. Three decisions make up the work. The investment decision chooses what to buy, the financing decision chooses how to pay, and the dividend decision chooses how much profit to pay out and how much to keep as retained earnings. The success of the business each day shows in its operating efficiency, liquidity and profitability.

THINK IT THROUGH

2.T1 A bakery has Tk 150,000 of spare profit and must weigh three options: a new oven, a payout to the owners, or repaying a loan. Name the decision type in each option.

2.T2 In the bakery, say what changes in its sources of funds if it repays the loan, and what changes if it pays the owners a dividend instead. Which choice leaves more profit inside the business?

WHERE PEOPLE GO WRONG

Some students treat buying an asset and paying for it as one decision. They are two. The first asks what to buy, and

The questions ask for each decision, each amount and the cost of the loan.

The first step is the dividend decision, which fixes how much profit stays.

The next steps separate what is bought from how it is paid for.

The last steps count the remaining profit, the loan and its yearly charge.

The conclusion names each decision and the resulting figures.

The check confirms that the money is fully accounted for.

The chapter in brief.

Mistake: investment and financing are the same decision.

Mistake: retained earnings are a pile of cash.

Mistake: a profitable business always has money to pay its bills.

Mistake: the financial manager and the accountant do the same work.

A London bank collapsed in 1995 because one trader's work was never checked.

Leeson held both the trading and the records of it, so no one checked him.

The inquiry found weak control and weak management, as well as one man's wrongdoing.

the second asks how to pay. A cheap loan does not make a poor purchase sensible.

Some students picture retained earnings as cash lying in the bank. Retained earnings are profit the business has kept, and it may already have spent that profit on stock or fittings. At Meghna and Sons, Tk 210,000 of retained earnings buys textbook stock.

A business can earn a profit and still lack cash on wage day, because customers may pay late or the profit may be tied up in stock. Profit and liquidity are different things.

The accountant records what has happened. The financial manager plans, raises and controls money to decide what should happen next. In a small shop one person may do both, but the two kinds of work stay different.

IN HISTORY

In 1762 a family founded Barings in London. It was a merchant bank, which means a bank that serves trade and large clients. By 1995 it was 233 years old. On 26 February 1995 it collapsed. The cause lay in Singapore, where a Barings subsidiary employed a trader named Nick Leeson. A trader buys and sells financial contracts for a bank. Leeson dealt in contracts called futures, which are agreements to buy or sell at a fixed price on a later date.

Leeson was allowed to control both the trading and the recording of those trades. Normally two people hold these tasks, so that each checks the other. With one man in both roles, his losses stayed hidden while they grew. Reports put the losses at about £827 million, more than the bank could absorb.

The Bank of England, which is Britain's central bank and supervises banks, had a board inquire into the collapse. Its report in July 1995 found that several people at Barings had failed to do their jobs properly. The Dutch group ING bought the bank for £1, and in December 1995 a Singapore court sentenced Leeson to six and a half years in prison. In this chapter's words, weak financial control, not bad luck, brought down the bank: the money moved without checks that would have shown the problem in time.

REVIEW QUESTIONS

2.1 Describe the roles of a financial manager in a small firm, such as a bookshop run by two brothers. Name the three duties and give one example of each.

2.2 A clinic decides to buy a scanner for Tk 2,000,000 and to pay for it with a bank loan. Name the investment decision and the financing decision in this choice. Which of the three core decisions has not yet been made?

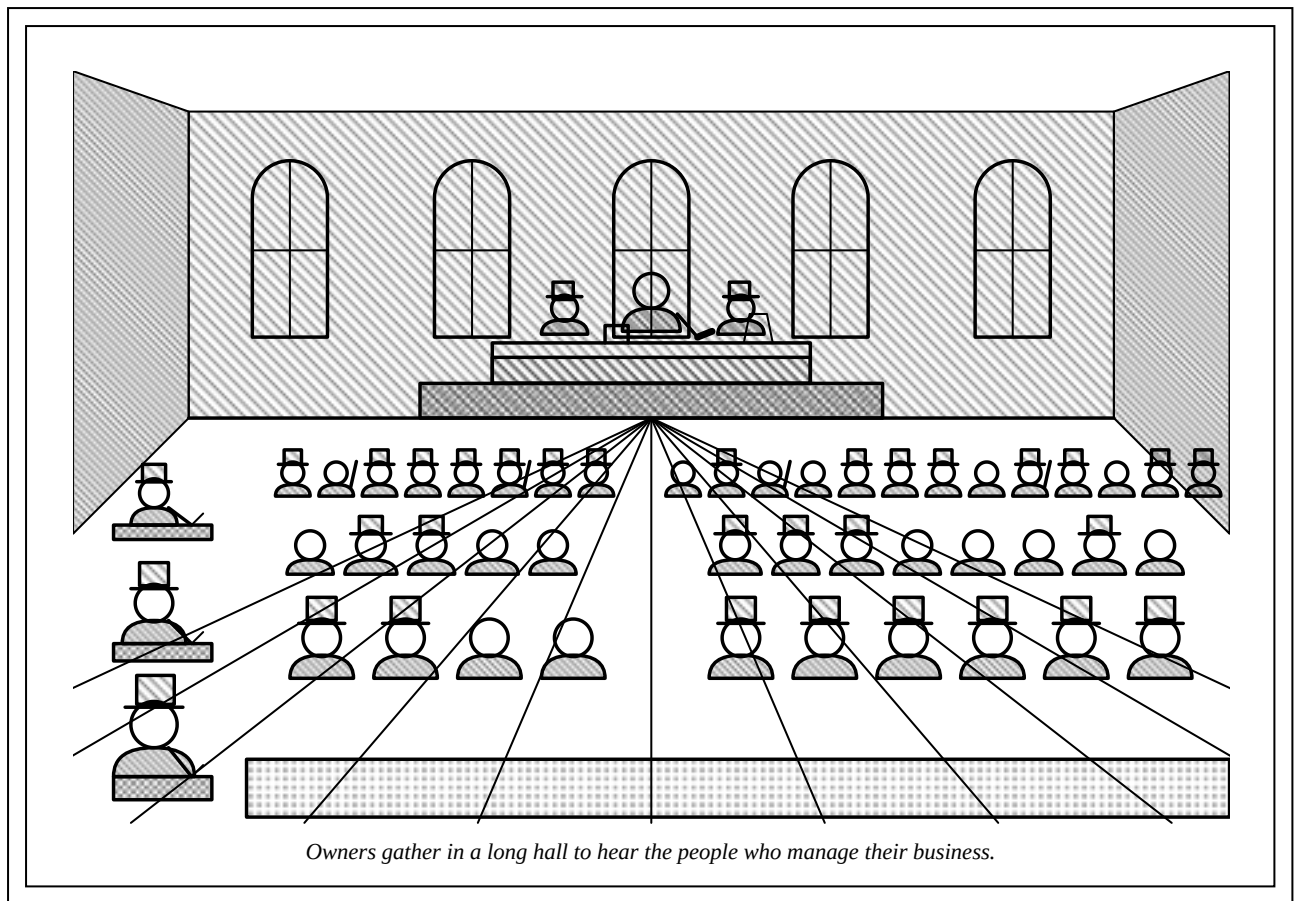
2.3 Explain the difference between financial planning and financial control, using a tailor who expects to pay Tk 30,000 for cloth.

2.4 A shop earns a profit every month but cannot pay its suppliers on time. Explain how weak liquidity disrupts the daily operations of the shop.

2.5 A household must decide between buying a new fridge with its savings and repaying a loan with the same savings. Name the decision type of each option.

2.6 A corner shop owner earns a profit of Tk 90,000. She spends Tk 40,000 of it on a new freezer and pays herself Tk 50,000. Name the decisions and state the dividend and the retained earnings.

- 2.7 A small manufacturer, in one year, buys a new machine, pays for it partly with a loan, and pays its owners a share of profit. List the three decisions and explain how the dividend decision affects the financing decision.
- 2.8 Tailor A spends Tk 70 for each Tk 100 of sales and Tailor B spends Tk 80. Which has the greater operating efficiency, and why does efficiency differ from profitability?



CHAPTER III

The Goal of the Firm and Agency

What should a business aim for, and who makes sure it does?



A business needs one clear aim. This chapter explains why the aim is the wealth of the owners, not only this year's profit, and why hired managers may follow a different aim. It names the clash that follows and the cost it brings.

What you should already know before you begin.

A puzzle: which of two shops was run better?

A second puzzle: what happens when the owner is not the manager?

The chapter answers two questions, one about aims and one about people.

You will need two ideas from earlier chapters. The first is *wealth*, first defined as a stock of value (Chapter 1, section 1.1).¹ The second is *profitability* (Chapter 2, section 2.3).² Terms such as dividend, retained earnings and financial manager are used here as they were taught earlier.

Two friends each run a small shop that sells school supplies. At the end of the year the first friend reports a larger profit than the second. Most people answer at once that the first friend ran the better business. Then we learn more. The first friend stopped buying new stock in the last months, and the shelves are nearly bare. The second friend kept the shelves full, and a large order from a school is already waiting for the new year. The answer is now less certain.

Now change the story. The owner of one shop moves to another city and hires a manager to run it. The owner wants a business that stays strong for many years. The manager is paid a bonus that depends on this year's results. Will the manager choose what the owner would choose? The manager may be honest and hard-working and still face a different set of rewards.

These two stories lead to two questions. What should a business aim for? And what happens when the people who own a business are not the people who run it? The first question is answered by the *goal of the business*. The second is answered by a body of ideas called agency theory. We take the first question in sections 3.1 and 3.2 and the second in section 3.3. A worked case then uses the facts of Meghna and Sons to test every idea.

* * *

3.1 PROFIT MAXIMISATION AND ITS LIMITS

GO SLOWLY HERE.

A market stall shows how a bigger profit today can hurt the stall tomorrow.

Profit maximisation. The aim of earning the largest profit possible in a period, usually one year.

The first and oldest answer to the question of aims is profit maximisation.

A business that never earns a profit cannot survive, so the aim has a true core.

Begin with a stall in a market. The stall holder sells tomatoes and could earn more this week by mixing in spoiled fruit and charging the usual price. This week the profit is larger. Next week the buyers who were cheated stay away, and the profit falls below its old level. The choice that gave the best result today gave a poor result over a longer time. Keep this picture in mind, because the whole of section 3.1 repeats it in business form.

The oldest answer to the question "what should a business aim for?" has a name. **Profit maximisation** is the aim of earning the largest profit possible in a period, usually one year. A bookshop that chooses between two plans for its stock, and picks the plan with the larger profit for the year, follows this aim. The aim has real strengths. It is easy to state, it is easy to measure from the records, and it is linked to profitability, the idea you met in Chapter 2.

Profit matters for every person connected with a business. For the owner, it is the reward for the money placed in the business. For the manager, it is a sign of good work. For the employee, it is a sign that the job is likely to last. For the lender, it is a source of the money needed to repay a loan. We do not reject profit. We ask whether the *largest profit in one year* is the right aim, and the answer is that it is not complete.

¹ Chapter 1, 'Money, Finance and the Firm', section 1.1 'What Finance Is': wealth was first defined there as a stock of value.

² Chapter 2, 'The Financial Manager's Work', section 2.3: profitability was taught there. Section 2.2 'The Three Core Decisions' shows what owners give up or gain through dividend and financing decisions.

THE PROBLEM OF TIMING

Profit is earned in a particular period, and a plan that raises profit in one period may lower it in another. The following case file shows the difficulty with figures.

Case file, illustrative. Date: 15 May 2024. Business: Meghna and Sons, a bookshop in Dhaka. Choice: cut restocking, meaning the buying of new books to replace those sold. Effect on this year's profit: an increase of Tk 80,000. Effect on next year's profit: a decrease of Tk 150,000.

A rule that says "earn the largest profit this year" favours the cut, because Tk 80,000 is a gain. Look at both years together and the picture changes: Tk 80,000 gained, then Tk 150,000 lost, which is a net fall of Tk 70,000. There is a second point. Money received this year can be used at once, for example to repay a loan or to buy stock, and money received next year cannot. So the two amounts are not equal in usefulness. Chapter 4 teaches how to compare money at different times with exact arithmetic. For now, remember only that *when* a profit arrives matters.

THE PROBLEM OF RISK

The second problem is risk, meaning the chance that the result will differ from what was expected. Imagine two plans for a small tailoring shop. Plan one promises Tk 50,000 from a school uniform order that has been signed and will be paid. Plan two promises Tk 50,000, but only if a large crowd comes to a festival. The promised profit is the same. Nobody who understood the difference would treat the plans as equal. The rule "largest profit" cannot see this difference, because it looks only at the number promised.

There is a third and smaller limit. Profit for a year is a figure produced by accounting for that period. Choices about when to spend, such as delaying the restocking of the shelves, change the figure for the year without making the business better. A person who is judged only by the figure may be tempted to change it.

The rule to remember is short. Profit maximisation is a useful first aim, but it ignores when profits arrive and how certain they are. A better aim must take both into account. Section 3.2 supplies it.

Before moving on. Profit maximisation means earning the largest profit in a period, usually one year. Its limits are timing, risk, and the fact that a yearly profit can be shaped by choices about spending. You should now be able to say why a larger profit this year is not always a better result for the owner.

The same profit has different usefulness at different times.

A choice that raises this year's profit and lowers the next year's.

Adding the two years shows a net loss, though the rule favours the cut.

Two plans may promise the same profit and still differ in how likely the profit is.

Profit is an accounting figure, and some choices change it without improving the business.

Profit maximisation is useful but incomplete.

A short summary of section 3.1 with a test sentence.

3.2 WEALTH MAXIMISATION

Suppose you own a used bicycle and a friend asks what it is worth. You do not answer with the price you paid two years ago, and you do not answer with how much you enjoy riding it. You think about what a buyer would hand over today. A shop nearby might say Tk 6,000. That answer is the idea we need first.

The name for this idea is **market value**. Market value is the price at which something could be sold today in a market. The used bicycle has a market value of Tk 6,000 if a buyer would pay that amount now. Notice that market value looks at what others will pay, so it reflects what buyers believe about the future. A bicycle that is likely to last many more years sells for more than one that is about to fail.

GO SLOWLY HERE.

Everyday items teach us what the words "worth" and "price" mean.

Market value. The price at which something could be sold today in a market.

Market value is the price a buyer would pay now.

Value. Word in use. In daily life, how much we like or need something. In finance, an amount of money that a market would pay for something.

The owners' share in a business also has a market value.

Wealth maximisation. The aim of making the market value of the owners' stake as large as possible over time.

Wealth maximisation is the improved answer to the question of aims.

The market looks forward, so it responds to timing and risk.

A table compares the two aims on four matters.

Stakeholder. Any person or group affected by a business, such as workers, customers, lenders and owners.

A stakeholder is anyone affected by the business, not only its owners.

The owners of a business hold a stake in it, meaning a share of what the business owns after its debts are paid. That stake also has a market value: the price at which it could be sold today. For a large business whose common stock is traded in a public market, the price can be read each day. For a small family business, no daily price exists, so owners estimate what a buyer would pay.

We can now state the improved answer. **Wealth maximisation** is the aim of making the market value of the owners' stake as large as possible over time. Remember that wealth, from Chapter 1, is a stock of value. Wealth maximisation says that the owners' stock of value should grow, and that the growth is measured by market value, not by the profit of one year. A bookshop that gives up a small profit this year to win a large loyal customer group has chosen this aim if buyers would pay more for the business afterwards.

IN SHORT

Wealth maximisation means making the market value of the owners' stake as large as possible over time.

PAUSE AND CHECK 3.P1 A shop owner can raise this year's profit by cutting the quality of goods, but customers will leave later. Explain in two sentences why wealth maximisation would reject the cut.

Why does this aim solve the two problems from section 3.1? A buyer who wants to purchase a business thinks about all the profits that the business is likely to earn, not the profit of the last year only. A buyer pays more for profit that comes sooner, because it can be used sooner. A buyer pays less for profit that is uncertain, because it may not arrive. So market value already contains an opinion about timing and about risk. When we aim at market value, we aim at what the owners could gain by selling their stake. Return to the restocking choice of 15 May 2024. A buyer would see that the cut lowers the next year's profit by more than it raises this year's, and would pay less for the business. The cut lowers the owners' wealth.

The table that follows compares the two aims. Read each row from left to right: the first column names a matter, and the next two columns say how each aim treats it.

THE TWO AIMS COMPARED

MATTER	PROFIT MAXIMISATION	WEALTH MAXIMISATION
What is measured	Profit of one period	Market value of the owners' stake
Time	Looks at one year	Looks at all future years
Risk	Not considered	Included in the price a buyer will pay
Main weakness	Can favour a short-term gain	Needs a market price, or a careful estimate

A business affects many people besides its owners. A **stakeholder** is any person or group affected by a business, such as workers, customers, lenders and owners. A bookshop's stakeholders include its staff, the schools that buy from it, the bank that lent money and the family that owns it. Wealth maximisation does not tell the business to ignore these people. Treating them badly usually lowers market value in the end, because staff leave, customers go elsewhere and lenders

refuse further loans. The law and honest dealing also set limits that no aim may break.

The aim has limits that you should know. A market price may be wrong for a time, and prices change as news arrives. A business whose common stock is not traded has no price to read, so its value must be estimated, and different people may estimate differently. Even so, the aim gives a more useful guide than profit alone, and it is the aim this book uses.

Before moving on. Market value is the price at which something could be sold today. Wealth maximisation aims to make the market value of the owners' stake as large as possible over time, and it takes account of timing and risk. Stakeholders are all the people affected by a business, and treating them well usually supports long-run value. You should now be able to say, in one sentence, why wealth maximisation is a better aim than profit maximisation.

Wealth maximisation has limits too.

A short summary of section 3.2 with a test sentence.

3.3 AGENCY THEORY

In 1776 the Scottish thinker Adam Smith (1723–1790) published *The Wealth of Nations*. In it he doubted that the directors of large companies would guard the money of their shareholders with the same care that owners give to their own. Directors, he wrote, manage other people's money. Two hundred years later, in 1976, the economists Michael Jensen and William Meckling gave this old worry a precise form and a name. This section explains their idea in layers, beginning with an everyday case.

A farmer cannot go to the market, so a neighbour sells the rice harvest and receives a payment for every sack sold. The farmer wants a good price for the whole harvest. The neighbour earns more by selling many sacks quickly, even at a low price. The two people want different things, though each may be honest. Write this picture down, because every idea in this section is a version of it.

The word for the farmer in the story is **principal**. A principal is a person who hires another person to act for them. The owners of a business are its principals. The owners of a bookshop, for example, are the principals when they hire someone to run a branch.

The word for the neighbour is **agent**. An agent is a person who acts for a principal and takes decisions in the principal's name. A hired branch manager is an agent. The relation between the two is called an agency relation, and it exists wherever one person decides on behalf of another: a taxi owner and a driver, a school principal and a bursar, a business owner and a manager.

Two facts make this relation difficult. First, the agent decides, and the principal often cannot see every decision. Second, the result of the decision falls mainly on the principal. A manager who spends the owner's money freely gives up nothing personally, while the owner gives up the money.

When the aims of the two people differ, a clash may follow. An **agency conflict** is a clash between what a principal wants and what an agent chooses to do. The owners of a business want wealth maximisation. The manager may want a large bonus, a quiet life or a bigger business to run. The conflict is not a sign of dishonesty. It arises because the agent's rewards differ from the principal's.

Before the figures, look at a picture of the idea. Read the figure from the bottom upwards. The two boxes at the bottom name the people, and each arrow shows the direction of that person's aim. The dotted space between the arrows, marked by a dashed line with two heads, is the gap between the aims. The wider the gap, the greater the risk of an agency conflict.

GO SLOWLY HERE.

A historical hook: in 1776 Adam Smith doubted that hired directors guard other people's money as owners do.

An everyday bridge: a neighbour paid for each sack sold.

Principal. A person who hires another person to act for them.

The person who hires is called the principal.

Agent. A person who acts for a principal and takes decisions in the principal's name.

The person hired is called the agent.

The agent decides, but the principal bears much of the result.

Agency conflict. A clash between what a principal wants and what an agent chooses to do.

The clash of aims has a name: agency conflict.

The figure shows two aims pointing in different directions with a gap between them.

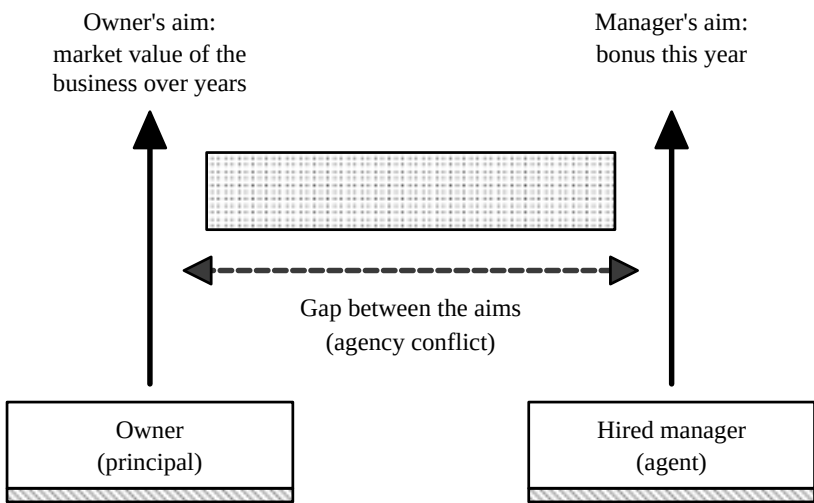


FIG. 3.1. Owner and hired manager with the gap between their aims

The case file shows two plans and an agent who gains while the owners lose.

A manager paid a share of sales is offered two plans.

The table shows the plans side by side.

The following case file shows an agency conflict with exact figures.

Case file, illustrative. Date: 1 April 2024. Business: Meghna and Sons opens a second branch and hires Mr Shafiq to run it. His pay is a yearly bonus of 5% of branch sales. Plan X (steady pricing): sales Tk 1,500,000; profit before bonus Tk 150,000; bonus Tk 75,000; owners keep Tk 75,000. Plan Y (deep discounts and credit to schools): sales Tk 2,000,000; profit before bonus Tk 70,000; bonus Tk 100,000; owners keep minus Tk 30,000.

The next table puts the figures in one place. Read down each plan's column: the sales come first, then the profit before the bonus, then the bonus, and last the amount the owners keep.

PLAN X AND PLAN Y FOR THE SECOND BRANCH (TK)

ITEM	PLAN X	PLAN Y
Sales	1,500,000	2,000,000
Profit before bonus	150,000	70,000
Bonus (5% of sales)	75,000	100,000
Owners keep	75,000	-30,000

Each side prefers a different plan.

Viewpoints: the same rule looks different to owner, manager and lender.

The manager gains Tk 25,000 more under Plan Y, because Tk 100,000 less Tk 75,000 is Tk 25,000. The owners are worse off by Tk 105,000, because Tk 75,000 less minus Tk 30,000 is Tk 105,000. The manager's bonus rises with sales, while the owners' share depends on profit. Deep discounts raise sales and lower profit, so the bonus rule rewards the plan that the owners would reject. The manager prefers Plan Y and the owners prefer Plan X.

PAUSE AND CHECK 3.P2 In the case file, which figure in the manager's pay rule causes the conflict, and why?

The viewpoints differ. The owner sees a bonus rule that pays for sales whatever the result for profit. The manager sees a rule to follow, and a manager who follows it has not broken any promise. A lender, who is a person that gives money to the business and expects repayment with interest, sees a business whose owners now keep less, and so has less safety.

Conflicts of this kind have a price, and the price has a name.

An **agency cost** is the loss in value, plus the cost of prevention, that comes from an agency conflict. In the case file the loss in value is Tk 105,000 if the manager chooses Plan Y. The cost of prevention is the money and time spent to stop such a choice, for example the owners' time spent reading branch records each month. Jensen and Meckling showed that the loss and the cost of prevention belong together, because the owners can reduce one only by paying for the other.

IN SHORT

An agency cost is the loss from an agency conflict plus the cost of preventing it.

Owners cannot remove agency cost completely, but they can reduce it by bringing the interests of the agent closer to their own. Common ways are these. They can pay the manager by a measure of long-term value, not only one year. They can give the manager a stake in the business, so that a loss is a loss for the manager too. They can check records and require reports. They can also rely on pressure from outside: a manager who does poor work finds it harder to be hired again. Each way costs something, and the owners choose where the cost of prevention is smaller than the loss prevented.

Return to the restocking choice of 15 May 2024. A manager who is judged only on this year's profit sees a gain of Tk 80,000 and does not feel the fall of Tk 150,000 that comes later. The owners, who own the business in both years, see a net fall of Tk 70,000. The conflict here is again a difference in the time that each person looks at.

Agency theory also has limits. It does not say that managers are selfish, and many managers act in the owners' interest without any rule that forces them. It does not say that every cost of prevention is worth paying. It describes a pressure that exists, so that owners will think about pay rules before the pressure does harm.

Before moving on. A principal hires an agent, who takes decisions in the principal's name. An agency conflict is a clash between what the principal wants and what the agent chooses. An agency cost is the loss from the conflict plus the cost of preventing it. You should now be able to say who the principal and the agent are in a simple case, and why their aims may differ.

WORKED CASE: THE SECOND BRANCH

A worked case applies the ideas of the chapter to one situation. We follow five steps: *Situation, Question, Reasoning, Conclusion, Check.*

Situation. Meghna and Sons opened a second branch on 1 April 2024 and hired Mr Shafiq, paid 5% of branch sales as a yearly bonus. Plan X gives sales of Tk 1,500,000 and owners keep Tk 75,000. Plan Y gives sales of Tk 2,000,000 and owners keep minus Tk 30,000. On 15 May 2024 a second choice arose: cutting restocking

Interest. Word in use. In daily life, a wish to learn more about something, or the benefit a person gains from a result. In finance, the price paid for the use of borrowed money.

Agency conflict has a cost.

Agency cost. The loss in value, plus the cost of prevention, that comes from an agency conflict.

The loss and the cost of preventing it are together the agency cost.

Owners reduce the conflict by lining up the rewards of the agent with their own aim.

The restocking choice shows how a short-term measure tempts an agent.

The theory has limits as well.

A short summary of section 3.3 with a test sentence.

The worked case follows five steps: situation, question, reasoning, conclusion, check.

The situation is the case file facts for the second branch and the restocking choice.

The question asks which aim each side follows and what the conflict costs.

Reasoning works through the plans and then through the restocking choice.

The conclusion names the preferences and the aim that favours the cut.

The check confirms the arithmetic and the sense of the result.

A ladder of situations shows the idea in new settings.

Familiar case: a tailor paid on monthly profit.

Reshaped case: a driver paid per trip.

New setting: a school principal and a bursar.

Combined case: three aims at once.

Summary of the chapter.

would raise this year's profit by Tk 80,000 and lower next year's profit by Tk 150,000.

Question. Which plan does each side prefer, and what is the loss in value from the conflict? Which aim, profit or wealth, favours the restocking cut?

Reasoning. First we find the bonus under each plan: 5% of Tk 1,500,000 is Tk 75,000, and 5% of Tk 2,000,000 is Tk 100,000. Second we find what the owners keep: Tk 150,000 less Tk 75,000 is Tk 75,000 under Plan X, and Tk 70,000 less Tk 100,000 is minus Tk 30,000 under Plan Y. Third we compare: the manager gains Tk 25,000 more under Plan Y and the owners lose Tk 105,000. Fourth we turn to the restocking choice. This year's profit rises by Tk 80,000, and next year's falls by Tk 150,000, so the two years together show a fall of Tk 70,000.

Conclusion. The manager prefers Plan Y and the owners prefer Plan X. The loss in value from the conflict is Tk 105,000, and the cost of preventing it, such as checking the records, is added to that loss to find the agency cost. A rule of one-year profit favours the restocking cut, while wealth maximisation rejects it, because the market would pay less for the owners' stake.

Check. Tk 75,000 plus Tk 30,000 is Tk 105,000, which agrees with the loss. Tk 100,000 less Tk 75,000 is Tk 25,000, which agrees with the manager's gain. The result is sensible, because a bonus on sales rewards sales and does not reward profit.

THE SAME IDEA IN FOUR SITUATIONS

The idea is useful only if you can recognise it elsewhere. Each situation below changes the setting while keeping the idea.

A familiar case. A tailor manages a shop for its owner and is paid on the monthly profit. The tailor skips the buying of fabric to show a larger profit this month. The owner loses the sales of later months. Here the tailor is the agent and the owner is the principal.

A reshaped case. A taxi owner pays a hired driver for each trip. The driver takes many short trips and drives hard, and the car wears out faster. The driver's reward is the number of trips, and the owner's reward is the condition and profit of the car.

A new setting. The principal of a school wants the best use of limited fees. The bursar, who keeps the school accounts, is judged on spending less than the budget. The bursar may delay repairs that the principal wants. The setting has no profit, yet the pattern is the same.

A combined case. A hired manager wants a larger bonus, the owners want a larger market value, and a lender wants safe repayment with interest. All three may prefer different plans, and the manager's choice affects the other two.

SUMMARY

The goal of the business is wealth maximisation, which is the aim of making the market value of the owners' stake as large as possible over time. Profit maximisation, the aim of the largest profit in a period, ignores timing and risk. Stakeholders are affected by the business and should not be treated badly. When owners hire managers, a principal and an agent exist, and an agency conflict may follow. An agency cost is the loss from the conflict plus the cost of prevention.

THINK IT THROUGH

3.T1 A manager is paid 4% of branch sales. Plan P gives sales of Tk 1,000,000 and profit before bonus of Tk 120,000. Plan Q gives sales of Tk 1,400,000 and profit before bonus of Tk 60,000. Find the bonus and the amount the owners keep under each plan, say which plan each side prefers, and find the loss in value to the owners if the manager chooses Plan Q.

3.T2 Suppose Mr Shafiq were paid 5% of the change in profit over the two years instead of 5% of sales. Using the restocking choice of 15 May 2024, find his gain or loss and say whether he would now cut restocking.

WHERE PEOPLE GO WRONG

The table lists four common mistakes. Read each row from left to right: the first column gives the mistake and the second gives the correction.

Four common misunderstandings are stated and corrected in the table.

COMMON MISTAKES AND CORRECTIONS

MISTAKE	CORRECTION
The plan with the largest profit this year is always best	It may lower later profit or carry more risk; wealth maximisation weighs both
Wealth maximisation means ignoring workers and customers	Stakeholders affect long-run value, and the law and honest dealing still apply
An agency conflict means the manager is dishonest	The conflict comes from different rewards; an honest manager may still follow the pay rule
Agency cost exists only in large companies	It exists wherever one person decides for another, even in a small shop

IN HISTORY

In 1776 Adam Smith, a Scottish thinker (1723–1790), published *The Wealth of Nations* in London. In the part on companies owned by many shareholders, he doubted that the directors, who manage other people's money, would watch it with the care that owners give to their own. He expected carelessness and waste.

In 1976 two economists of the University of Rochester in the United States, Michael C. Jensen and William H. Meckling, published "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure" in the *Journal of Financial Economics* (volume 3, pages 305–360). They defined agency costs, related them to the separation of ownership from control, and studied the agency costs created by debt and by outside owners. They also grouped the costs under three heads: the cost of watching the agent, the cost the agent bears to give assurance, and the loss that remains.

Smith in 1776 and Jensen and Meckling in 1976 form the history of agency.

Jensen and Meckling gave the worry a precise form exactly two hundred years later.

SOURCES

Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), is a founding work of economics. It matters here for its early warning about directors who manage the money of others.

Michael C. Jensen and William H. Meckling, "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure", *Journal of Financial Economics*, volume 3 (1976), is the landmark work. It matters because

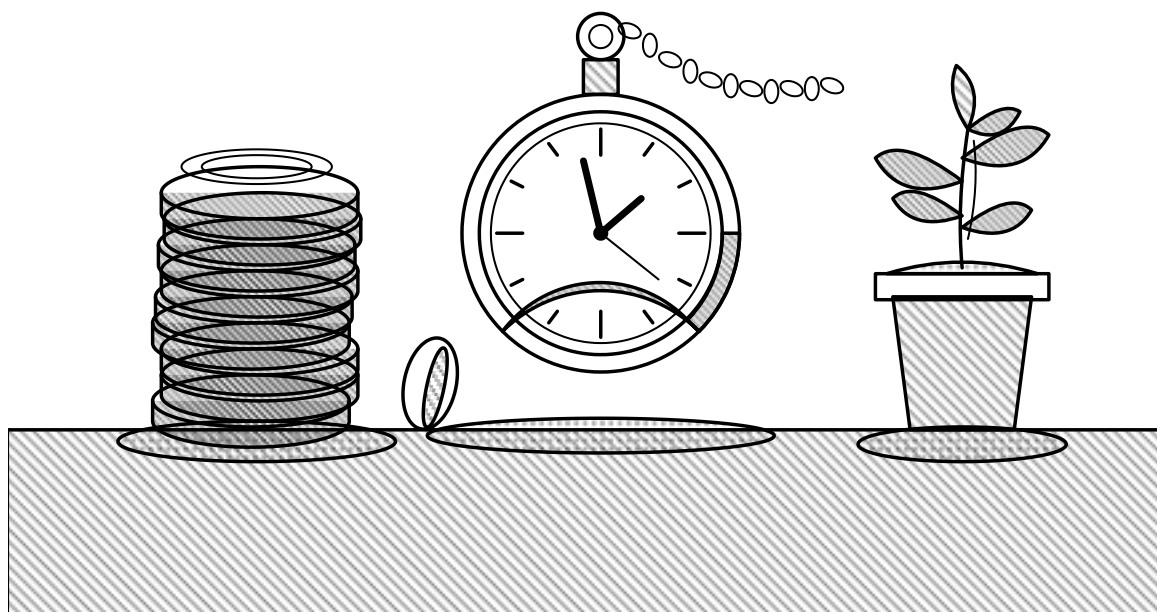
Smith's book is the early source for the worry about other people's money.

The paper by Jensen and Meckling is the founding work of agency theory in finance.

it defined agency cost and applied it to the way a business is owned and financed.

REVIEW QUESTIONS

- 3.1 Explain why profit maximisation can mislead when returns come at different times or carry different risk.
- 3.2 Explain wealth maximisation in plain words to a village shopkeeper who has never heard the term.
- 3.3 Describe an agency conflict between a taxi owner and a driver who is paid for each trip.
- 3.4 Name three ways in which owners can bring the interests of a manager closer to their own, and say what each costs.



A pocket watch, a stack of coins and a young plant: time works on money as it works on a seed.

CHAPTER IV

The Time Value of Money: Single Sums

Is a taka today worth the same as a taka next year?



Money has a price that depends on time. This chapter explains why a taka received sooner is worth more than a taka received later, and teaches two moves with one amount at a time: forward in time to find what it will grow to, and back in time to find what it is worth today.

What to bring from Chapter 1.

A friend offers the same amount today or a year from today.

Raising the later amount shows where people change their minds.

The chapter turns this feeling into arithmetic.

You will need two ideas from Chapter 1, section 1.1: *funds* and *cash flow*.¹ Debt, equity and wealth are used here as they were taught earlier.

A friend offers you a choice. You may take Tk 100,000 today, or you may take Tk 100,000 one year from today. The friend is honest, and you are certain you will be paid on the agreed day. Which do you choose? Nearly everyone chooses today. Yet the two amounts are equal. The choice seems to say that the same number of taka is not the same thing at two different times.

Now suppose the friend raises the later offer to Tk 105,000, then to Tk 110,000, then to Tk 120,000. At some amount you will change your answer and choose to wait. The amount at which you change measures how much you value time. It differs between people, but it is never zero. The habit is not foolish. Money in hand can be used, kept safe or put to work, and money promised for later can do none of these things until the day arrives.

This chapter turns that feeling into arithmetic. Section 4.1 explains the idea and the words around it. Section 4.2 moves an amount forward in time to find what it will grow to. Section 4.3 moves an amount back in time to find what it is worth today. Section 4.4 allows growth to happen more than once a year. A set of worked examples closes the chapter, and every one of them can be done by hand. We study one amount at a time here; payments that repeat are the subject of Chapter 5, 'Annuities'.

* * *

4.1 WHY TIME CHANGES THE VALUE OF MONEY

GO SLOWLY HERE.

A farmer's seed shows why money in hand early can become more later.

Time value of money. The idea that money received sooner is worth more than the same amount received later.

The farmer's lesson has a name: the time value of money.

Three reasons support the idea, and this chapter uses the first.

Begin with a farmer who has 10 kilograms of rice seed. The farmer can eat the seed today, or sow it and wait for the harvest. If the farmer sows it, the harvest in six months may be 40 kilograms. Ten kilograms today is therefore worth more than ten kilograms after the harvest, because the seed of today can become forty. Keep this picture in mind. Seed that is sown grows, and seed that is eaten is gone. Money in hand can be sown in the same way, as a deposit in a bank or as stock for a shop.

We can now name the idea. The **time value of money** is the idea that money received sooner is worth more than the same amount received later. Tk 100,000 received today is worth more than Tk 100,000 received in a year, because today's money can be put to work for that year. The idea does not depend on prices rising. It holds even when every price stays the same, because the reason is the earning power of funds, not the cost of goods.

Three reasons support the idea. First, money in hand can earn: it can be lent, deposited or used to buy stock that will be sold at a profit. Second, prices usually rise over time, so a taka buys a little less each year. Third, a promise for later may fail, while money in hand is certain. This chapter works with the first reason and treats every future amount as certain. Chapter 9, 'Return and Risk', deals with uncertainty.

¹ Chapter 1, 'Money, Finance and the Firm', section 1.1 'What Finance Is': funds and wealth define what is being timed.

The earning power of money has a price, and the price has a name. **Interest** is the price paid for using money for a period, received by the lender. If you lend Tk 100,000 to a bank for a year and the bank pays you Tk 8,000 for the use of it, the Tk 8,000 is interest. The borrower pays the interest and the lender receives it. A business that borrows through debt pays interest, and a business that deposits spare funds receives it.

The amount of interest depends on how much is lent, so we also need a way to compare loans of different sizes. The **interest rate** is the interest for one period, shown as a percentage of the amount lent or borrowed. Tk 8,000 earned on Tk 100,000 in one year gives $Tk\ 8,000 \div Tk\ 100,000 = 0.08$, which is 8% a year. When no period is named, this book means one year. In formulas we write the rate as a decimal, so 8% becomes 0.08, and we use the letter *i* for it.

The farmer who eats the seed gives something up. **Opportunity cost** is the value of the best choice given up when you pick another. The farmer's opportunity cost of eating the seed is the harvest that sowing would have given. In the same way, Tk 100,000 kept idle in a drawer for a year has an opportunity cost: the interest a bank would have paid, which is Tk 8,000 at 8%. The cost is real although no one sends a bill, because it is a gain that never arrived.

The words change with the viewpoint, but the idea stays the same. A lender sees interest as a reward for waiting. A borrower sees it as the price of using money before it is earned. An owner or manager sees it as the opportunity cost of the funds placed in the business. Chapter 3 explained why earlier wealth is preferred, because wealth maximisation looks at the market value of the owners' stake over time.² A manager who prefers wealth sooner, and an owner who asks what the funds could have earned elsewhere, both apply the time value of money.

The idea has limits. The size of the effect depends on the interest rate available: at a higher rate, waiting costs more. The idea also assumes that the later amount is certain. If the friend might not pay, the later promise is worth less for that reason as well. We set that matter aside until Chapter 9.

Before moving on. The time value of money is the idea that money received sooner is worth more than the same amount received later. Interest is the price of using money for a period, the interest rate is that price as a percentage per period, and opportunity cost is the value of the best choice given up. You should now be able to say why Tk 100,000 today is worth more than Tk 100,000 in a year.

4.2 FUTURE VALUE OF A SINGLE SUM

Take the smallest numbers first. You deposit Tk 10,000 in a bank that pays 10% a year, and you leave it for two years. After one year the bank pays Tk 1,000, which is 10% of Tk 10,000, and your account holds Tk 11,000. Now the second year begins. How much does the bank pay? There are two possible answers, and the difference between them is the centre of this section.

² Chapter 3, 'The Goal of the Firm and Agency', section 3.2 'Wealth Maximisation': why a manager prefers earlier wealth.

Interest. The price paid for using money for a period, received by the lender.

Interest. Word in use. In daily life, a wish to learn about something, or the benefit a person gains from a result. In finance, a payment for the use of money.

Interest is the price of using money for a period.

Interest rate. The interest for one period, shown as a percentage of the amount lent or borrowed.

Rate. Word in use. In daily life, a speed or a count in a period, such as a rate of travel. In finance, a percentage of an amount for a period, such as 8% a year.

The interest rate puts the price as a percentage, so loans of different sizes can be compared.

Opportunity cost. The value of the best choice given up when you pick another.

The farmer who eats the seed gives up the harvest, and that loss is an opportunity cost.

Lender, borrower and owner see the same idea in different words.

The effect depends on the rate and on the later amount being certain.

A short summary of section 4.1 with a test sentence.

GO SLOWLY HERE.

A small deposit shows interest added year by year.

Single sum. One payment or receipt made at one point in time.

A single sum is one payment at one moment.

Simple interest. Interest worked out on the original amount only.

The first answer: interest on the original amount only.

Compound interest. Interest worked out on the original amount plus interest already earned.

The second answer: interest on the amount plus earlier interest.

Simple interest ignores earlier interest; compound interest includes it.

Future value. What an amount today will grow to after a stated time at a stated rate.

Future value is the amount an account will hold later, found by multiplying once for each year.

The future value factor is a number that can be used for any amount.

First worked example: given, find and formula.

First worked example: steps, answer and check.

The business case uses the same method on a larger deposit.

The table shows the account at the end of each year under both kinds of interest.

We first need a name for the Tk 10,000. A **single sum** is one payment or receipt made at one point in time. The deposit of Tk 10,000 today is a single sum. Payments that repeat each year are not single sums; Chapter 5 handles them.

The first answer is **simple interest**, which is interest worked out on the original amount only. Under simple interest the second year's interest is again 10% of Tk 10,000, which is Tk 1,000, so the account holds Tk 12,000 after two years. Each year adds the same amount. In words: interest = original amount $\times$ interest rate $\times$ number of years.

The second answer is **compound interest**, which is interest worked out on the original amount plus interest already earned. In the second year the bank pays 10% of Tk 11,000, which is Tk 1,100, so the account holds Tk 12,100. The extra Tk 100 is interest earned on the first year's interest. Interest on interest is the source of the extra growth. Repeating this addition year after year is called compounding.

PAUSE AND CHECK 4.P1 Using the deposit of Tk 10,000 at 10% a year, find the amount in the account after three years under compound interest. Work year by year.

Before moving on to the shortcut. Simple interest uses the original amount every year. Compound interest uses the original amount plus the interest already earned, so it grows faster.

Year-by-year working becomes slow when the time is long, so a shortcut is useful. Each year the amount is multiplied by $(1 + i)$. In the example it is multiplied by 1.10, and after two years it has been multiplied twice. The **future value** is what an amount today will grow to after a stated time at a stated rate. In words: future value = single sum $\times (1 + \text{interest rate})$ to the power of the number of years. In symbols: $FV = S \times (1 + i)$ to the power of n , where FV is the future value, S is the single sum today, i is the interest rate for one period as a decimal, and n is the number of periods.

The number $(1 + i)$ to the power of n is called the future value factor. It does not depend on the size of the single sum, so it can be worked out once and used for any amount. For 10% and 2 years it is $1.10 \times 1.10 = 1.21$.

Given. $S = \text{Tk } 10,000$; $i = 10\% \text{ a year} = 0.10$; $n = 2 \text{ years}$. *Find.* The future value, FV . *Formula.* In words: future value equals the single sum times one plus the interest rate, to the power of the number of years. In symbols: $FV = S \times (1 + i)$ to the power of n .

Steps. (1) $1 + i = 1 + 0.10 = 1.10$. (2) 1.10 to the power of $2 = 1.10 \times 1.10 = 1.21$. (3) $FV = \text{Tk } 10,000 \times 1.21 = \text{Tk } 12,100$. *Answer.* The Tk 10,000 will grow to Tk 12,100 in two years. *Check.* Year by year, Tk 10,000 became Tk 11,000 and then Tk 12,100, so the two methods agree. The result is Tk 100 above the simple interest total of Tk 12,000, as expected.

Case file, illustrative. Date: 1 July 2024. Meghna and Sons deposits Tk 200,000 at 8% a year for 5 years. Simple interest gives $\text{Tk } 200,000 \times (1 + 0.08 \times 5) = \text{Tk } 280,000$. Compound interest gives $\text{Tk } 200,000 \times 1.08$ to the power of 5 = Tk 293,865.62, using a future value factor of 1.469328.

The table shows the account at the end of each year. Read across a row: the year, then the total under simple interest, then the total under compound interest.

TK 200,000 AT 8% A YEAR (TK)

YEAR	SIMPLE INTEREST	COMPOUND INTEREST
0	200,000.00	200,000.00
1	216,000.00	216,000.00
2	232,000.00	233,280.00
3	248,000.00	251,942.40
4	264,000.00	272,097.79
5	280,000.00	293,865.62

In year 1 the two totals agree, because there is no earlier interest to earn on. After that, compounding pulls ahead. By year 5 the compound total is Tk 13,865.62 above the simple total, and the gap grows each year.

The figure draws the same two lines. To read it, look at the horizontal axis first: it shows the years from today. The vertical axis shows the amount in thousands of taka, and each point shows what the account holds at the end of that year. The dashed line is simple interest and the solid line is compound interest. A straight line means the same amount is added each year. A line that bends upward means each year adds more than the year before.

In year 1 the two agree; after that compounding pulls ahead.

How to read the figure.

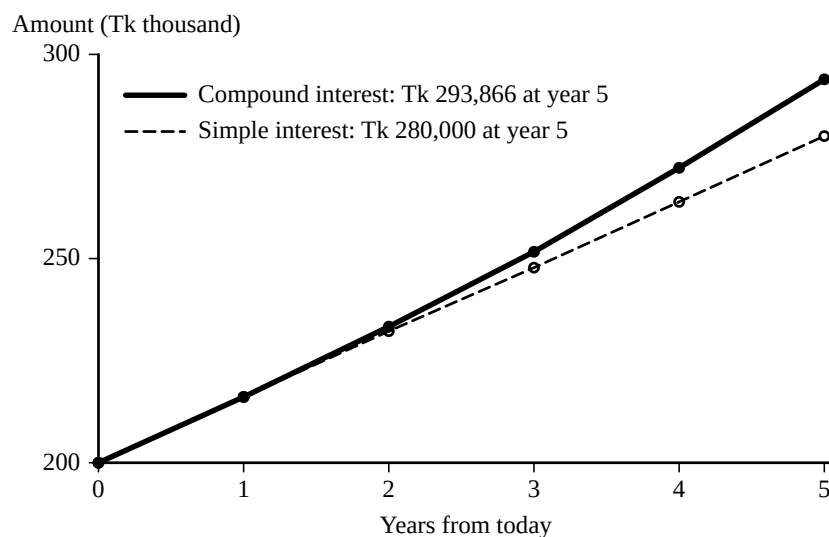


FIG. 4.1. Growth of Tk 200,000 at 8% a year over five years: simple interest against compound interest

Three cautions apply. The rate and the time must use the same period: a yearly rate needs the time counted in years. The rate enters the formula as a decimal, so 8% is 0.08. And the formula assumes interest is added once a year; section 4.4 changes that. Compounding also works against borrowers, because a debt grows in the same way.

Before moving on. A single sum is one payment at one point in time. Simple interest is worked out on the original amount only, and compound interest on the original amount plus interest already earned. Future value is what an amount today will grow to, and $FV = S \times (1 + i)^n$. You should now be able to say how much Tk 10,000 becomes in two years at 10% a year, and why the answer is above Tk 12,000.

Three cautions apply to the formula.

A short summary of section 4.2 with a test sentence.

4.3 PRESENT VALUE OF A SINGLE SUM

GO SLOWLY HERE.

A target amount shows the need to work backwards.

Present value. What an amount to be received later is worth today at a stated rate.

Present. Word in use. In daily life, the time now, or a gift. In finance, the worth today of an amount due later.

Present value is the worth today of a later amount.

Discounting. Finding a present value by working backwards from a future amount at a rate.

Discounting moves an amount back in time by dividing once for each year.

Discount rate. The rate used to turn a future amount into a present value.

The discount rate is the rate used when working backwards.

The formula mirrors the one in section 4.2.

Second worked example: given, find and formula.

Second worked example: steps, answer and check.

Tables of factors save repeated multiplication.

Now reverse the question. You know that you will need Tk 12,100 in two years, and a bank pays 10% a year with yearly compounding. How much must you deposit today? Section 4.2 gave a way from today to a later date. This section gives the way back. Think of one road: in section 4.2 we travelled forward along it, and here we travel the same road in the opposite direction.

The answer to the question has a name. The **present value** is what an amount to be received later is worth today at a stated rate. If Tk 12,100 is due in two years and the rate is 10%, its present value is the deposit that grows to Tk 12,100 in two years. Section 4.2 showed that Tk 10,000 grows to Tk 12,100 in two years at 10%. So the present value of Tk 12,100 is Tk 10,000.

The working has a name too. **Discounting** is finding a present value by working backwards from a future amount at a rate. Compounding multiplies by $(1 + i)$ for each year, and discounting divides by $(1 + i)$ for each year. Year by year: $\text{Tk } 12,100 \div 1.10 = \text{Tk } 11,000$, and $\text{Tk } 11,000 \div 1.10 = \text{Tk } 10,000$. Each step back removes one year of interest.

PAUSE AND CHECK 4.P2 Using the same rate of 10% a year, find the present value of Tk 13,310 due in three years. Divide by 1.10 once for each year.

The rate used in discounting has its own name. The **discount rate** is the rate used to turn a future amount into a present value. In our example it is 10%, the same number as the interest rate, because 10% is what the money could earn. One number serves in two directions: it is the interest rate when we move forward and the discount rate when we move back. Chapter 7, 'Capital Budgeting: Decisions and Cash Flows', and the chapter after it use the discount rate to value projects.

The shortcut mirrors the one in section 4.2. In words: present value = future amount $\div (1 + \text{interest rate})$ to the power of the number of years. In symbols: $PV = FV \div (1 + i)$ to the power of n . Dividing by a factor is the same as multiplying by its reciprocal, $1 \div (1 + i)$ to the power of n , which is called the present value factor. For 10% and 2 years it is $1 \div 1.21 = 0.826446$.

Given. $FV = \text{Tk } 12,100$; $i = 10\%$ a year $= 0.10$; $n = 2$ years. *Find.* The present value, PV . *Formula.* In words: present value equals the future amount divided by one plus the interest rate, to the power of the number of years. In symbols: $PV = FV \div (1 + i)$ to the power of n .

Steps. (1) $1 + i = 1.10$. (2) 1.10 to the power of $2 = 1.21$. (3) $PV = \text{Tk } 12,100 \div 1.21 = \text{Tk } 10,000$. *Answer.* The present value of Tk 12,100 due in two years is Tk 10,000. *Check.* Grow the answer forward: $\text{Tk } 10,000 \times 1.21 = \text{Tk } 12,100$. The present value is below the future amount, as it must be when the rate is above zero.

Tables of factors save repeated arithmetic. To read a table, find the row for the number of years, move across to the column for the rate, and read the factor. The first table gives future value factors and the second gives present value factors. A calculator can replace the tables, but the tables are useful for checking and for finding a missing rate or time, as the worked examples show.

FUTURE VALUE FACTORS, $(1 + i)$ TO THE POWER OF N

YEARS	6%	8%	10%
1	1.060000	1.080000	1.100000
2	1.123600	1.166400	1.210000
3	1.191016	1.259712	1.331000
4	1.262477	1.360489	1.464100
5	1.338226	1.469328	1.610510
6	1.418519	1.586874	1.771561

PRESENT VALUE FACTORS, $1 \div (1 + i)$ TO THE POWER OF N

YEARS	6%	8%	10%
1	0.943396	0.925926	0.909091
2	0.889996	0.857339	0.826446
3	0.839619	0.793832	0.751315
4	0.792094	0.735030	0.683013
5	0.747258	0.680583	0.620921
6	0.704961	0.630170	0.564474

Case file, illustrative. Date: 1 July 2024. Meghna and Sons needs Tk 500,000 on 30 June 2029 for a premises deposit. The present value at 8% a year is Tk 340,291.60, using a present value factor of 0.680583.

The time from 1 July 2024 to 30 June 2029 is 5 years. The table gives 0.680583 for 5 years at 8%, and $\text{Tk } 500,000 \times 0.680583 = \text{Tk } 340,291.50$. A calculator that keeps every digit of the factor gives Tk 340,291.60, which is the figure in the case file; the gap of Tk 0.10 comes from rounding the factor. So a deposit of Tk 340,291.60 today at 8% will grow to Tk 500,000 on the date needed. Check: $\text{Tk } 340,291.60 \times 1.469328$ is about Tk 500,000.

The figure shows a timeline. Time runs from left to right, and the first mark is today. The upper arrow points forward and is labelled compounding: moving along it multiplies by $(1 + i)$ for each year. The lower arrow points back, is drawn with a dashed line and is labelled discounting: moving along it divides by $(1 + i)$ for each year.

The business case needs a future amount to be met by a deposit today.

Working of the case file: five years, factor, answer and check.

How to read the timeline figure.

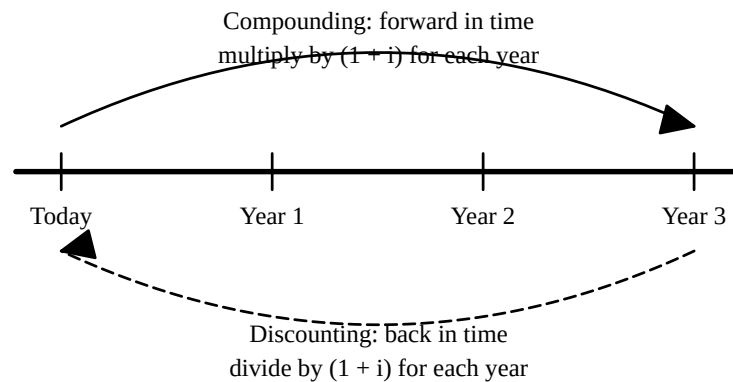


FIG. 4.2. Compounding moves an amount forward in time and discounting moves it back

Four cautions apply to present value.

A short summary of section 4.3 with a test sentence.

GO SLOWLY HERE.

A bank that adds interest twice a year pays more than one that adds it once.

Compounding period. The time between one addition of interest and the next.

The compounding period is the time between additions of interest.

Method: change the rate and the time to match the period.

Effective annual rate. The single yearly rate that gives the same growth as compounding within the year.

The effective annual rate lets us compare banks that add interest at different times.

Four cautions apply. A present value is below the future amount whenever the rate is above zero. A higher rate or a longer time makes the present value smaller: Tk 1 due in 5 years is worth 0.680583 at 8% but only 0.620921 at 10%. Rounding the factors can change the last digits of an answer. And the rate and the time must use the same period, as in section 4.2.

Before moving on. Present value is what an amount to be received later is worth today at a stated rate. Discounting finds it by dividing by $(1 + i)$ once for each year, and the rate used is the discount rate. You should now be able to say why Tk 12,100 due in two years at 10% is worth Tk 10,000 today.

4.4 COMPOUNDING WITHIN THE YEAR

A bank offers 12% a year on Tk 10,000 but adds interest every six months. Is that the same as adding 12% once at the end of the year? Work it through. After six months the bank adds half a year's interest, 6% of Tk 10,000, which is Tk 600, so the account holds Tk 10,600. After twelve months it adds 6% of Tk 10,600, which is Tk 636, so the account holds Tk 11,236. Adding once a year would give Tk 11,200. The bank that adds twice pays Tk 36 more.

The reason lies in the **compounding period**, which is the time between one addition of interest and the next. In the example it is six months. Because the bank adds interest twice a year, the second addition earns interest on the first. The shorter the compounding period, the sooner interest starts to earn interest.

To handle this by hand, change both the rate and the time to match the period. The rate for one period is the yearly rate divided by the number of periods in a year. The number of periods is the years times the number of periods in a year. In symbols: $FV = S \times (1 + i \div m)^{m \times t}$, where m is the number of periods in a year and t is the number of years. In the example, $i \div m = 0.12 \div 2 = 0.06$ and $m \times t = 2$, so $FV = \text{Tk } 10,000 \times 1.06 \text{ to the power of } 2 = \text{Tk } 11,236$.

To compare banks that add interest at different times, we turn each offer into one yearly figure. The **effective annual rate** is the single yearly rate that gives the same growth as compounding within the year. In symbols: effective annual rate $= (1 + i \div m)^m - 1$. In the example, $1.06 \times 1.06 - 1 = 0.1236$, so the effective annual rate is 12.36%. The advertised 12% is a yearly rate that ignores compounding within the year, and the effective rate is higher.

Case file, illustrative. Date: 1 July 2024. A deposit of Tk 200,000 at 8% a year is compounded half-yearly at 4% for 10 periods and grows to Tk 296,048.86. The effective annual rate is 8.16%.

The working is $Tk\ 200,000 \times 1.04$ to the power of 10 = $Tk\ 200,000 \times 1.480244 = Tk\ 296,048.86$. The effective annual rate is $1.04 \times 1.04 - 1 = 0.0816$, which is 8.16%. Yearly compounding gave Tk 293,865.62 in section 4.2, so compounding twice a year earns Tk 2,183.24 more over the five years. A common slip is to use 8% for each half-year, or to use 5 periods with the half-year rate. Always change both the rate and the number of periods.

Before moving on. The compounding period is the time between additions of interest. The rate per period is the yearly rate divided by the periods in a year, and the number of periods is the years times the periods in a year. The effective annual rate is the yearly rate that gives the same growth. You should now be able to say why 12% compounded half-yearly is the same as 12.36% compounded once a year.

The business case compounds half-yearly.

Working of the case file and a comparison with yearly compounding.

A short summary of section 4.4 with a test sentence.

WORKED EXAMPLES

These examples are a little harder than those in the sections. Each follows the same order: Given, Find, Formula, Steps, Answer and Check.

The examples here are a little harder than those in the sections.

A REALISTIC FUTURE VALUE

Given. $S = Tk\ 36,500$; $i = 7.5\%$ a year = 0.075; $n = 6$ years. *Find.* The future value, FV. *Formula.* In words: future value equals the single sum times one plus the interest rate, to the power of the number of years. In symbols: $FV = S \times (1 + i)$ to the power of n .

Third worked example: given, find and formula.

Steps. (1) $1 + i = 1.075$. (2) Multiply by 1.075 six times, keeping six decimal places: 1.075, 1.155625, 1.242297, 1.335469, 1.435629, 1.543302. So 1.075 to the power of 6 = 1.543302. (3) $FV = Tk\ 36,500 \times 1.543302 = Tk\ 56,330.52$. A calculator that keeps every digit gives Tk 56,330.51; the one paisa gap comes from rounding the factor. *Answer.* Tk 36,500 grows to about Tk 56,331 in six years. *Check.* Simple interest would give $Tk\ 36,500 \times (1 + 0.075 \times 6) = Tk\ 36,500 \times 1.45 = Tk\ 52,925$. The compound result is higher, as it should be.

Third worked example: steps, answer and check.

FINDING A MISSING RATE

Given. $S = Tk\ 25,000$; $FV = Tk\ 33,275$; $n = 3$ years. *Find.* The interest rate, i . *Formula.* Future value factor = $FV \div S$. Then find that factor in the row for n years. *Steps.* (1) $33,275 \div 25,000 = 1.331$. (2) In the row for 3 years of the future value table, the factors are 1.191016 at 6%, 1.259712 at 8% and 1.331000 at 10%. *Answer.* The interest rate is 10% a year. *Check.* $Tk\ 25,000 \times 1.331 = Tk\ 33,275$, which agrees with the given amount.

Fourth worked example: a rate to be found from the table.

FINDING A MISSING TIME

Given. $S = Tk\ 40,000$; $FV = Tk\ 58,773.12$; $i = 8\%$ a year. *Find.* The number of years, n . *Steps.* (1) $58,773.12 \div 40,000 = 1.469328$. (2) In the 8% column of the future value table, the factor 1.469328 appears in the row for 5 years. *Answer.* The time is 5 years. *Check.* $Tk\ 40,000 \times 1.469328 = Tk\ 58,773.12$.

Fifth worked example: a time to be found from the table.

COMPARING TWO OFFERS

Given. Offer A pays Tk 70,000 in 3 years. Offer B pays Tk 85,000 in 5 years. Money can earn 8% a year. Later we try 12%. *Find.* The better offer at each rate. *Formula.* $PV = FV \times$ present value factor. Offers received at different times can be compared fairly only after each is turned into its present value. *Steps at 8%.*

Sixth worked example: two offers compared by present value at two rates.

A: Tk 70,000 $\times$ 0.793832 = Tk 55,568.24. B: Tk 85,000 $\times$ 0.680583 = Tk 57,849.56. *Steps at 12%.* The factors are $1 \div 1.12$ to the power of 3 = 0.711780 and $1 \div 1.12$ to the power of 5 = 0.567427. A: Tk 70,000 $\times$ 0.711780 = Tk 49,824.60. B: Tk 85,000 $\times$ 0.567427 = Tk 48,231.30. *Answer.* At 8% Offer B is better, by about Tk 2,281. At 12% Offer A is better, by about Tk 1,593. *Check.* A higher rate shrinks every present value, and it shrinks the later amount by more. So the nearer offer gains ground as the rate rises, and the ranking can change.

COMMON SLIPS

Four slips appear often, and each has a simple defence.

Four slips appear again and again. The table gives each slip and a way to avoid it. Read each row from left to right.

COMMON SLIPS AND HOW TO AVOID THEM

SLIP	HOW TO AVOID IT
Using 8 in the formula instead of 0.08	Write the rate as a decimal and say it aloud: 8% is 0.08
Using a yearly rate with half-years	Divide the rate and multiply the years by the periods in a year
Multiplying when you should divide, or the reverse	Forward in time multiplies; back in time divides; a present value is below the later amount
Forgetting the 1 in $(1 + i)$	The factor is $(1 + i)$ to the power of n , never i to the power of n

SUMMARY

Summary of the chapter.

Money received sooner is worth more than the same amount received later: this is the time value of money. Interest is the price of using money, and the interest rate states it as a percentage of the amount per period. Compound interest earns interest on interest, and future value is $FV = S \times (1 + i)$ to the power of n . Present value is found by discounting, $PV = FV \div (1 + i)$ to the power of n , at the discount rate. When interest is added within the year, divide the rate and multiply the periods, and compare offers by the effective annual rate.

THINK IT THROUGH

4.T1 A student chooses between Tk 90,000 today and Tk 130,000 in 5 years. Money can earn 7% a year, compounded yearly. Find the present value of Tk 130,000, decide which offer is better, and show a check by finding the future value of Tk 90,000.

4.T2 Tk 15,000 grows to Tk 21,277.79 at 6% a year. Use the future value table to find the number of years.

WHERE PEOPLE GO WRONG

Five common misunderstandings are stated and corrected in the table.

The table lists five common mistakes. Read each row from left to right: the first column gives the mistake and the second gives the correction.

COMMON MISTAKES AND CORRECTIONS

MISTAKE	CORRECTION
Tk 100,000 today and Tk 100,000 in a year are worth the same because the numbers match	Money today can be put to work, so the later amount has a lower present value
Simple and compound interest are equal	They agree only in the first period; after that compound interest is larger
Present value is the amount you will receive	Present value is today's worth of that amount; the amount received later is the future value
The discount rate and the interest rate are unrelated	They can be the same number, used in opposite directions
A yearly rate can be used for half-years without change	Divide the rate by the periods in a year and multiply the years by the same number

IN HISTORY

In 1683 the Swiss mathematician Jacob Bernoulli (1654–1705) studied a question about compound interest. Suppose Tk 1 is lent at 100% a year. Added once, the interest gives Tk 2 at the end of the year. Added twice, at 50% each half-year, it gives Tk 2.25. Bernoulli asked what happens if interest is added more and more often. The amount keeps rising, but each step adds less than the one before.

Bernoulli showed that the amount never reaches Tk 3, and that it stays above Tk 2. The limit is the number now written e , about 2.71828; the letter was introduced later by Leonhard Euler, who first used it in a letter in 1731. In this chapter the same idea appears in a smaller form. At 12% a year, the effective annual rate rises from 12% to 12.36% with half-yearly compounding and to 12.55% with quarterly compounding, but it can never rise above about 12.75%, however often interest is added.

In 1683 Jacob Bernoulli asked what happens when interest is added more and more often.

Bernoulli showed the amount stays between 2 and 3, and the limit became the number e .

SOURCES

Jacob Bernoulli's study of compound interest in 1683 is the first known step towards the number e . It matters here because it showed that adding interest more often raises the total, but only up to a limit.

Leonhard Euler, *Introductio in analysin infinitorum* (1748), gave the full treatment of e and showed that it is the limit of Bernoulli's expression. It matters because it fixed the number and its notation for later use in finance and science.

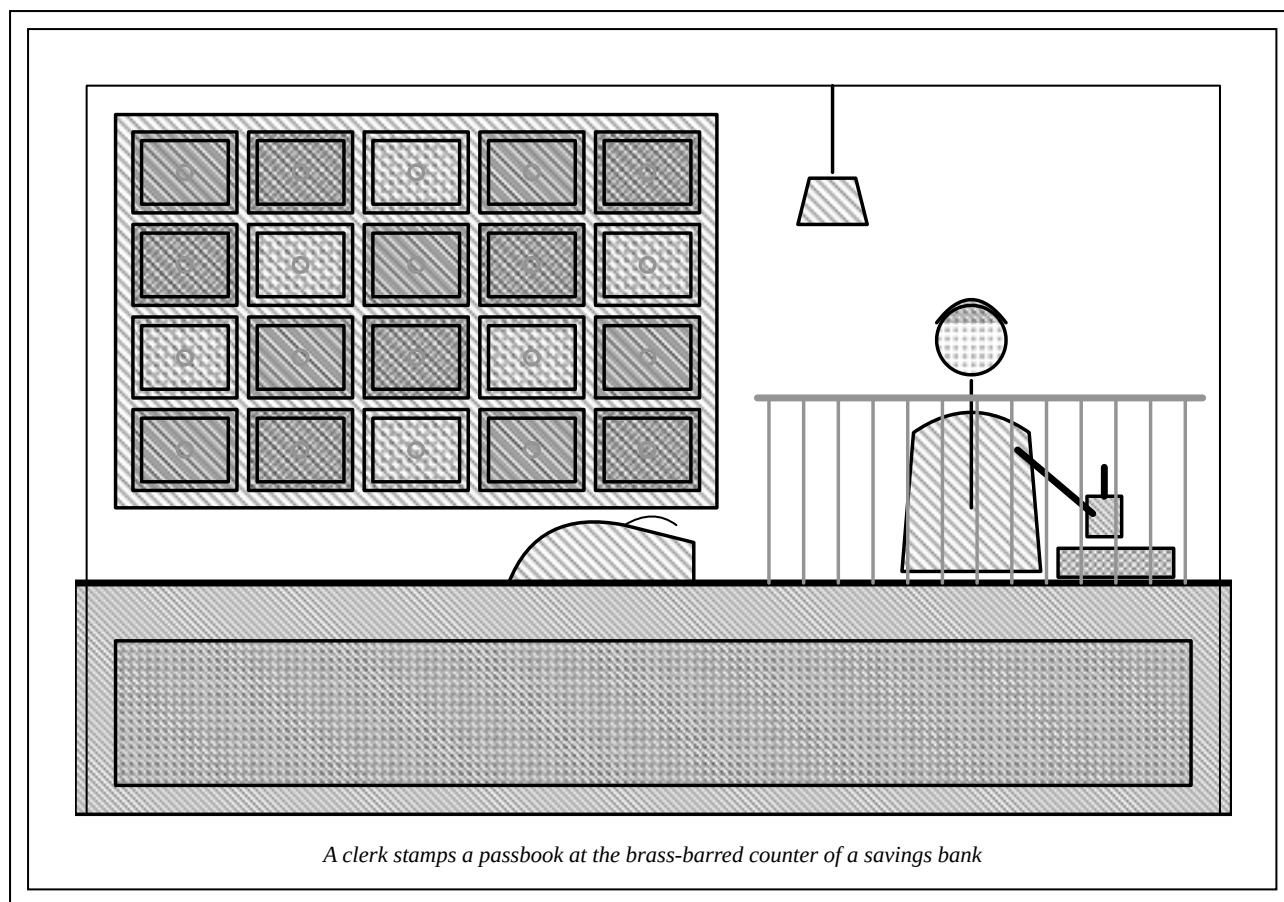
Bernoulli's 1683 problem is the early source of the limit that became the number e .

Euler's book of 1748 gave the number its full treatment.

REVIEW QUESTIONS

- 4.1 Find the future value of Tk 15,000 at 9% a year for 6 years.
- 4.2 Find the present value of Tk 80,000 due in 4 years at 11% a year.
- 4.3 Compare simple interest with compound interest on Tk 25,000 over 8 years at 6% a year, and state the difference.
- 4.4 Find the effective annual rate for 12% a year compounded quarterly.

4.5 Explain the opportunity cost of idle cash in plain words, with a small example.



A clerk stamps a passbook at the brass-barred counter of a savings bank

CHAPTER V

Annuities

How does one value a promise that arrives in equal instalments?



This chapter teaches equal payments made at equal intervals. It shows how to find their future value and their present value, how to read a factor table, and how to find an unknown payment for a savings target or a loan.

YOU WILL NEED

Chapter 4, sections 4.2 and 4.3: the future value and the present value of a single sum. Every idea in this chapter is built from those two.

A choice between Tk 3,000 now and three payments of Tk 1,000 cannot be settled by adding the payments.

The same shape appears in saving: equal deposits grow, because each one earns interest for a different length of time.

Single sums can answer both questions, but the labour grows with every extra payment.

A friend offers you a choice. You may take Tk 3,000 today, or you may receive Tk 1,000 at the end of each of the next three years. Which is the better offer? Many people notice that both offers total Tk 3,000 and conclude that the two are equal. They are not equal. Money received today can be deposited and can earn interest; money received next year cannot do so until next year. The Tk 3,000 offer is therefore worth more. But how much more? To answer, you must give each Tk 1,000 payment its own value today.

Now reverse the picture. Suppose that you deposit Tk 1,000 at the end of each of the next three years in a bank that pays 10% a year. How much will you hold at the end of the third year? The answer is not Tk 3,000. The first deposit earns interest for two years, the second for one year, and the last for no time at all. Each deposit is a single sum with its own length of time, and Chapter 4 showed how to treat a single sum.¹

You could answer both questions with the tools of Chapter 4. You would treat each payment as a separate single sum and then total the results. For three payments this is tolerable. For a loan repaid over twenty years, or a pension paid over thirty, it would fill pages, and one slip would spoil every line below it. Yet series of equal payments are everywhere in business: rent, loan instalments, savings plans, pensions. This chapter gives such a series a name and a shortcut. It also asks you to see time in a new way, as a row of equal steps with a payment placed on some of them. Go slowly; each idea is short, and each rests on the one before.

* * *

5.1 WHAT AN ANNUITY IS

GO SLOWLY HERE.

Rent, loan instalments and savings plans share one shape: equal payments at equal intervals.

The word annuity names any series of equal payments made at equal intervals for a set number of periods.

Annuity. A series of equal payments made at equal time intervals for a set number of periods.

Annuity. Word in use. In daily life, a yearly sum paid to a person, often as a pension. In finance, any series of equal payments for a set number of periods.

Begin with three ordinary households. One family pays Tk 8,000 rent on the last day of every month. A student deposits Tk 500 on the first day of every month. A father repays a loan with Tk 4,000 each month for two years. The purposes differ, but the payments share three features. Every payment has the same size. The same length of time separates each payment from the next. And the series stops after a known number of payments. Finance gives this shape a name.

That name is **annuity**. An annuity is a series of equal payments made at equal time intervals for a set number of periods. Three deposits of Tk 1,000, made one at the end of each of three successive years, form an annuity. So does the monthly loan repayment above. A period is the length of the interval between payments: a month, a quarter or a year. In this chapter the period is one year unless we say otherwise, and the interest rate is always stated for that same period.

¹ Chapter 4, 'The Time Value of Money: Single Sums', section 4.2 'Future Value of a Single Sum' and section 4.3 'Present Value of a Single Sum'.

The timing of the first payment needs care. Does it fall at the end of the first period or at its start? The answer matters, because a payment made one period earlier has one more period in which to earn interest. Finance therefore separates two kinds of annuity.

An **ordinary annuity** is an annuity whose payments fall at the end of each period. An **annuity due** is an annuity whose payments fall at the start of each period. Take three payments of Tk 1,000 over three years. In an ordinary annuity they are made at the end of year 1, year 2 and year 3. In an annuity due they are made at the start of year 1, year 2 and year 3. The start of year 1 is today, and the start of year 2 is the end of year 1. Loan instalments and bank interest are usually ordinary annuities, because they are paid after the period has run. Rent is often an annuity due, because landlords ask for it in advance. When no kind is stated, take the annuity to be ordinary.

The figure below shows both kinds on a timeline. Read each line from left to right. The ticks on a line are moments in time, numbered 0, 1, 2 and 3, where 0 is today and each later number is the end of a year. Each block is one payment of Tk 1,000. On the upper line the blocks sit at the ends of the years, at 1, 2 and 3. On the lower line they sit at the starts, at 0, 1 and 2. The lower blocks are shifted one step to the left, and that shift is the whole difference between the two kinds.

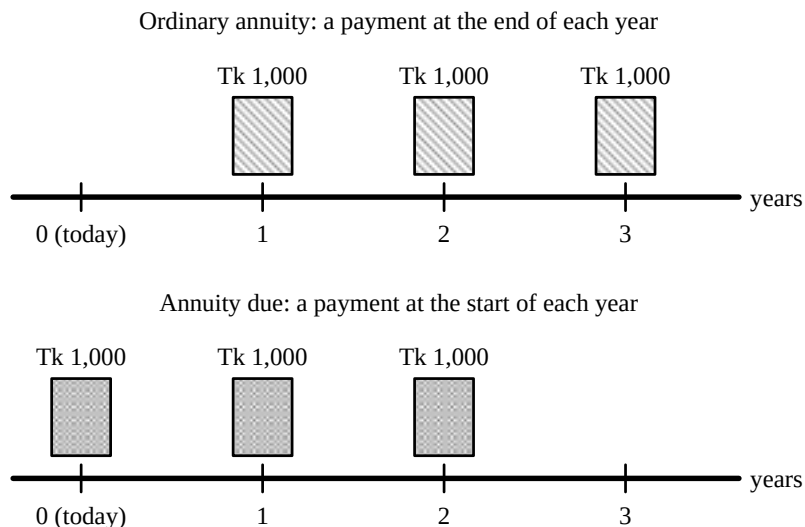


FIG. 5.1. Two timelines of three yearly payments: at the end of each year (ordinary annuity) and at the start of each year (annuity due)

Four viewpoints help here. The saver sees an annuity as a way to grow a sum by regular deposits. The borrower sees it as a series of instalments that must be met on time. A landlord, as owner of the property, sees it as income. The financial manager sees it as a pattern whose worth must be measured, so that rent, loans and savings can be compared fairly.

One case is unusual but useful. Suppose that the payments never stop. A series of equal payments with no final date is a **perpetuity**. A perpetuity is an annuity that never ends. A fund that pays a scholarship of Tk 50,000 every year for ever is an example. It may seem impossible to value an endless series, since the payments continue

Whether a payment falls at the start or at the end of a period changes its value.

An ordinary annuity pays at period ends; an annuity due pays at period starts.

Ordinary annuity. An annuity whose payments fall at the end of each period.

Annuity due. An annuity whose payments fall at the start of each period.

Due. Word in use. In daily life, owed or expected on a date, as in "the rent is due on Friday". In finance, the word "due" in annuity due means payment at the start of each period.

The figure reads from left to right: ticks on the line are moments in time, and blocks are payments.

To the saver an annuity grows a sum; to the borrower it is a burden; to the financial manager it is a pattern to value.

A perpetuity is an annuity with no final payment.

Perpetuity. An annuity that never ends.

Recap: an annuity has equal payments at equal intervals; ordinary means period ends, due means period starts.

GO SLOWLY HERE.

The future value of an annuity is the total of every payment grown to the date of the last payment.

Worked example 1 treats each deposit as a single sum and totals the results.

The multipliers 1.21, 1.10 and 1.00 depend only on the interest rate and the number of payments, so they can be totalled once and reused.

The formula multiplies the payment by one factor that depends on the interest rate and the number of payments.

The case file supplies realistic figures: Tk 60,000 a year for 5 years at 8%.

without limit. Section 5.3 shows that its present value is nevertheless a fixed, finite sum.

Before moving on. An annuity is a series of equal payments at equal intervals for a set number of periods. It is ordinary when the payments fall at period ends and due when they fall at period starts. A perpetuity never ends. *You should now be able to say where each payment of an ordinary annuity and of an annuity due falls on a timeline.*

* * *

5.2 FUTURE VALUE OF AN ANNUITY

The future value of an ordinary annuity is the sum that the payments reach on the date of the last payment. Each payment is first grown at compound interest to that date, and the grown amounts are then totalled. We begin with the smallest numbers, so that the method is the only new thing to learn.²

Given. Three deposits of Tk 1,000, one at the end of each year for three years; interest rate 10% a year. *Find.* The sum held at the end of year 3. *Formula.* Future value of a single sum = amount $\times$ (1 + interest rate) to the power of the number of years the amount earns interest. *Steps.* The first deposit earns interest for 2 years: Tk 1,000 $\times$ 1.10² = Tk 1,000 $\times$ 1.21 = Tk 1,210. The second earns for 1 year: Tk 1,000 $\times$ 1.10 = Tk 1,100. The third earns for no years: Tk 1,000. The total is Tk 1,210 + Tk 1,100 + Tk 1,000 = Tk 3,310. *Answer.* The deposits reach Tk 3,310 at the end of year 3. That is Tk 310 more than the Tk 3,000 deposited, and the extra Tk 310 is interest. *Check.* Interest on the first deposit is Tk 210 and on the second Tk 100, and Tk 210 + Tk 100 = Tk 310. The figures agree.

Look at the multipliers in the steps: 1.21, 1.10 and 1.00. Their total is 3.31, and 3.31 $\times$ Tk 1,000 = Tk 3,310. This total does not depend on the size of the payment. It depends only on the interest rate and the number of payments. We may therefore calculate it once and reuse it for any payment. It is called the future value factor of an ordinary annuity. For long series, a formula gives it in one step.

The rule, in words first: the future value of an ordinary annuity equals the payment multiplied by the future value factor. The factor equals (1 + interest rate) to the power of the number of payments, less 1, all divided by the interest rate. In symbols: $FVA = P \times [(1 + i)^n - 1] \div i$ Here FVA stands for the future value of the annuity, P for the payment made each period, i for the interest rate per period written as a decimal, and n for the number of payments. The expression in square brackets is the factor. It comes from totalling the multipliers, each of which is the one before it multiplied by 1 + i. For our example, $(1.10^3 - 1) \div 0.10 = (1.331 - 1) \div 0.10 = 0.331 \div 0.10 = 3.31$, the same total as before.

PAUSE AND CHECK 5.P1 Using the formula, find the future value factor for 10% and 4 years. Then find the future value of four deposits of Tk 1,000, one at the end of each year.

Case file, illustrative. Meghna and Sons, 15 August 2024. All calculations in this chapter's case use an interest rate of 8% a year. Tanvir deposits Tk 60,000 at the end of each of 5 years. The future value of this ordinary annuity is Tk 351,996.06, using the factor 5.866601.

² Recall: future value and compound interest, Chapter 4, 'The Time Value of Money: Single Sums', section 4.2 'Future Value of a Single Sum'.

Given. Tk 60,000 deposited at the end of each of 5 years; interest rate 8% a year. *Find.* The sum held at the end of year 5. *Formula.* $FVA = P \times [(1 + i)^n - 1] \div i$, with $P = \text{Tk } 60,000$, $i = 0.08$ and $n = 5$. *Steps.* First the power: $1.08^5 = 1.46932808$. Next the factor: $(1.46932808 - 1) \div 0.08 = 0.46932808 \div 0.08 = 5.866601$. Then the sum: $\text{Tk } 60,000 \times 5.866601 = \text{Tk } 351,996.06$. *Answer.* The deposits reach Tk 351,996.06 at the end of year 5. *Check.* The deposits total $5 \times \text{Tk } 60,000 = \text{Tk } 300,000$. The answer exceeds this by Tk 51,996.06, which is the interest earned. The table below confirms the figure year by year.

To see the factor at work, follow the balance through the five years. Each year the opening balance earns 8% interest, and then the deposit of Tk 60,000 arrives at the year end. Read each row from left to right: opening balance plus interest plus deposit gives the closing balance.

BALANCE OF THE DEPOSITS, YEAR BY YEAR (TK)

YEAR	OPENING BALANCE	INTEREST AT 8%	DEPOSIT AT YEAR END	CLOSING BALANCE
1	0.00	0.00	60,000.00	60,000.00
2	60,000.00	4,800.00	60,000.00	124,800.00
3	124,800.00	9,984.00	60,000.00	194,784.00
4	194,784.00	15,582.72	60,000.00	270,366.72
5	270,366.72	21,629.34	60,000.00	351,996.06

The last closing balance equals the answer from the formula, so the shortcut and the long method agree. The next figure draws the closing balances as bars. Each bar stands for the balance at the end of one year, and its height shows money in taka. The lower, line-hatched part of a bar is the total deposited so far; the upper, cross-hatched part is interest earned. Notice that the interest part grows faster each year, because interest is earned on earlier interest.

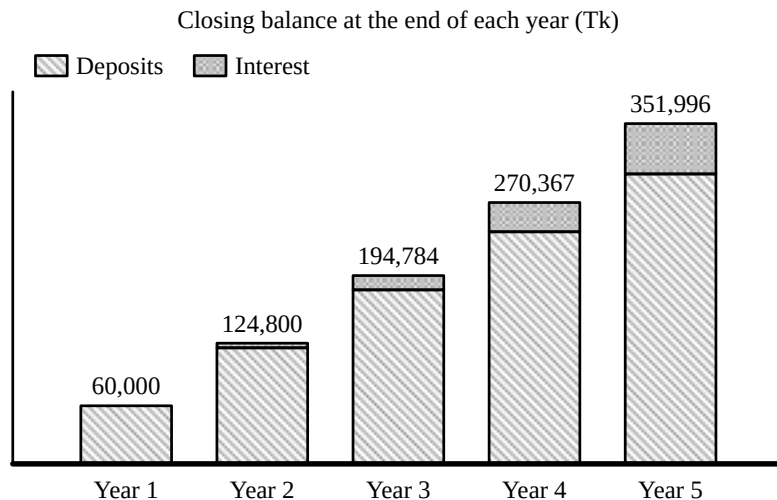


FIG. 5.2. Closing balance of the savings plan at the end of each year, split into deposits and interest

Now consider the annuity due. If each deposit is made one year earlier, it earns one more year of interest. Every deposit is therefore worth $(1 + i)$ times as much at the end. So the future value of an annuity due equals the future value of the ordinary annuity multiplied by $(1 + i)$. With the smallest numbers: Tk 3,310 ×

Worked example 2 repeats the method with the case figures, in the standard order from Given to Check.

The table follows the balance year by year: opening balance, interest, deposit, closing balance.

The last closing balance matches the formula, and the figure splits each balance into deposits and interest.

An annuity due earns one extra period of interest on every deposit, so its value is the ordinary value times $1 + i$.

The case file sets the annuity due beside the ordinary annuity.

Multiplying by 1.08 reproduces the case figure, and the difference equals one year of interest on the ordinary value.

Recap: future value of an ordinary annuity is the payment times a factor; for an annuity due, multiply by $1 + i$.

$1.10 = \text{Tk } 3,641$. Check by deposit: $\text{Tk } 1,000 \times (1.10^3 + 1.10^2 + 1.10) = \text{Tk } 1,331 + \text{Tk } 1,210 + \text{Tk } 1,100 = \text{Tk } 3,641$.

Case file, illustrative. Meghna and Sons, 15 August 2024. If the same Tk 60,000 is deposited at the start of each of the 5 years, at 8% a year, the future value is Tk 380,155.74.

The calculation is $\text{Tk } 351,996.06 \times 1.08 = \text{Tk } 380,155.74$. The check is the difference: $\text{Tk } 380,155.74 - \text{Tk } 351,996.06 = \text{Tk } 28,159.68$, which is 8% of Tk 351,996.06. In other words, depositing at the start of each year earns one extra year of interest on the final sum of the ordinary annuity.

PAUSE AND CHECK 5.P2 Find the future value of three deposits of Tk 1,000 at 10% a year when each deposit is made at the start of the year.

Before moving on. The future value of an ordinary annuity is the payment times the factor $[(1 + i)^n - 1] \div i$. For an annuity due, multiply that result by $1 + i$. *You should now be able to say why the annuity due is worth more at the end: each payment earns interest for one extra period.*

* * *

5.3 PRESENT VALUE OF AN ANNUITY

GO SLOWLY HERE.

The present value of an annuity is the total of every payment brought back to today.

Worked example 3 discounts each of the three payments in the puzzle at 10% and totals the results.

The puzzle is answered: at 10%, Tk 3,000 today is worth more than three payments of Tk 1,000.

The present value formula gives the factor for all payments in one step.

The case file offers four yearly rent payments of Tk 120,000, to be valued at 8%.

Return to the opening puzzle. The offer of three payments of Tk 1,000 has a value today, and we can now find it. The present value of an annuity is the sum today that is worth the same as all the payments together. Each payment is discounted back to today, and the discounted amounts are then totalled.³

Given. Three payments of Tk 1,000, at the end of years 1, 2 and 3; discount rate 10% a year. *Find.* The present value of the three payments. *Formula.* Present value of a single sum = amount $\div (1 + \text{discount rate})$ to the power of the number of years until the amount is received. *Steps.* Payment 1: $\text{Tk } 1,000 \div 1.10 = \text{Tk } 909.09$. Payment 2: $\text{Tk } 1,000 \div 1.21 = \text{Tk } 826.45$. Payment 3: $\text{Tk } 1,000 \div 1.331 = \text{Tk } 751.31$. The total is $\text{Tk } 909.09 + \text{Tk } 826.45 + \text{Tk } 751.31 = \text{Tk } 2,486.85$. *Answer.* The present value of the three payments is Tk 2,486.85. *Check.* A present value must be less than the Tk 3,000 received in total, and Tk 2,486.85 is. A second check: the future value Tk 3,310 from worked example 1, divided by 1.331, also gives Tk 2,486.85.

So the puzzle is answered. At 10% a year the three payments are worth Tk 2,486.85 today, and the offer of Tk 3,000 today is better by Tk 513.15. The payments are worth less than their total because each one arrives after a delay.

The rule, in words first: the present value of an ordinary annuity equals the payment multiplied by the present value factor. The factor equals 1 less $(1 + \text{interest rate})$ to the power of minus the number of payments, all divided by the interest rate. In symbols: $\text{PVA} = P \times [1 - (1 + i)^{-n}] \div i$ PVA stands for the present value of the annuity; P, i and n mean what they meant in section 5.2. The power of minus n is a short way to write division by $1 + i$, repeated n times. For worked example 3, $(1 - 1.10^{-3}) \div 0.10 = (1 - 0.751315) \div 0.10 = 0.248685 \div 0.10 = 2.48685$, and $2.48685 \times \text{Tk } 1,000 = \text{Tk } 2,486.85$.

Case file, illustrative. Meghna and Sons, 15 August 2024. The landlord offers 4 yearly rent payments of Tk 120,000, each at year end. The interest rate is 8% a year. The present value is Tk 397,455.22, using the factor 3.312127.

³ Recall: present value and discount rate, Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'. Here the discount rate is the interest rate that money could earn on a deposit.

Given. Tk 120,000 paid at the end of each of 4 years; discount rate 8% a year. Find. The present value of the four payments. Formula. $PVA = P \times [1 - (1 + i)^{-n}] \div i$, with $P = \text{Tk } 120,000$, $i = 0.08$ and $n = 4$. Steps. First the power: $1.08^{-4} = 0.73502985$. Next the factor: $(1 - 0.73502985) \div 0.08 = 0.26497015 \div 0.08 = 3.312127$. Then the value: $\text{Tk } 120,000 \times 3.312127 = \text{Tk } 397,455.24$. The case file gives Tk 397,455.22, because the factor shown is rounded to six places; the unrounded factor gives the case figure. Answer. The four payments are worth Tk 397,455.22 today, although they total Tk 480,000. Check. The table below discounts each payment alone, as in worked example 3.

Read each row from left to right: the payment, multiplied by the discount multiplier for that year, gives its present value. The multipliers are shown to six places, and the present values are computed from unrounded multipliers.

PRESENT VALUE OF EACH RENT PAYMENT AT 8% (TK)

YEAR	PAYMENT	DISCOUNT MULTIPLIER	PRESENT VALUE
1	120,000	0.925926	111,111.11
2	120,000	0.857339	102,880.66
3	120,000	0.793832	95,259.87
4	120,000	0.735030	88,203.58
Total	480,000		397,455.22

Before calculators, people did not calculate powers each time. They read the multiplier from a **factor table**. A factor table is a printed table of ready-made multipliers for given rates and numbers of periods. A reader finds the row for the number of periods, moves across to the column for the factor needed, and multiplies that factor by the payment. The table below lists the two annuity factors for 8% a year.

ANNUITY FACTORS FOR AN INTEREST RATE OF 8% A YEAR

NUMBER OF PAYMENTS	FUTURE VALUE FACTOR	PRESENT VALUE FACTOR
1	1.000000	0.925926
2	2.080000	1.783265
3	3.246400	2.577097
4	4.506112	3.312127
5	5.866601	3.992710

Row 5 of the first column holds 5.866601, the factor used for the savings plan in section 5.2. Row 4 of the second column holds 3.312127, the factor used for the rent offer. The same method serves for any rate, provided that the table lists it.

The present value of an annuity due follows the same reasoning as before. Each payment comes one period sooner, so it is discounted one period less and is worth $(1 + i)$ times as much. Therefore the present value of an annuity due equals the present value of the ordinary annuity multiplied by $(1 + i)$. For three payments of Tk 1,000 at 10%: $\text{Tk } 2,486.85 \times 1.10 = \text{Tk } 2,735.54$. Check by payment: the first payment, made today, is worth Tk 1,000, and the others are worth Tk 909.09 and Tk 826.45, which total Tk 2,735.54.

Now return to the perpetuity of section 5.1. The payments never stop, yet the present value is finite, because payments in the distant future are discounted to almost nothing. The rule is short: the present value of a perpetuity equals the payment divided by the interest rate. In symbols, $PV = P \div i$.

Worked example 4 values the landlord's offer in the standard order.

The table discounts each rent payment separately; the present values total the answer.

Printed tables store factors, so that no powers need to be calculated.

Factor table. A printed table of ready-made multipliers for given rates and numbers of periods.

Two cells of the table are the factors used in the case file.

For an annuity due, each payment is one period closer, so the present value is the ordinary value times $1 + i$.

A perpetuity is valued by one division, because a fund that pays out only its interest lasts for ever.

The case file gives a scholarship fund that pays Tk 50,000 for ever.

The division reproduces the case figure, and a check shows why the fund never runs out.

Four slips cause most errors: the wrong factor, mixed periods, wrong timing and early rounding.

Recap: present value is the payment times a factor; for a due multiply by $1 + i$; for a perpetuity divide by the interest rate.

Sometimes the payment is the unknown, and the same factors answer by division.

Sinking fund. Money put aside in equal deposits to reach a target sum by a set date.

A loan repaid in equal instalments is an annuity whose present value is the amount borrowed.

Amortisation. Repaying a loan by equal payments that cover the interest and part of the amount borrowed.

Worked example 5 uses a small loan with illustrative figures that are not part of the case file.

Each year the interest share of the instalment falls and the loan share rises, until the balance is zero.

Case file, illustrative. Meghna and Sons, 15 August 2024. A scholarship fund pays Tk 50,000 a year for ever. The interest rate is 8% a year. The present value is Tk 625,000.

The calculation is $\text{Tk } 50,000 \div 0.08 = \text{Tk } 625,000$. The check: a fund of Tk 625,000 earns $8\% \times \text{Tk } 625,000 = \text{Tk } 50,000$ each year. If the fund pays out only that interest, the fund itself stays whole, and the payments can continue for ever. The rule has limits. It holds only while the interest rate stays at 8%, which no one can promise for ever. It also assumes that the first payment comes at the end of the first period; if the first payment is made today, add one payment to the result.

Common slips are few. Using the future value factor when a present value is wanted gives a sum that grows instead of shrinks. Mixing periods, such as monthly payments with a yearly interest rate, spoils every figure. Treating an annuity due as ordinary forgets the multiplication by $1 + i$. Rounding a factor to two or three places loses whole taka on large payments, so keep six places.

Before moving on. The present value of an ordinary annuity is the payment times the factor $[1 - (1 + i)^{-n}] \div i$. For an annuity due, multiply by $1 + i$. For a perpetuity, divide the payment by the interest rate. A factor table stores the factors ready made. *You should now be able to say why a series of payments is worth less today than its total.*

* * *

5.4 FINDING THE PAYMENT

So far the payment was given and the value was unknown. Often the position is reversed. A business knows the sum it needs, or the sum it has borrowed, and wishes to find the payment. The same two factors serve, but we divide instead of multiplying. The first case is a **sinking fund**. A sinking fund is money put aside in equal deposits to reach a target sum by a set date. The rule is: $\text{deposit} = \text{target sum} \div \text{future value factor}$. The worked example that follows solves the van problem of the case file in full.

The second case is a loan. The lender receives equal payments, usually at period ends, so the lender holds an ordinary annuity whose present value is the amount borrowed. **Amortisation** means repaying a loan by equal payments that cover the interest and part of the amount borrowed. The rule is: $\text{instalment} = \text{amount borrowed} \div \text{present value factor}$. Each instalment pays that year's interest first, and the rest repays part of the loan.

Given. A loan of Tk 100,000, repaid by 3 equal year-end instalments; interest rate 10% a year. *Find.* The yearly instalment. *Formula.* $\text{Instalment} = \text{amount borrowed} \div \text{present value factor}$. The factor for 10% and 3 payments is 2.486852. *Steps.* $\text{Tk } 100,000 \div 2.486852 = \text{Tk } 40,211.48$. *Answer.* The yearly instalment is Tk 40,211.48. *Check.* The table below shows the loan falling to zero after the third instalment.

Read each row from left to right. The opening balance earns 10% interest. The rest of the instalment, Tk 40,211.48 less the interest, repays part of the loan, and the closing balance is what remains.

AMORTISATION OF A LOAN OF TK 100,000 AT 10% (TK)

YEAR	OPENING BALANCE	INTEREST	PART OF LOAN REPAID	CLOSING BALANCE
1	100,000.00	10,000.00	30,211.48	69,788.52
2	69,788.52	6,978.85	33,232.63	36,555.89
3	36,555.89	3,655.59	36,555.89	0.00

Before moving on. To find an unknown payment, divide the known sum by the right factor: the target by the future value factor for a sinking fund, the loan by the present value factor for amortisation. *You should now be able to say which factor belongs to saving and which to borrowing.*

* * *

Recap: a target sum divided by the future value factor gives the deposit; a loan divided by the present value factor gives the instalment.

WORKED EXAMPLE: SAVING FOR A VAN

Case file, illustrative. Meghna and Sons, 15 August 2024. A van replacement needs Tk 500,000 in 5 years. The interest rate is 8% a year, and the deposits are equal and made at each year end. The equal year-end deposit is Tk 85,228.23.

Given. Target Tk 500,000 at the end of year 5; equal ordinary deposits; interest rate 8% a year; $n = 5$. *Find.* The equal year-end deposit P . *Formula.* Deposit = target $\div$ future value factor, so $P = FVA \div \{(1 + i)^n - 1\} \div i$. *Steps.* The factor for 8% and 5 payments is 5.866601 (section 5.2). Then $P = Tk\ 500,000 \div 5.866601 = Tk\ 85,228.23$. *Answer.* Meghna and Sons must deposit Tk 85,228.23 at the end of each of 5 years. *Check.* Reverse the step: $Tk\ 85,228.23 \times 5.866601 = Tk\ 500,000.02$. The two poisha of difference arise because the deposit was rounded to the nearest poisha.

Compare the parts. Five deposits total $5 \times Tk\ 85,228.23 = Tk\ 426,141.15$. Interest therefore supplies about Tk 73,859 of the Tk 500,000, which is about 15%. A business that saves over a longer period needs smaller deposits, because interest has more time to work.

* * *

The van problem asks for an unknown payment and then checks it by future value.

Worked example 6 divides to find the deposit and then reverses the step as a check.

Deposits supply most of the target and interest supplies the rest.

SUMMARY

An annuity is a series of equal payments at equal intervals for a set number of periods. It is ordinary when payments fall at period ends and due when they fall at period starts; a perpetuity never ends. The future value of an ordinary annuity is $P \times [(1 + i)^n - 1] \div i$, and its present value is $P \times [1 - (1 + i)^{-n}] \div i$. For an annuity due, multiply either result by $1 + i$. A perpetuity is worth $P \div i$. A target sum divided by the future value factor gives the deposit of a sinking fund, and a loan divided by the present value factor gives the instalment under amortisation.

An annuity is a series of equal payments; two factors value it, and division finds an unknown payment.

THINK IT THROUGH

5.T1 A supplier offers machine parts for Tk 400,000 today or Tk 90,000 at the end of each of the next 5 years. The interest rate is 10% a year, and the present value factor for 10% and 5 payments is 3.790787. Which offer costs less in present value, and why?

5.T2 Explain in plain words why an annuity due is worth more than an ordinary annuity with the same payment, interest rate and number of payments.

Multiplying the payment by the number of payments ignores the timing of the money.

A factor must match the question: future value factors grow money and present value factors shrink it.

The period of the payment and the period of the interest rate must be the same.

An annuity due needs the extra multiplication by $1 + i$, and a perpetuity rule needs a steady rate.

In 1693 Edmond Halley used records from Breslau to price life annuities.

Halley's table showed how many people survive to each age, and he used it to price annuities.

WHERE PEOPLE GO WRONG

Mistake: "Four payments of Tk 25,000 are worth Tk 100,000." That is their total, not their value today. Payments in the future are worth less than money now, so their present value is lower than the total.

Mistake: using the future value factor to value an offer today. The future value factor answers "what will the payments grow to?" The present value factor answers "what are the payments worth now?" Choose by the question asked.

Mistake: using a yearly interest rate with monthly payments. Either convert the rate to a monthly rate and count months, or use yearly payments. Mixed periods give wrong answers that look reasonable.

Mistake: treating a due as ordinary, or trusting $P \div i$ for any rate. The first loses one period of interest. The second holds only while the interest rate stays the same for ever.

IN HISTORY

Edmond Halley (1656–1742) is best known for predicting the return of the comet that now bears his name. In 1693 he also published, in the *Philosophical Transactions of the Royal Society of London*, a paper on the mortality of mankind. His data were the birth and death records of the city of Breslau, now Wrocław in Poland, which a pastor of the city, Caspar Neumann, had sent to the Royal Society.

From these records Halley made a table showing how many people of a group born in the same year would survive to each age. He then used the table to compute the price of life annuities, which pay a sum each year for as long as the buyer lives. In outline, the fair price is the present value of each yearly payment, weighted by the chance that the buyer is alive to receive it. The calculation joins two ideas of this chapter: equal payments over time, and discounting.

REVIEW QUESTIONS

5.1 Find the future value of Tk 12,000 deposited each year for 6 years at 9% a year, first as an ordinary annuity and then as an annuity due.

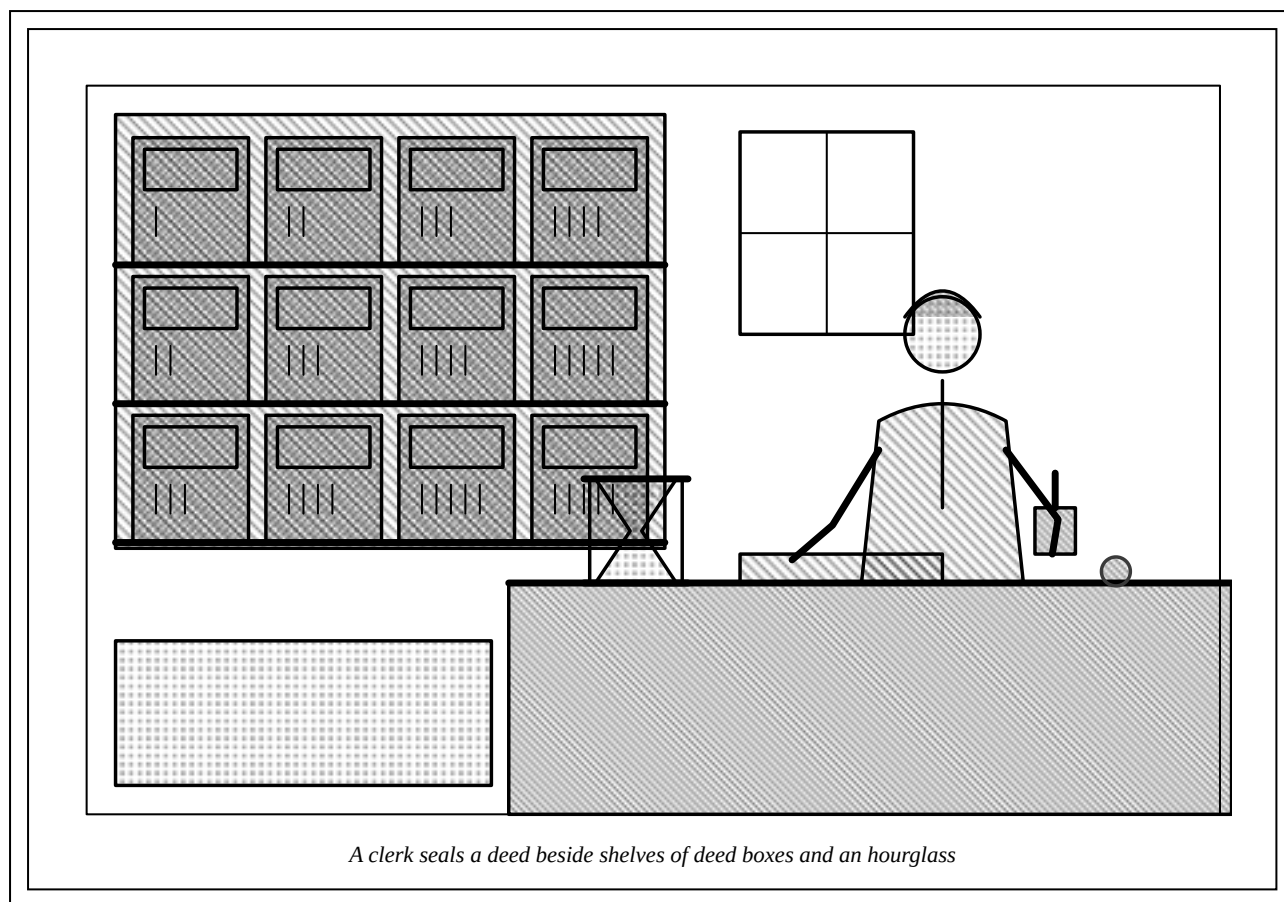
5.2 Find the present value of a yearly payment of Tk 45,000 for 5 years at 10% a year, paid at each year end.

5.3 Find the equal yearly deposit that reaches Tk 300,000 in 4 years at 7% a year, with deposits at each year end.

5.4 Find the level yearly payment that repays a loan of Tk 250,000 over 3 years at 12% a year.

5.5 Explain the difference between an ordinary annuity and an annuity due, and say which has the higher future value.

5.6 Find the present value of a perpetuity of Tk 20,000 a year at an interest rate of 5% a year, and explain why a fund of that size can pay for ever.



CHAPTER VI

Uneven Cash Flows

Can two plans that pay the same total be worth different amounts?



This chapter extends present value and future value to payments that change in size from year to year. It shows how to value such a series, how to split it into a level part and an odd part, and how to compare two series at a common rate.

YOU WILL NEED

Chapter 5, section 5.3: the present value of an annuity. Chapter 4, section 4.3: the present value of a single sum.

Two plans that pay the same total may still differ in value, because the order of the payments differs.

The shortcut for equal payments cannot be used here, because these payments are not equal.

The remedy is to return to single sums, and to learn a tidy way of working with them.

Two stalls at a book fair each promise you Tk 6,000 over three years, paid at the end of each year. The first stall pays Tk 1,000, then Tk 2,000, then Tk 3,000. The second pays the same amounts in the opposite order: Tk 3,000, then Tk 2,000, then Tk 1,000. Both plans total Tk 6,000. Are they worth the same to you? Most people feel that the second plan is better, because the largest payment arrives soonest. The feeling is correct, but a feeling is not a number. How much better is it?

Chapter 5 gave a fast way to value equal payments. That way cannot be used directly now, because the payments differ from year to year. Yet real receipts seldom stay equal. A shop's sales grow as customers learn of it. A machine earns less as it ages. A loan may be repaid in large amounts first and small amounts later. If we could value only equal payments, we could value very little of business life.

The remedy lies in Chapter 4. Each payment, taken alone, is a single sum received on one date, and a single sum can be discounted. Once every payment has been discounted, the results can be added. This chapter shows the method, shows where the shortcut of Chapter 5 still helps, and shows how to compare two series fairly.¹

* * *

6.1 WHAT A MIXED STREAM IS

Wages that rise, fees that change and takings that follow the seasons all have payments of different sizes.

A mixed stream is a series of payments or receipts that differ in size from one period to the next.

Mixed stream. A series of payments or receipts that differ in size from one period to the next.

Stream. Word in use. In daily life, a small river or a flow of water. In finance, a series of payments or receipts arriving over time.

A single sum, an annuity and a mixed stream differ in how many payments there are and whether they are equal.

Begin with everyday life. A worker's wage may rise by a fixed amount each year. School fees change when a child moves to a higher class. The takings of a shop are larger before a festival and smaller after it. In each case a series of money amounts arrives over time, and the amounts are not equal. Finance has a name for this shape.

That name is **mixed stream**. A mixed stream is a series of payments or receipts that differ in size from one period to the next. Receipts of Tk 1,000, Tk 2,000 and Tk 3,000 at the ends of three successive years form a mixed stream. A receipt is money coming in; a payment is money going out. The valuing method is the same for both, and in this chapter we use receipts. The word *mixed* tells us that the sizes are mixed, not equal.

Three shapes of cash flow now stand side by side. A single sum is one payment on one date. An annuity is several equal payments at equal intervals. A mixed stream is several payments whose sizes may differ. The table lists them. Read a row from left to right: the shape, the number of payments, the sizes, and the tool for valuing it.

¹ Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'; Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'.

THREE SHAPES OF CASH FLOW

SHAPE	NUMBER OF PAYMENTS	SIZES	TOOL FOR VALUING IT
Single sum	One	Not applicable	One discounting step
Annuity	Several	All equal	The annuity factor
Mixed stream	Several	May differ	Discount each, then total

Notice that the shapes are nested. An annuity is a mixed stream whose sizes happen to be equal, and a single sum is a mixed stream with one payment. The method for a mixed stream therefore works for every shape, although it is slower. The annuity factor is a shortcut for the equal case.

The shape appears from several viewpoints. The owner of a shop sees a mixed stream in yearly profit that grows as the business matures. The financial manager sees it in the cash that a new machine will bring. A lender sees it in a loan whose repayments are uneven. Each of them must turn the stream into one number before deciding anything.

Before moving on. A mixed stream is a series of payments or receipts that differ in size from one period to the next. A single sum and an annuity are special cases of it. Because the sizes differ, we value each payment alone and then total the results.

* * *

6.2 PRESENT AND FUTURE VALUE OF A MIXED STREAM

The method has a name. **Discounted cash flow** is a way of valuing a series of future cash flows by turning each one into a present value. The present values share one date, today, so they can be added. We begin with the smallest numbers, so that the method is the only new thing to learn.²

Given. Receipts of Tk 1,000, Tk 2,000 and Tk 3,000 at the ends of years 1, 2 and 3; discount rate 10% a year. *Find.* The present value of the stream. *Formula.* Present value of a single sum = amount $\div$ (1 + discount rate) to the power of the number of years until it is received. *Steps.* Year 1: Tk 1,000 $\div$ 1.10 = Tk 909.091. Year 2: Tk 2,000 $\div$ 1.21 = Tk 1,652.893. Year 3: Tk 3,000 $\div$ 1.331 = Tk 2,253.944. The total is Tk 909.091 + Tk 1,652.893 + Tk 2,253.944 = Tk 4,815.928, or Tk 4,815.93. *Answer.* The present value of the stream is Tk 4,815.93. *Check.* The present value must be less than the Tk 6,000 received in total, and it is.

PAUSE AND CHECK 6.P1 Value the opposite stream, Tk 3,000, Tk 2,000 and Tk 1,000 at the ends of years 1, 2 and 3, at 10%. Is it worth more or less than Tk 4,815.93, and why?

The rule, in words first: the present value of a mixed stream equals the total of each receipt divided by (1 + discount rate) to the power of its year. In symbols: $PV = C_1 \div (1 + r) + C_2 \div (1 + r)^2 + C_3 \div (1 + r)^3 + \dots$ Here PV stands for the present value, C_1 , C_2 and C_3 for the receipts of years 1, 2 and 3, and r for the discount rate per year, written as a decimal. The three dots mean that the pattern

A mixed stream is the general shape; the other two are special cases of it.

The owner, the manager and the lender each meet mixed streams in different places.

Recap: a mixed stream has payments of different sizes, and single sums and annuities are special cases.

Discounted cash flow turns each future amount into a present value and totals the results.

Discounted cash flow. A way of valuing a series of future cash flows by turning each one into a present value.

Worked example 1 discounts three uneven receipts at 10% and totals the present values.

The rule for a mixed stream: present value is the total of each receipt divided by its own growth multiplier.

² Recall: present value and discounting, Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'.

The future value of a mixed stream grows each receipt to the last date and totals the results.

The case file gives two stall plans, and their figures are valued at 12% in the tables.

The first table discounts every receipt of both plans at 12%; the totals are the present values.

The entries of B, rounded to the poisha, total two poisha less than the case figure, because of rounding.

The second table grows each receipt to year 5 at 12% and totals the results.

continues for as many years as the stream lasts. Each term is a single sum from Chapter 4, and the formula only totals them.

The future value of a mixed stream is the sum that all receipts reach on the date of the last receipt, when each has grown at compound interest. A receipt in year 1 grows for two years, one in year 2 grows for one year, and one in year 3 does not grow. In the clean example: Tk 1,000 $\times$ 1.21 = Tk 1,210; Tk 2,000 $\times$ 1.10 = Tk 2,200; and Tk 3,000 stays Tk 3,000. The future value is Tk 1,210 + Tk 2,200 + Tk 3,000 = Tk 6,410. Check: Tk 4,815.93 $\times$ 1.331 = Tk 6,410, because the future value equals the present value grown for three years.

Case file, illustrative. Meghna and Sons, 10 October 2024. Two book-fair stall plans, with receipts at year end for 5 years. Stream A: Tk 40,000; Tk 55,000; Tk 60,000; Tk 70,000; Tk 80,000. Stream B: Tk 80,000; Tk 70,000; Tk 60,000; Tk 55,000; Tk 40,000. Both total Tk 305,000. At 12% a year, the present value of A is Tk 212,147.18 and of B Tk 227,589.53. The future value at year 5 of A is Tk 373,875.81 and of B Tk 401,090.51.

Worked example 2 repeats the method with realistic figures. Read each row of the first table from left to right: the receipt, multiplied by the discount multiplier for its year, gives its present value. The multipliers are shown to six places, and the present values are calculated from unrounded multipliers.

PRESENT VALUE OF EACH RECEIPT AT 12% (TK)

YEAR	DISCOUNT MULTIPLIER	RECEIPT A	PRESENT VALUE A	RECEIPT B	PRESENT VALUE B
1	0.892857	40,000	35,714.29	80,000	71,428.57
2	0.797194	55,000	43,845.66	70,000	55,803.57
3	0.711780	60,000	42,706.81	60,000	42,706.81
4	0.635518	70,000	44,486.27	55,000	34,953.49
5	0.567427	80,000	45,394.15	40,000	22,697.07
Total		305,000	212,147.18	305,000	227,589.53

The present value of A is Tk 212,147.18, and of B Tk 227,589.53, as in the case file. The five rounded entries of B total Tk 227,589.51, two poisha less, because each entry was rounded separately; the unrounded sum is the case figure. A poisha is one hundredth of a taka.

For the future value, each receipt is grown to the end of year 5. A receipt of year 1 grows for four years, so its growth multiplier is $1.12^4 = 1.573519$. The multiplier for year 2 is $1.12^3 = 1.404928$, for year 3 it is $1.12^2 = 1.2544$, for year 4 it is 1.12, and year 5 has none.

FUTURE VALUE AT YEAR 5 OF EACH RECEIPT, AT 12% (TK)

YEAR	GROWTH MULTIPLIER	FUTURE VALUE A	FUTURE VALUE B
1	1.573519	62,940.77	125,881.55
2	1.404928	77,271.04	98,344.96
3	1.254400	75,264.00	75,264.00
4	1.120000	78,400.00	61,600.00
5	1.000000	80,000.00	40,000.00
Total		373,875.81	401,090.51

Check. The future value equals the present value grown for five years, and $1.12^5 = 1.762342$. For stream A, Tk 212,147.18 $\times$ 1.762342 is about Tk 373,876. For stream B, Tk 227,589.53 $\times$ 1.762342 is about Tk 401,091. Both agree with the tables to within the rounding of the multiplier.

Many streams contain a stretch of equal receipts with one or two odd amounts. Then the stream can be split into an annuity part and an odd part. The annuity part is valued with the annuity factor of Chapter 5, and the odd part as a single sum. The two present values are added. Chapter 5 gave the factor for 10% and 3 payments as 2.486852.

Given. Receipts of Tk 20,000, Tk 20,000 and Tk 30,000 at the ends of years 1, 2 and 3; discount rate 10% a year. *Find.* The present value of the stream. *Formula.* Present value = annuity payment $\times$ annuity factor + odd amount $\times$ discount multiplier of its year. *Steps.* The level part is Tk 20,000 in each of 3 years: Tk 20,000 $\times$ 2.486852 = Tk 49,737.04. The odd part is the extra Tk 10,000 in year 3: Tk 10,000 $\div$ 1.331 = Tk 7,513.15. The total is Tk 49,737.04 + Tk 7,513.15 = Tk 57,250.19. *Answer.* The present value of the stream is Tk 57,250.19. *Check.* Discount each receipt alone: Tk 18,181.82 + Tk 16,528.93 + Tk 22,539.44 = Tk 57,250.19. The two methods agree.

Three things fix the value of a stream: the size of each receipt, the date of each receipt, and the discount rate. Nothing else matters to the arithmetic. The name of the project, the person who pays and the total of the receipts do not enter the formula. This is why two plans that look very different can have the same present value, and two plans that look alike can differ. When you meet a new stream, list the receipts with their years first, as in the tables of this chapter, and only then begin to discount.

A table is the safest way to value a stream by hand. Write the years down the left side. Next to each year write the receipt, then the multiplier, then the product. The total of the last column is the present value. If a year has no receipt, write zero and keep the row, so that no year is skipped. Check the table by reading the multipliers down the column: they must fall steadily, because each later year has a smaller multiplier.

Before moving on. Discounted cash flow values a stream by turning each receipt into a present value and adding. The future value grows each receipt to the last date and adds. A level part may use the annuity factor. *Common slip:* using the wrong year for a receipt, so that it is discounted too much or too little.

* * *

6.3 COMPARING STREAMS WITH A COMMON RATE

A buyer who must choose between two streams needs one number for each. Totals do not serve, because they ignore timing. The fair way is to value both at the same date, usually today, and at the same discount rate. The stream with the higher present value is worth more. The same ranking appears if both are grown to the same later date, because every future value is the present value grown by the same multiplier.

The figure shows the two stall plans of the case file. Its horizontal axis lists the years, and each year has two bars. The line-hatched bar is stream A, which rises; the cross-hatched bar is stream B, which falls. The height of a bar is the receipt of that year, and the number above it gives the receipt in thousands of taka.

Both

streams

The check: a present value grown for five years gives the future value, to within a few poisha.

Streams that contain a level part can use the annuity factor for that part.

Worked example 3 splits a stream of Tk 20,000, Tk 20,000 and Tk 30,000 into a level part and an odd part.

A stream's value depends only on the sizes of the receipts, their dates and the discount rate.

A tidy table prevents most errors: one row for each year, one column for each quantity.

Recap: discount each receipt, or split the stream into an annuity part and an odd part, and total the present values.

Two streams can be compared fairly only at the same date and the same discount rate.

The figure draws the receipts of the two stall plans side by side, year by year.

total Tk 305,000, but the bars of B are taller at the left, where money is worth more.

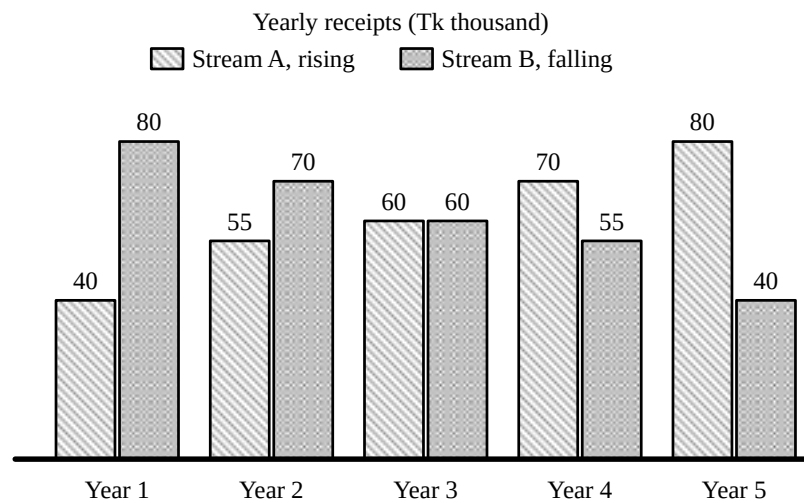


FIG. 6.1. Yearly receipts of two five-year streams: A rising, B falling, with the same total

The case file shows that B is worth more at 12%, although both streams total the same.

The gap of Tk 15,442.35 in present value is the price of receiving the large amounts later.

The ranking of two streams may depend on the discount rate, so it is wise to test more than one rate.

The ranking reverses because a higher rate punishes the distant payment of F more than the early payments of E.

To test a rate, repeat the whole comparison at a lower and a higher rate and watch whether the ranking changes.

Case file, illustrative. Meghna and Sons, 10 October 2024. At 12% a year, B is worth more than A because larger receipts come sooner. The present value of A is Tk 212,147.18 and of B Tk 227,589.53. The future value at year 5 of A is Tk 373,875.81 and of B Tk 401,090.51.

The difference in present value is Tk 227,589.53 – Tk 212,147.18 = Tk 15,442.35. The difference in future value is Tk 401,090.51 – Tk 373,875.81 = Tk 27,214.70. Neither gap comes from the totals, which are equal. Both come from timing alone. Stream B delivers Tk 80,000 in year 1, and that money can start earning interest at once.

A discount rate is rarely known with certainty, so a careful reader tests the ranking at more than one rate. Sometimes the ranking holds at every rate, as with A and B. Sometimes it reverses. Take stream E, which pays Tk 30,000 at the end of each of 5 years, and stream F, which pays Tk 200,000 once, at the end of year 5. At 5% a year, E is worth Tk 30,000 × 4.329477 = Tk 129,884, using the annuity factor of Chapter 5, and F is worth Tk 200,000 × 0.783526 = Tk 156,705. At 15% a year, E is worth Tk 30,000 × 3.352155 = Tk 100,565 and F is worth Tk 200,000 × 0.497177 = Tk 99,435. At 5% F wins; at 15% E wins.

The reason is plain. A higher discount rate shrinks distant money faster than near money. The single large payment of F lies five years away, so it loses most. The yearly payments of E are nearer on average, so they lose less. The lesson is not to fear the reversal but to expect it: when two streams have different totals and different timing, test the rate. When they have the same total, as A and B do, the stream with earlier receipts wins at every positive rate.

The test has three steps. First choose a low rate and a high rate that bracket the likely one, for example 5% and 15%. Second, value each stream at both rates, using the same method each time. Third, compare the rankings. If the same stream wins at both rates, the decision is secure. If the winner changes, there is a rate between the two at which the streams are worth the same, and the decision depends on which side of that rate the true rate lies. A reader who meets such a case should say so plainly and name the rates, instead of hiding the dependence.

The viewpoint of the decision-maker also counts. An owner who must repay a loan next year may prefer the stream with early receipts even at a low rate, because the cash is needed on that date. A saver who will not need the money for ten years may care little about timing. The discount rate expresses these needs in one number, so choosing the rate is itself a decision, to be made with care and stated openly.

Before moving on. Compare streams by present value at one common rate. Test the ranking at several rates, because it can reverse when totals differ. With equal totals, the stream with the larger early receipts has the higher value.

* * *

WORKED EXAMPLE: TWO STREAMS AT THREE RATES

Here we compare a rising stream R of Tk 1,000, Tk 2,000 and Tk 3,000 with a falling stream F of Tk 3,000, Tk 2,000 and Tk 1,000, all at year ends. Both total Tk 6,000. The multipliers below are the discount multipliers for years 1, 2 and 3.

Given. Stream R: Tk 1,000; 2,000; 3,000. Stream F: Tk 3,000; 2,000; 1,000. Discount rates 9%, 12% and 15% a year. *Find.* The present value of each stream at each rate, and which is worth more. *Formula.* Present value = total of (receipt $\times$ discount multiplier of its year), where the multiplier is $1 \div (1 + \text{rate})$ to the power of the year. *Steps.* At 12%: $R = \text{Tk } 892.857 + \text{Tk } 1,594.388 + \text{Tk } 2,135.341 = \text{Tk } 4,622.59$. $F = \text{Tk } 2,678.571 + \text{Tk } 1,594.388 + \text{Tk } 711.780 = \text{Tk } 4,984.74$. The steps at 9% and 15% use the multipliers in the table. *Answer.* F is worth more than R at all three rates. The results are in the second table. *Check.* At a rate of 0% both streams are worth Tk 6,000, so the gap must vanish as the rate falls. The gap in the table shrinks as the rate falls, which agrees.

Read the first table by choosing a rate in the left column and moving right to the multiplier for each year.

DISCOUNT MULTIPLIERS

RATE	YEAR 1	YEAR 2	YEAR 3
9%	0.917431	0.841680	0.772183
12%	0.892857	0.797194	0.711780
15%	0.869565	0.756144	0.657516

Read the second table by rate. For each rate it gives the present value of R, of F, and the difference between them.

PRESENT VALUES OF THE TWO STREAMS (TK)

RATE	PRESENT VALUE OF R	PRESENT VALUE OF F	GAP (F - R)
9%	4,917.34	5,207.84	290.50
12%	4,622.59	4,984.74	362.15
15%	4,354.40	4,778.50	424.10

The table repeats the lesson of section 6.3. The falling stream wins at every rate, and the gap grows with the rate, because a higher rate makes early money more valuable compared with late money. From the viewpoint of an owner choosing between two plans with equal totals, the safe choice is the plan that pays sooner.

A viewpoint matters: an owner who needs cash early values early receipts more than the arithmetic alone shows.

Recap: compare streams at the same date and rate, test the rate, and expect the earlier-heavy stream to win at equal totals.

The worked example compares a rising and a falling stream, with equal totals, at 9%, 12% and 15%.

The steps follow the standard order from Given to Check, with the work at 12% shown in full.

The first table lists the discount multipliers for the three rates and three years.

The second table shows that the gap between the falling and rising streams widens as the rate rises.

Both patterns agree with the rule: earlier receipts matter more, and more so at higher rates.

* * *

A mixed stream is valued by discounting each receipt; streams are compared at a common rate.

SUMMARY

A mixed stream is a series of payments or receipts that differ in size from one period to the next. Discounted cash flow values it by turning each receipt into a present value and adding. The future value grows each receipt to the last date and adds. A level part of a stream may use the annuity factor, and the rest is valued as single sums. Streams are compared at one date and one discount rate; the ranking may change with the rate, so test more than one.

THINK IT THROUGH

6.T1 A supplier offers two payment plans for the same machine. Plan X pays Tk 90,000, Tk 60,000 and Tk 30,000 at the ends of years 1, 2 and 3. Plan Y pays Tk 30,000, Tk 60,000 and Tk 90,000 at the same dates. At 10% a year, which plan costs less in present value, and why?

6.T2 Explain in plain words why two streams with the same total can have different present values.

WHERE PEOPLE GO WRONG

The total of a stream is not its value, because the total ignores timing.

Each receipt needs the multiplier for its own year, not the multiplier of another year.

The annuity factor applies only to the level part of a stream, never to the whole stream.

A ranking found at one rate may not hold at another.

Mistake: "The two plans total the same, so they are worth the same." The total adds money from different dates as if the dates did not matter. Timing changes value, and the present value shows it.

Mistake: using one multiplier for all years. The multiplier for year 3 is smaller than that for year 1. Using one multiplier for every year treats the stream as if every receipt came on the same date.

Mistake: using the annuity factor on unequal receipts. The factor assumes equal payments. Split the stream into a level part and an odd part first, or discount each receipt alone.

Mistake: ranking streams at one rate and then trusting the result. Where totals differ, the ranking can reverse. Test a low rate and a high rate before deciding.

IN HISTORY

Benjamin Franklin's bequest of 1790 grew for two centuries with withdrawals at set dates.

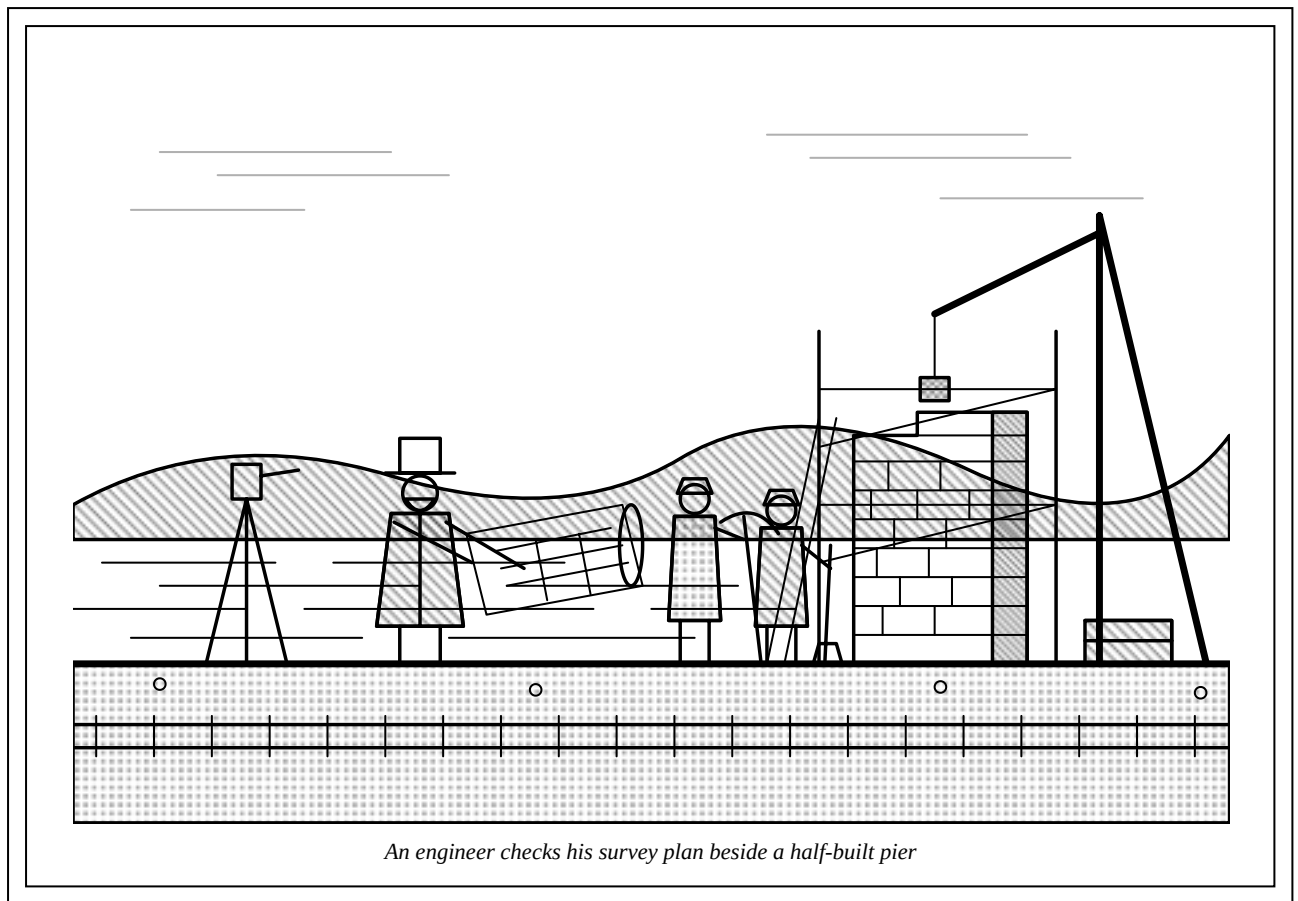
The bequest is a mixed stream: nothing for a century, then a withdrawal, then the remainder.

Benjamin Franklin (1706–1790) died in Philadelphia on 17 April 1790. In a codicil to his will he left 1,000 pounds each to Boston, where he was born, and to Philadelphia, where he lived most of his adult life. The money was to be lent to young married craftsmen who had served an apprenticeship. Part of it could be used after 100 years, and the rest after 200 years, in 1990.

Franklin's plan is a mixed stream seen from the owner's side. No money left the fund for a century, one part was then withdrawn, and the remainder was withdrawn two centuries after his death. Compound interest worked on the fund during all that time. Accounts of the sums reached in 1990 differ; they are usually given as a few million dollars for each city, with the Boston fund the larger.

REVIEW QUESTIONS

- 6.1 Find the present value of receipts of Tk 20,000, Tk 35,000 and Tk 50,000 at the ends of years 1, 2 and 3, at 10% a year.
- 6.2 Find the future value at the end of year 3 of the stream in question 6.1, at 10% a year.
- 6.3 A stream pays Tk 15,000 at the end of each of years 1 to 4 and Tk 40,000 at the end of year 5. Split it into an annuity part and an odd part, and find its present value at 10% a year. The annuity factor for 10% and 5 payments is 3.790787.
- 6.4 Compare the rising stream of question 6.1 with the falling stream Tk 50,000, Tk 35,000 and Tk 20,000, at 9% and at 15% a year. Which is worth more at each rate?
- 6.5 Explain how a mixed stream differs from an annuity, and say why the annuity factor cannot be used for a mixed stream as a whole.



CHAPTER VII

Capital Budgeting: Decisions and Cash Flows

Which purchase will still be paying for itself in five years?



Some decisions tie up a business's money for many years and are hard to undo. This chapter explains what such decisions are called, the steps a business follows to make them, and how to measure the cash that one project spends and receives. We follow one machine from its first payment to its last.

A rickshaw owner faces a choice that is hard to undo.

You will need: Chapter 2, section 2.2, the investment decision.

The dealer's promise leaves many questions open.

Three questions turn hope into a method.

This chapter measures the cash; the next chapter judges it.

Imagine a man in Dhaka who hires out three rickshaws by the day. He has saved Tk 90,000, and a dealer offers him a fourth rickshaw for exactly that sum. The dealer promises that the new rickshaw will repay its price within a year. Should the owner agree?

The owner should hesitate, and the reasons are worth listing. The rickshaw will not earn money on every day. Some days bring rain, some weeks bring repairs, and a rival may begin work on the same road. The Tk 90,000 leaves his hands today, but the money it earns returns slowly, over years. And once the rickshaw is bought, he cannot easily recover his money, because a used rickshaw sells for much less than a new one.

So the owner needs more than hope. He must ask which payments will really happen, and when. He must ask which of them exist only because he buys the rickshaw. And he must ask whether the money coming in is worth more than the money going out, remembering that money received later is worth less than money received today. You met that idea in Chapters 4 to 6. Each question has a method, and the method has a name.

This chapter teaches the first half of that method. It explains what a decision of this size is called, why it deserves careful work, and which steps a business follows. Then it shows how to measure the cash that a project spends and receives. Chapter 8, 'Capital Budgeting: Evaluation Techniques', shows how to judge whether those cash flows are good enough. Our running example is a print-on-demand machine for Meghna and Sons, the family bookshop in Dhaka. The shop and its figures are illustrative.

* * *

7.1 WHAT CAPITAL BUDGETING IS AND WHY IT MATTERS

Spending on onions and spending on a new room are different kinds of choice.

Businesses face the same difference between small and large choices.

Recall: the investment decision, Chapter 2, 'The Financial Manager's Work', section 2.2 'The Three Core Decisions'.

Capital. Word in use. In daily life, money or wealth that a person owns. In finance, the long-term funds and assets that a business uses to earn profit.

Capital budgeting. The process of planning, evaluating and choosing long-term investments.

Capital budgeting is the planning of long-term investments.

Three marks identify a capital budgeting decision.

Begin with a choice from home. A family spends Tk 3,000 a week on food. If it buys too many onions, the harm is small, because the onions are eaten or wasted within days. Now suppose the same family decides to build a second room for Tk 400,000. The room cannot be returned to the builder, and it will serve the family for thirty years. Both choices are spending, but they are not the same kind of choice. The second is large, slow to repay and hard to reverse.

A business meets the same difference. In Chapter 2 you learned that the investment decision is the choice of which assets the business should hold, and that the financial manager shares in that choice. Most investment decisions are small and repeat every week, such as restocking a shelf. A few are large and rare. The large ones need a method of their own, and the name of that method is capital budgeting.

Capital budgeting means the process of planning, evaluating and choosing long-term investments. When Meghna and Sons asks whether to buy a print-on-demand machine that will work for five years, the shop is doing capital budgeting. The word *budgeting* appears because the business plans in advance how much money to commit and what it expects to receive. The word *capital* needs care, as the side note explains: here it means long-term funds and the assets bought with them, not a person's savings.

How do you recognise a decision that belongs to capital budgeting? Look for three marks. First, the money is committed for a long time, usually many years.

Second, the amount is large compared with the size of the business. Third, the decision is hard to reverse, because a used machine sells for far less than a new one. At 1 January 2024, a year before the machine was considered, the assets of Meghna and Sons were Tk 2,400,000. A machine with an initial cost of Tk 900,000 is therefore more than a third of everything the shop owned. All three marks are present.

Notice that size is measured against the business, not in taka alone. A sum of Tk 900,000 is large for a bookshop but small for a national bank. The marks also work together. A cheap item that lasts for years is not a capital budgeting decision, because a mistake costs little. A large payment that the seller returns in full is easier to manage, because the mistake can be undone. The decisions that need the most care are those with all three marks.

The money paid out has a name too. **Capital expenditure** means spending on assets that will serve the business for many years. A machine, a delivery van and a new shop floor are capital expenditure. A month of electricity is not, because it is used up at once. Capital budgeting is the planning; capital expenditure is the payment.

Why spend time on weighing a proposal before accepting it? There are four reasons. The first is size: one mistake can use a large part of the business's funds. The second is length: the machine works for years, so a wrong decision is paid for in every one of those years. The third is that the asset is hard to sell, so the owner cannot escape a poor choice cheaply. The fourth is that funds spent on the machine cannot also be spent on the next proposal, so every decision closes other doors.

Careful work has limits. The future is uncertain, and forecasts will sometimes be wrong. What careful work does is to put every assumption on paper, so that others can question it and the business can check it later. A forecast that is written down can be improved; a hope that is never written down cannot.

Different people see the same decision differently. The owner asks whether the funds will earn more than they cost. The financial manager must prepare the forecasts and defend them. An employee may gain work when a machine arrives, or lose it if the decision goes wrong. A customer gains a new service if the decision is good, and may face higher prices if it is not.

Before moving on. Capital budgeting is the planning of long-term investments, and capital expenditure is the money paid. A decision belongs here when it is long, large for the business and hard to reverse. Careful appraisal matters because errors are costly and last for years.

* * *

Size is relative, and the marks work together.

Capital expenditure. Spending on assets that will serve the business for many years.

The money paid out on such assets has its own name.

Four reasons make careful appraisal worth the effort.

Careful work cannot remove error, but it exposes assumptions.

Each viewpoint sees the decision differently.

Recap: capital budgeting is planning for large, long and hard-to-reverse choices.

7.2 THE STEPS OF THE PROCESS

A builder who must cross a river does not begin by pouring concrete. She first asks where a bridge might go, then measures the ground, then chooses a design, then builds, and later inspects the work. A business facing a large choice also works in order. Following the same steps each time keeps the decision orderly and makes mistakes easier to find.

A large task becomes manageable when it is split into steps.

Project. Word in use. In daily life, a piece of work such as a school project. In capital budgeting, one proposed investment, such as a machine or a branch, treated as a single choice.

The capital budgeting process has five steps.

The figure shows the five steps in order.

The capital budgeting process has five steps, and the word *project* in them has the special meaning shown in the side note. Step one is to find proposals: ideas come from owners, staff, customers and changes in the market. Step two is to estimate the cash flows of each proposal, year by year. Step three is to evaluate the cash flows and choose. Step four is to implement the choice by buying, installing and training. Step five is to review the results by comparing real cash flows with the forecast.

Figure 7.1 shows the sequence. Read it from left to right. Each box is one step, and each solid arrow means that the next step begins when the last one ends. The dashed arrow below the boxes returns from the last step to the first: lessons from the review improve the next search for proposals. The labels under two boxes show where this book teaches that step.

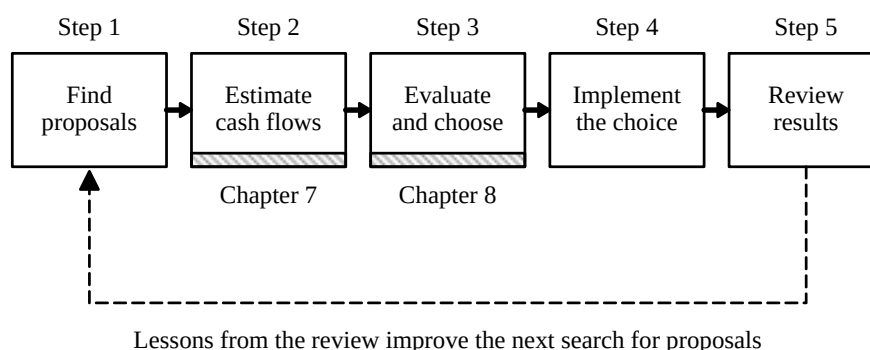


FIG. 7.1. *The five steps of the capital budgeting process*

Each step has an owner and a question.

The steps are a guide, not a rigid rule.

Case: two proposals compete for the same funds.

The case shows why every proposal needs measured cash flows.

Each step answers a question. Step one asks what we could do. Step two asks what money will move, and when. Step three asks whether the money is worth the cost of the funds; Chapter 8 teaches the tools. Step four asks how to do it well. Step five asks what we learn. The financial manager is most active in steps two, three and five, because they need figures and honest comparison.

The steps have exceptions. In a small business one person may perform several steps in a week, and the steps may overlap. A proposal may return to step two when new facts appear. Step two is usually the hardest, because it forecasts the future; an error there passes into every later step. This is why the rest of this chapter concentrates on it.

Case file, illustrative. 5 January 2025: Meghna and Sons holds two proposals. One is a print-on-demand machine with an initial investment of Tk 900,000. The other is a branch fit-out that needs Tk 900,000. The two cannot be run together. A feasibility survey for the machine, costing Tk 15,000, was paid in December 2024.

The case file shows why the rest of the process matters. Two proposals compete, and each needs a measured set of cash flows before any choice is made. Two new names describe how proposals relate to each other.

Independent projects are projects where accepting one has no effect on whether another is accepted. For example, the print-on-demand machine and a new set of shelves for the front room are independent: the shop may accept both, either or neither.

Mutually exclusive projects are projects where accepting one means rejecting the others. The machine and the branch fit-out are mutually exclusive, because the case file says they cannot be run together. Two clinics choosing between two scanners, when only one will be bought, face the same relation.

Before moving on. The process has five steps: find proposals, estimate cash flows, evaluate and choose, implement, and review. Projects are independent when one does not affect another, and mutually exclusive when choosing one rejects the rest.

* * *

7.3 CASH FLOW FROM AN INVESTMENT

Return to the rickshaw owner. He pays Tk 90,000 for the rickshaw today. Last month he paid Tk 2,000 for a mechanic's report about rickshaws, and that money cannot be recovered. He also pays Tk 500 a month for the garage where his three rickshaws stand, and he pays the same whether or not he buys a fourth. Which of these amounts should he count against the new rickshaw? The answer decides whether his forecast is honest.

RELEVANT CASH FLOW

A **relevant cash flow** is a cash flow that changes only because a project is accepted. If the garage rent stays at Tk 500 a month with or without the new rickshaw, it is not relevant. Three tests help. The item must be cash, meaning money that really moves, not an accounting entry. It must lie in the future, because money already spent cannot be recovered. And it must change if the project goes ahead; an item that would be the same either way is not relevant.

The second test needs a name. A sunk cost is money already spent that cannot be recovered, whatever the owner decides. Sunk costs do not change with the decision, so they are not relevant. Many students find this hard to accept, because the money feels lost if the project is rejected. But the money is lost in both cases, so it cannot help to choose between them.

Case file, illustrative. 5 January 2025: a feasibility survey for the print-on-demand machine, costing Tk 15,000, was paid in December 2024. It is a sunk cost and is not a relevant cash flow.

PAUSE AND CHECK 7.P1 A friend says the Tk 15,000 survey makes the machine less attractive, because the shop has already paid for it. Is the friend right? Answer in one or two sentences.

INITIAL INVESTMENT

Before a machine earns anything, someone must pay for it. The **initial investment** is the cash spent at the start to get a project going. It includes the price of the asset and everything needed to make it ready for work, such as delivery and installation. It also includes working capital where the project needs it. In this book, working capital means the money tied up in stock and unpaid customer bills that a project needs in order to run; Chapter 15, 'Working Capital and the Cash Cycle', teaches it in full. The

Independent projects. Projects where accepting one has no effect on whether another is accepted.

Independent projects do not affect each other.

Mutually exclusive projects.

Projects where accepting one means rejecting the others.

Mutually exclusive projects compete: one only.

Recap: five steps, and two kinds of relation between projects.

Measuring cash flow means deciding which money to count.

Relevant cash flow. Cash flow that changes only because a project is accepted.

Only cash that changes because of the project counts.

A cost already paid is not a reason for or against a project.

Case: the survey is sunk, so it is left out.

The first cash flow is what must be paid to start.

Initial investment. Cash spent at the start to get a project going.

Case: the initial investment adds three payments.

Operating cash flow is not the same as profit.

Cost. Word in use. In daily life, the price paid for something. Here, a cash cost is money paid out in a year to run the project; depreciation is also called a cost in accounts, but no cash leaves.

Depreciation. The part of an asset cost charged as an expense in each year of its use.

Depreciation spreads the cost of an asset over its years of use.

EBIT. Earnings before interest and tax: the profit from operations before paying lenders or the tax office.

EBIT is the profit from operations before interest and tax.

Operating cash flow. Cash a project produces each year from running it, after tax.

Operating cash flow is the cash a project makes each year after tax.

A second route gives the same answer, which is a useful check.

initial investment is paid at time zero, which means today, before year one begins.

Case file, illustrative. 5 January 2025: the print-on-demand machine for Meghna and Sons needs an initial investment of Tk 900,000. It is made up of the machine at Tk 780,000, installation at Tk 60,000 and working capital of Tk 60,000. The machine and installation together, Tk 840,000, are the cost of the asset.

OPERATING CASH FLOW

Once the machine works, it brings in cash each year. Take a tailor's machine that adds Tk 100,000 of revenue in a year, meaning extra money from customers, and costs Tk 40,000 in cash to run. Is the yearly cash flow Tk 60,000? Not quite, for two reasons: depreciation changes the tax, and the tax itself is a payment. We take them in turn, using small clean figures so that the method is the only thing to learn.

The first reason is **depreciation**. Depreciation means the part of an asset cost charged as an expense in each year of its use. A machine that costs Tk 100,000, works for five years and then has no resale value is charged at Tk 20,000 in each year. Charging the same amount each year is called the straight-line method. If a resale value is expected, subtract it first: the yearly charge is the cost, less the resale value, divided by the number of years. No cash leaves the business in those years, because the cash was paid at the start. But the charge lowers the profit on which tax is calculated, so it still changes the tax paid.

The second tool is **EBIT**, which stands for earnings before interest and tax. EBIT is the profit from operations before paying lenders or the tax office. For the tailor, EBIT equals revenue, less cash costs, less depreciation: $100,000 - 40,000 - 20,000 = \text{Tk } 40,000$. Some writers call EBIT operating profit, but this book uses EBIT from here on. Interest is left out on purpose. The cost of the funds is allowed for in the discount rate, which Chapter 8 teaches, so counting interest here as well would count the same cost twice.

The **operating cash flow** is the cash a project produces each year from running it, after tax. Calculate it in three moves. First find EBIT. Second subtract the tax on EBIT. Third add depreciation again, because depreciation was subtracted to reach EBIT but no cash was paid. For the tailor, with tax at 25%, the tax is $0.25 \times 40,000 = \text{Tk } 10,000$. Then $40,000 - 10,000 = 30,000$, and adding depreciation of 20,000 gives an operating cash flow of Tk 50,000.

IN SHORT

Operating cash flow = EBIT – tax on EBIT + depreciation. In symbols, $\text{OCF} = \text{EBIT} - (T \times \text{EBIT}) + D$, where OCF is operating cash flow, T is the tax rate and D is depreciation.

There is a second route to the same answer: revenue, less cash costs, less tax. For the tailor, $100,000 - 40,000 - 10,000 = \text{Tk } 50,000$. The two routes agree, and a good habit is to run both. If they differ, the error is usually a forgotten step. Tax rules also differ by country and change over time, and the rate of 25% used in the running case is assumed for illustration only; always check the current rules of the place concerned.

TERMINAL CASH FLOW

When the project ends, some cash may return. The **terminal cash flow** is cash received at the end of a project life, such as the resale value of equipment. It has two parts: the sale of the asset, and the working capital that comes back as stock is sold and customers pay what they owe. These amounts are not earned by operating, so they are kept apart from the operating cash flow.

Case file, illustrative. End of year 5 of the print-on-demand machine: resale Tk 120,000, equal to book value, so no tax on the sale; working capital of Tk 60,000 returned. Terminal cash flow Tk 180,000.

Here book value means the cost of the asset less the depreciation charged so far. If an asset sells for more than its book value, the gain is usually taxed. For example, a machine with a book value of Tk 120,000 that sells for Tk 140,000 has a gain of Tk 20,000, and tax at 25% is Tk 5,000, so the business keeps Tk 135,000. Rules for taxing such gains differ by country, so a student must always read the local rule.

Before moving on. A relevant cash flow changes only because the project is accepted; sunk costs do not count. The initial investment is paid at the start, the operating cash flow each year, and the terminal cash flow at the end. Operating cash flow equals EBIT, less tax, plus depreciation.

* * *

WORKED EXAMPLE: CASH FLOWS OF THE PRINT-ON-DEMAND MACHINE

The tailor's figures were small so that the method stood out. Now we use the same method on the machine for Meghna and Sons, in the standard order: Given, Find, Formula, Steps, Answer and Check.

Case file, illustrative. *Given.* 5 January 2025. Machine Tk 780,000; installation Tk 60,000; working capital Tk 60,000. Each year for five years: added revenue Tk 700,000; cash costs Tk 380,000; depreciation Tk 144,000, straight line over 5 years; tax at 25%, assumed for illustration only. End of year 5: resale Tk 120,000, equal to book value; working capital Tk 60,000 returned. The Tk 15,000 survey is a sunk cost.

Find. The net cash flow of each year from year 0 to year 5, counting relevant cash flows only. *Formula.* In words, net cash flow in a year equals the initial investment (year 0 only, shown as a negative number), plus the operating cash flow, plus the terminal cash flow (year 5 only). In symbols, $NCF = -I + OCF + TCF$, where NCF is net cash flow, I is the initial investment and TCF is terminal cash flow.

Steps. Step 1, the initial investment: $780,000 + 60,000 + 60,000 = \text{Tk } 900,000$. The survey of Tk 15,000 is left out, because it is sunk. Step 2, a check on depreciation: the asset cost is $780,000 + 60,000 = \text{Tk } 840,000$, so $(840,000 - 120,000) \div 5 = 720,000 \div 5 = \text{Tk } 144,000$, which matches the case. After five years the book value is $840,000 - (5 \times 144,000) = \text{Tk } 120,000$, equal to the resale value, so the sale brings no tax.

Step 3, the operating cash flow. $EBIT = 700,000 - 380,000 - 144,000 = \text{Tk } 176,000$. Tax = $0.25 \times 176,000 = \text{Tk } 44,000$. EBIT less tax = $176,000 - 44,000 = \text{Tk } 132,000$. Adding depreciation gives $132,000 + 144,000 = \text{Tk } 276,000$ each year. Step 4, the terminal cash flow: resale 120,000 + working capital 60,000 = Tk 180,000, so year 5 brings $276,000 + 180,000 = \text{Tk } 456,000$.

Terminal cash flow. Cash received at the end of a project life, such as the resale value of equipment.

At the end of its life a project may return cash.

Case: resale and working capital make up the terminal cash flow.

Book value decides whether a sale creates a taxable gain.

Recap: four kinds of cash flow, and which ones count.

The same method now meets realistic figures.

Given: the facts of the case, as recorded.

Find and Formula: the question and the rule.

Steps 1 and 2: the initial investment and depreciation.

Steps 3 and 4: operating and terminal cash flow.

Table 7.1 collects the results: read by column and by row.

Table 7.1 gathers the results. Read each column as one year, and each row as one kind of cash flow. A negative number is cash paid out; a positive number is cash received.

NET CASH FLOW OF THE MACHINE, IN TAKA

ITEM	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Initial investment	-900,000	0	0	0	0	0
Operating cash flow	0	276,000	276,000	276,000	276,000	276,000
Terminal cash flow	0	0	0	0	0	180,000
Net cash flow	-900,000	276,000	276,000	276,000	276,000	456,000

Answer: the shop pays Tk 900,000 now and receives Tk 276,000 a year, with Tk 456,000 in year 5.

Answer. The machine needs Tk 900,000 at the start, brings Tk 276,000 in each of years 1 to 4, and brings Tk 456,000 in year 5. *Check.* Run the second route for operating cash flow: $700,000 - 380,000 - 44,000 = \text{Tk } 276,000$, the same as before. Adding the net cash flows gives $4 \times 276,000 + 456,000 - 900,000 = \text{Tk } 660,000$. That sum ignores the time value of money, so it does not yet show whether the machine is worth buying; Chapter 8 completes that step. For the present value of level cash flows, which it will use, see Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'.

Three slips are common in this calculation.

Watch for three common slips. The first is to forget the sign: the initial investment is cash paid out, so it is negative. The second is to forget depreciation in the second move, so that EBIT is reported as the cash flow. The third is to mix periods, such as placing the working capital at the end of year 1 instead of at year 0. Writing the units, taka, on every line catches many errors.

* * *

SUMMARY

The chapter's ideas in a few lines.

Capital budgeting is the process of planning, evaluating and choosing long-term investments. Such decisions are large, long and hard to reverse, so they follow five steps: find proposals, estimate cash flows, evaluate and choose, implement, and review. Projects may be independent or mutually exclusive. Only relevant cash flows count. The initial investment is paid at the start, the operating cash flow equals EBIT less tax plus depreciation, and the terminal cash flow arrives at the end.

THINK IT THROUGH

Two situations to test the chapter's ideas.

7.T1 A shop is considering a delivery van. Price Tk 560,000; registration and fitting Tk 40,000; working capital Tk 50,000. The van works for 4 years and then sells for Tk 100,000, equal to its book value. Each year it adds revenue of Tk 500,000 and cash costs of Tk 260,000. Depreciation is straight line on the asset cost less the resale value. Tax is 25%. Build the table of net cash flows for years 0 to 4.

7.T2 A clinic must choose between two scanners and can buy only one. Are the proposals independent or mutually exclusive, and what is the first step of the process that the clinic should take?

WHERE PEOPLE GO WRONG

A common belief is that profit and cash flow are the same. They are not. Depreciation reduces profit but no cash leaves; so operating cash flow is higher than EBIT less tax. The correction is to add depreciation again.

Profit and cash flow are different things.

Another belief is that money already spent must count against a project. It cannot, because the money is gone either way. Only cash that changes with the decision is relevant.

A sunk cost is not a reason to continue or to stop.

A third mistake is to forget working capital, or to treat it as lost. It is paid at the start and returns at the end of the project, so it appears twice, with opposite signs.

Working capital is both paid and returned.

A fourth mistake is to subtract loan interest in the yearly cash flow. In this method the cost of funds is allowed for in the discount rate, taught in Chapter 8, so interest is left out of the cash flows. This also keeps one cost from being counted twice.

Interest on loans is allowed for elsewhere.

IN HISTORY

The Padma Bridge in Bangladesh carries road traffic over the Padma River near Dhaka. It was inaugurated on 25 June 2022 and opened to traffic the next day. According to the Bangladesh Bridge Authority, as reported in the press, it cost Tk 307.70 billion, paid from the government's own funds. In 2012 the World Bank decided not to fund the project after it found signs of corruption involving a Canadian construction firm, and the government then decided to build with its own money.

A very large public project, judged on long-run cash and benefit.

The bridge shows all three marks of this chapter: it was large, long-lived and impossible to reverse. Its early estimate, near Tk 101.62 billion in 2007, rose after several revisions. The finance ministry gave the funds to the Bridge Authority as a loan, to be repaid over 35 years in 140 quarterly instalments from toll revenue and other income. This is not a full analysis, but it shows the central question of capital budgeting: will the cash and the wider benefit over a long life justify the money spent today?

The bridge shows all three marks and a loan repaid from tolls.

REVIEW QUESTIONS

7.1 What is capital budgeting, and why does a business need it? Give the three marks of such a decision.

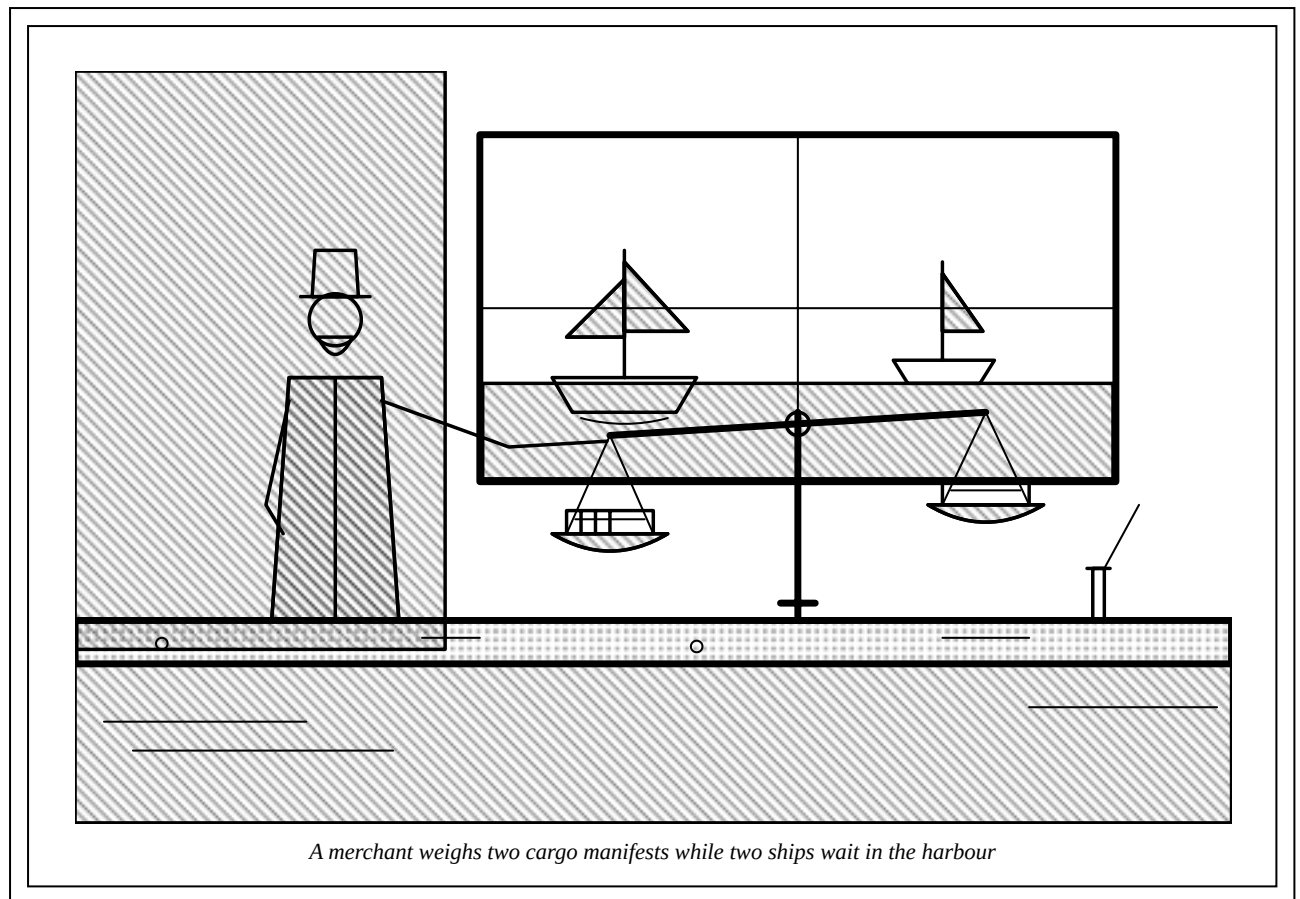
7.2 A bakery plans to buy a second oven. Apply the five steps of the capital budgeting process to this decision, with one sentence for each step.

7.3 Explain independent projects and mutually exclusive projects, with two examples of each.

7.4 A shop is considering a freezer. Which of these items are relevant cash flows, and why? (a) The freezer price, Tk 200,000. (b) Tk 5,000 paid last month for a report on freezers. (c) Tk 8,000 a year of extra electricity. (d) Tk 30,000 of extra stock needed at the start. (e) Shop rent of Tk 40,000 a year, paid with or without the freezer.

7.5 A project adds revenue of Tk 240,000 a year with cash costs of Tk 120,000 and depreciation of Tk 40,000. Tax is 25%. Find EBIT, tax and the operating cash flow.

7.6 A machine with a book value of Tk 50,000 sells for Tk 70,000 at the end of its life. Tax is 25%, and Tk 20,000 of working capital returns. Find the terminal cash flow.



CHAPTER VIII

Capital Budgeting: Evaluation Techniques

Two ships sail tomorrow: which cargo deserves the merchant's money?



A business that has measured the cash flows of a project must still decide whether the project is worth its money. This chapter teaches four tools for that judgement: payback period, net present value, internal rate of return and the profitability index. It also shows what to do when the tools give different answers.

A merchant must choose one cargo, and the two manifests point in different directions.

You will need: Chapter 7, section 7.3, and Chapter 6, section 6.2.

Speed, size and timing each pull the choice a different way.

This chapter gives four tools for the judgement.

A merchant in a busy port has money for one cargo only. Two captains bring him lists of goods, called manifests, and each captain asks for the same sum. The first captain promises that the merchant will have his money back within a year, but the cargo then earns little. The second captain needs longer, but his cargo earns far more in the end. Which captain should the merchant choose?

The merchant can argue for either captain. Speed is attractive, because money returned early can be used again, and because the future may disappoint. Size is attractive, because the larger total is worth more. Timing complicates both: a coin received in a year's time is worth less than a coin received today, so the totals cannot be compared as they stand.

In Chapter 7 you learned to measure the cash flows of a project. This chapter teaches how to judge them. We study four tools in turn, and we apply each one to the same pair of projects at Meghna and Sons, the family bookshop in Dhaka, so that you can compare the answers. Where the tools disagree, the last sections explain which one to trust and why.

* * *

8.1 PAYBACK PERIOD

The first question most owners ask is how soon the money returns.

Payback period. The time a project takes to recover its initial investment from its cash flows.

Payback period is the time needed to recover the initial investment.

Equal inflows allow a one-step formula.

Unequal inflows need a running total.

Case: two projects with the same initial investment compete.

The calculation for Press goes year by year.

An owner who has paid Tk 10,000 for a machine soon asks a plain question: when will I have my money back? Suppose the machine brings in Tk 5,000 at the end of each year for three years. After year 1 the owner has recovered Tk 5,000, after year 2 the full Tk 10,000. The answer is two years, and that answer has a name.

The **payback period** is the time a project takes to recover its initial investment from its cash flows. For the machine above, the payback period is 2 years. A shorter payback period means the money is tied up for less time, which is why many owners like the tool. It also means the business is exposed to the unknown future for less time.

Given. Initial investment Tk 10,000; inflows of Tk 5,000 at the end of each of 3 years. *Find.* The payback period. *Formula.* With equal inflows, payback period (years) equals initial investment divided by the yearly inflow. *Steps.* $10,000 \div 5,000 = 2$. *Answer.* The payback period is 2 years. *Check.* Two years of Tk 5,000 equal Tk 10,000, so the money is recovered exactly at the end of year 2.

Real projects rarely give equal inflows, so we need a second method. Add the inflows year by year to make a running total, called the cumulative cash flow, starting from the negative initial investment. Find the year in which the total changes from negative to positive. Then use this rule: payback period equals the number of full years before recovery, plus the amount still unrecovered at the start of the recovery year, divided by the inflow of that year. This assumes the inflows arrive evenly through the year.

Case file, illustrative. 2 February 2025: Meghna and Sons compares two mutually exclusive projects. Project Press: initial investment Tk 900,000; inflows Tk 276,000 in years 1 to 4 and Tk 456,000 in year 5. Project Fit-out: initial investment Tk 900,000; inflows Tk 400,000; 300,000; 250,000; 200,000; 150,000. Required return 12%.

Press. After year 1 the cumulative cash flow is $-900,000 + 276,000 = -624,000$; after year 2, $-348,000$; after year 3, $-72,000$; after year 4, $+204,000$. Recovery therefore falls in year 4. Unrecovered at the start of year 4: Tk 72,000.

Fraction of year 4: $72,000 \div 276,000 = 0.26$. Payback period = $3 + 0.26 = 3.26$ years.

Fit-out. The running total of inflows is 400,000 after year 1 and 700,000 after year 2, still short of 900,000. In year 3 the inflow is 250,000. Unrecovered at the start of year 3: $900,000 - 700,000 = \text{Tk } 200,000$. Fraction of year 3: $200,000 \div 250,000 = 0.80$. Payback period = $2 + 0.80 = 2.80$ years. *Check.* After 3 years the inflows total 950,000, which is above 900,000, so recovery does fall in year 3.

Figure 8.1 shows the cumulative cash flow of Press. The horizontal axis shows years from the start; the vertical axis shows the running total in thousands of taka. Each point is the total at the end of a year. The line starts at -900 and rises with each inflow. The point where it crosses the zero line is the payback period, here at 3.26 years.

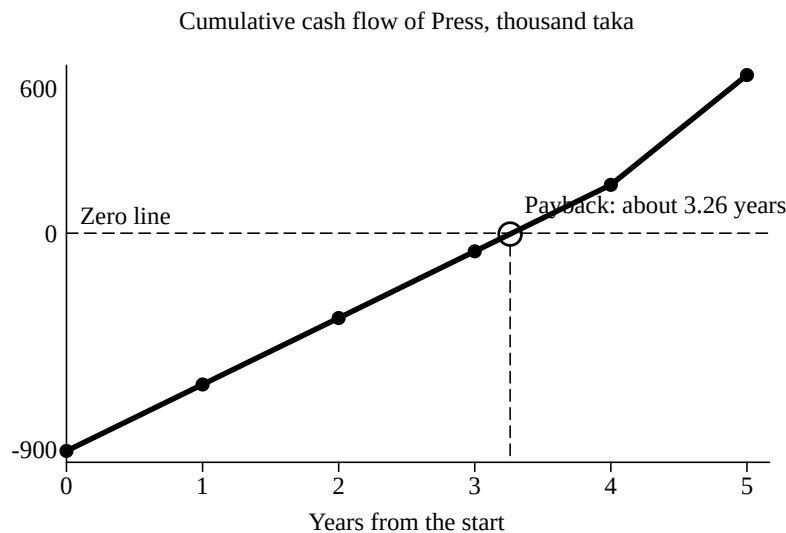


FIG. 8.1. Cumulative cash flow of Project Press, crossing zero at the payback point

To turn the tool into a decision rule, managers set a maximum acceptable payback period, for example 3 years. A project is accepted if its payback period is no longer than the limit. With a limit of 3 years, Fit-out passes at 2.80 years and Press fails at 3.26 years. The limit is a judgement, not a fact, and a different limit gives a different answer.

The tool has real limits. It ignores all cash flows after the payback point: Press earns Tk 456,000 in year 5, and payback never looks at it. It treats a taka in year 1 and a taka in year 3 as equal, so it ignores the time value of money. And the limit has no firm basis. Payback remains useful as a quick check of how long money is at risk, but it cannot be the only test.

The owner may like payback because it speaks of risk: the sooner the money returns, the less time the future has to spoil the plan. The financial manager values it as a quick screen, which removes weak proposals before slower work begins. A lender may care about it too, since a short payback suggests that loan instalments will be met. An employee or customer rarely sees the figure, but both feel its effects when a business cannot recover its money and must cut costs.

Fit-out recovers its money sooner.

The figure shows the running total rising through zero.

A decision rule needs a limit set by management.

Payback ignores later cash and the timing of money.

Different people use payback for different reasons.

PAUSE AND CHECK 8.P1 Project A has an initial investment of Tk 60,000 and inflows of Tk 30,000 in each of years 1 and 2, then nothing. Project B has the same initial investment and inflows of Tk 20,000 in each of years 1 to 6. Find each payback period. What does payback miss?

Recap: payback measures the speed of recovery only.

Before moving on. The payback period is the time to recover the initial investment. Add the inflows into a running total, find the recovery year, and add the fraction of that year. It is simple but blind to later cash and to timing.

* * *

8.2 NET PRESENT VALUE

A promise of more taka later is worth less than it seems.

Return to the machine of Tk 10,000 with inflows of Tk 5,000 for three years. The inflows total Tk 15,000, so the machine seems to gain Tk 5,000. But money received later is worth less than money received today, because today's money can be used or invested at once. To compare fairly we convert every inflow into its worth today, using the method of Chapter 6, section 6.2, in which each receipt of a mixed stream is discounted separately.

Net present value. The present value of a project's cash inflows minus its initial investment.

Net present value is the worth today of the inflows, less what we pay.

Return. Word in use. In daily life, the act of giving something back. In finance, the gain from an investment, shown as a percentage of the money invested.

Required return. The lowest return a project must earn to be worth accepting.

The required return is the discount rate and the test.

The decision rule follows from the meaning of zero.

Net present value is the present value of a project's cash inflows minus its initial investment. Net present value is often shortened to NPV. For a project with a negative result, the inflows are worth less today than the initial investment. The discount rate used to find each present value is chosen next.

The **required return** is the lowest return a project must earn to be worth accepting. Other books call it the required rate of return or the hurdle rate. In this chapter we use it as the discount rate. It depends on the risk of the project and on the cost of the funds; Chapter 10 and Chapter 17 explain how it is found. For Meghna and Sons it is 12%.

A clean example shows the method.

The rule is: accept the project when its net present value is zero or positive, and reject it when it is negative. A zero net present value means the project earns exactly the required return. A positive value means it earns more, and the amount is the extra worth added, in taka today.

Given. Initial investment Tk 10,000; inflows of Tk 5,000 at the end of years 1, 2 and 3; discount rate 10%. *Find.* The net present value. *Formula.* The present value of an inflow equals the inflow divided by $(1 + r)$ to the power of t , where r is the discount rate and t the year. Net present value equals the sum of these present values minus the initial investment. *Steps.* Year 1: $5,000 \div 1.10 = 4,545$. Year 2: $5,000 \div 1.21 = 4,132$. Year 3: $5,000 \div 1.331 = 3,757$. Sum = 12,434. NPV = $12,434 - 10,000 = \text{Tk } 2,434$. *Answer.* Net present value is Tk 2,434, so the machine should be accepted. *Check.* The inflows are equal, so Chapter 5 gives a shortcut: $5,000 \times 2.4869 = 12,434$, the same.

PAUSE AND CHECK 8.P2 A project has an initial investment of Tk 20,000 and inflows of Tk 12,000 at the end of each of years 1 and 2. The discount rate is 10%. Find the net present value and say whether to accept.

Net present value has its own limits.

Net present value also has limits, and an honest student should know them. Its answer depends on the forecast cash flows, and forecasts can be wrong. It depends on the chosen required return too: a higher rate lowers every present

value, so a project that passes at 12% might fail at 18%. Because of this, a careful business tests a project at more than one rate. Many students find this uncomfortable, because the method looks exact. The arithmetic is exact; the inputs are estimates.

Now take Fit-out at 12%. Table 8.1 sets out the work. Read each row as one year: the inflow, the divisor (1.12 to the power of the year number) and the present value, which is the inflow divided by the divisor. The last row adds the present values.

PRESENT VALUE OF PROJECT FIT-OUT AT 12%, IN TAKA

YEAR	INFLOW	DIVISOR	PRESENT VALUE
1	400,000	1.120000	357,143
2	300,000	1.254400	239,158
3	250,000	1.404928	177,945
4	200,000	1.573519	127,104
5	150,000	1.762342	85,114
Total			986,464

NPV = 986,464 – 900,000 = Tk 86,464. The value is positive, so Fit-out is acceptable at 12%. *Check.* The inflows total 1,300,000, and the present value is much lower, as we expect from discounting.

For Press the inflows in years 1 to 4 are equal, so they form an annuity. The present value factor for 4 years at 12% is $(1 - 1.12 \text{ to the power of } -4) \div 0.12 = 3.0373493$. Then $276,000 \times 3.0373493 = 838,308$. The year 5 inflow is $456,000 \div 1.762342 = 258,747$. The total is $838,308 + 258,747 = 1,097,055$, so NPV = $1,097,055 - 900,000 = \text{Tk } 197,055$. Press is also acceptable.

Press and Fit-out are mutually exclusive, so the shop can accept only one. When both pass the test, choose the larger net present value, because it measures the extra worth in taka today: Tk 197,055 against Tk 86,464. Press is the better project on this measure.

Before moving on. Net present value equals present value of inflows minus the initial investment. The required return is the discount rate and the hurdle. Accept when the value is zero or positive; among exclusive projects, choose the highest.

* * *

8.3 INTERNAL RATE OF RETURN AND PROFITABILITY INDEX

Net present value answers in taka. Many owners prefer an answer in percent, which is easy to compare with a loan rate. Two further tools give that kind of answer, and both build on net present value.

A realistic example uses five unequal inflows.

Fit-out has a positive net present value.

Press uses the annuity shortcut for years 1 to 4.

When only one project can be accepted, the larger net present value wins.

Recap: discount every inflow, subtract the initial investment, accept if the result is not negative.

Sometimes owners prefer one percentage to a sum of taka.

Rate. Word in use. In daily life, speed or price per unit. Here, the rate in internal rate of return is a yearly return on the money invested, not a charge set by a lender.

Internal rate of return. The discount rate at which a project's net present value is exactly zero.

The internal rate of return is the discount rate that makes the net present value zero.

The **internal rate of return** is the discount rate at which a project's net present value is exactly zero. Often shortened to IRR, it is the yearly return the project itself earns. Figure 8.2 shows how it is found. The horizontal axis shows the discount rate; the vertical axis shows the net present value of Press in thousands of taka. As the discount rate rises, the value falls. The point where the line crosses zero, near 19.9%, is the internal rate of return.

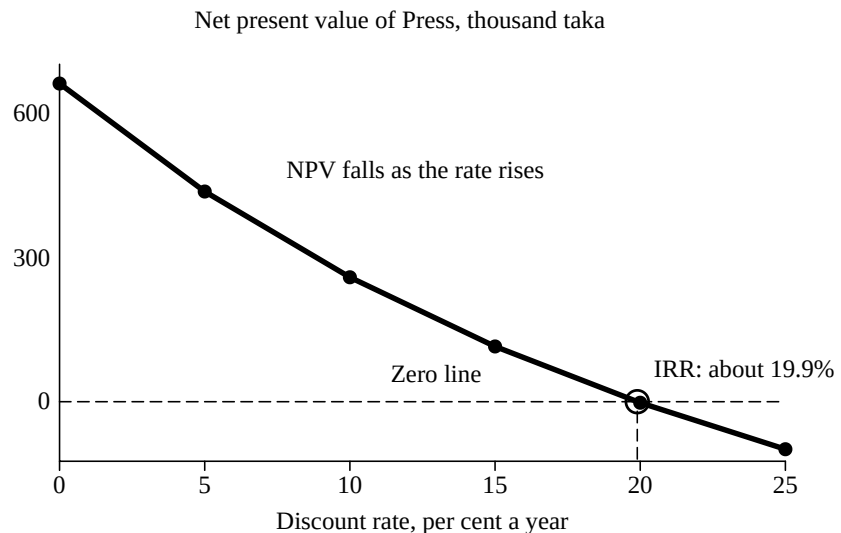


FIG. 8.2. Net present value of Project Press at different discount rates, crossing zero at the internal rate of return

Trial and interpolation find the rate by hand.

To find it by hand, calculate the net present value at two rates, one giving a positive value and one a negative value, then interpolate. For Press, NPV at 15% is +114,687 and at 20% is -2,253. The rate is $15 + 5 \times 114,687 \div (114,687 + 2,253) = 15 + 4.90 = 19.9\%$.¹ The decision rule is to accept when the internal rate of return is at least the required return: 19.9% exceeds 12%. For Fit-out, NPV at 15% is +27,975 and at 20% is -56,925, giving about 16.6%, which also exceeds 12%.

Profitability index. The present value of a project's inflows divided by its initial investment.

The profitability index shows the present value earned for each taka invested.

Both tools have limits.

The **profitability index** is the present value of a project's inflows divided by its initial investment. Press: $1,097,055 \div 900,000 = 1.22$. Fit-out: $986,464 \div 900,000 = 1.10$. A value above 1 means each taka invested returns more than a taka in present value. Accept when the index is at least 1, which is the same test as a net present value of zero or more.

Both tools have limits. The internal rate of return can mislead when projects differ much in size or timing. The profitability index ignores the size of the gain: a small project can have a high index and still add little. Here the two projects have equal initial investments, so neither problem arises.

* * *

¹ A spreadsheet finds the rate with a built-in function. It works by the same repeated trial, only faster.

8.4 WHEN THE METHODS DISAGREE

Table 8.2 collects the results for Press and Fit-out. Read each row as one tool and each column as one project. The last column names the project that tool prefers.

RESULTS FOR THE TWO PROJECTS

TOOL	PRESS	FIT-OUT	PREFERS
Payback period	3.26 years	2.80 years	Fit-out
Net present value at 12%	Tk 197,055	Tk 86,464	Press
Internal rate of return	19.9%	16.6%	Press
Profitability index	1.22	1.10	Press

Payback prefers Fit-out; the other three prefer Press. The reason is the shape of the cash flows. Fit-out is front-loaded, so it recovers its money first. Press brings large cash in year 5, which payback ignores and the discounted tools count.

When rankings conflict for mutually exclusive projects, let net present value decide. It uses all cash flows, respects timing and states the result as a gain in taka. Payback still has a role as a second check, for example when the business is short of cash. Even net present value and the internal rate of return can disagree when projects differ in size or timing; again, net present value is the guide.

Situation matters as well. An owner with plenty of funds will weigh the extra Tk 110,591 of value heavily. An owner who must repay a loan soon may value the early cash of Fit-out. Even so, this is a reason to look at the cash plan, not to change the ranking by value. The ranking stays the same; only the owner's readiness to accept a lower value for speed changes.

* * *

WORKED EXAMPLE: CHOOSING BETWEEN PRESS AND FIT-OUT

We now combine the tools and justify a choice, in the standard order: Given, Find, Formula, Steps, Answer and Check.

Given. The case file of 2 February 2025: Press and Fit-out, each with an initial investment of Tk 900,000, a required return of 12% and the inflows stated earlier. *Find.* Which project Meghna and Sons should accept. *Formula.* Payback period as above; net present value equals present value of inflows minus initial investment; accept if the value is at least zero; choose the highest value among exclusive projects.

Steps. Press by year: $246,429 + 220,026 + 196,451 + 175,403 + 258,747 = 1,097,056$. This differs from the earlier 1,097,055 by Tk 1, because each term was rounded; the unrounded total is 1,097,055.07. Both agree that NPV is Tk 197,055 for Press, and Table 8.1 gives Tk 86,464 for Fit-out. The profitability indices are 1.22 and 1.10.

The same two projects now receive four different tests.

Payback disagrees because it ignores late cash.

Net present value decides because it measures worth in taka.

The choice also depends on the business's situation.

The four tools meet in one decision.

Given and Find: the data and the question.

Steps: check each project on the discounted tools.

IN SHORT

Among mutually exclusive projects that pass the test, choose the highest net present value. Use payback only as a second look at how long the money is tied up.

Answer and check: accept Press.

Answer. Accept Press. Both projects pass the net present value test, but Press adds Tk 197,055 against Tk 86,464, and the internal rate of return and the profitability index agree. The only tool that prefers Fit-out is payback, and its advantage is a recovery 0.46 years earlier, in exchange for Tk 110,591 less value. *Check.* $197,055 - 86,464 = 110,591$. Press passes a stricter test as well: its internal rate of return is 19.9%, well above 12%. A common slip is to compare payback figures first and stop; another is to compare projects at different discount rates.

* * *

The four tools and the rule for conflict.

The payback period is the time to recover the initial investment; it is simple but ignores later cash and timing. Net present value discounts the inflows at the required return and subtracts the initial investment; accept when it is zero or more. The internal rate of return is the discount rate that makes the net present value zero. The profitability index is the present value of inflows divided by the initial investment. When rankings conflict for mutually exclusive projects, net present value decides.

Two questions to test your judgement.

8.T1 Two mutually exclusive projects each have an initial investment of Tk 500,000 and a required return of 10%. Project X has inflows of 300,000, 200,000, 100,000 and 100,000 in years 1 to 4. Project Y has inflows of 100,000, 150,000, 200,000 and 400,000. Find each payback period and net present value, rank the projects by each tool, and decide.

8.T2 In plain words, explain why a project's net present value is exactly zero when the discount rate equals its internal rate of return.

A short payback is not the same as a good project.

Many students believe the project with the shorter payback is the better one. It is only the faster one. Project A in the pause question recovers its money in 2 years but never earns more than its cost.

Discount rates must stay the same when comparing.

Another mistake is to compare net present values worked out at different discount rates. Use one required return for all projects of similar risk. Otherwise the comparison measures the choice of rates, not the projects.

The internal rate of return is not always the whole story.

A third mistake is to treat the internal rate of return as a guaranteed yearly gain. It is the rate at which the net present value reaches zero. It also does not state the size of the gain in taka.

Cash flows, not accounting profit, go into every tool.

A fourth mistake is to use accounting profit instead of the cash flows built in Chapter 7. Depreciation, for example, must be added again, as that chapter showed.

Two writers shaped the idea of discounting in business.

Irving Fisher (1867–1947) was an American economist at Yale University. In *The Theory of Interest* (1930) he set out the idea behind net present value: the worth of an investment is the present value of the cash it will bring. For many years, however, most firms judged projects by simpler rules such as payback.

SUMMARY

THINK IT THROUGH

WHERE PEOPLE GO WRONG

IN HISTORY

Joel Dean (1906–1979) was an American economist who wrote *Capital Budgeting* (Columbia University Press, 1951). Later writers often credit this book, and his consulting work, with moving discounted cash flow methods into company practice. The tools of this chapter descend from their work.

Dean carried discounting into company practice.

SOURCES

Irving Fisher, *The Theory of Interest* (1930). It matters because it explains why a sum due later is worth less today and shows how to value a stream of future receipts.

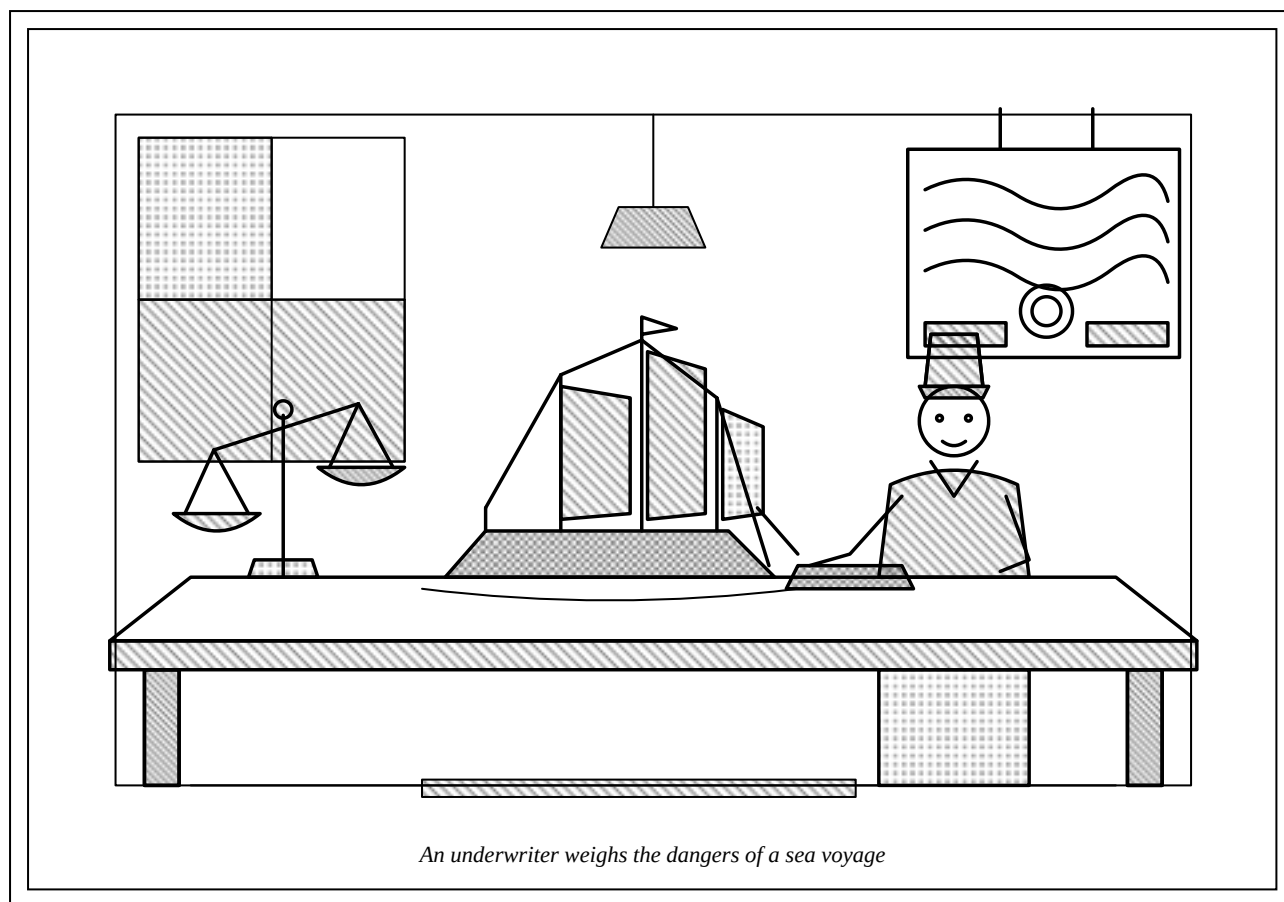
Fisher gave the theory.

Joel Dean, *Capital Budgeting* (1951). It matters because it is an early book that treats the choice of investments as a management method for companies, and it helped to spread discounting.

Dean gave the practice.

REVIEW QUESTIONS

- 8.1 Find the payback period of a project with an initial investment of Tk 400,000 and inflows of 120,000, 150,000, 180,000 and 200,000 in years 1 to 4.
- 8.2 Find the net present value of the same project at a required return of 10%. Should it be accepted?
- 8.3 State the accept-or-reject rule for payback period, net present value, internal rate of return and the profitability index.
- 8.4 For the same project, calculate the net present value at 20% and at 25%, then estimate the internal rate of return by interpolation.
- 8.5 Project M has inflows with a present value of Tk 330,000 and an initial investment of Tk 300,000. Project N has a present value of Tk 540,000 and an initial investment of Tk 500,000. Find the profitability index and net present value of each. Which ranks first by each measure?
- 8.6 In one sentence each, say what the payback period, net present value, internal rate of return and profitability index measure.
- 8.7 When two mutually exclusive projects rank differently by payback and by net present value, which should decide, and why?



CHAPTER IX

Return and Risk

Which is the better bet: a small sure gain, or the chance of a large one?



This chapter shows how to state what an investment pays and how uncertain that payment is. You will describe a gain as a percentage, list the outcomes that might happen, find their average and measure how widely they scatter. Meghna and Sons uses these tools to compare two product lines.

Two jobs, one certain and one uncertain: which would you take?

Neither choice is foolish, because the question is about how much uncertainty you will accept.

This chapter gives three tools: a way to state a gain, a way to describe uncertainty, and a way to compare choices.

The chapter relies on one earlier idea, required return, from Chapter 8.

Money put to work earns a gain or suffers a loss, and we need a name for it.

Return. the total gain or loss from an investment over a period, income plus any change in value

Return. Word in use. In daily life, to give back or to come back. In finance, the gain or loss from an investment.

Return is everything gained or lost on an investment during a period.

Return as an amount cannot be compared between investments of different size.

Rate of return. the return shown as a percentage of the amount first invested

Rate of return divides the return by the amount first invested.

The case file gives the first full example of both ideas.

The same example set out in the standard order shows each step.

Imagine that you are offered two jobs selling books at a fair. The first pays Tk 12,000 every month, whatever happens. The second pays Tk 2,000 in a poor month, Tk 14,000 in an ordinary month and Tk 26,000 in a busy month, and nobody can tell you in advance which kind of month will come. Which job would you take?

Many people choose the first job, although the second pays more over a long run of months. Others choose the second. Both groups answer a question that the amount of money cannot settle alone: how much uncertainty will you accept for the chance of a larger gain? A business meets the same question each time it buys stock, borrows money or starts a project.

Three smaller questions follow from the puzzle. How do we state what an investment pays, in one number that can be compared? How do we describe uncertainty in numbers rather than in feelings? How do we compare two choices when one pays more and also varies more? By the end of the chapter you will answer each question with a figure that you can calculate by hand.

You will need the idea of required return from Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value'. Required return is the least return that makes a project worth accepting. Here we learn to measure the returns that are compared with it.

* * *

9.1 WHAT RETURN IS

Suppose you lend Tk 1,000 to a cousin, who repays Tk 1,100 after one year. You have gained Tk 100. Suppose instead that you buy a goat for Tk 1,000 and sell it a year later for Tk 900. You have lost Tk 100. In both cases you placed funds in something and later counted the result.

Finance gives the name **return** to this result. Return means everything you gain or lose from an investment in a period: the payments you receive, plus any rise or fall in its price. An *investment* is money placed in something in the hope of gain. If a bond pays Tk 60 in interest and its price rises by Tk 40, the return is Tk 100.

A return stated in taka has a weakness. A return of Tk 100 is large on an investment of Tk 500 and small on an investment of Tk 50,000. To compare fairly, we scale the return by the size of the investment.

The **rate of return** is the return shown as a percentage of the amount first invested. In words: rate of return equals return divided by amount first invested, multiplied by 100. In symbols, with R for the return and P for the amount first invested, the rate is $(R \div P) \times 100\%$. The Tk 100 gained on the loan of Tk 1,000 is a rate of return of 10%.

Case file, illustrative. Meghna and Sons, 2 March 2025. A savings bond bought for Tk 100,000 pays Tk 6,000 interest in the year and could be sold at Tk 104,000. Return = Tk 6,000 + (Tk 104,000 – Tk 100,000) = Tk 10,000. Rate of return = Tk 10,000 ÷ Tk 100,000 = 0.10, or 10%.

Given. Amount first invested Tk 100,000; interest received Tk 6,000; price at the end of the year Tk 104,000. *Find.* The return and the rate of return. *Formula.* Return equals income received plus the change in value, where the change in value is the end price less the start price. *Steps.* The change in value is 104,000 –

100,000 = Tk 4,000. The return is $6,000 + 4,000 = \text{Tk } 10,000$. The rate is $10,000 \div 100,000 \times 100\% = 10\%$. *Answer.* The bond gave a return of Tk 10,000, a rate of return of 10% for the year. *Check.* Ten percent of Tk 100,000 is Tk 10,000, so the two answers agree.

Three limits deserve care. First, a return can be negative: if the bond could be sold only at Tk 96,000, the change in value would be -Tk 4,000 and the return would be Tk 2,000. Second, a rate means little unless the period is stated, so we say "10% a year". Third, an unsold gain is a gain on paper only, because the price may change before the sale. Different people also read return differently: the owner sees a reward for waiting, the financial manager sees a figure to set against the required return, and a lender sees only the interest.

Before moving on. Return is the total gain or loss over a period, income plus any change in value. The rate of return divides it by the amount first invested, so that investments of different size can be compared.

* * *

Return can be negative, depends on the period and may exist only on paper.

The section's two ideas are return and rate of return.

9.2 WHAT RISK IS

You can walk to a shop by two routes. On average each takes 25 minutes. Route A takes between 20 and 30 minutes, whatever the day. Route B takes 10 minutes on a quiet day and 40 minutes when the traffic is heavy. If you must catch a bus at a fixed hour, which route do you choose? Most people choose route A, although the averages are equal. The difference between the routes is not the average. It is the uncertainty of the arrival time.

Finance gives the name **risk** to this uncertainty. Risk means the chance that actual results will differ from those we expected, especially for the worse. A savings bond that pays a fixed amount has very little risk. A shop that hopes to sell Tk 7,200,000 of books in a year, but may sell less, has more. The word does not mean certain loss. A certain loss has no uncertainty, so it carries no risk.

GO SLOWLY HERE.

Two routes with the same average time still feel different.

Risk. the chance that actual results differ from those expected, especially for the worse

Risk. Word in use. In daily life, the danger of harm, such as the risk of a fall. In finance, the chance that results differ from what was expected.

Risk is the chance that results differ from what was expected, especially for the worse.

The more widely the possible results are scattered, the greater the risk.

How do we recognise more risk? Look at the possible results. If every possible result lies close to the expected one, the risk is small. If some results lie far above and others far below, the risk is large. Route B has more risk than route A for exactly this reason. The rest of this chapter turns that idea into numbers.

Test. To judge the risk of a choice, ask how far the actual result could fall from the expected result, and how likely that is. A choice whose results all lie near the expected one has low risk.

The reader gains a plain test for the amount of risk.

The puzzle at the start of the chapter shows why risk matters to a business. Owners and lenders can place their funds in safe or in uncertain uses. They accept the uncertain use only if they expect a higher return. This link between a higher expected return and greater risk is the main idea of Chapter 10, 'Risk, Required Return and Market Efficiency'. Here we must first learn how to measure each side.

People bear risk only when they expect something in return.

A business meets risk from two different directions. The first comes from the work it does. The second comes from the way it pays for that work.

Risk has more than one source, and two sources matter most to a business.

Business risk. the risk that operating profit varies because of sales, costs and trade conditions

Business risk comes from the trade itself.

Financial risk. the extra risk to owners that comes from funding part of a business by borrowing

Financial risk comes from borrowing.

The two kinds differ in cause and in who bears the extra risk.

To read the table, move along a row and compare the two columns.

The section ends with a recap and a test sentence.

GO SLOWLY HERE.

A list of possible outcomes needs a number for how likely each one is.

Probability. a number from 0 to 1 showing how likely an outcome is

Probability puts a chance on a scale from 0 (cannot happen) to 1 (certain).

Probability distribution. a list of possible outcomes with the probability of each, adding up to one

A probability distribution lists every possible outcome with its probability.

Business risk is the risk that operating profit varies because of sales, costs and trade conditions. A bookshop shows the idea. Sales rise before school terms and fall in the holidays. A rival may open nearby, and the price of paper may rise. Every business in the same trade shares these dangers, whatever the way it is funded. Business risk grows when a business has many fixed operating costs, such as rent, which must be paid even when sales are poor. Chapter 11, 'Operating Leverage and Break-even Analysis', section 11.1 'Fixed and Variable Costs', teaches this link.

Financial risk is the extra risk to owners that comes from funding part of a business by borrowing. Suppose a bookshop borrows Tk 600,000 at 10% a year. The interest is Tk 60,000 every year, in a busy year and in a poor one. After the interest is paid, what remains for the owners rises and falls more sharply than operating profit does. Chapter 12, 'Financial and Total Leverage', section 12.1 'Borrowing and Fixed Financing Charges', teaches this link in full.

HOW THE TWO KINDS OF RISK DIFFER

POINT	BUSINESS RISK	FINANCIAL RISK
Source	Sales, costs and trade conditions	Borrowing, which brings fixed interest
Who meets it	Every business in the trade	Owners of a business that borrows
Can the business avoid it?	Not by changing its funding	Yes, by borrowing less

Read each row from left to right. In the first row the source differs: the trade creates one kind and borrowing creates the other. In the last row the difference is choice: the owners decide how much to borrow, but they cannot choose the conditions of the trade.

Before moving on. Risk is the chance that results differ from those expected, especially for the worse. Business risk comes from the trade, and financial risk comes from borrowing. *You should now be able to say what risk means in finance, and why a certain loss is not risk.*

* * *

9.3 DESCRIBING UNCERTAIN OUTCOMES

Return to the two routes. On any day the traffic is clear, usual or heavy, and we cannot know which in advance. We can say how often each one happens. Suppose that over 100 days the traffic is heavy on 25 of them. Then heavy traffic has a chance of 25 in 100. Finance writes such a chance as one number.

Probability is a number from 0 to 1 showing how likely an outcome is. Zero means the outcome cannot happen, and one means it is certain. A fair coin landing heads has a probability of 0.5. Heavy traffic on 25 days in 100 has a probability of $25 \div 100 = 0.25$.

A probability distribution is a list of possible outcomes with the probability of each, adding up to one. The probabilities add up to one because exactly one of the outcomes must happen. The distribution for route B is: 10 minutes with probability 0.25, 25 minutes with 0.50, and 40 minutes with 0.25. The sum is $0.25 + 0.50 + 0.25 = 1.00$.

PAUSE AND CHECK 9.P1 A list of outcomes gives probabilities of 0.2, 0.5 and 0.4. Is it a probability distribution? Say why.

Where do probabilities come from? A business may use its past records or the judgement of people who know the trade. Either way they are estimates, not promises. A careful manager states them openly so that other people can question them.

Three outcomes with three probabilities are hard to compare with another set. We want one number that sums up the whole distribution: the average outcome. The plain average of 10, 25 and 40 minutes treats each outcome as equally likely, which is wrong, because 25 minutes happens twice as often as each of the others. We must weight each outcome by its probability.

Expected value is the average outcome, weighting each outcome by its probability. In words: multiply each outcome by its probability, then add the products. In symbols, with x for an outcome and p for its probability, expected value = $(p_1 \times x_1) + (p_2 \times x_2) + (p_3 \times x_3)$. Many students find this idea confusing, because *expected* sounds like a prediction. It is a weighted average and promises nothing about any one day.

EXPECTED TIME OF TWO ROUTES

TRAFFIC	PROBABILITY	ROUTE A MINUTES	P × MINUTES	ROUTE B MINUTES	P × MINUTES
Clear	0.25	20	5.0	10	2.5
Usual	0.50	25	12.5	25	12.5
Heavy	0.25	30	7.5	40	10.0
Total	1.00		25.0		25.0

Given. The distributions of the two routes in the table. *Find.* The expected time of each route. *Formula.* Expected value equals the sum of probability × outcome. *Steps.* Route A: $0.25 \times 20 = 5.0$; $0.50 \times 25 = 12.5$; $0.25 \times 30 = 7.5$; the sum is 25.0 minutes. Route B: $0.25 \times 10 = 2.5$; $0.50 \times 25 = 12.5$; $0.25 \times 40 = 10.0$; the sum is 25.0 minutes. *Answer.* Both routes have an expected time of 25 minutes. *Check.* An expected value must lie between the smallest and the largest outcome: 25 lies between 20 and 30, and between 10 and 40.

Two limits matter. First, the expected value is the average over many repeats, not a forecast for one day. A coin that pays Tk 0 or Tk 10 with equal probability has an expected value of Tk 5, which can never be paid on one toss. Second, the answer is only as good as the probabilities. Common slips are probabilities that do not add up to one, and a plain average used in place of the weighted one.

Before moving on. A probability is a number from 0 to 1. A probability distribution lists the outcomes with their probabilities, which add up to one. The expected value weights each outcome by its probability. *You should now be able to say what an expected value is and why both routes have the same one.*

* * *

9.4 MEASURING THE SPREAD OF OUTCOMES

The routes have the same expected time, yet we chose route A. The difference lies in how far the outcomes lie from 25 minutes. Route A never misses by more than 5 minutes, while route B misses by 15. To measure risk we need a number for this scatter.

Probabilities come from records or from judgement, so they are estimates.

A whole distribution needs to be summed up in one number.

Expected value. the average outcome, weighting each outcome by its probability

Expected value multiplies each outcome by its probability and adds the products.

The table sets out the weighted calculation for both routes, one row for each kind of day.

The worked steps repeat the table in the standard order.

Expected value is a long-run average, and it need not be an outcome that can happen.

The section's ideas are probability, the distribution and the expected value.

GO SLOWLY HERE.

Equal expected values hide the difference between the two routes.

Spread. Word in use. In daily life, to cover a surface thinly, such as butter on bread. In finance, how far outcomes lie from the expected value.

Standard deviation. a measure of how widely outcomes spread around the expected value

Standard deviation measures the typical distance of outcomes from the expected value.

The four steps turn distances into one number.

The table works through the four steps for both routes.

Each deviation is squared, weighted, added and then the root is taken.

The figure shows the spread as the width of a bell-shaped outline.

The **standard deviation** is a measure of how widely outcomes spread around the expected value. A small standard deviation means the outcomes huddle near the expected value. A large one means they lie far from it. We build it in four small steps, each in plain arithmetic.

Step one: find each outcome's deviation, which is the outcome less the expected value. *Step two:* square each deviation, because some are below the expected value and some above, and without squaring they would cancel to zero. *Step three:* multiply each squared deviation by its probability and add the results; this total is called the variance. *Step four:* take the square root, which returns the answer to the units of the outcomes. In symbols, standard deviation = $\sqrt{[\text{sum of } p \times (x - \text{expected value})^2]}$.

STANDARD DEVIATION OF TWO ROUTES (EXPECTED TIME 25 MINUTES)

TRAFFIC	PROBABILITY	A: DEVIATION	A: P × DEVIATION ²	B: DEVIATION	B: P × DEVIATION ²
Clear	0.25	−5	6.25	−15	56.25
Usual	0.50	0	0.00	0	0.00
Heavy	0.25	+5	6.25	+15	56.25
Total (variance)	1.00		12.50		112.50

Given. The distributions and the expected time of 25 minutes. *Find.* The standard deviation of each route. *Steps.* Route A: $(-5)^2 = 25$, weighted $0.25 \times 25 = 6.25$, twice, so the variance is 12.50 and the square root is 3.54 minutes. Route B: $(-15)^2 = 225$, weighted $0.25 \times 225 = 56.25$, twice, so the variance is 112.50 and the square root is 10.61 minutes. *Answer.* Route A has a standard deviation of 3.54 minutes, route B 10.61 minutes. *Check.* Route B's deviations are three times larger than route A's, and 10.61 is about three times 3.54.

PAUSE AND CHECK 9.P2 Why does the calculation square each deviation before adding?

To read the next figure, look at the horizontal line: it shows the possible outcomes, from low to high. The height of an outline above any point shows how likely outcomes near that point are. Both outlines are centred on the same expected value, marked by the dashed vertical line. The narrow outline is tall because most outcomes lie near the centre. The wide outline is low because the outcomes are scattered far from it.

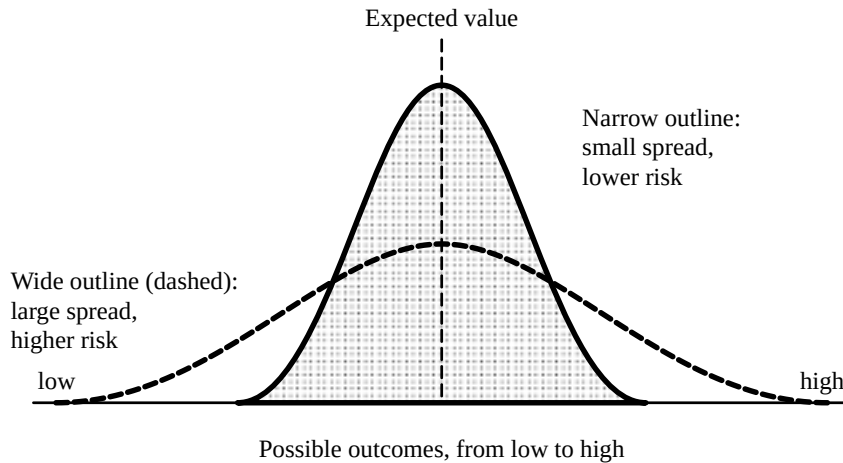


FIG. 9.1. Two sets of outcomes with the same expected value but different spread

Standard deviation is stated in the units of the outcomes, so it fits a comparison only when the expected values are equal. A standard deviation of Tk 8,000 is small beside an expected value of Tk 500,000 and large beside one of Tk 10,000. We need risk for each unit of return.

The **coefficient of variation** is the standard deviation divided by the expected value, showing risk for each unit of return. In symbols, coefficient of variation = standard deviation ÷ expected value. For route A it is $3.54 \div 25 = 0.14$, and for route B it is $10.61 \div 25 = 0.42$. A larger figure means more risk for each unit of the expected result. The coefficient cannot be used when the expected value is zero or negative.

Before moving on. The standard deviation measures spread around the expected value in four steps: deviate, square, weight and add, then take the root. The coefficient of variation divides it by the expected value so that choices of different size can be compared. Common slips are forgetting the root and forgetting to weight by probability. *You should now be able to say what the standard deviation measures and why we divide it by the expected value.*

* * *

WORKED CASE: TWO PRODUCT LINES

The case file gives the facts. We will find the expected value, the standard deviation and the coefficient of variation for each line, and then judge.

Case file, illustrative. Meghna and Sons, 2 March 2025. Two product lines are compared on Tk 100,000 invested in each. Trade is slow with probability 0.25, normal with probability 0.50 and busy with probability 0.25. Study guides return 9%, 12% and 15% in those states. Art supplies return 2%, 14% and 26%.

To read the figure, look at the three groups of bars: slow, normal and busy trade. In each group, the left bar is study guides and the right bar is art supplies. The height of a bar is the return in that kind of trade, and the label under the group gives its probability.

Standard deviation has a limit when the expected values of two choices differ.

Coefficient of variation. the standard deviation divided by the expected value, showing risk for each unit of return

The coefficient of variation divides the standard deviation by the expected value.

The section ends with a recap and a test sentence.

Meghna and Sons compares two product lines, using every tool of the chapter.

The case file gives the trade states, their probabilities and the returns of each line.

The figure shows the returns of each line in each kind of trade.

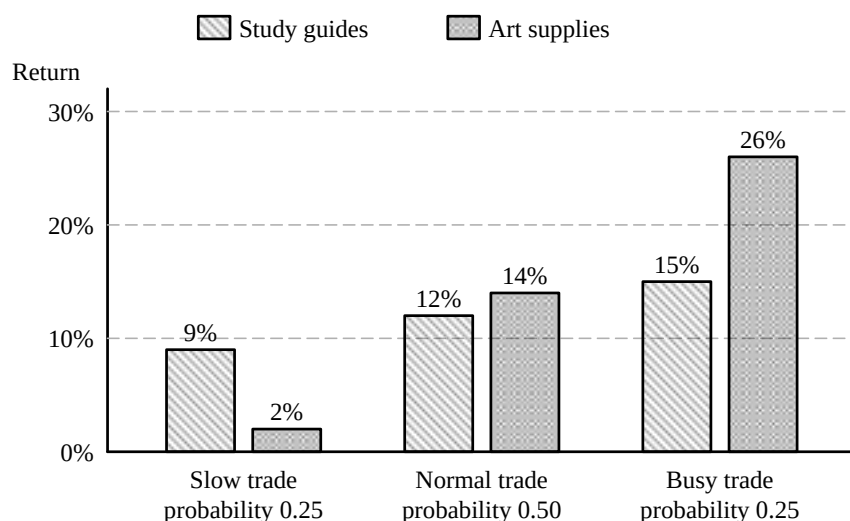


FIG. 9.2. Returns of two product lines in slow, normal and busy trade

The first step weights each return by its probability.

The second step finds the spread of each line around its own expected value.

The third step divides each spread by its expected value and a judgement follows.

A reshaped case tests the same method on takings in taka.

A combined case ranks three assets by average outcome, spread and spread per unit of return.

Given. The distributions above. *Find.* The expected value, standard deviation and coefficient of variation of each line. *Steps for expected value.* Study guides: $0.25 \times 9 = 2.25$; $0.50 \times 12 = 6.00$; $0.25 \times 15 = 3.75$; the sum is 12%. Art supplies: $0.25 \times 2 = 0.50$; $0.50 \times 14 = 7.00$; $0.25 \times 26 = 6.50$; the sum is 14%.

Steps for standard deviation. Study guides: the deviations are -3 , 0 and $+3$. The squares are 9 , 0 and 9 . Weighted: $2.25 + 0 + 2.25 = 4.50$, the variance. The square root is 2.12% . Art supplies: the deviations are -12 , 0 and $+12$. The squares are 144 , 0 and 144 . Weighted: $36 + 0 + 36 = 72$, the variance. The square root is 8.49% .

Steps for the coefficient. Study guides: $2.12 \div 12 = 0.18$. Art supplies: $8.49 \div 14 = 0.61$. *Answer.* Art supplies have the higher expected return, 14% against 12% , and much more risk for each unit of return, 0.61 against 0.18 . *Check.* Art supplies' deviations are four times as large as those of study guides, and 8.49 is about four times 2.12 . Whether the extra return is worth the extra risk depends on the required return and on how much risk the owners will accept.

Reshaped case. Two market stalls have weekly takings with the same probabilities, 0.25 , 0.50 and 0.25 . Stall X takes Tk 8,000, Tk 10,000 or Tk 12,000. Stall Y takes Tk 6,000, Tk 12,000 or Tk 18,000. Stall X: expected value $2,000 + 5,000 + 3,000 = \text{Tk } 10,000$; deviations $\pm 2,000$; variance $0.25 \times 4,000,000 \times 2 = 2,000,000$; standard deviation Tk 1,414; coefficient 0.14 . Stall Y: expected value $1,500 + 6,000 + 4,500 = \text{Tk } 12,000$; deviations $\pm 6,000$; variance $0.25 \times 36,000,000 \times 2 = 18,000,000$; standard deviation Tk 4,243; coefficient 0.35 . Stall Y takes more and carries more risk for each taka.

Combined case. Three assets have expected values and standard deviations as shown. The table introduces them in a row each, with the coefficient calculated as standard deviation $\div$ expected value.

THREE ASSETS COMPARED

ASSET	EXPECTED VALUE	STANDARD DEVIATION	COEFFICIENT OF VARIATION
P	8%	2%	0.25
Q	15%	6%	0.40
R	11%	5.5%	0.50

Read each row across. Q has the highest expected value. P has the lowest risk for each unit of return, at 0.25. R has the highest, at 0.50, yet its expected value lies below Q's, so R is the weakest choice of the three on both counts.

* * *

Asset Q has the highest return, and asset R has the most risk for each unit of return.

SUMMARY

Return is the gain or loss on an investment, and the rate of return scales it by the amount invested. Risk is the chance that results differ from those expected. A probability distribution lists outcomes with probabilities adding to one. The expected value is its weighted average, the standard deviation measures the spread around it, and the coefficient of variation shows the risk for each unit of return.

The chapter's tools are return, risk, probability, expected value, standard deviation and the coefficient of variation.

THINK IT THROUGH

9.T1 Project X has an expected value of Tk 50,000 and a standard deviation of Tk 10,000. Project Y has an expected value of Tk 80,000 and a standard deviation of Tk 24,000. Which carries more risk for each unit of return? Show the figures.

WHERE PEOPLE GO WRONG

Error: "Risk is the chance of losing money." Risk is the chance that results differ from those expected. A result better than expected is also a difference, and the standard deviation counts it. A certain loss carries no risk at all.

A common error is to think that risk means a loss.

Error: "The expected value is what will happen." It is a weighted average over many repeats. In one trial the actual result may differ widely, and may never equal the expected value.

A second error is to treat the expected value as a forecast.

Error: "The line with the smaller standard deviation is the safer one." A smaller standard deviation is safer only when the expected values are equal. When they differ, compare the coefficients of variation. Also check that the probabilities add up to one, and do not forget the square root.

A third error is to compare standard deviations when the expected values differ.

IN HISTORY

Edward Lloyd kept a coffee house in Tower Street, in the City of London near the River Thames. The London Gazette first mentioned it in 1688. Lloyd did not insure ships himself. He provided premises and reliable news of shipping, and the ship owners, merchants and captains who met there arranged insurance among themselves.¹ Each wealthy person took a share of the risk of a ship and its cargo, and signed his name beneath the policy with the amount he would cover. For this reason such people were called underwriters. No single storm could ruin one of them, because each had placed only a part of the cargo at risk.

Merchants who shared the risk of ships began the great insurance market of London.

In 1952 Harry Markowitz, an American economist, published a paper called "Portfolio Selection". He measured the risk of an investment by the spread of its returns around their expected value, which is the idea of this chapter, and he showed that investors should weigh expected return against spread.

Harry Markowitz was the first to measure investment risk by spread.

¹ The date 1688 is the first recorded mention of the coffee house. Its exact opening date is not known.

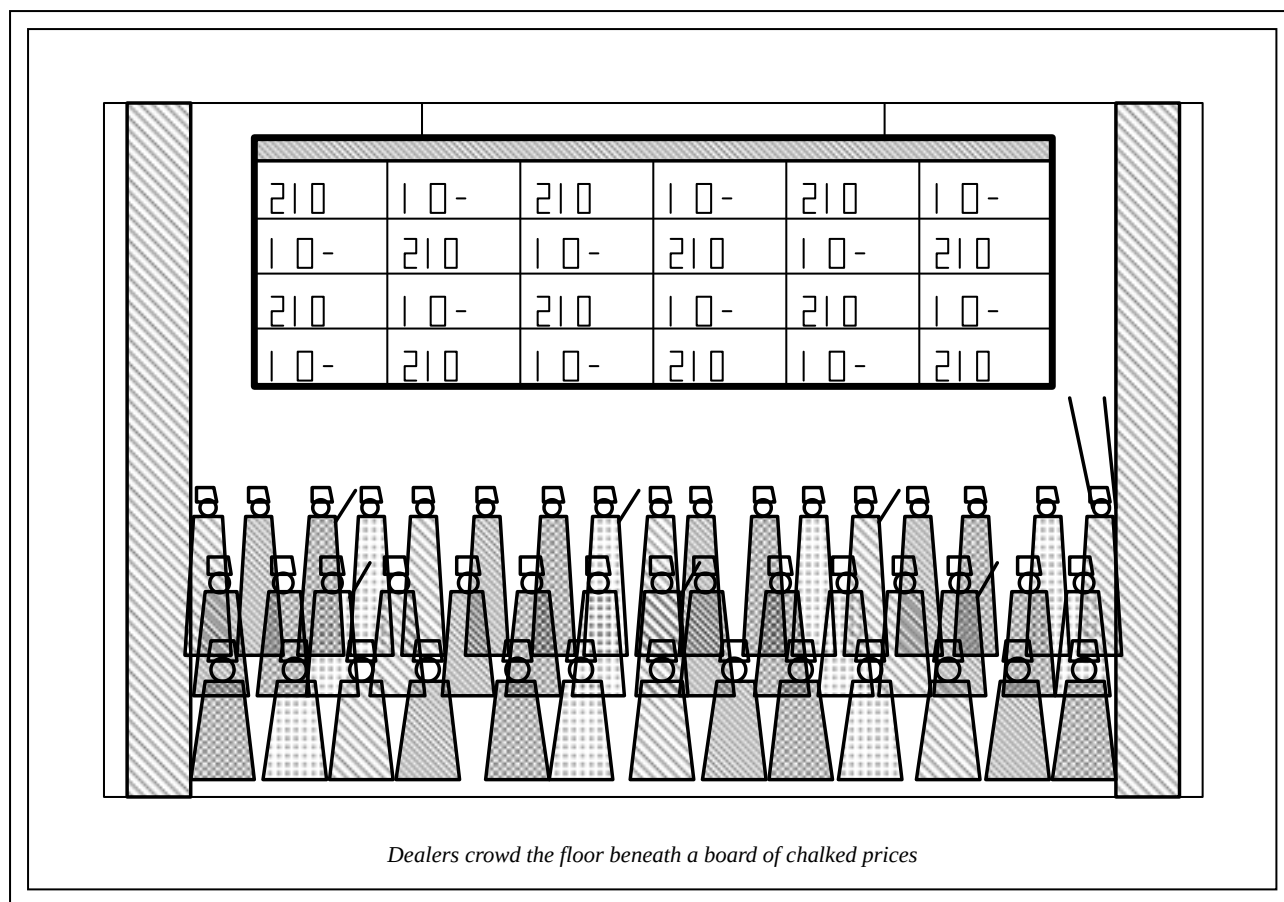
The landmark works behind the chapter are Knight, 1921, and Markowitz, 1952.

SOURCES

Frank Knight, in *Risk, Uncertainty and Profit* (1921), argued that profit is the reward for bearing uncertainty, and he separated risks that can be measured from those that cannot. Harry Markowitz, in "Portfolio Selection" (*Journal of Finance*, 1952), first treated the spread of returns as the measure of investment risk. The Lloyd's account draws on the history pages published by Lloyd's of London.

REVIEW QUESTIONS

- 9.1 A bond costs Tk 80,000, pays Tk 5,600 and could be sold for Tk 82,000. Find the return and the rate of return.
- 9.2 Explain the difference between business risk and financial risk, with one example of each.
- 9.3 An asset returns 6%, 10% or 14%, with probabilities 0.25, 0.50 and 0.25. Find the expected value and the standard deviation.
- 9.4 Asset M has an expected return of 10% and a standard deviation of 3%. Asset N has an expected return of 18% and a standard deviation of 9%. Find each coefficient of variation and say which carries more risk for each unit of return.



CHAPTER X

Risk, Required Return and Market Efficiency

Why must an uncertain gain promise more than a certain one?



This chapter explains why a risky investment must promise more than a safe one, and how much more. You will separate risk that can be removed from risk that cannot, measure how strongly an investment moves with the whole market, and turn that measure into a required return. The chapter ends with the question of how markets use information.

A certain gain and an uncertain gain raise one question: how much more must the second promise?

Different people give different answers, so we need a principle rather than a feeling.

The chapter gives a rule for the extra return, a way to separate removable danger from the rest, and a view of how markets use information.

The chapter relies on earlier ideas about risk, return and the benchmark for projects.

Lenders ask for more from a borrower who is less likely to repay.

Risk-free rate. the return on an investment with no chance of loss, such as a short government bill

The risk-free rate is the reward for waiting alone, with no uncertainty in it.

Premium. Word in use. In daily life, a payment for insurance, or an extra charge added to a price. In finance, the extra return demanded for bearing risk.

Risk premium. the extra return investors demand for taking on risk above the risk-free rate

The extra return above the risk-free rate is the price of bearing risk.

Turned round, the same relation builds the required return from its two parts.

Suppose a relative offers you a choice. You may lend Tk 10,000 to the government for three months and receive 8% a year with certainty. Or you may lend the same sum to a stranger who plans to sell tea from a cart, and the stranger offers 9%. Would you accept the stranger's offer? Most people say no. One extra percentage point seems too little for the chance that the stranger repays nothing. What if the stranger offered 12%? Or 20%?

Two people may answer differently, and neither is foolish. Yet a business meets this question every day, with real money. It must decide whether a project that might earn 14% is worth the danger that it earns far less. If managers answered by feeling, a business would accept a good project in one month and refuse it in the next, for no reason that anyone could explain.

This chapter builds the principle in four steps. First: how much extra return does a given danger demand? Second: is all danger paid for, or can some of it be removed at no cost? Third: can we turn the answer into one formula that gives a required return? Fourth: how quickly do markets take in what is known? By the end you will calculate a required return by hand and use it to accept or reject an investment.

You will need Chapter 9, 'Return and Risk', sections 9.2 to 9.4, which define risk and show how to measure it with the expected value and the standard deviation. You will also need Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value', where the required return is the benchmark for a project.

* * *

10.1 RISK PREMIUM

Return to the tea seller and the government. Both want to borrow Tk 10,000 for three months. The government will almost certainly repay. The tea seller may have a poor season and repay only part. A sensible lender asks the tea seller for a higher rate. The extra is the price of the added uncertainty, and its size depends on how much danger the lender sees.

To measure the extra, we need a starting point that has none. The **risk-free rate** is the return on an investment with no chance of loss, such as a short government bill. A *government bill* is a loan to the government, repaid within a year or less. No investment is perfectly free of risk, but a short loan to a stable government comes very close. The risk-free rate pays only for waiting, which is the time value of money. In the running case the natural choice would be a Bangladesh government bill; the 8% used below is assumed for illustration.

Now the extra has a name. A **risk premium** is the extra return investors demand for taking on risk above the risk-free rate. Risk, as Chapter 9 defined it, is the chance that results differ from those expected.¹ In words: risk premium equals required return less risk-free rate. If the risk-free rate is 8% and the tea seller must offer 12% before you will lend, the risk premium is $12\% - 8\% = 4\%$.

The relation can be turned round. Required return equals the risk-free rate plus the risk premium. Here $8\% + 4\% = 12\%$. The required return is the least

¹ Chapter 9, 'Return and Risk', section 9.2 'What Risk Is', defines risk.

return at which the investor accepts the danger. A return below it does not pay enough for the risk; a return above it leaves a gain.

Given. Risk-free rate 8%; the lender demands 12% from the tea seller. *Find.* The risk premium. *Formula.* Risk premium equals required return less risk-free rate. *Steps.* $12\% - 8\% = 4\%$. *Answer.* The risk premium is 4 percentage points. *Check.* Adding the premium back to the risk-free rate, $8\% + 4\%$, gives 12%, which is the required return we began with.

The table sets three loans in order of rising danger. Read each row from left to right: the borrower, the premium the lender demands, and the required return, which is the risk-free rate of 8% plus that premium. The figures are illustrative.

REQUIRED RETURN RISING WITH RISK (ILLUSTRATIVE FIGURES)

BORROWER	RISK PREMIUM	REQUIRED RETURN
Government, three-month bill	0%	8%
Long-established bookshop	2%	10%
New tea seller with a cart	4%	12%

The viewpoints differ. The lender asks what extra return would make the loan worth the danger. The borrower sees the premium as a cost of funds. The financial manager sees both, because the required return is the benchmark for judging a project, as Chapter 8 taught,² and the manager must therefore know what premium to add.

Two limits matter. First, the premium is what investors *demand*, not what they receive. An investment that promises 12% may pay 12%, more, or nothing; the premium is the reason investors accept the danger, not a guarantee. Second, the premium is not the same for every investor or in every period. It rises when people grow fearful and falls when they grow confident. A common slip is to call the premium itself the required return, and to forget to add the risk-free rate.

Before moving on. The risk-free rate pays for waiting alone. The risk premium is the extra that investors demand for risk, and the required return is the sum of the two. *You should now be able to say why a loan to a tea seller must offer more than a government bill.*

* * *

10.2 DIVERSIFICATION AND BETA

Imagine a trader who sells umbrellas. In a wet week the profit is Tk 20,000, and in a dry week it is nothing. Each kind of week is equally likely, so the expected value is Tk 10,000 a week, with a wide spread around it. A long dry spell could ruin the trader. Now suppose the trader puts half the money into a sunshade stall. In a wet week the umbrellas earn Tk 10,000 and the sunshades nothing; in a dry week the reverse. Either way the profit is Tk 10,000.

The expected value is still Tk 10,000, but the spread has gone: the standard deviation has fallen from Tk 10,000 to zero. Real investments never cancel each other so perfectly. The idea still holds. When one investment does badly for a reason that helps another, the pair is steadier than either alone.

A short worked example shows the two steps in the standard order.

The table lists three loans in order of rising danger, with the premium rising too.

Each person who sees the same loan sees a different side of the premium.

The premium is a demand and not a promise, and it changes with the mood of the market.

The section's ideas are the risk-free rate, the risk premium and the required return.

GO SLOWLY HERE.

Two stalls that suffer in opposite weather make a steadier living than one.

Holding both kinds of stall removes the swings, although real life is never so tidy.

² Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value', uses the required return as the benchmark for a project.

Portfolio. a collection of investments held together by one investor

A portfolio is the whole set of investments that one investor holds.

Diversification. spreading money across different investments so that one loss hurts less

Diversification spreads money across different investments.

Diversifying removes some risk but not all, and the two kinds of risk need names.

Firm-specific risk. risk that affects one business only and can be reduced by diversifying

Firm-specific risk strikes one business and can be diversified away.

Market. Word in use. In daily life, a place where people buy and sell goods, such as a bazaar. In finance, all the buyers and sellers of one kind of investment, taken together.

Market risk. risk that affects all investments in a market and cannot be removed by diversifying

Market risk reaches every investment and remains however many are held.

The figure shows risk falling as different assets are added, until only market risk is left.

The trader's two stalls form a **portfolio**. A portfolio is a collection of investments held together by one investor. A person who owns common stock in five businesses holds a portfolio of five holdings. So does a business that owns a shop, a warehouse and a loan to a customer. What matters about a portfolio is the combined result, not the result of each part on its own.

Building such a portfolio has its own name. **Diversification** is spreading money across different investments so that one loss hurts less. The trader's move from one stall to two is diversification. The word *different* matters. Ten stalls that all sell umbrellas spread the money but not the danger, because one wet season hurts all ten.

Does diversifying remove all risk? It does not. Think of two kinds of bad news. A fire at one factory, a strike at one shop or a failed new product harms one business and leaves the others alone. A rise in the cost of borrowing, a fall in demand across the whole country or a war reaches almost every business at once.

The first kind is **firm-specific risk**: risk that affects one business only and can be reduced by diversifying. An investor who holds many businesses feels one failure as a small part of the portfolio, and good news at some businesses offsets bad news at others.

The second kind is **market risk**: risk that affects all investments in a market and cannot be removed by diversifying. Here *market* means all the buyers and sellers of a kind of investment, taken together. A fall in demand across the whole country lowers the profit of nearly every business, so no mixture of businesses escapes it.

To read the next figure, look first at the two axes. The horizontal axis counts how many different assets the portfolio holds. The vertical axis shows the total risk of the portfolio. The curve falls steeply at first, because each new asset removes much firm-specific risk, and then flattens at a floor. The shaded area above the floor is firm-specific risk; the floor itself is market risk.

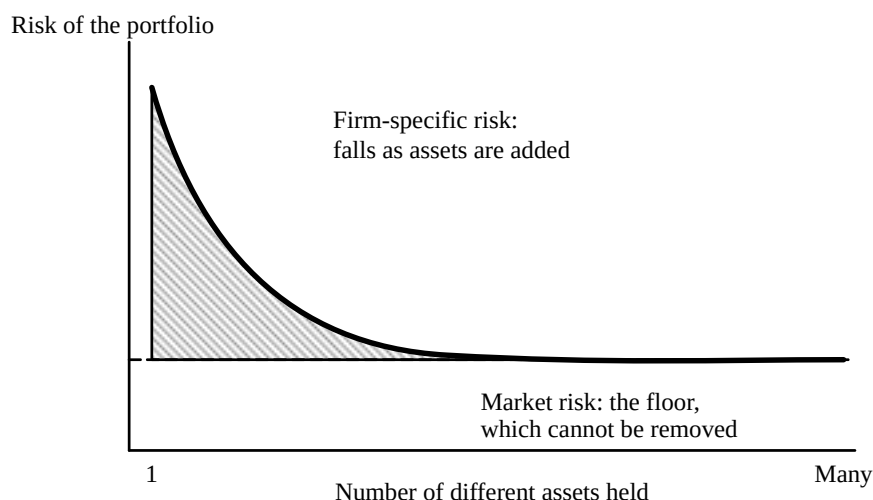


FIG. 10.1. Risk of a portfolio falling as different assets are added, then levelling at a floor

Here is the idea that carries the rest of the chapter. Because an investor can remove firm-specific risk by diversifying, the market does not pay extra for bearing it. Only market risk earns a premium. To find a fair required return for one investment, we must therefore measure how much market risk it carries. There is one exception to remember: a family that owns a single bookshop cannot diversify its wealth, so for that family firm-specific risk is real. The theory fits best an investor who holds many different assets.

Picture ten shops in one bazaar. When the number of visitors rises by 10%, a shop selling festival goods may see its sales rise by 15%, and a shop selling rice may see a rise of only 6%. Each shop follows the crowd, but by a different amount. The crowd plays the part of the market, and the amount of following is what we want to measure.

The **beta** of an investment is a number showing how strongly it moves with the whole market. By agreement the whole market has a beta of exactly 1. An investment with a beta of 1.5 tends to move one and a half times as far as the market: when the return of the market rises by 10%, the return of this investment tends to rise by about 15%, and when the market falls by 10%, it tends to fall by about 15%. An investment with a beta of 0.6 moves about six-tenths as far.

The table lists four betas. Read each row from left to right: the beta, what it means, and the usual movement of the investment when the return of the market changes by 10%.

HOW TO READ A BETA

BETA	MEANING	USUAL MOVEMENT FOR A 10% MARKET CHANGE
0	Does not move with the market	0%
0.6	Moves less than the market	6%
1.0	Moves as the market does	10%
1.5	Moves more than the market	15%

PAUSE AND CHECK 10.P1 An investment has a beta of 2. The market return rises by 5%. By about how much would you expect its return to rise?

Investors are paid for market risk and not for risk that anyone can remove.

Several shops in one bazaar follow the crowd by different amounts.

GO SLOWLY HERE.

Beta. a number showing how strongly an investment moves with the whole market

Beta compares the movement of an investment with the movement of the whole market.

The table gives four betas and what each means.

Beta is found from past records, and it measures market risk only.

The section's ideas are the portfolio, diversification, the two kinds of risk and beta.

After Markowitz measured risk, one question remained: which risk should the market pay for?

GO SLOWLY HERE.

Capital asset pricing model. a formula setting required return as risk-free rate plus beta times the market risk premium

The model builds the required return from the risk-free rate and a premium scaled by beta.

The first example uses small, clean numbers so that only the method must be learned.

Two more betas show the formula at the ends of its range.

Security market line. the straight line that shows required return rising with beta

The security market line draws the formula as a picture.

Two limits apply. First, beta is found by comparing the past returns of an investment with those of the market, and data services publish it; this book uses betas that are given and does not teach the estimation. Past movement may not repeat. Second, beta measures only market risk. The standard deviation of Chapter 9 measures total risk, which holds both kinds,³ so an investment can have a high standard deviation and a low beta when most of its risk is firm-specific.

Before moving on. A portfolio is the set of investments one investor holds, and diversification spreads money across different investments. It removes firm-specific risk but not market risk, and beta shows how strongly an investment moves with the market. *You should now be able to say why diversifying cannot remove market risk.*

* * *

10.3 THE CAPITAL ASSET PRICING MODEL

In 1952 Harry Markowitz showed how to measure the risk of an investment by the spread of its returns. A question stayed open. If investors diversify, which part of that spread should the market pay for? In 1964 William Sharpe, an American economist, gave the answer: only market risk, measured by beta. John Lintner in 1965 and Jan Mossin in 1966, working separately, reached similar results. Sharpe later shared the 1990 Nobel Memorial Prize in Economic Sciences for this line of work.

The **capital asset pricing model** is a formula setting required return as risk-free rate plus beta times the market risk premium. The *market risk premium* is the premium for the whole market: the return of the market less the risk-free rate. In words: required return equals the risk-free rate plus beta times (market return less risk-free rate). In symbols, with R_f for the risk-free rate, R_m for the market return and β for beta: required return = $R_f + \beta \times (R_m - R_f)$.

Given. Risk-free rate 6%; market return 10%; beta 1.0. *Find.* The required return. *Formula.* Required return = $R_f + \beta \times (R_m - R_f)$. *Steps.* First, the market risk premium: $10\% - 6\% = 4\%$. Second, beta times the premium: $1.0 \times 4\% = 4\%$. Third, add the risk-free rate: $6\% + 4\% = 10\%$. *Answer.* The required return is 10%. *Check.* A beta of 1 means the investment moves as the market does, so it must require the market return, 10%, and it does.

Try two other betas with the same rates. With a beta of 0 the required return is $6\% + 0 \times 4\% = 6\%$, the risk-free rate, because an investment that does not move with the market carries no market risk. With a beta of 2 it is $6\% + 2 \times 4\% = 14\%$, because the investment carries twice the market's risk and earns twice the market's premium.

PAUSE AND CHECK 10.P2 With a risk-free rate of 6% and a market return of 10%, find the required return for an investment with a beta of 0.5.

The formula can be drawn. The **security market line** is the straight line that shows required return rising with beta. Beta runs along the horizontal axis and required return up the vertical axis. At a beta of 0 the line starts at the risk-free rate. For each whole step of beta, the line climbs by the market risk premium. At a beta of 1 it passes through the market return.

³ Chapter 9, 'Return and Risk', section 9.4 'Measuring the Spread of Outcomes', shows that standard deviation measures total risk.

To read the next figure, find an investment by its beta on the horizontal axis. Go up to the line, then across to the vertical axis. The height of the dot is the required return for that beta. An investment whose expected return lies above the line promises more than it must; one below the line promises less.

The figure places two investments on the line by their betas.

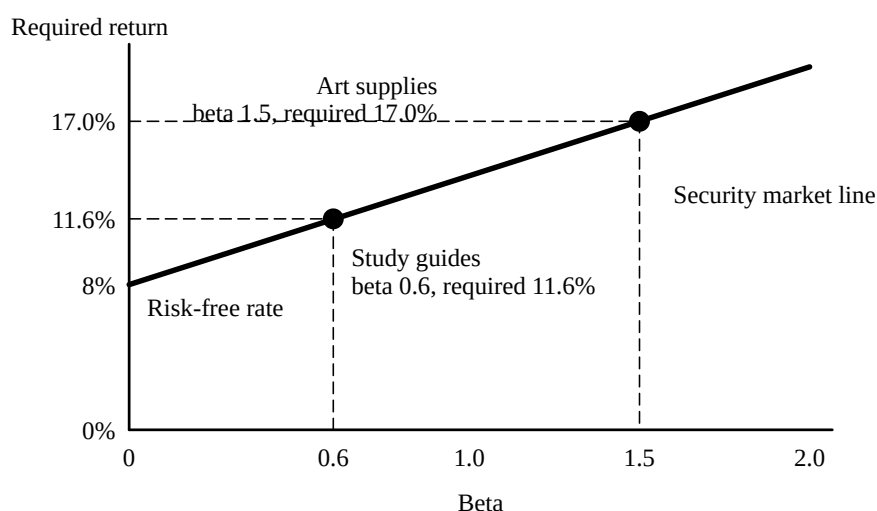


FIG. 10.2. The security market line, with the required returns of two product lines

The model gives a rule for decisions. Accept an investment when its expected return is equal to or above its required return, which means it lies on or above the line. Reject it when the expected return is below the required return. This is the same test as Chapter 8 used with net present value: a project is worth accepting when it earns at least the benchmark.

The decision rule compares the expected return with the line.

The model has limits. It assumes that investors hold diversified portfolios and that one market return and one beta explain required returns. Betas change over time, and the market return and risk-free rate are estimates, so different estimates give different required returns. Even so, it is the standard first tool, because it turns a feeling about risk into a number that can be argued about.

The model rests on simplifying assumptions, so it guides and does not prove.

Before moving on. The capital asset pricing model sets required return as the risk-free rate plus beta times the market risk premium. The security market line draws it as a straight line. *You should now be able to say how a higher beta raises the required return.*

The section's ideas are the pricing formula and the line that draws it.

* * *

10.4 EFFICIENT MARKETS

Suppose a business announces a large new order. When should the price of its common stock rise: before the news, when the news is announced, or a week later? If buyers and sellers are quick and well informed, the price should move at the moment of the announcement, and not after. A reader who waits a week to buy finds that the gain has gone.

If a business announces good news, when should its share price change?

An **efficient market** is a market in which prices already reflect the information available. If a market is efficient, the market value of an investment is a fair summary of what is known about it. New information moves the price, and since new information cannot be predicted, price changes look random. Efficient does not mean that every price is right, or that nobody ever loses money.

Efficient market. a market in which prices already reflect the information available

In an efficient market the price already contains what is known.

The strength of the claim depends on which information is meant.

Weak-form efficiency. prices already reflect all past price information

The weak form covers past prices only.

Semi-strong-form efficiency. prices reflect all public information, not only past prices

The semi-strong form covers everything that is public.

Strong-form efficiency. prices reflect all information, public and private

The strong form covers private information too.

The table sets the three forms side by side.

How much information do prices contain? There are three answers, each stronger than the one before. They differ in the kind of information that prices already reflect.

Weak-form efficiency means that prices already reflect all past price information. If it holds, studying old price charts cannot show which investment will rise next, because the pattern is already in the price.

Semi-strong-form efficiency means that prices reflect all public information, not only past prices. Annual reports, news and announcements are public. If this form holds, reading them after they appear brings no gain, because the price has already moved.

Strong-form efficiency means that prices reflect all information, public and private. Even a manager's secret knowledge would be in the price. Few people believe this form holds, and in many countries trading on secret information is a crime.

The table lists the three forms. Read each row from left to right: the form, the information already in the price, and an example of what would not help an investor if the form holds.

THE THREE FORMS OF MARKET EFFICIENCY

FORM	INFORMATION IN THE PRICE	WOULD NOT HELP
Weak	Past prices	Studying last year's price chart
Semi-strong	All public information	Reading a report published this morning
Strong	Public and private information	Using a secret known only to managers

Efficiency varies between markets and periods, so claims must be made with care.

Efficiency is a matter of degree. Studies of large, busy markets have found support for the weak form and mixed results for the others. Smaller markets, where fewer people trade and less information is published, may be less efficient. A student should say which market and which period a claim concerns. For a financial manager, the lesson is practical: where prices reflect information quickly, the market value of the business is a fair guide to the quality of the manager's decisions.

Before moving on. An efficient market is one whose prices already reflect the information available. The weak form covers past prices, the semi-strong form all public information and the strong form all information. *You should now be able to say why new information, and not old information, moves prices.*

* * *

WORKED CASE: TWO PRODUCT LINES ON THE LINE

The case file gives the market rates and the beta and expected return of each line.

Case file, illustrative. Meghna and Sons, 20 March 2025. The risk-free rate is 8%, a short government bill, assumed for illustration. The market return is 14%, so the market risk premium is 6%. Study-guide line: beta 0.6, expected return 12%, found in Chapter 9. Art-supplies line: beta 1.5, expected return 14%.

The first line is worked in full, in the standard order.

Given. $R_f = 8\%$; $R_m = 14\%$; study-guide beta 0.6; expected return 12%. *Find.* The required return, and whether to accept the line. *Formula.* Required return $= R_f + \beta \times (R_m - R_f)$. *Steps.* Market risk premium: $14\% - 8\% = 6\%$. Beta times premium: $0.6 \times 6\% = 3.6\%$. Required return: $8\% + 3.6\% = 11.6\%$. Compare: expected return 12% is above 11.6%. *Answer.* The required return is

11.6%. The expected return is higher, so accept. *Check.* A beta between 0 and 1 must give a required return between 8% and 14%, and 11.6% does.

Given. Art-supplies beta 1.5; expected return 14%; same rates. *Find.* The required return, and whether to accept. *Steps.* Beta times premium: $1.5 \times 6\% = 9.0\%$. Required return: $8\% + 9.0\% = 17.0\%$. Compare: expected return 14% is below 17.0%. *Answer.* The required return is 17.0%. The expected return is lower, so reject. *Check.* A beta above 1 must give a required return above the market return of 14%, and 17.0% does.

The second line is worked the same way and reaches the opposite decision.

Chapter 9 found that art supplies had the higher expected return, 14% against 12%. Here the higher return does not pay for the higher market risk. The two lines also agree with the standard deviation of Chapter 9, which found art supplies the riskier, although beta and standard deviation measure different things. The owner sees a line that fails the test, and the financial manager sees the reason: its return is too low for its beta.

Ranking by expected return alone would have chosen the wrong line.

The table combines the two results. Read each row from left to right: the beta, the required return from the line, the expected return, and the line's position.

The table places both lines against the security market line.

BOTH LINES AGAINST THE SECURITY MARKET LINE

LINE	BETA	REQUIRED RETURN	EXPECTED RETURN	POSITION AND DECISION
Study guides	0.6	11.6%	12%	0.4 points above the line: accept
Art supplies	1.5	17.0%	14%	3.0 points below the line: reject

Watch for three slips. Some students multiply beta by the market return and not by the premium. Some forget to add the risk-free rate at the end. Some compare the expected return with the market return and not with the required return.

Common slips arise at each step of the formula.

* * *

SUMMARY

The required return equals the risk-free rate plus a risk premium. Diversification removes firm-specific risk but not market risk, so only market risk earns a premium. Beta shows how strongly an investment moves with the market. The capital asset pricing model sets required return as the risk-free rate plus beta times the market risk premium, and the security market line draws it. An efficient market is one whose prices already reflect the information available.

The chapter's rule is that only market risk earns a premium, and beta sets its size.

THINK IT THROUGH

10.T1 The risk-free rate is 8% and the market return is 14%. A stationery line has a beta of 1.0 and an expected return of 13%. Find the required return, say whether the line lies above or below the security market line, and decide whether to accept it.

WHERE PEOPLE GO WRONG

Error: "A large portfolio has no risk." Diversification removes firm-specific risk. Market risk remains, however many different assets are held.

A common error is to think that diversifying removes all risk.

A second error is to think that a high beta marks a bad investment.

Error: "A high beta means a bad investment." A high beta means that the investment moves strongly with the market, in both directions. It raises the required return; it does not make the investment good or bad. The test is whether the expected return meets the required return.

A third error is to think that an efficient market is always right.

Error: "An efficient market never makes mistakes." Efficiency means prices reflect the information available, which may be incomplete. Prices can still fall sharply, and investors can still lose.

A fourth error is to treat the required return as a forecast.

Error: "The required return is what the investment will earn." It is the benchmark: the least return that pays for the market risk. The expected return is the estimate of what the investment may earn.

IN HISTORY

On one day in 1987 the market fell by about a fifth, and no single business was to blame.

On Monday 19 October 1987, the Dow Jones Industrial Average, an index that follows the share prices of 30 large United States businesses, fell by 508 points, a little over 22% in one day.⁴ Markets in many other countries fell as well. Nearly every business lost value together, whatever news it had announced, which is what market risk means: an investor who held many different businesses could not escape the fall.

The crash tested how markets take in information.

The crash also tested the idea of an efficient market. No single piece of public news seemed large enough to explain so great a fall in a day, and economists still debate the causes; a presidential commission later pointed to trading by computer programs among them. Yet the market absorbed the shock within a day, and the index regained its loss in about two years.

SOURCES

Sharpe, 1964, and Fama, 1970, are the landmark works for this chapter.

William Sharpe, "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk" (*Journal of Finance*, 1964), set out the model that links required return to beta. Eugene Fama, "Efficient Capital Markets: A Review of Theory and Empirical Work" (*Journal of Finance*, 1970), gave the three forms of efficiency their names and reviewed the evidence. Harry Markowitz, 1952, whose work is described in Chapter 9, supplied the idea of measuring risk by spread.

REVIEW QUESTIONS

10.1 The risk-free rate is 7%, the market return is 13% and an investment has a beta of 1.2. Find the required return.

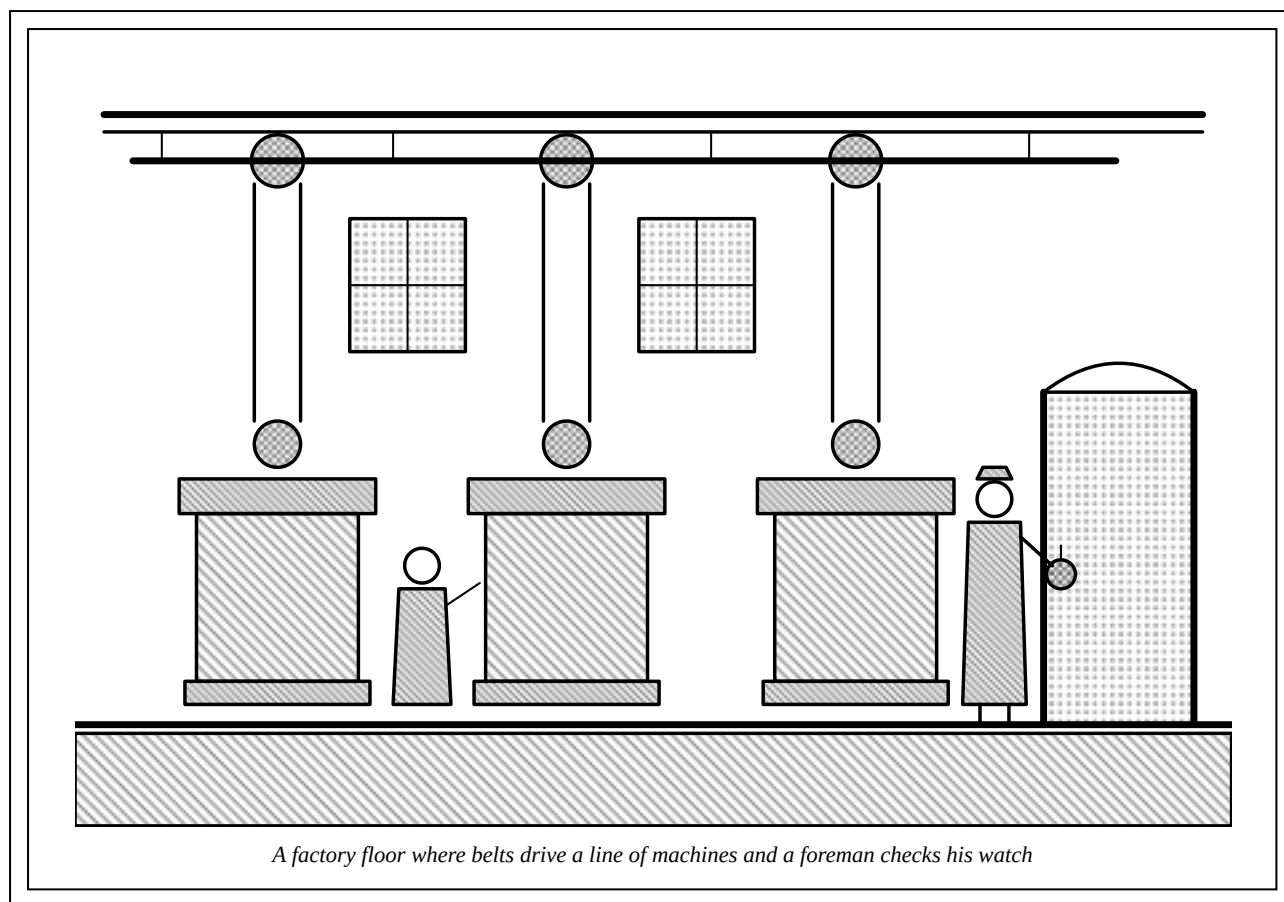
10.2 Explain why diversification lowers some risk but not all.

10.3 Draw a security market line with a risk-free rate of 6% and a market return of 10%, with beta from 0 to 2. Read from it the required return at a beta of 0.5 and at a beta of 1.5.

10.4 Name the three forms of market efficiency and give one example for each of information that would not help an investor if the form holds.

10.5 An investor finds an asset whose expected return sits below the security market line. Should the investor accept it? Explain in plain words.

⁴ Sources round the figures differently; the index closed at about 1,739, a fall of about 22.6%.



CHAPTER XI

Operating Leverage and Break-even Analysis

Why can a small rise in sales bring a large rise in profit?



Some costs stay the same however much a business sells, and others grow with every unit. This chapter separates the two kinds, finds the sales volume at which a business neither gains nor loses, and measures how strongly a change in sales moves profit. It closes by linking these ideas to risk.

Sales rise by one eighth, yet profit rises by one half; the reason is the subject of this chapter.

A tea seller shows the same effect: once the rent is paid, each extra cup adds mostly profit.

The chapter answers three questions: where the loss ends, how strongly profit responds to sales, and what this means for risk.

A business pays some costs whatever it sells and other costs only as it sells.

Consider two numbers from the case file of the printing unit of Meghna and Sons, the family-run bookshop in Dhaka that appears throughout this book. In one scenario the unit sells 8,000 packs in a year and earns an EBIT of Tk 80,000. EBIT is the profit of a business before interest and tax are taken away.¹ In the next scenario the unit sells 9,000 packs. Sales have grown by one eighth, which is 12.5%. A reader might expect EBIT to grow by about the same share. It does not. EBIT rises from Tk 80,000 to Tk 120,000, a rise of 50%.

Think of a tea seller who rents a stall. The rent is the same on a quiet day and on a busy day. The tea leaves and milk, however, are used up one cup at a time. On a quiet day the seller may barely cover the rent. On a busy day, once the rent is covered, almost every extra cup adds to the seller's profit. A rise in cups sold of one tenth can therefore raise profit by far more than one tenth. The printing unit behaves in the same way, and so does every business that has costs which do not move with sales.

This chapter answers three questions. First, how many units must a business sell before it stops making a loss? Second, by what percentage does profit change when sales change by one percent? Third, why does the answer matter to anyone who owns, lends to or works for the business? We shall build each answer one idea at a time, starting with the way costs behave.

YOU WILL NEED

Chapter 9, 'Return and Risk', section 9.2 'What Risk Is': business risk. Chapter 7, 'Capital Budgeting: Decisions and Cash Flows', section 7.3 'Cash Flow from an Investment': EBIT.

* * *

11.1 FIXED AND VARIABLE COSTS

Start with a small business that you could run yourself. A seller rents a table in a market for Tk 8,000 a year and sells notebooks. Each notebook is bought from a wholesaler for Tk 12 and sold for Tk 20. The seller soon notices that two kinds of money leave the business. One kind is the same whether the table sells ten notebooks or a thousand. The other kind grows with every notebook sold. Businesses of every size, from this table to a large factory, contain both kinds, and separating them is the first step of this chapter.

¹ Recall: EBIT, Chapter 7, 'Capital Budgeting: Decisions and Cash Flows', section 7.3 'Cash Flow from an Investment'. From here on the book uses the short form EBIT.

An operating cost is money spent on the everyday work of the business, before any interest or tax is paid. A **fixed operating cost** means an operating cost that stays the same whatever the number of units sold. The market rent of Tk 8,000 a year is a fixed operating cost, because it is the same for ten notebooks or for a thousand. The depreciation of a machine and the salary of a shop manager are further examples.

A **variable operating cost** means an operating cost that rises and falls with the number of units sold. The Tk 12 paid for each notebook is variable. If the seller sells 100 notebooks, this cost is Tk 1,200; if the seller sells 200, it is Tk 2,400. Each unit adds the same amount. Materials, packing and sales commission are other examples.

We can now write a rule for the whole bill. In words: total operating cost equals fixed operating cost plus variable operating cost per unit multiplied by units sold. In symbols, $TC = F + v \times Q$. Here TC is total operating cost, F is fixed operating cost, v is variable operating cost per unit, and Q is the number of units sold. Suppose the seller sells 500 notebooks. Then $TC = 8,000 + 12 \times 500 = 8,000 + 6,000 = \text{Tk } 14,000$. Sales revenue is $500 \times 20 = \text{Tk } 10,000$, so EBIT is $10,000 - 14,000$, a loss of Tk 4,000.

The **contribution margin** is the selling price per unit minus the variable operating cost per unit. Each notebook gives $20 - 12 = \text{Tk } 8$. We call it a contribution because those Tk 8 go first towards the fixed cost and, once that is paid, towards profit.

Case file, illustrative. Meghna and Sons, 6 April 2025, printing unit. Selling price Tk 90 per pack. Variable operating cost Tk 50 per pack. Contribution margin Tk 40 per pack. Fixed operating cost Tk 240,000 a year.

Real costs seldom divide so neatly. An electricity bill may have a standing charge, which is fixed, and a charge for each unit of power used, which is variable; such a cost is part fixed and part variable. A cost is also fixed only within a range of volume. If the seller doubles sales, one table may no longer suffice, and a second table brings a second rent. In this chapter we assume that the selling price, the variable cost per unit and the fixed cost stay the same across the volumes we study. The manager must judge whether this holds in practice, and the owner, who bears the fixed costs in a poor year, has most reason to ask.

Before moving on. A fixed operating cost stays the same whatever the units sold, and a variable operating cost grows with each unit. The contribution margin is the price less the variable cost per unit, which for the printing unit is Tk 40.

Cost. Word in use. In daily life, the price you pay for something. In business, any money given up to run the business, such as rent, wages or materials.

Fixed. Word in use. In daily life, fastened in place or repaired. Here, describing a cost that does not change with the number of units sold.

Fixed operating cost. an operating cost that stays the same whatever the number of units sold

The table rent is a fixed operating cost: it does not change with the notebooks sold.

Variable operating cost. an operating cost that rises and falls with the number of units sold

The Tk 12 paid for each notebook is a variable operating cost, because it grows with each unit sold.

Total operating cost equals the fixed cost plus the variable cost per unit multiplied by the units sold.

Contribution margin. selling price per unit minus variable operating cost per unit

Margin. Word in use. In daily life, the empty edge of a page or a small safety gap. In business, a difference between a price and a cost, as here.

Each unit sold contributes its price less its variable cost towards the fixed cost.

The printing unit's record of 6 April 2025 gives the price, costs and contribution margin used later.

Real costs are only roughly fixed or variable, and a fixed cost holds only within a range of volume.

A fixed cost stays the same as units sold change, a variable cost grows with them, and the contribution margin links the two.

* * *

The seller wants to know how many notebooks must be sold before the loss ends.

Operating break-even point. the sales volume at which EBIT is exactly zero

The operating break-even point is the volume at which EBIT is exactly zero.

The first example uses small round numbers so that the method is the only new thing.

The formula divides the fixed cost by the contribution margin, in words and in symbols.

The steps find the contribution margin first and then divide the fixed cost by it.

A table of three volumes shows loss below break-even, zero at break-even and profit above it.

11.2 THE OPERATING BREAK-EVEN POINT

Return to the seller who lost Tk 4,000 on 500 notebooks. Each notebook gives Tk 8 of contribution, and the fixed cost of Tk 8,000 must be covered first. How many notebooks are needed? Each one pays Tk 8 towards Tk 8,000, so we need $8,000 \div 8 = 1,000$ notebooks. At that volume the seller neither gains nor loses. Below it the seller has a loss, and above it a profit.

The **operating break-even point** means the sales volume at which EBIT is exactly zero. For the seller it is 1,000 notebooks. It is measured in units, such as notebooks or packs, and not in taka. The word *operating* reminds us that we look at EBIT, before interest and tax.

THE BREAK-EVEN RULE

EBIT equals contribution minus fixed cost. EBIT is zero when contribution equals fixed cost, so break-even volume equals fixed operating cost divided by contribution margin.

CLEAN NUMBERS FIRST

Given. Selling price Tk 20 per notebook, variable operating cost Tk 12 per notebook, fixed operating cost Tk 8,000 a year. *Find.* The operating break-even point.

Formula. In words: break-even volume equals fixed operating cost divided by contribution margin. In symbols, $Q = F \div CM$. Here Q is the number of units at break-even, F is the fixed operating cost, and CM is the contribution margin per unit, which is the price minus the variable cost per unit.

Steps. First, $CM = 20 - 12 = \text{Tk } 8$ per notebook. Second, $Q = 8,000 \div 8 = 1,000$ notebooks. *Answer.* The seller reaches break-even at 1,000 notebooks a year. *Check.* At 1,000 notebooks, revenue is $20 \times 1,000 = \text{Tk } 20,000$. Total operating cost is $8,000 + 12 \times 1,000 = \text{Tk } 20,000$. EBIT is $20,000 - 20,000 = 0$, as it should be.

PAUSE AND CHECK 11.P1 A stall has a contribution margin of Tk 25 per unit and a fixed operating cost of Tk 5,000 a year. How many units must it sell to reach the operating break-even point?

EBIT AT DIFFERENT VOLUMES

The break-even point is one volume out of many, so it helps to see EBIT on both sides of it. The table below uses the notebook seller. Each row multiplies the units sold by the contribution margin of Tk 8, then subtracts the fixed cost of Tk 8,000. Read across a row to follow that calculation.

EBIT OF THE NOTEBOOK SELLER AT THREE VOLUMES

NOTEBOOKS SOLD	CONTRIBUTION (TK)	FIXED COST (TK)	EBIT (TK)
800	6,400	8,000	-1,600
1,000	8,000	8,000	0
1,200	9,600	8,000	1,600

Notice the pattern. Moving from 800 to 1,000 notebooks adds Tk 1,600 to EBIT, and so does moving from 1,000 to 1,200. Each step of 200 notebooks adds $200 \times 8 = \text{Tk } 1,600$, because the fixed cost does not grow. Below 1,000 notebooks the EBIT is negative, which means a loss.

REALISTIC NUMBERS

Case file, illustrative. Meghna and Sons, 6 April 2025, printing unit. Break-even: $240,000 \div 40 = 6,000$ packs. EBIT at 7,000 packs: Tk 40,000. EBIT at 8,000 packs: Tk 80,000. EBIT at 9,000 packs: Tk 120,000.

We now follow the arithmetic behind these lines. *Given.* Price Tk 90, variable cost Tk 50, fixed cost Tk 240,000 a year. *Find.* The break-even point, and EBIT at 7,000, 8,000 and 9,000 packs. *Formula.* The same two rules as before: $Q = F \div \text{CM}$, and $\text{EBIT} = Q \times \text{CM} - F$. *Steps.* $\text{CM} = 90 - 50 = \text{Tk } 40$. Break-even $Q = 240,000 \div 40 = 6,000$ packs. At 7,000 packs, $\text{EBIT} = 7,000 \times 40 - 240,000 = 280,000 - 240,000 = \text{Tk } 40,000$. At 8,000 packs, $\text{EBIT} = 320,000 - 240,000 = \text{Tk } 80,000$. At 9,000 packs, $\text{EBIT} = 360,000 - 240,000 = \text{Tk } 120,000$. *Answer.* The unit breaks even at 6,000 packs, and each extra 1,000 packs adds Tk 40,000 to EBIT. *Check.* 1,000 packs $\times$ Tk 40 = Tk 40,000, which matches each step in the list.

Figure 11.1 shows the same unit as a chart. The horizontal axis shows packs sold in a year, and the vertical axis shows taka a year. Each point on a line shows the money at that volume. The revenue line starts at zero and rises by Tk 90 per pack. The total cost line starts at the fixed cost of Tk 240,000 and rises by Tk 50 per pack. The lines cross at 6,000 packs, which is revenue of $6,000 \times 90 = \text{Tk } 540,000$. To the left of the crossing the cost line lies above revenue, and the gap is the loss. To the right, revenue lies above cost, and the gap is profit.

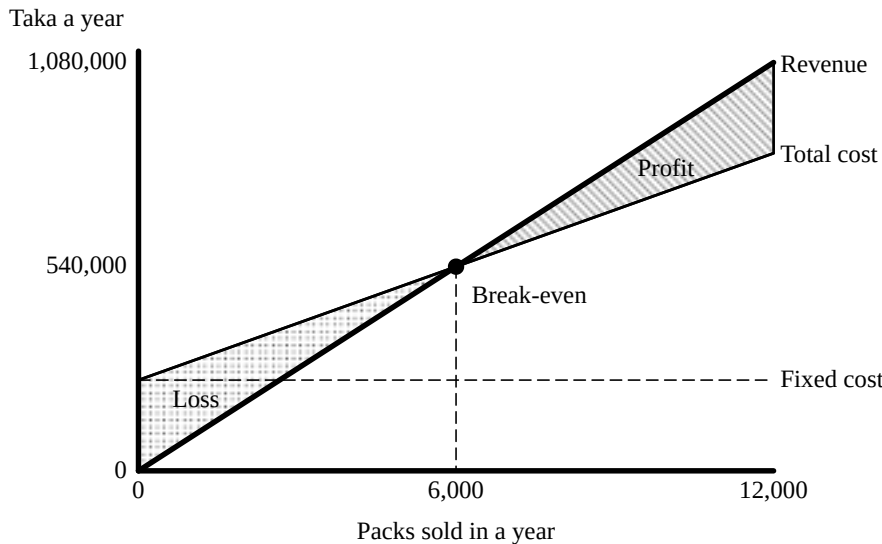


FIG. 11.1. Revenue and total cost of the printing unit, crossing at the break-even point

Three slips are common. The first is dividing the fixed cost by the price instead of the contribution margin; $240,000 \div 90$ gives about 2,667 packs, far too low, because it ignores the variable cost. The second is mixing periods, such as a fixed cost for a year with a volume for a month. The third is giving the answer in taka. If the division leaves a fraction, such as 2,666.7, round up to the next whole unit, because part of a unit cannot be sold.

Each extra 200 notebooks adds the same Tk 1,600 to EBIT, because the fixed cost does not grow.

The printing unit applies the same method to larger, realistic figures.

The steps behind the case file repeat the method: contribution first, then subtract the fixed cost.

The chart shows revenue and total cost as two lines, with loss and profit as the gaps between them.

Common slips are dividing by the price, mixing the periods, and forgetting that break-even is a number of units.

Break-even analysis assumes that price, variable cost and fixed cost stay the same across the volumes studied.

The break-even point is found by dividing the fixed cost by the contribution margin, and EBIT rises with every unit above it.

At the printing unit, a rise in sales of one eighth lifts EBIT by one half; we can now explain why.

Operating leverage. the effect of fixed operating costs in making EBIT change by more than sales change

Operating leverage means fixed costs make EBIT move by a larger percentage than sales.

Degree of operating leverage. the percentage change in EBIT caused by a one percent change in sales

The degree of operating leverage puts a number on the effect: the percentage change in EBIT per one percent change in sales.

The worked steps find the degree of operating leverage at 8,000 packs and at 9,000 packs.

The case file records both values, and the check agrees with the percentages in the puzzle.

The check multiplies the sales change by the degree of operating leverage and compares it with the direct change.

The break-even rule has limits. It assumes that price and variable cost per unit do not change as volume grows, and that the fixed cost stays the same across the range studied. A manager who plans to sell far more than today may meet a new fixed cost, such as a second machine, and then the rule must be applied again with the new figures.

Before moving on. The operating break-even point equals the fixed operating cost divided by the contribution margin. For the printing unit it is 6,000 packs. Below it EBIT is negative, and above it EBIT rises by the contribution margin with each extra pack.

* * *

11.3 DEGREE OF OPERATING LEVERAGE

Return to the puzzle that opened the chapter. At 8,000 packs EBIT is Tk 80,000, and at 9,000 packs it is Tk 120,000. Why does a rise of 12.5% in sales give a rise of 50% in EBIT? Each extra pack adds Tk 40 of contribution, and no extra fixed cost. The extra 1,000 packs therefore add Tk 40,000, and all of it reaches EBIT. Measured against a base EBIT of only Tk 80,000, that is a rise of 50%.

The word comes from the lever, where a small push at one end moves a heavy load at the other. **Operating leverage** is the effect of fixed operating costs in making EBIT change by more than sales change. A business with no fixed operating cost has none, because its EBIT would change by the same percentage as its sales. The larger the fixed cost, the stronger the effect.

We want one number that measures the strength of this effect. The **degree of operating leverage** is the percentage change in EBIT caused by a one percent change in sales. A value of 4.00 means that each 1% change in sales changes EBIT by 4%. It is found from contribution and EBIT at one starting volume. In words: degree of operating leverage equals contribution divided by EBIT. In symbols, $DOL = (Q \times CM) \div EBIT$. The formula arises because a 1% change in sales changes contribution by 1% of $Q \times CM$, all of which reaches EBIT, and dividing by EBIT states that change as a percentage of EBIT.

THE PRINTING UNIT AT TWO VOLUMES

Given. Price Tk 90, variable cost Tk 50, contribution margin Tk 40, fixed cost Tk 240,000. *Find.* The degree of operating leverage at 8,000 packs and at 9,000 packs. *Formula.* $DOL = (Q \times CM) \div EBIT$, where Q is the starting volume. *Steps.* At 8,000 packs, contribution = $8,000 \times 40 =$ Tk 320,000, EBIT = $320,000 - 240,000 =$ Tk 80,000, and $DOL = 320,000 \div 80,000 = 4.00$. At 9,000 packs, contribution = $9,000 \times 40 =$ Tk 360,000, EBIT = $360,000 - 240,000 =$ Tk 120,000, and $DOL = 360,000 \div 120,000 = 3.00$.

Case file, illustrative. Meghna and Sons, 6 April 2025, printing unit. Degree of operating leverage at 8,000 packs = $320,000 / 80,000 = 4.00$. At 9,000 packs = $360,000 / 120,000 = 3.00$. A 12.5% rise in sales from 8,000 to 9,000 packs lifts EBIT by 50%.

Answer. The degree of operating leverage is 4.00 at 8,000 packs and 3.00 at 9,000 packs. *Check.* The rise in sales is $1,000 \div 8,000 = 12.5\%$. Multiplying, $12.5\% \times 4.00 = 50\%$. Directly, $(120,000 - 80,000) \div 80,000 = 50\%$. The two results agree. The value must be taken at the starting volume, which is 8,000 packs for this rise.

Figure 11.2 shows the result in pictures. Each pair of bars compares a value before and after the rise in sales. Both starting values are scaled to 100, so the height of the second bar shows the new value as a share of the first. The sales pair rises from 100 to 112.5, and the EBIT pair rises from 100 to 150. The second pair climbs four times as far as the first.

Figure 11.2 shows the two changes side by side, scaled so that each starting value equals 100.

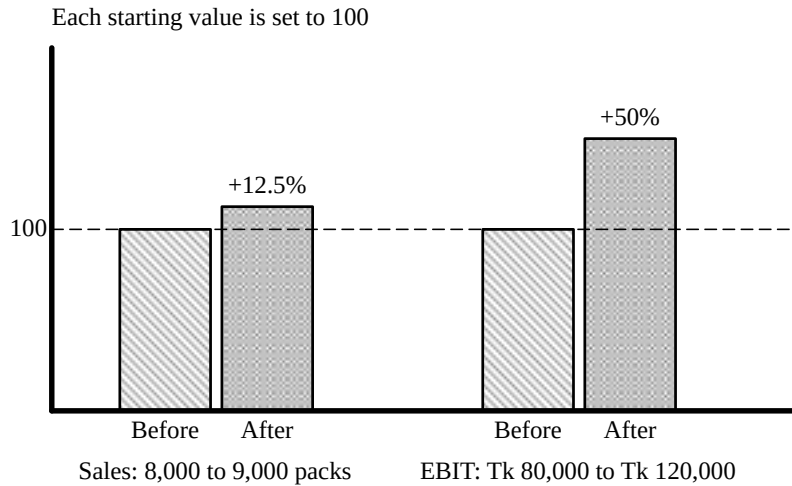


FIG. 11.2. Sales rise by 12.5% while EBIT rises by 50%, with each starting value set to 100

PAUSE AND CHECK 11.P2 A business has a contribution of Tk 60,000 and an EBIT of Tk 20,000. What is its degree of operating leverage, and by what percentage would EBIT change if sales fell by 10%?

The lever works downward as well. If sales fall from 8,000 to 7,000 packs, a fall of 12.5%, EBIT falls from Tk 80,000 to Tk 40,000. That is a fall of 50%, which is $12.5\% \times 4.00$. The case file lists this EBIT of Tk 40,000 for 7,000 packs. The same fixed cost that magnified the gain now magnifies the loss.

The lever works in both directions, so a fall in sales cuts EBIT by the same multiple.

Four points limit the use of the number. First, it changes with volume: it falls from 4.00 to 3.00 as the unit moves from 8,000 to 9,000 packs. Second, it is highest close to break-even, where EBIT is small. Third, at break-even EBIT is zero, so the division cannot be performed and the value is undefined. Fourth, below break-even EBIT is negative and the value has no useful meaning. Common slips include dividing EBIT by contribution, which turns the answer upside down, and using the starting volume of a different rise.

The degree of operating leverage is highest near break-even, falls as volume grows, and cannot be used at break-even.

Before moving on. Operating leverage is the effect of fixed cost in making EBIT move more than sales. The degree of operating leverage equals contribution divided by EBIT at the starting volume, which is 4.00 at 8,000 packs for the printing unit.

The degree of operating leverage is contribution divided by EBIT, and it is large when EBIT is small relative to contribution.

* * *

11.4 OPERATING LEVERAGE AND BUSINESS RISK

Is high operating leverage good? If sales were certain to rise, the answer would be yes, because EBIT would rise faster. Sales are not certain. Business risk is the uncertainty about a business's EBIT.² The printing unit shows both sides: a rise of 12.5% in sales at 8,000 packs lifts EBIT by 50%, and a fall of the same size

Leverage that magnifies gains also magnifies losses, which raises business risk.

² Recall: business risk, Chapter 9, 'Return and Risk', section 9.2 'What Risk Is'.

The more of a business's costs are fixed, the higher its leverage, the larger its swings in EBIT, and the higher its business risk.

Fixed costs raise risk because they cannot be reduced when sales fall.

The notebook seller's numbers are used to find break-even and leverage at two volumes in one problem.

The steps compute the contribution margin, then the break-even point, then EBIT and leverage at each volume.

The answer and check show leverage is higher close to break-even and agree with a direct calculation.

Fixed and variable costs lead to a break-even point and a degree of operating leverage, which together describe business risk.

A printer, separate from Meghna and Sons, is deciding between two cost structures for the same price.

cuts it by 50%. The fixed cost of Tk 240,000 must be paid in a poor year and in a good one alike.

The rule follows. The larger the share of fixed cost in total cost, the higher the degree of operating leverage, the larger the swings in EBIT, and the higher the business risk. Owners may accept this in return for larger gains in good years. Lenders worry that EBIT may fall too low to pay what is owed. Employees may fear lost jobs when sales fall. Operating leverage is only one source of business risk, because demand and input prices also change.

Before moving on. Higher fixed cost means higher leverage and a larger swing in EBIT in both directions. That swing is why operating leverage raises business risk.

* * *

WORKED EXAMPLE: BREAK-EVEN AND LEVERAGE TOGETHER

Given. Selling price Tk 20, variable cost Tk 12, fixed cost Tk 8,000 a year. *Find.* The break-even point, and the EBIT and degree of operating leverage at 1,200 and at 2,000 notebooks, then test the result with a rise of 10% in sales from 2,000. *Formula.* $Q = F \div CM$, $EBIT = Q \times CM - F$, and $DOL = (Q \times CM) \div EBIT$.

Steps. $CM = 20 - 12 = \text{Tk } 8$. Break-even $Q = 8,000 \div 8 = 1,000$ notebooks. At 1,200 notebooks, contribution = $1,200 \times 8 = \text{Tk } 9,600$, $EBIT = 9,600 - 8,000 = \text{Tk } 1,600$, and $DOL = 9,600 \div 1,600 = 6.00$. At 2,000 notebooks, contribution = $2,000 \times 8 = \text{Tk } 16,000$, $EBIT = 16,000 - 8,000 = \text{Tk } 8,000$, and $DOL = 16,000 \div 8,000 = 2.00$.

Answer. Break-even is 1,000 notebooks. The degree of operating leverage is 6.00 at 1,200 notebooks and 2.00 at 2,000 notebooks. *Check.* A rise of 10% in sales from 2,000 gives 2,200 notebooks. EBIT is then $2,200 \times 8 - 8,000 = 17,600 - 8,000 = \text{Tk } 9,600$. The rise is 1,600 on 8,000, which is 20%, and $10\% \times 2.00 = 20\%$. The volume closer to break-even has the higher leverage.

* * *

SUMMARY

A fixed operating cost does not change with units sold, and a variable operating cost does. The contribution margin is price minus variable cost per unit. The operating break-even point equals fixed cost divided by contribution margin. EBIT equals units sold times contribution margin, minus fixed cost. The degree of operating leverage equals contribution divided by EBIT; it is highest near break-even and shows how strongly EBIT responds to sales. Greater leverage raises business risk.

THINK IT THROUGH

A printer sells at Tk 90 per pack. Today the variable operating cost is Tk 50 per pack and the fixed operating cost is Tk 240,000 a year. A new machine would cut the variable cost to Tk 30 per pack and raise the fixed cost to Tk 480,000 a year.

11.T1 Find the operating break-even point before and after the new machine.

11.T2 At 9,000 packs, find the EBIT and the degree of operating leverage before and after the new machine. Which cost structure carries more business risk?

WHERE PEOPLE GO WRONG

Some students say that a fixed cost is the same for every unit. This is wrong. The total fixed cost stays the same, so the fixed cost per unit falls as more units are sold. The printing unit spreads Tk 240,000 over 6,000 packs at Tk 40 each, but over 8,000 packs at Tk 30 each.

Some students divide the fixed cost by the selling price. The price includes the variable cost, which must be paid for each unit. Only the contribution margin is left to cover the fixed cost, so it is the divisor.

Some students think a high degree of operating leverage is always an advantage. It multiplies gains when sales rise and multiplies losses when sales fall. Whether it helps depends on the likely path of sales.

Some students quote one degree of operating leverage for a whole business. It depends on the starting volume. The printing unit has 4.00 at 8,000 packs and 3.00 at 9,000 packs, and no value at break-even.

Fixed cost per unit is not fixed: only the total fixed cost stays the same.

The break-even point divides by the contribution margin, not by the price.

High leverage is not always good, because it magnifies losses as well as gains.

The degree of operating leverage is not one fixed number for a business: it changes with the volume.

IN HISTORY

Henry Ford (1863–1947) was an American car maker who founded the Ford Motor Company in 1903 in Detroit, Michigan. The Model T went on sale in 1908. From 1913, at the Highland Park plant, on the edge of Detroit, Ford's factory used a moving assembly line, in which the car moved past the workers instead of the workers moving to the car. The line needed heavy spending on plant and machinery, which is fixed cost, and it cut the labour needed per car, which is variable cost.

The price of a Model T was about 850 dollars at its launch in 1908 and had fallen to 260 dollars for the Runabout model by 1925. Ford sold the car in far greater numbers than any maker before him. Large fixed costs are carried well only at high volume, which is the lesson of this chapter. A business that raises its fixed costs must be sure its sales will stay above its break-even point.

Henry Ford's moving assembly line raised fixed cost, lowered variable cost per car, and led to a fall in the Model T's price.

The price of the Model T fell from about 850 dollars to 260 dollars, as high volume spread the fixed cost.

REVIEW QUESTIONS

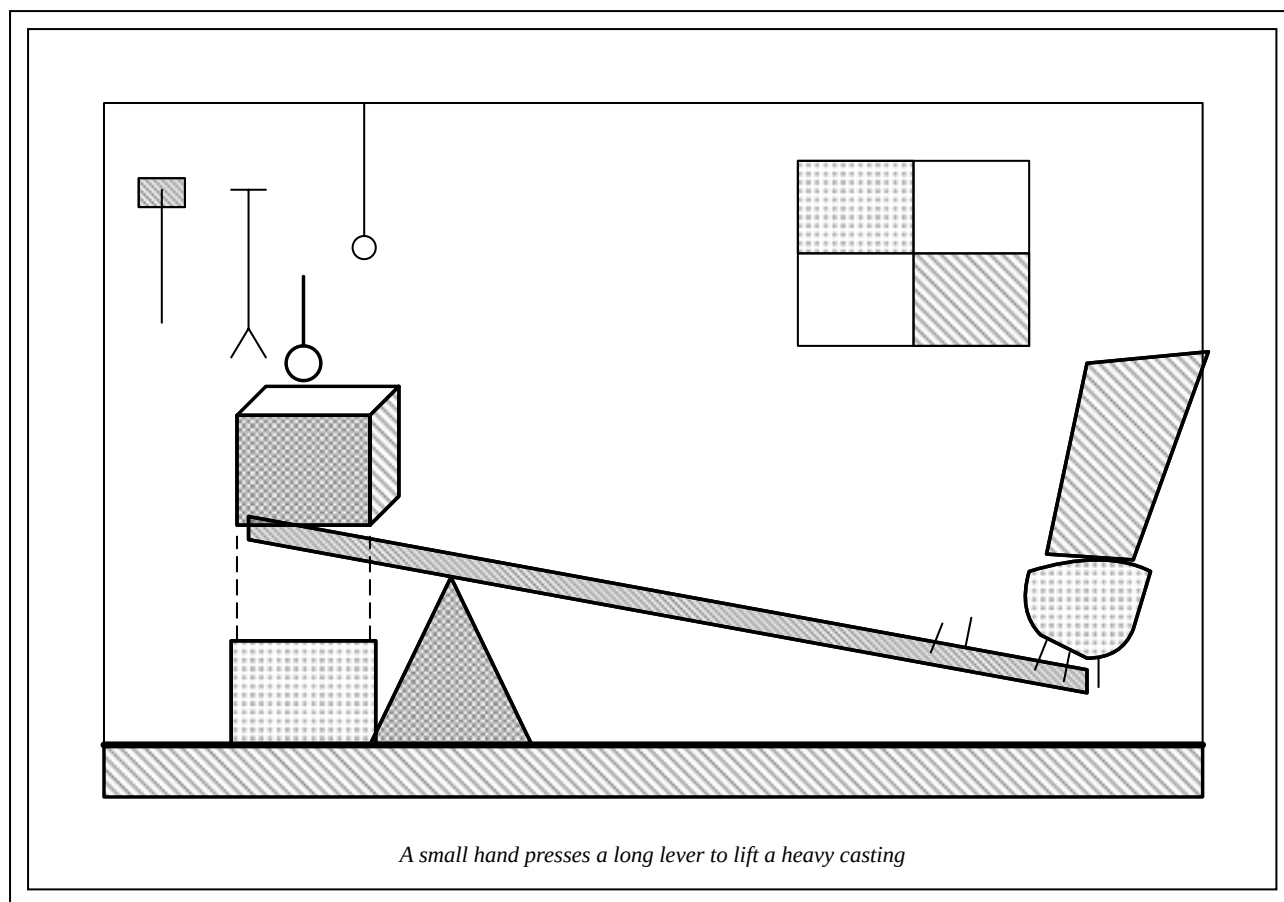
11.1 Explain the difference between a fixed operating cost and a variable operating cost, and give one example of each from a bookshop.

11.2 A business sells at Tk 150 per unit, with a variable operating cost of Tk 95 per unit and a fixed operating cost of Tk 330,000 a year. Find the contribution margin and the operating break-even point.

11.3 For the business in Question 11.2, find EBIT at 5,000, 6,000 and 7,000 units.

11.4 For the same business, find the degree of operating leverage at 8,000 units.

11.5 Explain why the degree of operating leverage is higher near the break-even point than far above it. Use the printing unit's figures.



CHAPTER XII

Financial and Total Leverage

What happens to the owners when a business borrows to push its profit higher?



This chapter shows how borrowing and preferred stock make the profit for each owner's share change by more than the operating profit does. You will calculate that profit for each share, then two measures of how strongly a change in sales or in operating profit moves it. The printing unit data of Chapter 11 is used again, now with its financing added.

Two printing units with the same operating profit react very differently when sales rise.

The answer is a lever: a fixed payment that does not change while profit does.

This chapter measures the lever in three steps.

The chapter extends the data set of Chapter 11 and the idea of financial risk.

A tea stall shows the effect of borrowing in the smallest numbers.

In a poor year the same loan turns the gain into a loss.

Charge. Word in use. In daily life, a price asked for a service, or a claim made against a person. In finance, a cost that a business must pay.

Fixed financing charge. interest or preferred dividend that must be paid whatever the level of EBIT

A fixed financing charge is a payment to those who funded the business, owed at any level of EBIT.

Interest is paid before tax is calculated, but a preferred dividend is paid after it.

Picture two printing units. Both sell the same number of packs, both have the same costs, and both make an operating profit of Tk 80,000. The first was paid for entirely with the owners' own money. The second was paid for partly with a bank loan. Sales rise by 12.5%, and in each unit the operating profit rises by 50%. The owners of the first unit see their profit rise by half. The owners of the second see theirs nearly double. Why?

The difference lies in what the second unit must pay to the bank. The bank wants its interest in a good year and in a poor year alike. When operating profit rises, the owners keep all of the rise, because the interest stays the same. A small hand pressing a long lever can lift a heavy weight, and a fixed payment works in the same way. But a lever works downward as well as upward, and the owners of the second unit would feel a fall equally sharply.

We take three steps. First, we see why fixed payments to lenders and to a second kind of owner make the profit for each share swing more widely. Second, we measure that swing with one number. Third, we join it to the swing from operating costs that Chapter 11 measured, and read the combined number as a sign of risk.

You will need Chapter 11, 'Operating Leverage and Break-even Analysis', sections 11.2 and 11.3, which set out the printing unit data and the degree of operating leverage. You will also need Chapter 9, 'Return and Risk', section 9.2 'What Risk Is', where financial risk was defined.

* * *

12.1 BORROWING AND FIXED FINANCING CHARGES

Suppose you plan a tea stall that costs Tk 10,000. Option one: you pay the whole sum from your savings. Option two: you pay Tk 5,000 and borrow Tk 5,000, and the loan costs Tk 500 in interest each year. In a good year the stall makes Tk 2,000 before any interest. Under option one you keep Tk 2,000, which is 20% of your Tk 10,000. Under option two you keep Tk 2,000 – Tk 500 = Tk 1,500, which is 30% of your own Tk 5,000. Borrowing has lifted your gain, measured on your own money.

Now suppose a poor year, when the stall makes only Tk 400 before interest. Under option one you keep Tk 400, which is 4% of your Tk 10,000. Under option two you owe Tk 500 and receive only Tk 400, so you are Tk 100 short: a loss of 2% on your own Tk 5,000. The loan helped in the good year and hurt in the poor year, because the Tk 500 did not change. The rest of this section gives that fixed payment its name.

A **fixed financing charge** is interest or preferred dividend that must be paid whatever the level of EBIT. *Financing* means paying for a business with funds from outside the day-to-day trade. The tea stall's Tk 500 is a fixed financing charge. Do not confuse it with the fixed operating cost of Chapter 11, such as rent. Both stay the same when sales change. The operating cost is paid to run the trade, and the financing charge is paid to those who supplied the funds.

Two facts about timing matter later. Interest is an expense: the business subtracts it before it calculates tax, so interest lowers the tax bill. A preferred dividend is a share of profit given to a class of owners, so it comes out after tax

and saves no tax. We will need this difference when we calculate. A business that cannot pay its interest risks action by its lender, which is why the charge is called fixed.

Common stock means ownership shares that receive what remains after all other claims, with no fixed dividend. Some countries call it ordinary shares or equity shares; this book uses common stock. The owners of common stock take the most risk and have the best chance of gain. If the business has Tk 33,000 left after all other claims, all of it belongs to them.

Preferred stock means ownership shares with a fixed dividend that is paid before any dividend on common stock. A business that issues Tk 100,000 of preferred stock at 12% promises Tk 12,000 each year to those owners, ahead of the common stock owners. Preferred stock owners do not usually vote, and they cannot ask for more than the fixed dividend. Terms differ between issues and between countries, and Chapter 14 teaches the rights of shareholders in full.¹

The table sets the three claims side by side. Read each row from left to right: who holds the claim, what the holder receives, whether the amount is fixed, and when it is paid.

THREE CLAIMS ON THE PROFIT OF A BUSINESS

CLAIM	WHAT IT RECEIVES	FIXED?	PAID
Lender	Interest	Yes	Before tax
Preferred stock	Preferred dividend	Yes	After tax, before common stock
Common stock	What remains	No	Last

Now we repeat the tea stall in business form, with small round numbers. *Given.* EBIT Tk 10,000; interest Tk 2,000; 1,000 common shares; ignore tax for now. *Find.* The profit for each share, at EBIT of Tk 10,000, Tk 12,000 and Tk 8,000. *Formula.* Profit for each share equals EBIT less interest, divided by the number of shares. The book gives this its proper name, earnings per share, in section 12.2. *Steps.* At Tk 10,000: $(10,000 - 2,000) \div 1,000 = \text{Tk } 8.00$. At Tk 12,000: $(12,000 - 2,000) \div 1,000 = \text{Tk } 10.00$. At Tk 8,000: $(8,000 - 2,000) \div 1,000 = \text{Tk } 6.00$. *Answer.* A 20% rise in EBIT lifts the profit for each share from Tk 8.00 to Tk 10.00, a rise of 25%; a 20% fall lowers it to Tk 6.00, a fall of 25%. *Check.* Without interest the profit for each share would be Tk 10.00, Tk 12.00 and Tk 8.00, and the change would be only 20% each way.

The extra swing has a name. **Financial leverage** is the effect of fixed financing charges in making earnings per share change by more than EBIT. In our example EBIT moved 20% and the profit for each share moved 25%, so financial leverage was at work. If a business has no fixed financing charges, the two percentages are equal. The effect cuts both ways, and the owner, the lender and the financial manager see it differently: the owner gains in good years, the lender gains security from a profitable business, and the manager must judge how much swing the owners can bear.

Share. Word in use. In daily life, a part of something divided among people, such as a share of a meal. In finance, one unit of ownership of a business.

Common stock. ownership shares that receive what remains after all other claims, with no fixed dividend

Common stock owners receive whatever is left, so their gain rises and falls with profit.

Preferred stock. ownership shares with a fixed dividend that is paid before any dividend on common stock

Preferred stock owners are paid a fixed dividend before common stock owners receive anything.

The table compares the three claims on the profit of a business.

A clean example shows the lever with numbers small enough to follow.

Financial leverage. the effect of fixed financing charges in making earnings per share change by more than EBIT

Financial leverage is the extra swing in profit for each share that fixed charges cause.

¹ Chapter 14, 'Valuing Common and Preferred Stock', section 14.1 'Rights of Shareholders', teaches the rights of common and preferred shareholders.

Financial leverage raises financial risk, and it helps only while EBIT covers the charges.

The section's ideas are fixed financing charges, the two kinds of stock and financial leverage.

Meghna and Sons Limited provides a full data set for measuring the swing.

Earnings per share. profit left for common stock owners divided by the number of common shares

Earnings per share is the profit for the common stock owners, shared out per common share.

Five subtractions and one division take EBIT to earnings per share.

A rise in EBIT shows the swing: earnings per share moves by much more.

Degree of financial leverage. the percentage change in earnings per share caused by a one percent change in EBIT

The degree of financial leverage gives the swing as one number.

Financial leverage is the cause of the financial risk that Chapter 9 defined: the extra risk to owners that comes from borrowing.² Two limits follow. First, the lever helps only while EBIT is high enough to cover the charges. If EBIT falls below the interest, the owners make a loss on the unit. Second, borrowing is not wrong in itself; the question is how much swing the owners can bear.

Before moving on. A fixed financing charge is interest or preferred dividend owed at any level of EBIT. Common stock receives what remains, and preferred stock receives a fixed dividend first. Financial leverage is the extra swing in profit for each share that these charges cause.

* * *

12.2 DEGREE OF FINANCIAL LEVERAGE

Case file, illustrative. Meghna and Sons Limited, 1 July 2025. The business was registered to own the shop and the printing unit, and the 8,000-pack data of Chapter 11 is repeated. Sales are 8,000 packs at Tk 90 each. Variable cost is Tk 50 a pack. Fixed operating cost is Tk 240,000. EBIT is therefore Tk 80,000. The printing unit is financed by a bank loan of Tk 200,000 at 10%, giving interest of Tk 20,000, and by Tk 100,000 of preferred stock at 12%, giving a dividend of Tk 12,000. There are 5,000 common shares. The tax rate is 25%.

The owners of common stock care about what is left for them. **Earnings per share** is the profit left for common stock owners divided by the number of common shares. If Tk 33,000 is left and there are 5,000 shares, each share has a claim on Tk 6.60. To find the profit left, we work down from EBIT, taking away each claim in the order of the table above.

Given. The case file above. *Find.* Earnings per share. *Formula.* In words: take EBIT, subtract interest to reach profit before tax, subtract tax, subtract the preferred dividend, and divide by the number of common shares. In symbols, with I for interest, T for the tax rate, Dp for the preferred dividend and N for the number of common shares: earnings per share = $[(EBIT - I) \times (1 - T) - Dp] \div N$. *Steps.* Profit before tax: $80,000 - 20,000 = \text{Tk } 60,000$. Tax at 25%: $0.25 \times 60,000 = \text{Tk } 15,000$. Profit after tax: $60,000 - 15,000 = \text{Tk } 45,000$. Less preferred dividend: $45,000 - 12,000 = \text{Tk } 33,000$. Divide by shares: $33,000 \div 5,000 = \text{Tk } 6.60$. *Answer.* Earnings per share is Tk 6.60. *Check.* The profit left, Tk 33,000, is below profit after tax, Tk 45,000, as it must be, because the preferred owners were paid first.

Now raise EBIT. At 9,000 packs, Chapter 11 gives EBIT of Tk 120,000. Profit before tax is $120,000 - 20,000 = \text{Tk } 100,000$; tax is Tk 25,000; profit after tax is Tk 75,000; less the preferred dividend of Tk 12,000 leaves Tk 63,000. Earnings per share is $63,000 \div 5,000 = \text{Tk } 12.60$. EBIT rose by 50%, from Tk 80,000 to Tk 120,000, but earnings per share rose from Tk 6.60 to Tk 12.60, which is a rise of $6.00 \div 6.60 = 90.9\%$.

The **degree of financial leverage** is the percentage change in earnings per share caused by a one percent change in EBIT. Here a 50% rise in EBIT caused a 90.9% rise in earnings per share, so a one percent rise in EBIT causes about 1.82 percent. In words: degree of financial leverage equals the percentage change in earnings per share divided by the percentage change in EBIT. Here $90.9\% \div 50\% = 1.82$.

² Chapter 9, 'Return and Risk', section 9.2 'What Risk Is', defined financial risk.

We can find the same number from the base figures alone. Interest comes out before tax, but the preferred dividend comes out after it, so we first convert the dividend to its before-tax equivalent by dividing by $(1 - T)$. In symbols: degree of financial leverage = $EBIT \div [EBIT - I - D_p \div (1 - T)]$. Given. EBIT Tk 80,000; I Tk 20,000; D_p Tk 12,000; T 25%. Steps. $12,000 \div 0.75 = \text{Tk } 16,000$. The bottom line is $80,000 - 20,000 - 16,000 = \text{Tk } 44,000$. Then $80,000 \div 44,000 = 1.8182$. Answer. The degree of financial leverage is 1.82. Check. It matches the $90.9\% \div 50\%$ found from the two EBIT levels, and $1.8182 \times 50 = 90.9$.

The same number can be found from one set of figures, without changing EBIT.

PAUSE AND CHECK 12.P1 A business has EBIT of Tk 50,000, interest of Tk 10,000 and no preferred stock. Find its degree of financial leverage.

Watch for these slips. Forgetting to divide the preferred dividend by $(1 - T)$ gives a wrong answer, because the dividend is paid after tax. Using after-tax interest is also wrong, because interest already stands before tax in the formula. The number holds near the EBIT at which it was found, and it changes at other levels of EBIT. With no fixed financing charges the degree is exactly 1.00, and as EBIT falls toward the charges the degree grows without limit.

Common slips come from the preferred dividend and from using the number far from its base.

Before moving on. Earnings per share is the profit left for common stock owners divided by the number of common shares. The degree of financial leverage shows how many percent earnings per share moves for each one percent move in EBIT. For the printing unit it is 1.82.

The section's ideas are earnings per share and the degree of financial leverage.

* * *

12.3 DEGREE OF TOTAL LEVERAGE

Think of two levers joined end to end. A push on the first moves the second, and the second lifts the weight. The printing unit has two such levers. Operating leverage turns a change in sales into a larger change in EBIT. Financial leverage turns a change in EBIT into a larger change in earnings per share. Chapter 11 measured the first lever with the degree of operating leverage.³ Section 12.2 measured the second. What does the whole chain do?

Two levers in a row multiply their effects, and each has been measured.

The table follows the rise from 8,000 to 9,000 packs. Read each row from left to right: the item, its value at 8,000 packs, its value at 9,000 packs, and the percentage change. Sales rise by 12.5%, EBIT by 50%, and earnings per share by 90.9%. Each change is larger than the one before.

The table follows one rise in sales through both levers.

ONE RISE IN SALES THROUGH BOTH LEVERS

ITEM	AT 8,000 PACKS	AT 9,000 PACKS	CHANGE
Sales	Tk 720,000	Tk 810,000	12.5%
EBIT	Tk 80,000	Tk 120,000	50.0%
Earnings per share	Tk 6.60	Tk 12.60	90.9%

To read the next figure, look at the three bars from left to right. The height of each bar is the percentage change in sales, in EBIT and in earnings per share. The lines between the bars show the multiplier from one to the next: 4.00 from sales to EBIT, and 1.82 from EBIT to earnings per share.

The figure shows the three changes as connected bars that grow.

³ Chapter 11, 'Operating Leverage and Break-even Analysis', section 11.3 'Degree of Operating Leverage', measured the first link in total leverage.

Sales to earnings per share: $\times 7.27$ in all
Bar height = percentage change

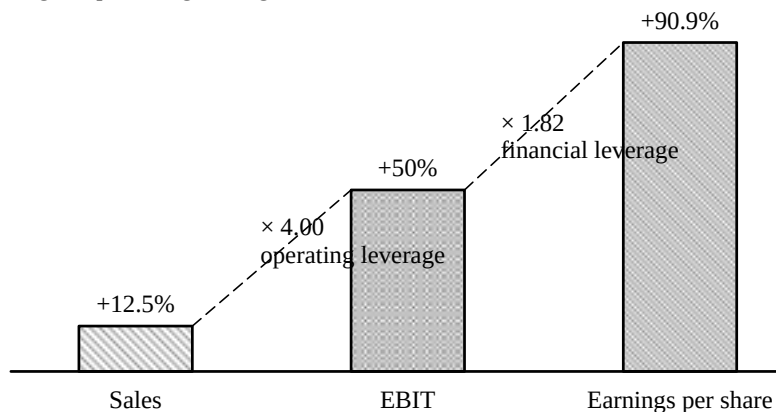


FIG. 12.1. Percentage changes in sales, EBIT and earnings per share, joined by the two degrees

Degree of total leverage. the percentage change in earnings per share caused by a one percent change in sales

The degree of total leverage measures the whole chain from sales to earnings per share.

Because the levers are joined, the two degrees multiply.

The lever works downward too, and the fall is equally sharp.

The section's ideas are the product rule and the downward effect.

The **degree of total leverage** is the percentage change in earnings per share caused by a one percent change in sales. In words: degree of total leverage equals the percentage change in earnings per share divided by the percentage change in sales. From the table, $90.9\% \div 12.5\% = 7.27$. A one percent rise in sales therefore lifts earnings per share by about 7.27 percent.

The same number comes from the two degrees. The degree of operating leverage was 4.00, since the contribution margin of Tk 320,000 (8,000 packs at Tk 40) divided by EBIT of Tk 80,000 is 4.00. The degree of financial leverage was 1.8182. *Formula.* Degree of total leverage = degree of operating leverage $\times$ degree of financial leverage. *Steps.* $4.00 \times 1.8182 = 7.2727$. *Answer.* The degree of total leverage is 7.27. *Check.* A second route gives the same result: contribution margin $\div$ [EBIT - I - Dp $\div$ (1 - T)] = $320,000 \div 44,000 = 7.27$. Also $7.27 \times 12.5\% = 90.9\%$, which agrees with the table.

Now apply the lever to a fall. At 7,000 packs, sales fall by 12.5%. The contribution margin is $7,000 \times \text{Tk } 40 = \text{Tk } 280,000$, and EBIT is $280,000 - 240,000 = \text{Tk } 40,000$, a fall of 50%. Profit before tax is $40,000 - 20,000 = \text{Tk } 20,000$; tax is Tk 5,000; profit after tax is Tk 15,000; less the preferred dividend of Tk 12,000 leaves Tk 3,000. Earnings per share is $3,000 \div 5,000 = \text{Tk } 0.60$, a fall of 90.9%. The same 12.5% change in sales that nearly doubled earnings per share has almost wiped it out.

Before moving on. The degree of total leverage is the percentage change in earnings per share for a one percent change in sales, and it equals the degree of operating leverage times the degree of financial leverage. For the printing unit it is 7.27, and it works downward as strongly as upward.

* * *

12.4 LEVERAGE AND THE RISK OF A FIRM

Business risk is the risk that operating profit varies, and operating leverage raises it. Financial risk is the extra risk to owners from borrowing, and financial leverage raises it. Total leverage joins them, so the degree of total leverage is a rough measure of the total risk that owners of common stock carry. A

The two kinds of risk map onto the two levers.

higher number means a wider swing in earnings per share for the same change in sales.

The two levers are not equally easy to change. The trade fixes much of a business's operating leverage: a printing unit needs machines whatever the owners prefer. The financing is a choice. A business with a high degree of operating leverage may therefore choose to borrow little, to keep the total from growing too large. A business with steady sales and little operating leverage can bear more borrowing.

Before moving on. Total leverage joins business risk and financial risk. Because the financing is a choice, owners can lower the total by borrowing less.

A business can choose its financing more freely than its trade.

The section's idea is that total leverage is a measure of total risk.

* * *

WORKED PROBLEM: DEBT REPLACED BY EQUITY

An illustrative business, not the case file, has EBIT of Tk 80,000, a contribution margin of Tk 240,000 and no preferred stock; the tax rate is 25%. Under Plan A it borrows Tk 200,000 at 10% (interest Tk 20,000) and has 5,000 common shares. Under Plan B it replaces the loan by selling 2,000 more common shares at Tk 100 each, giving 7,000 shares and no interest. We compare the plans.

Given. The data above. *Find.* Earnings per share, the degree of financial leverage and the degree of total leverage for each plan, then the effect of a 50% change in EBIT. *Steps for Plan A.* Earnings per share = $(80,000 - 20,000) \times 0.75 \div 5,000 = 45,000 \div 5,000 = \text{Tk } 9.00$. Degree of financial leverage = $80,000 \div 60,000 = 1.33$. The degree of operating leverage is $240,000 \div 80,000 = 3.00$, so the degree of total leverage is $3.00 \times 1.33 = 4.00$. *Steps for Plan B.* Earnings per share = $80,000 \times 0.75 \div 7,000 = 60,000 \div 7,000 = \text{Tk } 8.57$. The degree of financial leverage is $80,000 \div 80,000 = 1.00$, and the degree of total leverage is $3.00 \times 1.00 = 3.00$.

The table lists earnings per share at three levels of EBIT. Read each row from left to right: the level of EBIT, then the earnings per share and the change for each plan.

A reshaped problem tests the same method on two financing plans.

Both plans are worked from EBIT down to earnings per share.

The table shows what each plan does when EBIT changes by half.

EARNINGS PER SHARE UNDER THE TWO PLANS

EBIT	PLAN A	PLAN A CHANGE	PLAN B	PLAN B CHANGE
Tk 40,000	Tk 3.00	-66.7%	Tk 4.29	-50.0%
Tk 80,000	Tk 9.00	0%	Tk 8.57	0%
Tk 120,000	Tk 15.00	+66.7%	Tk 12.86	+50.0%

Answer. Plan A pays more at the base level, Tk 9.00 against Tk 8.57, and swings by 66.7% for a 50% change in EBIT, against 50.0% for Plan B. *Check.* Each swing equals the degree of financial leverage times 50%: $1.33 \times 50\% = 66.7\%$ and $1.00 \times 50\% = 50\%$. Neither plan is right in every case. Plan A suits owners who accept more swing for a higher base figure, and Plan B suits owners who value steadiness.

Plan A gives more in good years and less in poor years, so it carries more risk.

* * *

Fixed financing charges magnify earnings per share, and two degrees measure the effect.

A common error is to think that borrowing always raises profit for owners.

A second error is to treat a preferred dividend like interest.

A third error is to add the two degrees.

A fourth error is to treat a degree as a fixed property of the business.

A record debt-financed buyout showed how borrowing magnifies gains and losses.

The heavy debt forced the business to sell parts of itself.

SUMMARY

A fixed financing charge is interest or preferred dividend owed at any level of EBIT. Financial leverage is the extra swing in earnings per share that such charges cause, and the degree of financial leverage measures it. The degree of total leverage equals the degree of operating leverage times the degree of financial leverage. For the printing unit the three degrees are 4.00, 1.82 and 7.27, and all three work downward as well as upward.

THINK IT THROUGH

12.T1 A business has a high degree of operating leverage. Explain in plain words why its owners might choose to borrow little.

WHERE PEOPLE GO WRONG

Error: "Borrowing always increases earnings per share." It does so only while EBIT is high enough to cover the charges. In a poor year the same borrowing lowers earnings per share more than EBIT falls.

Error: "Preferred dividend is deducted before tax." Interest is deducted before tax, but the preferred dividend is paid out of profit after tax. That is why the formula divides it by $(1 - T)$.

Error: "Total leverage is operating leverage plus financial leverage." The degrees multiply: $4.00 \times 1.82 = 7.27$, and not 5.82.

Error: "The degree of financial leverage never changes." It is found at one level of EBIT and one financing plan, and it changes when either changes.

IN HISTORY

In 1988 Kohlberg Kravis Roberts, an American investment firm usually called KKR, won a bidding contest to buy RJR Nabisco, a large American maker of tobacco and food products. The price was about 25 billion dollars, or 109 dollars a share, the largest takeover of its time, and most of it was paid for with borrowed money.⁴ Such a purchase is called a leveraged buyout. The new owners hoped that profit from the business would exceed the interest, which would give them large gains on the small part they had paid themselves.

The lever turned out to be heavy. With so much interest to pay every year, the business sold parts of itself in the following years to reduce its debt. The episode is told in the book *Barbarians at the Gate*, by Bryan Burrough and John Helyar. It remains a plain lesson in the two edges of financial leverage: the owners gain when profit is high and carry a heavy burden when it is not.

REVIEW QUESTIONS

12.1 A business has EBIT of Tk 150,000, interest of Tk 30,000, a preferred dividend of Tk 18,000, a tax rate of 25% and 10,000 common shares. Find earnings per share.

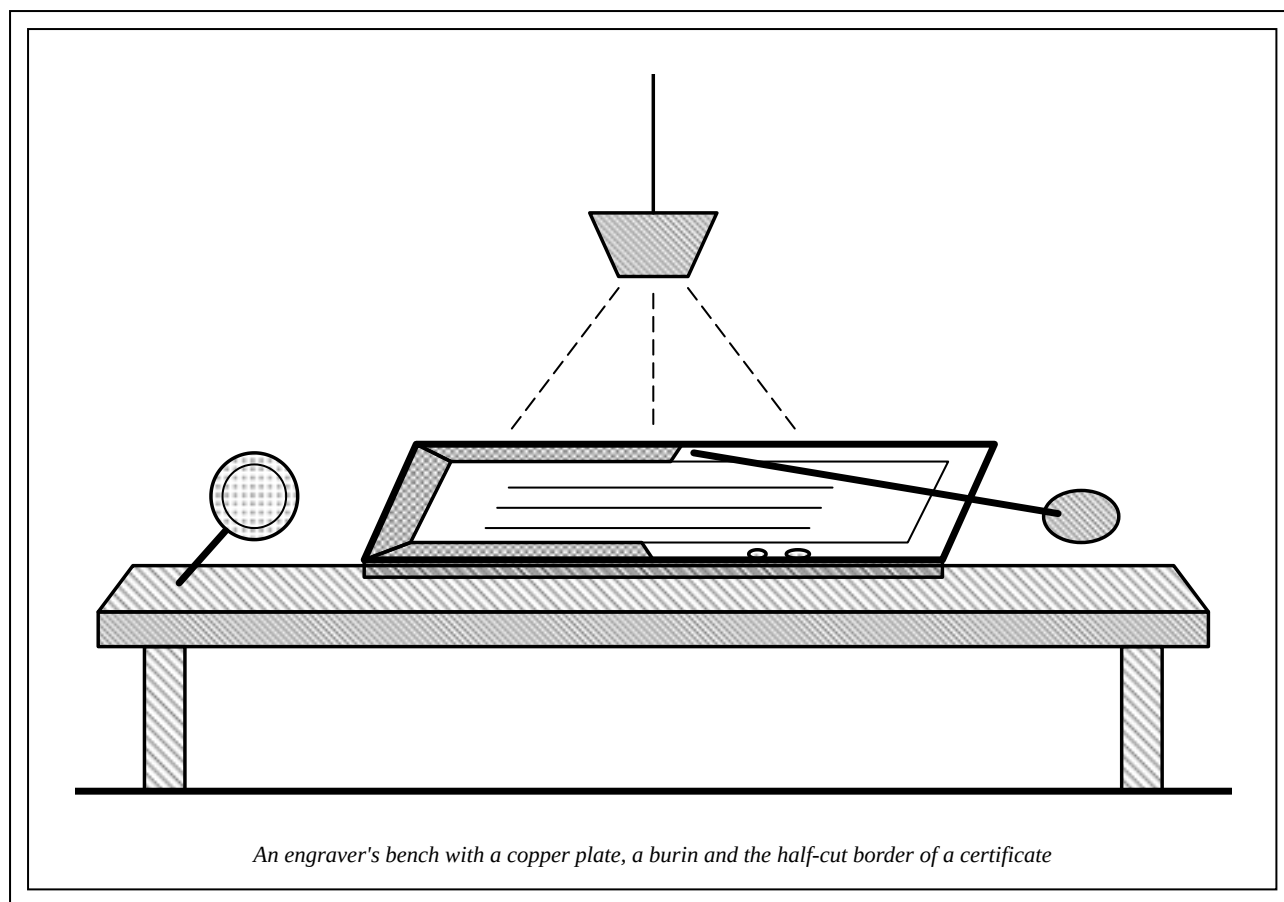
12.2 For the business in question 12.1, find the degree of financial leverage.

⁴ Sources describe the price as about 25 billion dollars; the amount differs slightly with the way debt is counted.

12.3 A business has a degree of operating leverage of 2.50 and a degree of financial leverage of 1.60. Find its degree of total leverage.

12.4 The degree of total leverage is 4.00 and sales rise by 10%. By about what percentage does earnings per share rise? If earnings per share was Tk 5.00, what is it now?

12.5 Two plans give the same EBIT of Tk 100,000 and ignore tax. Plan X pays interest of Tk 40,000 and has 4,000 shares. Plan Y pays no interest and has 8,000 shares. Compare earnings per share at EBIT of Tk 100,000 and Tk 60,000, and say which plan is riskier.



CHAPTER XIII

Valuing Bonds

What is a promise of future payments worth today?



A certificate can promise payments every year and a larger payment at the end. This chapter shows how to find what such a certificate is worth, why its worth rises and falls as market conditions change, and how to measure the return an investor earns by buying it at a given price.

A certificate promises payments for five years, and the puzzle is how much a sensible person would pay for it.

A taka received later is worth less than a taka today, so the price depends on timing and on the return available elsewhere.

The chapter names the certificate, defines worth, calculates it, and then measures an investor's return; the pace is slow because two kinds of payment are valued together.

Many small certificates let a business borrow a large sum from many lenders.

GO SLOWLY HERE.

Bond. a loan sold to investors, repaid at a set date with fixed interest along the way

A bond is a loan sold to investors, repaid at a set date with fixed interest along the way.

Suppose a friend offers you a certificate. It promises Tk 90 at the end of each of the next five years, and Tk 1,000 at the end of the fifth year. The friend asks Tk 1,000 for it today. Should you pay? The promises total $Tk\ 90 \times 5 + 1,000 = Tk\ 1,450$, so the certificate seems a bargain. Many students find this step convincing, because totalling the promises feels natural. Yet the reasoning has a fault, and a second look will show it.

A taka received in five years is worth less than a taka held today, because the taka held today can earn a return in the meantime.¹ So the fair price depends on when each payment arrives and on what return you could earn elsewhere. A person who could earn 12% elsewhere would pay less than a person who could earn 7%. Both could be sensible. The task of this chapter is to turn that idea into a number.

We shall go in four steps. First we name the certificate and its parts. Second we ask what it means for any asset to have a value. Third we calculate the value, using tools from earlier chapters. Fourth we compare the value with the price and measure the return. The pace is slow, because two kinds of payment must be valued at once, and each idea is finished before the next begins.

YOU WILL NEED

Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'. Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'. Chapter 8, 'Capital Budgeting: Evaluation Techniques', sections 8.2 'Net Present Value' and 8.3 'Internal Rate of Return and Profitability Index'.

* * *

13.1 WHAT A BOND IS

Imagine a business that needs Tk 5,000,000 to buy equipment. Few lenders can lend so large a sum alone, and one lender would carry all the risk. The business can instead divide the loan into 5,000 pieces of Tk 1,000 and sell each piece to a different person. Each buyer lends Tk 1,000 and receives a certificate that states what the business has promised to pay. This arrangement is common, and the certificate has a name.

A **bond** is a loan sold to investors, repaid at a set date with fixed interest along the way. The business that borrows is the issuer, and the person who lends is the investor, also called the bondholder. A bond is therefore debt: the investor is a lender and not an owner of the business.² In the example above, the 5,000 certificates together form one loan of Tk 5,000,000.

¹ Recall: present value, Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'.

² Preview: bonds and stock are compared in Chapter 14, 'Valuing Common and Preferred Stock', section 14.4 'Comparing Bonds and Stock'.

A bond has three facts printed or agreed at the start. The first is the **par value**. It means the amount printed on a bond that is repaid at maturity, a date explained below. Par value is also called face value. If the investor lends Tk 1,000 and the bond is repaid with Tk 1,000, then Tk 1,000 is the par value.

The second fact is the **coupon interest**. It is the fixed interest a bond pays each period, set as a percentage of par value. The percentage is called the coupon rate. A bond with a par value of Tk 1,000 and a coupon rate of 9% pays $9\% \times 1,000 = \text{Tk } 90$ each period. In the past, bond certificates carried printed coupons that the investor cut off and presented for payment, and the name has remained.

The third fact is **maturity**, the date on which a bond is repaid. A bond with five years to maturity makes its last coupon payment and repays its par value at the end of year 5. In this chapter, coupon interest is paid at the end of each year, unless we say otherwise. Section 13.3 treats interest paid twice a year.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. A corporate bond: par value Tk 1,000, coupon 9% a year (Tk 90), 5 years to maturity.

Figure 13.1 shows the payments that this bond promises. Read the line from left to right: it is a timeline, and each mark on it is the end of a year, from year 0 (today) to year 5. An arrow pointing up is money received by the investor. The five small arrows are the coupon payments of Tk 90. The one large arrow at year 5 is the repayment of par value, Tk 1,000. The investor receives the last coupon and the par value on the same date.

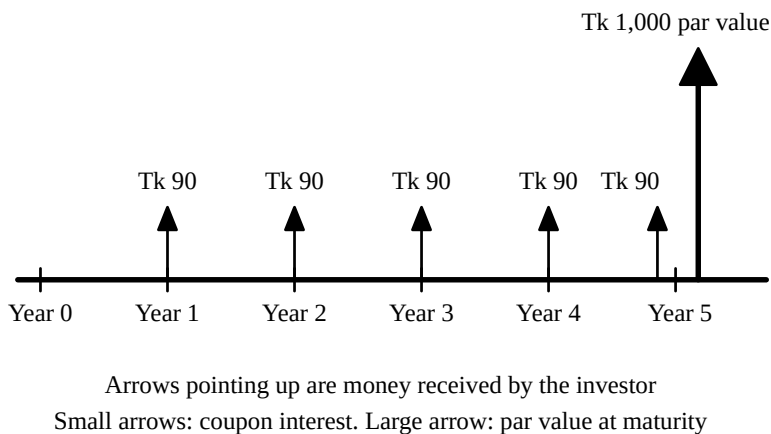


FIG. 13.1. A five-year bond: five coupon payments and a larger payment of par value at maturity

Look at the bond from two viewpoints. For the issuer, the payments are a fixed duty: the Tk 90 each year must be paid whether sales are good or poor. For the investor, the payments are known in advance, which makes the bond easier to

Par value. the amount printed on a bond that is repaid at maturity

Par. Word in use. In daily life, par means an equal level, as in "on a par with". In bonds, par is the amount printed on the bond.

Face. Word in use. In daily life, the front of the head or the front of an object. In bonds, face value is another name for par value.

The par value is the amount printed on the bond and repaid at the end.

Coupon interest. the fixed interest a bond pays each period, set as a percentage of par value

Coupon interest is the fixed interest paid each period, found by applying the coupon rate to par value.

Maturity. the date on which a bond is repaid

Maturity is the date on which the bond is repaid.

The case file records the terms of the corporate bond used throughout the chapter.

The timeline shows five small coupon payments and one larger payment of par value at maturity.

The issuer must pay the fixed amounts in every year; the investor knows them in advance, but may sell before maturity.

A bond has a par value, a coupon interest and a maturity, and together these fix every payment.

To value a bond we first need the idea of value itself.

GO SLOWLY HERE.

Valuation. the work of finding what an asset is worth from the cash it is expected to bring

Valuation is the work of finding what an asset is worth from the cash it is expected to bring.

Intrinsic value. the worth of an asset worked out from the present value of its expected cash flows

Intrinsic value is the present value of the expected cash flows, discounted at the required return.

Intrinsic value is a calculation, while market value is a price, and a gap between the two may guide a decision.

Valuation turns expected cash and a required return into one number, the intrinsic value.

A bond brings two kinds of payment, and each kind is valued with a tool from an earlier chapter.

value than most assets. Two limits apply. An investor need not hold the bond to maturity, and may sell it to another investor at a price that differs from par. And the issuer may fail to pay, which is a risk the investor must judge. A comparison of bonds with stock follows later in the book.

PAUSE AND CHECK 13.P1 A bond has a par value of Tk 500 and a coupon rate of 8% a year. How much coupon interest does it pay each year?

Before moving on. A bond is a loan sold to investors. Its par value is repaid at maturity, and its coupon interest is paid each period. For the case bond these are Tk 1,000, Tk 90 a year and five years. *You should now be able to say what a bond is and what its three facts are.*

* * *

13.2 THE IDEA OF VALUE

A neighbour offers to sell you a mango tree. What is it worth? The cost of planting it does not decide the answer, and neither does its age. The tree is worth what it will bring: the mangoes of the coming seasons, counted by when they will arrive. Ten mangoes next year are worth more to you than ten mangoes in ten years. The same thinking applies to any asset, and to a bond in particular.

Valuation is the work of finding what an asset is worth from the cash it is expected to bring. The word asset here means anything that brings cash, such as a tree, a shop or a bond. Valuation needs two inputs: the cash the asset is expected to bring, and the required return, which is the return the investor needs from an asset of that risk.

The result of valuation has a name. **Intrinsic value** is the worth of an asset worked out from the present value of its expected cash flows. Take the smallest case. An asset will pay Tk 1,100 in one year, and the required return is 10%. Its intrinsic value is $1,100 \div 1.10 = \text{Tk } 1,000$. Today's Tk 1,000, invested at 10%, would grow to the same Tk 1,100.

Do not confuse intrinsic value with market value, the price at which the asset trades today. Intrinsic value is a calculation from expected cash and a required return. Market value is observed in the market. If the intrinsic value is above the market value, the asset looks cheap to the investor who made the calculation; if below, it looks costly. The two may differ because investors disagree about the cash to come or about the required return.

Before moving on. Valuation finds what an asset is worth from the cash it is expected to bring. The intrinsic value is the present value of that cash at the required return. *You should now be able to say why Tk 1,100 due in one year has an intrinsic value of Tk 1,000 at a required return of 10%.*

* * *

13.3 THE VALUE OF A BOND

Look again at Figure 13.1. The bond brings two kinds of payment. The five coupon payments are equal and arrive at equal intervals, which is the pattern of an ordinary annuity.³ The par value is one sum, paid once at maturity, which is

³ Recall: the present value of an ordinary annuity, Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'. The par value is a single sum discounted as in Chapter 4, section 4.3.

the pattern of a single sum. We already know how to find the present value of each pattern. The value of the bond is the two present values added together, using the required return as the discount rate.

Here is the rule. In words: the value of a bond equals coupon interest times the annuity factor, plus par value times the single-sum factor. In symbols, $V = C \times a + P \times f$. Here V is the value of the bond, C is the coupon interest per period, and P is the par value. The annuity factor a equals $[1 - 1 \div (1 + r)^n] \div r$, and the single-sum factor f equals $1 \div (1 + r)^n$. In both, r is the required return per period, and n is the number of periods. The factors can be read from tables or found with a calculator.

THE SMALLEST CASE

Begin with one year, so that only one division is needed. A bond has a par value of Tk 1,000, a coupon rate of 10% and one year to maturity. The required return is 10%. The investor receives the coupon of Tk 100 and the par value of Tk 1,000 at the end of the year, a total of Tk 1,100. The value is $1,100 \div 1.10 = \text{Tk } 1,000$. Here the value equals the par value, and Section 13.4 explains why.

PAUSE AND CHECK 13.P2 A bond has a par value of Tk 1,000, a coupon rate of 5% and one year to maturity. The required return is 10%. What is its value?

CLEAN NUMBERS

Given. Par value Tk 1,000, coupon rate 10%, so $C = \text{Tk } 100$, three years to maturity, required return 10%. *Find.* The value of the bond. *Formula.* $V = C \times a + P \times f$, with $r = 10\%$ and $n = 3$.

Steps. First the single-sum factor: $(1.10)^3 = 1.10 \times 1.10 \times 1.10 = 1.331$, so $f = 1 \div 1.331 = 0.751315$. Second the annuity factor: $a = (1 - 0.751315) \div 0.10 = 2.486852$. Third the present value of the coupons: $100 \times 2.486852 = \text{Tk } 248.685$. Fourth the present value of the par value: $1,000 \times 0.751315 = \text{Tk } 751.315$. Fifth the sum: $V = 248.685 + 751.315 = \text{Tk } 1,000.000$. *Answer.* The bond is worth Tk 1,000.00. *Check.* The coupon rate equals the required return, so the value equals par, as in the one-year case.

REALISTIC NUMBERS

The case bond has a coupon rate of 9%, but investors in bonds of this risk require 12%. *Given.* Par value Tk 1,000, $C = \text{Tk } 90$, five years, required return 12%. *Find.* The value. *Formula.* $V = C \times a + P \times f$. *Steps.* $(1.12)^5 = 1.762342$, so $f = 1 \div 1.762342 = 0.567427$. Then $a = (1 - 0.567427) \div 0.12 = 3.604776$. The present value of the coupons is $90 \times 3.604776 = \text{Tk } 324.430$. The present value of the par value is $1,000 \times 0.567427 = \text{Tk } 567.427$. The sum is $324.430 + 567.427 = \text{Tk } 891.857$.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. Same corporate bond, required return 12%: value Tk 891.86.

Answer. The bond is worth Tk 891.86. *Check.* The bond pays Tk 90 a year while the market requires 12%, so an investor will pay less than par. The result is below Tk 1,000, as expected, and it is more than the par value's present value alone, Tk 567.43, because the coupons add value.

INTEREST PAID TWICE A YEAR

Many bonds pay interest every half-year. Three changes follow. The coupon of each payment is half the yearly coupon. The number of periods is twice the number of years. The required return per period is half the yearly rate. Take the case bond with interest paid half-yearly: the coupon is Tk 45,

GO SLOWLY HERE.

The formula adds the present value of the coupons to the present value of the par value.

A one-year bond is solved first, because it needs only one division.

The first full example uses round numbers so that the method is the only thing to learn.

The steps find the two factors, multiply each by its payment, and add the results.

The case bond is valued at a required return of 12%, which differs from its coupon rate.

The case file records the value at 12%, which is below par.

The value is below the par value of Tk 1,000, and a check by estimate confirms that this is sensible.

Many bonds pay interest twice a year, which changes the coupon, the number of periods and the rate.

The half-yearly value uses 6% for ten periods and gives Tk 889.60.

The case file records the half-yearly value of Tk 889.60.

The half-yearly value is slightly lower because 6% per half-year is the same as a higher return over a full year.

Common slips are using the coupon rate as the discount rate, counting years instead of periods, and forgetting to halve the rate.

The value of a bond is the present value of its coupons plus the present value of its par value, at the required return.

The same bond can be worth more than, equal to, or less than its par value, depending on the required return.

GO SLOWLY HERE.

At a required return of 9% the bond equals par, and at 7% it is worth more than par.

The case file records the three values, and the table sorts them into three kinds of bond.

The table names each case by comparing the value with par.

there are 10 periods, and the required return of 12% a year becomes 6% per half-year.

Steps. With $r = 6\%$ and $n = 10$, $f = 1 \div (1.06)^{10} = 0.558395$, and $a = (1 - 0.558395) \div 0.06 = 7.360087$. The present value of the coupons is $45 \times 7.360087 = \text{Tk } 331.204$. The present value of the par value is $1,000 \times 0.558395 = \text{Tk } 558.395$. The value is $331.204 + 558.395 = \text{Tk } 889.599$.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. Same corporate bond with half-yearly interest (coupon Tk 45, 10 periods) at 12% a year (6% per half-year): Tk 889.60.

Answer. The value is Tk 889.60. *Check.* It is a little below Tk 891.86. Half-yearly coupons arrive earlier, which would raise the value, but 6% each half-year compounds to $1.06 \times 1.06 = 1.1236$, which is a return of 12.36% over a year. This higher yearly return lowers the value by more.

Four slips are common. The first is discounting at the coupon rate instead of the required return. The second is counting years when interest is paid twice a year, where periods are needed. The third is halving the coupon but not the rate, or the reverse. The fourth is adding the present values of the coupons and the par value without checking that both use the same rate and the same number of periods.

Before moving on. The value of a bond equals the present value of its coupons plus the present value of its par value, both at the required return. For half-yearly interest, halve the coupon and the rate, and double the periods. *You should now be able to say why the case bond is worth Tk 891.86 at a required return of 12%.*

* * *

13.4 DISCOUNT, PAR AND PREMIUM BONDS

The case bond has a par value of Tk 1,000. At a required return of 12% it is worth Tk 891.86. What would it be worth if the market required 9%, or 7%? Before calculating, guess the direction. The bond pays a fixed Tk 90 a year. If other investments offer more than 9%, this bond is less attractive and its price should fall. If they offer less, it is more attractive and its price should rise.

Steps at 9%. With $r = 9\%$ and $n = 5$, $a = 3.889651$ and $f = 0.649931$. The present value of the coupons is $90 \times 3.889651 = \text{Tk } 350.069$, and of the par value $1,000 \times 0.649931 = \text{Tk } 649.931$. The sum is Tk 1,000.000. *Steps at 7%.* Here $a = 4.100197$ and $f = 0.712986$. The coupons give $90 \times 4.100197 = \text{Tk } 369.018$, and the par value gives $1,000 \times 0.712986 = \text{Tk } 712.986$. The sum is Tk 1,082.004, which is Tk 1,082.00 to the nearest paisa.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. Same corporate bond. Required return 12%: value Tk 891.86. Required return 9%: Tk 1,000.00. Required return 7%: Tk 1,082.00.

The table below collects the three results. Read each row from left to right: the required return, the value that follows from it, and the name of the kind of bond.

VALUE OF THE CASE BOND AT THREE REQUIRED RETURNS

REQUIRED RETURN	VALUE (TK)	KIND OF BOND
12%	891.86	Discount
9%	1,000.00	Par
7%	1,082.00	Premium

The names come from the price. A **discount bond** is a bond that sells for less than its par value. At 12%, the case bond sells for Tk 891.86, a discount of Tk 108.14. A **premium bond** is a bond that sells for more than its par value. At 7% it sells for Tk 1,082.00, a premium of Tk 82.00. A bond that sells at exactly par, as at 9%, is a par bond.

The rule is this. If the required return is above the coupon rate, the bond sells at a discount. If it equals the coupon rate, the bond sells at par. If it is below the coupon rate, the bond sells at a premium. The reason is that the coupon is fixed. When the market requires more than the coupon pays, the price must fall until the buyer's return reaches the required level. When the market requires less, buyers bid the price up.

Figure 13.2 draws the idea. The horizontal axis shows the required return, and the vertical axis shows the value of the bond. Each point on the curve is the value at one required return. The dashed horizontal line marks par value, Tk 1,000. The curve slopes downward: a higher required return always gives a lower value. It crosses the par line at 9%, the coupon rate. To the left lie premium bonds, and to the right lie discount bonds.

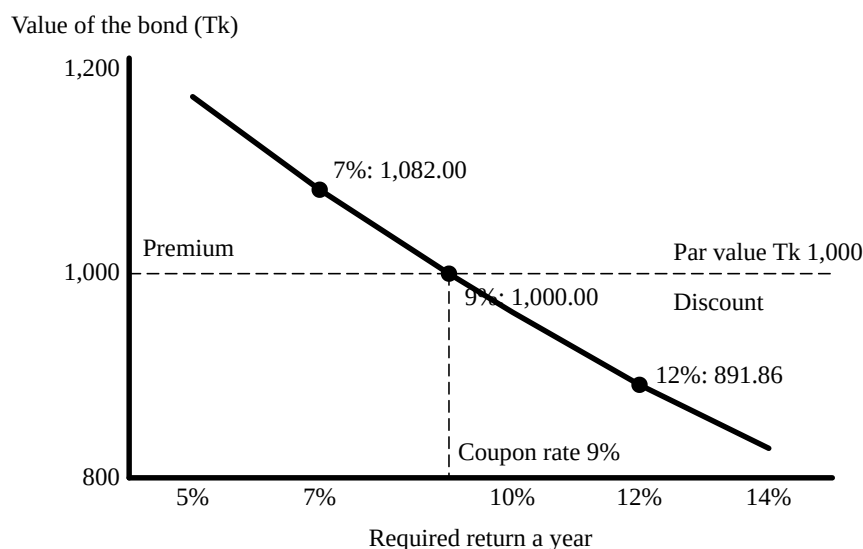


FIG. 13.2. Value of the case bond falling as the required return rises, with par value as a horizontal line

One limit should be noted. A discount or premium does not last. As maturity approaches, the value moves toward par, because the par value is repaid on that date.

Discount bond. a bond that sells for less than its par value

Premium bond. a bond that sells for more than its par value

A discount bond sells below par, and a premium bond sells above par.

The rule links the required return to the coupon rate: higher means discount, equal means par, lower means premium.

Figure 13.2 shows the value falling as the required return rises, crossing the par line at the coupon rate.

A bond's value moves toward par as maturity approaches, because the par value is then repaid.

GO SLOWLY HERE.

Yield to maturity. the yearly return an investor earns by buying a bond at its price and holding it to maturity

Yield. Word in use. In daily life, to yield is to give way, or the amount a crop produces. In bonds, yield is a return stated as a yearly percentage.

Yield to maturity is the yearly return from buying at the market price and holding to maturity.

The case file gives a price of Tk 920 and a yield to maturity of about 11.2%.

The method of trial tries two rates, sees that the price lies between the values, and interpolates.

The answer is checked by calculating the value at the estimated rate.

The yield to maturity differs from the coupon rate whenever the price differs from par.

A discount bond has a higher yield than its coupon rate, a par bond equal, and a premium bond lower.

A new bond is valued with half-yearly interest, and then its yield is found from a market price.

The first part converts to half-years, then values the coupons and the par value.

So far the required return was given, and we found the value. Now reverse the question. The price is given, and we want the return. The **yield to maturity** is the yearly return an investor earns by buying a bond at its price and holding it to maturity. It is found by the same idea as the internal rate of return: the rate that makes the present value of the payments equal the price paid.⁴

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. A bond priced at Tk 920 with the same coupon and 5 years left has yield to maturity about 11.2%.

Given. Price Tk 920, $C = \text{Tk } 90$, par value Tk 1,000, five years. *Find.* The yield to maturity. *Formula.* There is no direct formula, so we use trial. Choose a rate, calculate the value, and compare it with the price. A value above the price means the rate is too low, because a higher rate lowers the value. *Steps.* At 11% the value is Tk 926.08, above the price. At 12% the value is Tk 891.86, below the price. The yield lies between 11% and 12%. The price is $(926.08 - 920) = 6.08$ below the value at 11%, out of a total gap of $926.08 - 891.86 = 34.22$. The fraction is $6.08 \div 34.22 = 0.178$, so the yield is about $11\% + 0.178 \times 1\% = 11.18\%$.

Answer. The yield to maturity is about 11.2%. *Check.* At 11.2% the value is about Tk 919.10, within Tk 1 of the price of Tk 920. The straight-line step is an estimate, and the exact figure is a little lower, at 11.17%.

Compare the three rates for this bond. The coupon rate is 9% and never changes. The price of Tk 920 is below par, so the yield to maturity, 11.2%, is above the coupon rate: the investor gains the coupon and also the rise from Tk 920 to Tk 1,000. A bond at par has a yield equal to its coupon rate. A premium bond has a yield below it.

Before moving on. A bond sells at a discount when the required return is above the coupon rate, at par when equal, and at a premium when below. The yield to maturity is the return from buying at the price and holding to maturity, found by trial. *You should now be able to say why the case bond at Tk 920 has a yield to maturity above 9%.*

* * *

WORKED EXAMPLE: HALF-YEARLY INTEREST AND YIELD BY TRIAL

Given. A bond has a par value of Tk 1,000, a coupon rate of 8% a year paid half-yearly, and three years to maturity. *Find.* First, its value at a required return of 10% a year. Second, the yield to maturity if its price is Tk 940. *Formula.* $V = C \times a + P \times f$, with the half-yearly changes of Section 13.3.

Steps, part one. The coupon is 40, the periods are 6, and the rate is $10\% \div 2 = 5\%$ per half-year. At 5% and 6 periods, $f = 0.746215$ and $a = 5.075692$. The coupons are worth $40 \times 5.075692 = \text{Tk } 203.028$, and the par value is worth $1,000 \times 0.746215 = \text{Tk } 746.215$. The value is Tk 949.243.

⁴ Recall: internal rate of return, Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.3 'Internal Rate of Return and Profitability Index'.

Steps, part two. The price of Tk 940 is below Tk 949.24, so the rate must exceed 5%. Try 6%: $f = 0.704961$ and $a = 4.917324$, so the value is $40 \times 4.917324 + 1,000 \times 0.704961 = 196.693 + 704.961 = \text{Tk } 901.65$. The price lies between the two values. The fraction is $(949.24 - 940) \div (949.24 - 901.65) = 9.24 \div 47.59 = 0.194$, so the half-yearly rate is about $5\% + 0.194 \times 1\% = 5.19\%$. Bond markets state the yearly figure by doubling the half-yearly rate, which gives about 10.4%.

Answer. The value at 10% is Tk 949.24, and the yield to maturity at a price of Tk 940 is about 10.4% a year. *Check.* The bond is a discount bond, so its value is below par, and the coupon rate of 8% is below the required return of 10%, as the rule requires. A price of Tk 940 is below Tk 949.24, so the yield must exceed 10%, and 10.4% does.

* * *

SUMMARY

A bond is a loan sold to investors, with a par value, coupon interest and a maturity. Its intrinsic value is the present value of its coupons plus the present value of its par value, at the required return: $V = C \times a + P \times f$. For half-yearly interest, halve the coupon and the rate and double the periods. The value falls as the required return rises. A bond sells at a discount when the required return exceeds the coupon rate, at par when equal, and at a premium when below. The yield to maturity is the return from buying at the price and holding to maturity, found by trial.

The second part tries 5% and 6% per half-year, then interpolates and doubles the result.

The answers are Tk 949.24 and about 10.4%, and each is tested against the other.

A bond is valued by discounting its coupons and par value at the required return, and its price and yield move in opposite directions.

THINK IT THROUGH

The case bond has a par value of Tk 1,000, a coupon rate of 9% and five years to maturity. It now sells for Tk 940.

13.T1 Does the market require more or less than the coupon rate? Show the check by calculating the value at 10% and at 11%.

13.T2 Is this a discount bond, a par bond or a premium bond? What would the price become if the required return fell to 9%?

A bond price of Tk 940 is compared with the case bond to learn what the market requires.

WHERE PEOPLE GO WRONG

Some students treat the coupon rate as the investor's return. The coupon rate is fixed on par value. An investor who pays Tk 920 earns more than 9%, and one who pays Tk 1,082 earns less. The return depends on the price.

Some students expect a higher required return to give a higher value. The reverse is true. A higher required return means a larger discount on the fixed payments, so the present value falls.

Some students discount the coupons at the coupon rate. This always gives a value equal to par, whatever the market requires, and so answers nothing. The discount rate is the required return.

Some students think the yield to maturity is guaranteed. It is the return the investor earns if every payment is made and the bond is held to maturity. If the issuer fails to pay, or the investor sells early at another price, the return differs.

The coupon rate is not the return an investor earns, because the return depends on the price paid.

A higher required return lowers the value of a bond, and does not raise it.

The coupons are discounted at the required return, never at the coupon rate.

The yield to maturity is a promise only if the investor holds to maturity and the issuer pays in full.

In 1751 Henry Pelham merged several government debts into one stock that paid a fixed rate with no date for repayment.

Consols had no maturity, so the investor received interest forever and could sell the stock for a price that moved with the market.

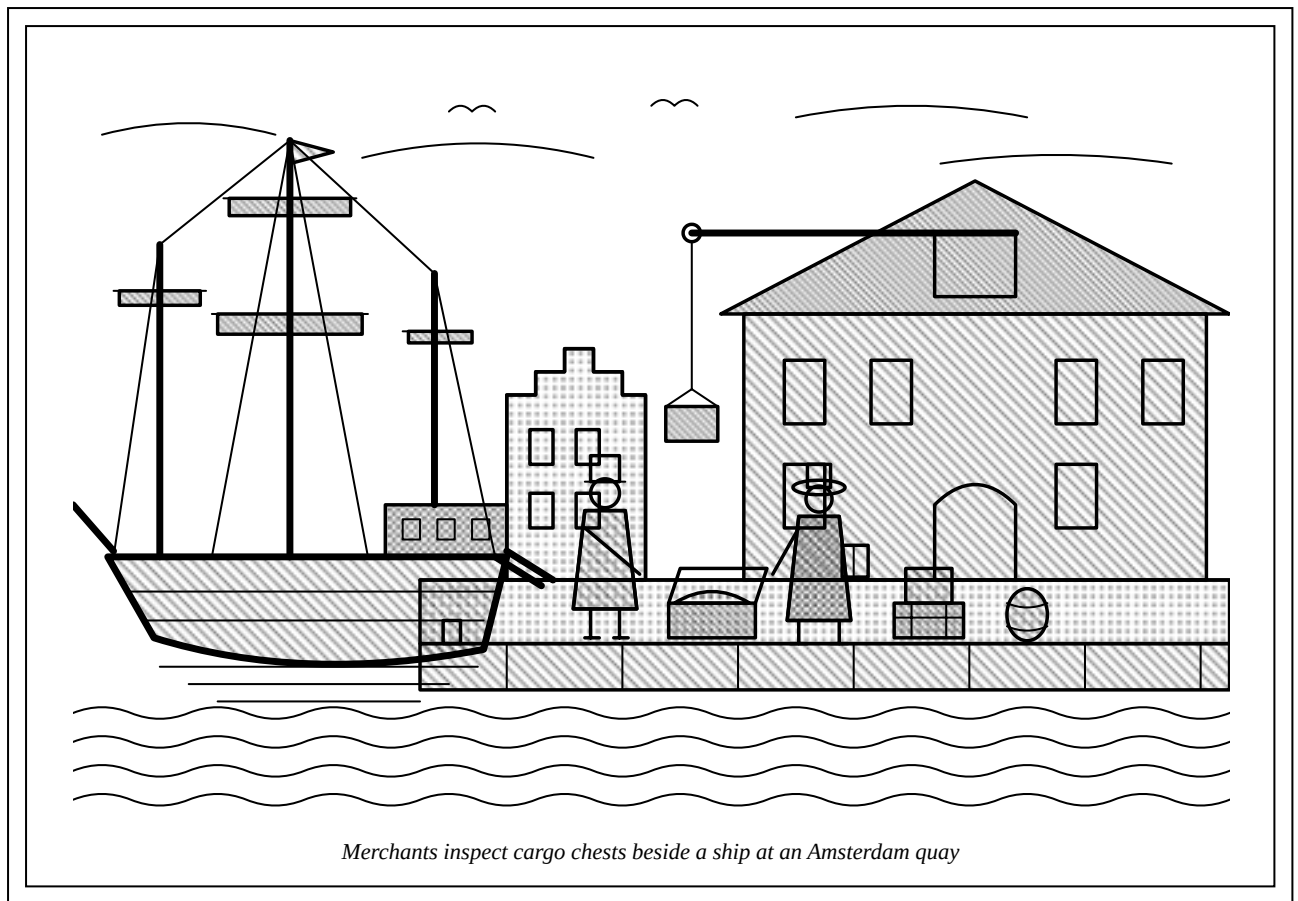
IN HISTORY

In the middle of the eighteenth century, the British government carried many separate debts. Henry Pelham, who was Prime Minister of Great Britain from 1743 until his death in 1754, took advantage of low market rates of interest to reduce the burden. In 1751 he merged several of these debts into one stock, later known as Consols, short for consolidated annuities. Its interest rate was brought down to 3 per cent.

Consols set no date at which the government had to repay. An investor received interest each year and, to turn the holding into cash, sold it to another investor at the market price. Because the interest was fixed, the price rose when market rates fell and fell when they rose, exactly as this chapter describes. The stock created a long-lived market in bonds, and later governments issued similar stocks. They have since been repaid.

REVIEW QUESTIONS

- 13.1 A bond has a par value of Tk 1,000, a coupon rate of 8% and four years to maturity. The required return is 10%. Find its value.
- 13.2 State how the required return compares with the coupon rate for a discount bond, a par bond and a premium bond, and give the reason for each.
- 13.3 A new bond has a par value of Tk 1,000 and a coupon rate of 6% paid half-yearly. It has two years to maturity, and the required return is 8% a year. Find its value.
- 13.4 The case bond (coupon 9%, par value Tk 1,000, five years) is priced at Tk 950. Using trial rates of 10% and 11%, estimate its yield to maturity.
- 13.5 Explain in plain words why the value of a bond falls when the required return rises.



CHAPTER XIV

Valuing Common and Preferred Stock

*What is a piece of a business worth when you cannot see
the profit it will make?*



*This chapter explains what the owners of a business hold and how
to give that holding a price. You will value preferred stock and
common stock with level dividends, then common stock whose
dividends grow at a steady rate. You will end by comparing stock
with bonds.*

YOU WILL NEED

Chapter 12, section 12.1 'Borrowing and Fixed Financing Charges'; Chapter 5, section 5.1 'What an Annuity Is'; and Chapter 13, section 13.2 'The Idea of Value'.

A neighbour offers a piece of her business for Tk 80. Is that fair?

The fair price depends on the payments and on the return you require.

Three questions remain, and the chapter answers each in turn.

Suppose a neighbour offers to sell you a small piece of her business. The piece pays its owner Tk 5 at the end of every year, and she expects the payments to continue for ever. She asks Tk 80 for it. Is that a fair price? You cannot see the future payments, and you cannot ask the business to return your money on demand. All you would hold is a paper that says you own a part. How would you decide what the paper is worth?

Most people begin by asking what they could earn elsewhere. Suppose a safe deposit pays 10% a year. Then Tk 50 in that deposit gives you Tk 5 each year. A price of Tk 80 now looks too high, because Tk 80 in the deposit would give Tk 8. A price of Tk 40 looks like a bargain. This reasoning carries most of the chapter: the worth of a piece of a business is linked to the payments it makes and to the return that you require from your money.

The neighbour's offer still hides three questions. What exactly do you own when you hold the paper? What if the yearly payment is not level but grows? And how does the paper differ from a bond, which also pays regular amounts? We take the questions in that order, beginning with the rights that come with ownership.

* * *

14.1 RIGHTS OF SHAREHOLDERS

Owning stock makes you an owner, and ownership brings a voice and a claim.

Stock. Word in use. In a shop, the goods kept ready for sale. In finance, ownership of a business, held in units called shares.

Share. Word in use. In daily life, a part of something divided among people, such as a share of a meal. In finance, one unit of ownership in a business.

The first right is a vote on how the business is guided.

Voting right. the right of a shareholder to vote on matters such as choosing directors

The second right is to whatever is left after everyone else is paid.

Residual claim. the right to what is left of a business's income and assets after all others are paid

Chapter 12 introduced common stock and preferred stock as the two kinds of stock that a business can issue.¹ Here we ask a new question: what does an owner actually hold? Picture a market stall owned by three cousins. Each cousin may speak and vote when the family meets to decide how to run the stall, and each is entitled to a part of whatever remains after the bills are paid. A voice and a claim: these two ideas are the centre of ownership, and each has a name. A person who owns shares is a shareholder.

The first right is the voice. A **voting right** means the right of a shareholder to vote on matters such as choosing directors. Directors are the people who guide a business on behalf of its owners. A holder of common stock usually receives one vote for each share held, so a holder of 100 shares casts 100 votes at the yearly meeting. The details differ between countries and between businesses, because the law of each country and the rules of each business set them.

The second right is the claim. Owners of common stock hold a **residual claim**, which means the right to what is left of a business's income and assets after all others are paid. The word *residual* means left over. Suppose a business has Tk 100 available after paying its suppliers and staff. It pays Tk 30 of interest to its lender and Tk 20 of dividends to its preferred stockholders. The Tk 50 that

¹ Chapter 12, 'Financial and Total Leverage', section 12.1 'Borrowing and Fixed Financing Charges': common stock and preferred stock were first defined there.

remains belongs to the common stockholders. The directors may pay all of it as dividends or keep some of it inside the business.

The residual claim has a main limit: it has no fixed size. A lender is promised a fixed amount, but an owner is promised nothing. When the business does well, the remainder can be large and the owner gains. When the business does badly, the remainder can be small or nothing. If the business ends, the common stockholders are paid last, after lenders and preferred stockholders, and they may receive nothing. This is why common stock carries more risk than a bond, and why owners expect a higher return for holding it.

Dividends are not the only reward. A **capital gain** is the profit made when an asset is sold for more than it cost. If you buy stock for Tk 50 and later sell it for Tk 56, your capital gain is Tk 6. Dividends and capital gain together form your return from holding the stock. If you sell for less than you paid, you have a loss instead.

Preferred stock is a different bargain. Its holders usually receive a fixed dividend, for example Tk 8 on each share every year. The business must pay the preferred dividend before it pays any dividend to common stockholders. If the business ends, preferred stockholders are paid before common stockholders, though after lenders. In return for this priority, preferred stockholders usually give up the voting right, and their dividend does not rise when the business prospers. Terms differ from one issue to another. Some preferred stock is cumulative, which means that a skipped dividend is still owed and must be paid before any common dividend. Unlike interest on a loan, a preferred dividend is not a legal debt, so the directors can skip it, although doing so harms the business's standing with investors.

Before moving on. Stock makes you an owner. Common stock carries a voting right and a residual claim, and its owner may also gain from selling it at a higher price. Preferred stock pays a fixed dividend ahead of common stock, usually without a vote.

* * *

14.2 VALUING PREFERRED STOCK AND LEVEL DIVIDENDS

How much should you pay for a stream of payments that never ends? The answer comes from an idea that you have used before: the value of any asset is the present value of the cash that it will bring. For stock, the cash is the dividends. A fixed dividend paid at the end of every year for ever is a perpetuity.² Because the dividend is level, the present values of all the payments add up to a short formula.

The rule is short. The value of stock with a level dividend equals the yearly dividend divided by the required return. In symbols, $P = D \div r$. Here P is the value of one share today, D is the dividend paid each year, and r is the required return written as a decimal, so 12% becomes 0.12. Preferred stock fits this rule directly, because its dividend is fixed. Common stock fits it only in the special case where the dividend stays level.

Given. A preferred share pays a fixed Tk 6 each year for ever. The required return is 12%, which is 0.12. *Find.* The value of one share today. *Formula.* Value equals the yearly dividend divided by the required return, or $P = D \div r$. *Steps.* $P =$

The claim has no fixed size: it can be large, or it can be nothing.

A second reward is selling stock for more than you paid.

Capital gain. the profit made when an asset is sold for more than it cost

Preferred stock pays a fixed dividend ahead of common stock, usually without a vote.

Recap of the section on rights.

The value of stock is the present value of the dividends it will pay.

For a level dividend, divide the yearly dividend by the required return.

A clean example: a Tk 6 dividend and a 12% required return.

² Chapter 5, 'Annuities', section 5.1 'What an Annuity Is': a fixed dividend for ever is a perpetuity.

Tk $6 \div 0.12 = \text{Tk } 50.00$. *Answer.* One share of this preferred stock is worth Tk 50.00. *Check.* Tk 50.00 placed in a deposit that pays 12% gives Tk 6 each year, which equals the dividend. An investor who requires 12% should therefore pay no more than Tk 50.00.

The same rule fits common stock only if its dividend really stays level.

The rule has limits. A preferred dividend is fixed by the terms of the issue, so the rule suits it well. A common dividend is decided by the directors year by year, so a level dividend is only an assumption. A business that earns more profit may raise its dividend, and a business that earns less may cut it. The rule also takes the required return as steady, although investors change their minds. The next section removes the first limit by letting the dividend grow.

Recap of the section on level dividends.

Before moving on. Stock is worth the present value of its dividends. When the dividend is level, the value is the dividend divided by the required return. This rule fits preferred stock well and common stock only when the dividend is level.

* * *

14.3 VALUING STOCK WITH GROWING DIVIDENDS

A growing dividend has no last payment, yet its value is not endless.

A publishing business paid a dividend of Tk 4.00 on each share this year, and its directors expect to raise the dividend by 6% every year for ever. Each payment is larger than the one before, and there is no final payment. Must we add up an endless list of present values? The answer is no, because a short formula does the work. First we must see how the dividends behave.

Each dividend is the one before it multiplied by one plus the growth rate.

The dividend paid most recently is the last dividend, written D_0 . The next dividend, D_1 , is the last dividend increased by the growth rate g : $D_1 = D_0 \times (1 + g)$. With $D_0 = \text{Tk } 4.00$ and $g = 6\%$, $D_1 = 4.00 \times 1.06 = \text{Tk } 4.24$. The dividend after that is $4.24 \times 1.06 = \text{Tk } 4.49$, and so on. The figure below shows the first five payments. To read it, look at the horizontal axis, which shows the year, and at the height of each bar, which shows the dividend paid in that year in taka. The first bar is the dividend already paid. The rising bars are the dividends to come.

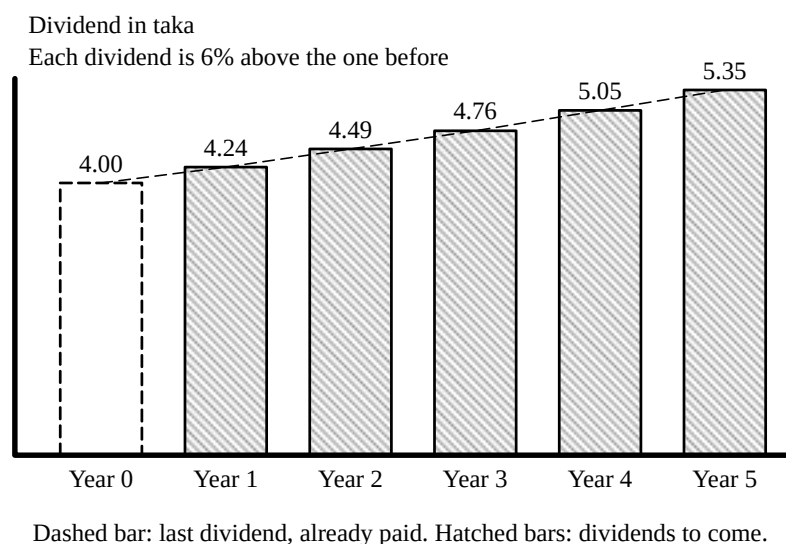


FIG. 14.1. Dividends growing at 6% a year, from Tk 4.00 to Tk 5.35

The pattern gives us the tool. The **dividend growth model** is a formula valuing stock as next dividend divided by required return minus growth rate. In words: value today equals the next dividend divided by the difference between the required return and the growth rate. In symbols, $P_0 = D_1 \div (r - g)$, where P_0 is the value today, D_1 is the next dividend, r is the required return and g is the growth rate, both written as decimals. The formula comes from the endless list of present values. Each dividend grows by the factor $1 + g$, yet is discounted by the factor $1 + r$. When r is larger than g , each present value is a little smaller than the one before, and the endless total settles at $D_1 \div (r - g)$.

IN SHORT

Value today = next dividend $\div$ (required return – growth rate). The next dividend is the last dividend $\times$ (1 + growth rate).

PAUSE AND CHECK 14.P1 A stock's next dividend is Tk 10, its required return is 15% and its growth rate is 5%. What is its value today?

The formula has firm limits. First, the growth rate must stay below the required return. If g equals r , the bottom of the fraction is zero. If g is larger than r , the answer is negative, which has no meaning. No business can keep raising its dividends faster than the required return for ever. Second, the formula assumes steady growth for ever, so it suits mature businesses better than young ones. Third, the answer is very sensitive when r and g are close. With $D_0 = \text{Tk } 4.00$ and $r = 12\%$, a growth rate of 11% gives $4.44 \div 0.01 = \text{Tk } 444.00$, though a growth rate of 6% gives only Tk 70.67. Fourth, a growth rate of zero returns us to the level formula of the last section.

Where does r come from? A stock is risky, so its required return is higher than that of a safe deposit. The capital asset pricing model, which you met earlier, builds r in steps: start with the return on a safe investment, then add the stock's beta multiplied by the extra return that the market pays above the safe rate. Take a safe return of 6%, a market return of 11% and a beta of 1.2. Then $r = 6\% + 1.2 \times (11\% - 6\%) = 6\% + 6\% = 12\%$. Every input is an estimate, so the value we find is an estimate too. It is the stock's intrinsic value, which can differ from the price at which the stock trades.

Before moving on. A growing dividend is valued by the dividend growth model: next dividend divided by the required return minus the growth rate. The growth rate must stay below the required return. The required return comes from the capital asset pricing model.

* * *

14.4 COMPARING BONDS AND STOCK

Both a bond and a stock bring cash to their holder, and both are valued by discounting that cash at the required return.³ The difference lies in the promise. A bond holder is a lender who is promised fixed interest and repayment of the amount lent. A stockholder is an owner who is promised nothing, but who holds a vote and a residual claim. The table below sets out six features. Read each row from left to right: the first cell names a feature, and the next two cells describe the bond and the common stock.

³ Chapter 13, 'Valuing Bonds', section 13.4 'Discount, Par and Premium Bonds': bond value is worked in the same way.

The dividend growth model values stock as next dividend over required return minus growth.

Dividend growth model. a formula valuing stock as next dividend divided by required return minus growth rate

The formula fails when growth equals or exceeds the required return.

The required return on stock reflects its risk and comes from the capital asset pricing model.

Recap of the section on growing dividends.

A bond holder is a lender, and a stockholder is an owner.

BOND AND COMMON STOCK COMPARED

FEATURE	BOND	COMMON STOCK
Holder is	A lender	An owner
Payment	Interest fixed by contract	Dividend decided by the directors
Voting right	None	Usually one vote a share
Claim if the business ends	Paid before owners	Residual claim, paid last
Reward	Interest and repayment	Dividends and capital gain
Risk to the holder	Lower	Higher

From the investor's viewpoint, the choice trades certainty for possible reward.

Recap of the section comparing bonds and stock.

Viewpoint matters. A cautious investor values certain payments and chooses the bond. An investor who accepts risk for a chance of greater reward may choose the stock. The business has its own view. Interest on a bond must be paid on time, whatever the business earns, but a dividend can be reduced when profit falls. Preferred stock sits between the two: its dividend is fixed like interest, yet it is not a legal debt.

Before moving on. A bond holder lends and is paid first, in fixed amounts. A stockholder owns and is paid last, in amounts that can vary. Both are valued as the present value of the cash they bring.

* * *

WORKED CASE: VALUING THREE STOCKS

The case file records three valuations made on one date.

Case file, illustrative. Meghna and Sons Limited, 7 September 2025. Preferred stock paying a fixed Tk 8 a year, required return 10%: value Tk 80.00. Common stock with a level dividend of Tk 5 and required return 10%: value Tk 50.00. Common stock of a publishing business: last dividend Tk 4.00, growth 6% a year for ever, required return 12%: next dividend Tk 4.24; value $4.24 / (0.12 - 0.06) = \text{Tk } 70.67$.

The first two entries use the level formula; each step is shown.

We now check each entry, beginning with the two that use the level formula. *Given.* For the preferred stock, $D = \text{Tk } 8$ and $r = 10\% = 0.10$. For the common stock, $D = \text{Tk } 5$ and $r = 0.10$. *Find.* The value of each. *Formula.* $P = D \div r$. *Steps.* Preferred: $P = 8 \div 0.10 = \text{Tk } 80.00$. Common: $P = 5 \div 0.10 = \text{Tk } 50.00$. *Answer.* The preferred stock is worth Tk 80.00 and the common stock Tk 50.00. *Check.* Tk 80.00 at 10% gives Tk 8.00 a year, and Tk 50.00 at 10% gives Tk 5.00 a year. Both match their dividends. The common result also answers the opening puzzle: a piece paying Tk 5 each year is worth Tk 50.00 to an investor who requires 10%, so Tk 80 was too much.

Now the growing stock: data, formula, steps, answer and check.

Given. Last dividend $D_0 = \text{Tk } 4.00$; growth rate $g = 6\% = 0.06$; required return $r = 12\% = 0.12$. *Find.* The value of one share today, P_0 . *Formula.* Value today equals the next dividend divided by the required return minus the growth rate, or $P_0 = D_1 \div (r - g)$, with $D_1 = D_0 \times (1 + g)$. *Steps.* First, $D_1 = 4.00 \times 1.06 = \text{Tk } 4.24$. Second, $r - g = 0.12 - 0.06 = 0.06$. Third, $P_0 = 4.24 \div 0.06 = \text{Tk } 70.67$, rounded to two decimal places. *Answer.* One share of this common stock is worth Tk 70.67. *Check.* With no growth, the value would be $4.24 \div 0.12 = \text{Tk } 35.33$, so a higher value for a growing dividend is sensible. A second check: the next dividend is $4.24 \div 70.67 = 0.06$, which is 6% of the value, and the value itself

grows 6% a year as a capital gain. Together they give 12%, equal to the required return.

Many students make the same slips here, so check for them. First, writing the percentage as a whole number: $8 \div 10$ gives 0.8, not 80.00, so convert 10% to 0.10 before dividing. Second, using the last dividend where the formula needs the next one: $4.00 \div 0.06$ gives Tk 66.67, which is wrong because D_1 is Tk 4.24. Third, adding the growth rate to the required return instead of subtracting it. Fourth, using the formula when the growth rate is not below the required return. Always write the unit, Tk, with the answer.

Four slips are common when using these formulas.

* * *

SUMMARY

Stock makes its holder a part-owner. Common stock carries a voting right and a residual claim, and its owner can also make a capital gain. Preferred stock pays a fixed dividend ahead of common stock. Stock is valued as the present value of its dividends: a level dividend gives $P = D \div r$, and a dividend growing at a steady rate g gives $P_0 = D_1 \div (r - g)$, provided g is below r . A bond holder lends and a stockholder owns, so the stockholder carries more risk for a chance of greater reward.

The chapter's main ideas in a few lines.

THINK IT THROUGH

14.T1 A cautious investor with Tk 10,000 wants steady yearly income and fears losing money. She can buy a bond or common stock in a growing business. Which would you advise, and why? Give two reasons from this chapter.

WHERE PEOPLE GO WRONG

Mistake: "All stock pays a fixed amount, as a bond does." Correction: only preferred stock has a fixed dividend. The directors decide each common dividend, and they may raise it, lower it or skip it.

Mistake: believing all stock pays a fixed amount.

Mistake: "The formula value is the price I must pay." Correction: the formula gives intrinsic value, which rests on estimates of growth and required return. The price in the market may be higher or lower.

Mistake: treating the formula value as the price.

Mistake: "The model counts dividends only, so it ignores capital gain." Correction: the model values the whole stream of dividends, and a rising price follows from rising dividends. The check in the worked case found a 6% dividend part and a 6% capital gain part.

Mistake: thinking the growth model ignores capital gain.

Mistake: "A fast-growing business can be given any growth rate." Correction: g must stay below r . When it does not, the formula breaks down or gives a negative value, and the assumption of such growth for ever must be rejected.

Mistake: letting growth reach or pass the required return.

IN HISTORY

Amsterdam in 1602 was a busy trading city in the Dutch Republic. In that year the Vereenigde Oost-Indische Compagnie, known in English as the Dutch East India Company, was formed to carry on trade with Asia. A voyage to Asia took many months each way, and the cost of ships and cargo was high. The company therefore raised money by selling shares to the public.

In 1602 the Dutch East India Company's shares were traded in Amsterdam.

Shares that could be sold gave each investor a way out.

The new feature was that the shares could be transferred. An investor who did not want to wait for the voyages to finish could sell the shares to another person, and buyers began to trade them in Amsterdam within days of the launch (Petram, 2014). Every sale needed a price, and a price needs a way to value a part of a business that cannot be seen. Buyers asked the questions of this chapter: what will the share pay, and what return do I require? Many historians treat the company as the first widely traded joint-stock company, meaning a business whose capital is divided into shares held by many owners. The historian Fernand Braudel cautioned that Amsterdam was not strictly the first stock market, so the claim is best made with care.

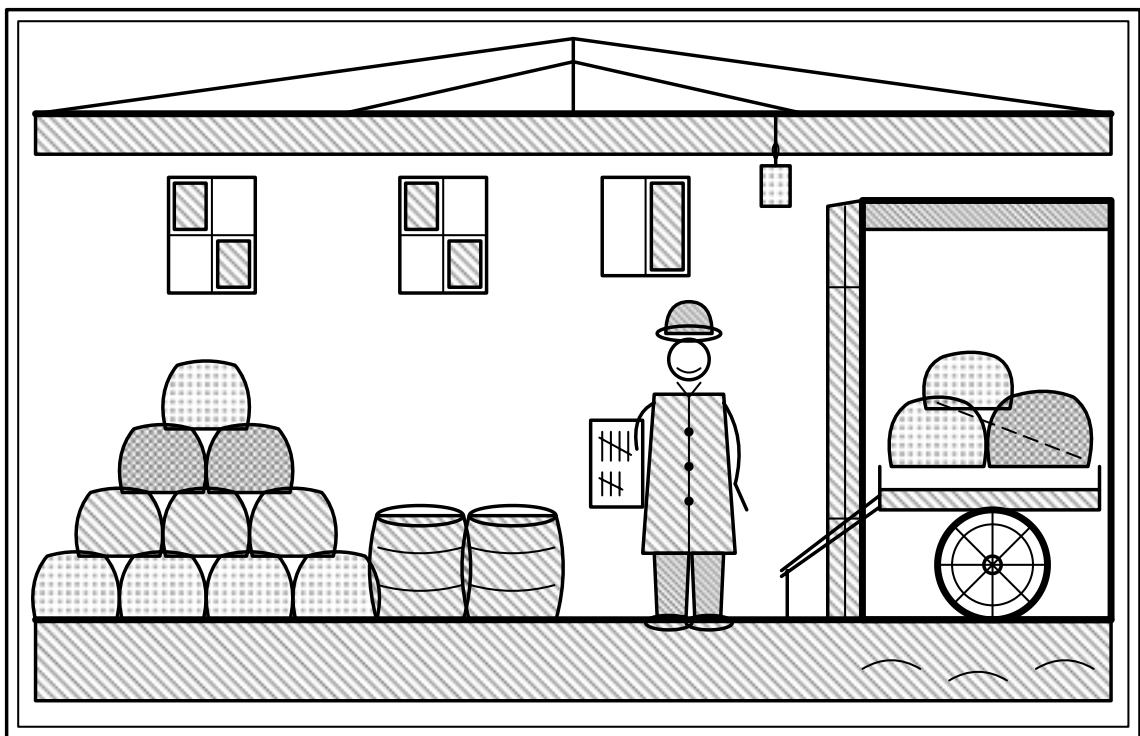
SOURCES

Three works shaped how stock is valued.

John Burr Williams, in *The Theory of Investment Value* (1938), argued that a stock is worth the present value of the dividends it will pay. This is the idea of the whole chapter. Myron Gordon and Eli Shapiro (1956) set out the growth formula that is often called the Gordon model, and that this book calls the dividend growth model. William Sharpe (1964) built the capital asset pricing model, which supplies the required return used in section 14.3. Lodewijk Petram (2014) tells the history of trade in the Dutch East India Company's shares.

REVIEW QUESTIONS

- 14.1 Preferred stock pays Tk 9 a year for ever. The required return is 12%. What is the value of one share?
- 14.2 Common stock has a last dividend of Tk 3.50, a growth rate of 5% and a required return of 11%. Find its value today.
- 14.3 Explain why the growth rate must stay below the required return in the dividend growth model.
- 14.4 Give three differences between a bond and a stock.
- 14.5 A common stock pays a level dividend of Tk 6 for ever. The required return rises from 12% to 15%. Find the new value and the fall in value.
- 14.6 Two stocks each have a last dividend of Tk 4.00 and a required return of 12%. Stock A grows at 4% a year and stock B at 7%. Find the value of each and explain the gap.
- 14.7 An investor requires 10%. She is offered a bond paying Tk 80 a year for 3 years plus Tk 1,000 at the end of year 3, and a common stock with a last dividend of Tk 2.00 and growth of 5% a year. Value one bond and ten shares, and give the total.



A warehouse, a clerk with a tally board and a loaded cart at the door

CHAPTER XV

Working Capital and the Cash Cycle

*How can a business that sells well still have no money to
pay its bills?*



*This chapter follows the money a business holds in goods and in
customers' promises to pay. It measures the delay in days, shows
what a slow cheque costs, and teaches you to estimate the cash a
business must keep working.*

A good year of sales and a record profit, yet an empty cash drawer: how can all three be true?

You will need: Chapter 2, 'The Financial Manager's Work', section 2.3 'How Finance Shapes Business Success'; Chapter 1, 'Money, Finance and the Firm', section 1.3 'Where Funds Come From and Where They Go'.

The answer lies in time: the shop paid before it was paid.

This chapter names the gap, measures it in days, and estimates the money needed to cross it.

A rice stall, with small numbers, shows the idea before any new word appears.

GO SLOWLY HERE.

The table lists what she holds and what she owes on Wednesday evening.

Imagine a bookshop that has had its best year. Sales rose in every month, and the profit is the largest in the history of the business. On the last evening of the year, the owner opens the cash drawer. It is almost empty. A supplier's bill falls due on Monday, and the staff must be paid on Thursday. The owner has a profit but no money. How can both be true?

The answer is not found in the sales. It is found in time. The shop paid for its books before it sold them. It then sold many of them to schools on credit, which means the schools promised to pay later. Between paying and being paid, the shop's money lay in two places where it could not be spent. Some lay on the shelves, in the form of books. The rest lay in the promises of customers. The profit was real, but it was not yet cash.

This gap between paying out and collecting is the subject of this chapter. It matters to every financial manager, because many businesses have failed for lack of cash while their sales were strong. We shall name the parts of the gap and measure it in days. We shall learn why even a cheque needs days to become money that can be used. Then we shall estimate how much money a business must hold to cross the gap. We begin with a market stall, because the idea is the same at any size.

* * *

15.1 WHAT WORKING CAPITAL IS

Before any new word, take the smallest picture we can find. A woman keeps a rice stall in a market. On Monday morning she buys ten sacks of rice from a wholesaler at Tk 1,000 a sack, Tk 10,000 in all. The wholesaler trusts her and lets her pay on Friday. By Wednesday evening she has sold six sacks at Tk 1,200 each. A tea stall and a canteen took Tk 1,500 of that rice on credit, which means they will pay next week. The rest was paid in cash, and the cash is in her tin. Four sacks remain on the stall.

Let us list what she holds and what she owes at that moment. Each row is one item and its value in taka. The first three rows are what she holds, and the fourth row adds them. The fifth row is what she owes. The last row is the difference.

WEDNESDAY EVENING AT THE RICE STALL

ITEM	TK
Cash in the tin	5,700
Four sacks of rice, valued at cost	4,000
Owed to her by the tea stall and the canteen	1,500
Total she holds	11,200
Owed to the wholesaler	10,000
Total held minus total owed	1,200

The cash, the rice and the customers' promises are what the woman uses to trade from day to day. Finance gives the money held in them a name. **Working capital** means the money a business has tied up in short-term assets such as stock and money owed by customers. "Tied up" means held in a form that cannot be spent at once. Her Tk 4,000 of rice cannot pay a bill until it is sold. The word *working* is used because this money keeps moving: from cash into goods, from goods into customers' promises, and back into cash.

Which assets belong in working capital? The short-term ones. A **current asset** is an asset expected to turn into cash within about a year. Her cash is already cash. Her rice will become cash when it is sold. The tea stall's promise will become cash when the tea stall pays. Her cart, which she will use for many years, is not a current asset.

The other side of the table is what she owes. A **current liability** is an amount owed that must be paid within about a year. The Tk 10,000 she owes the wholesaler is one, because it falls due on Friday. A loan that she may repay over five years is not.

The four sacks on her stall are her **inventory**. Inventory means goods held for sale or for use in making goods. The books on a bookshop's shelves are inventory. So are the cloth a tailor holds and the flour a baker holds before baking. Shopkeepers often say stock for the same thing. This book says inventory, and keeps the word stock for shares in a business, as in common stock.

The Tk 1,500 owed by the tea stall and the canteen is called **accounts receivable**. Accounts receivable is money owed to a business by customers who bought on credit. *Credit* here means permission to pay later. *Receivable* means "able to be received". A school that takes books in March and pays in April creates accounts receivable for a bookshop.

The mirror image is the Tk 10,000 owed to the wholesaler. **Accounts payable** is money a business owes to suppliers for goods bought on credit. *Payable* means "to be paid". The same debt appears on both sides of a trade. The wholesaler counts the Tk 10,000 as accounts receivable, and the stallholder counts it as accounts payable.

Look again at the last row of the table. Tk 11,200 held and Tk 10,000 owed leaves Tk 1,200. This is the stall's net working capital. **Net working capital** is current assets minus current liabilities. Here, Tk 11,200 – Tk 10,000 = Tk 1,200. If she owed more than she held in current assets, the answer would be negative. That

Working capital is the money a business keeps in the short-term things it uses to trade every day.

Working capital. the money a business has tied up in short-term assets such as stock and money owed by customers

Stock. Word in use. In daily life and in a shop, the goods on the shelves. In finance, also a share in a business, as in common stock. This book calls goods inventory.

A current asset is an owned item expected to become cash within about a year.

Recall: asset and liability, Chapter 1, section 1.3 'Where Funds Come From and Where They Go'.

Current. Word in use. In daily life, happening now, as in current news. In finance, due or expected within about a year, as in current asset.

Current asset. an asset expected to turn into cash within about a year

A current liability is an amount owed that must be paid within about a year.

Current liability. an amount owed that must be paid within about a year

Inventory is the goods a business holds, either to sell or to turn into other goods.

Inventory. goods held for sale or for use in making goods

Accounts receivable is what customers owe the business for goods taken on credit.

Account. Word in use. In daily life, a record of money kept at a bank. In accounting, a named record of one kind of item, such as the money owed by one customer.

Accounts receivable. money owed to a business by customers who bought on credit

Accounts payable is what the business owes its suppliers for goods taken on credit.

Accounts payable. money a business owes to suppliers for goods bought on credit

Net working capital is the short-term assets left after the short-term debts are subtracted.

Net working capital. current assets minus current liabilities

The case file gives the trading days and the yearly amounts for Meghna and Sons Limited.

Cost of sales is what the goods sold in the year cost the business to buy.

A current asset is only a hope of cash, so a large working capital does not make payment certain.

Recall: liquidity, Chapter 2, section 2.3 'How Finance Shapes Business Success'.

Owner, supplier and financial manager each look at working capital differently.

Recap: working capital is short-term money; net working capital is what is left after short-term debts.

would warn her that she could not pay her short-term debts from her short-term assets.

Case file, illustrative. Meghna and Sons Limited, 1 September 2025 (figures for the 12 months to 31 August 2025). Sales were Tk 9,000,000. Cost of sales was Tk 6,300,000. Inventory was held for 75 days. Customers (school accounts) paid in 20 days. Suppliers were paid in 45 days.

Cost of sales is the price the business paid for the goods it sold in the year. We use it here only as a measure. The shop's inventory and its accounts payable are valued at cost, so we compare them with cost of sales and not with sales.

Working capital has limits. A current asset is a hope of cash, not cash. Rice may not sell. A customer may pay late, or not at all. So a high net working capital does not guarantee that bills can be paid on Monday, if most of it is unsold goods. This is a question of liquidity, which you met in Chapter 2.

The owner sees working capital as money that cannot be taken out of the business. The supplier sees it as a sign that the business can pay its bills. The financial manager asks two questions: is the amount enough to keep trading, and is it larger than it needs to be?

Before moving on. Working capital is the money a business holds in short-term assets. Current assets turn into cash within about a year, and current liabilities fall due within about a year. Net working capital is the difference. Inventory, accounts receivable and accounts payable are the three items we follow in the next section. *You should now be able to say* how the stall's net working capital is found, and why it is Tk 1,200.

* * *

15.2 THE CASH CONVERSION CYCLE

Follow one parcel of goods from arrival to cash, counting the days.

GO SLOWLY HERE.

The operating cycle is the time from buying goods to receiving the customer's cash.

Operating cycle. the days from buying stock to collecting cash from the customer

The shop did not pay its supplier on day 0, so the supplier's money was in the goods for a time.

Follow one parcel of books through Meghna and Sons. On day 0 the books arrive from the supplier. They stand on the shelves for some time. Then a school buys them on credit, and some days later the school pays. Each stage takes days, and we can count them. In the case file, the books stand for 75 days, and the school takes 20 days to pay.

The time from the day the goods arrive to the day the customer's cash is collected has a name. The **operating cycle** is the days from buying stock to collecting cash from the customer. It has two parts. The inventory days are the days the goods wait for a buyer. The receivable days are the days a customer takes to pay. In words: operating cycle = inventory days + receivable days. For the shop, $75 + 20 = 95$ days.

PAUSE AND CHECK 15.P1 A business holds its goods for 40 days before they are sold, and its customers pay in 15 days. How long is its operating cycle?

But the shop did not pay the supplier on day 0. It paid on day 45. For those 45 days the supplier's money, not the shop's own, was in the goods. The operating cycle leaves this out. The shop must find its own money only after day 45, and it needs that money until the customer pays on day 95. That gap lasts $95 - 45 = 50$ days.

The gap has a name. The **cash conversion cycle** is the operating cycle minus the days a business takes to pay its suppliers. In symbols: $C = I + R - P$. Here C is the cash conversion cycle, I the inventory days, R the receivable days and P the payable days, which are the days the business takes to pay its suppliers. For the shop, $75 + 20 - 45 = 50$ days. Read it this way: for 50 days the shop has paid for goods and has not yet been paid by its customers.

IN SHORT

Cash conversion cycle = inventory days + receivable days – payable days. A shorter cycle means the business needs less of its own money.

The figure below is a timeline, and you read it from left to right. Each unit along the line is one day. Day 0 is the day the goods arrive, and day 95 is the day the cash arrives. The bracket at the top is the operating cycle. The three lower bars show inventory days, receivable days and payable days, and each bar starts and ends above the day marked beneath it. The outlined bar at the bottom is the gap, the cash conversion cycle.

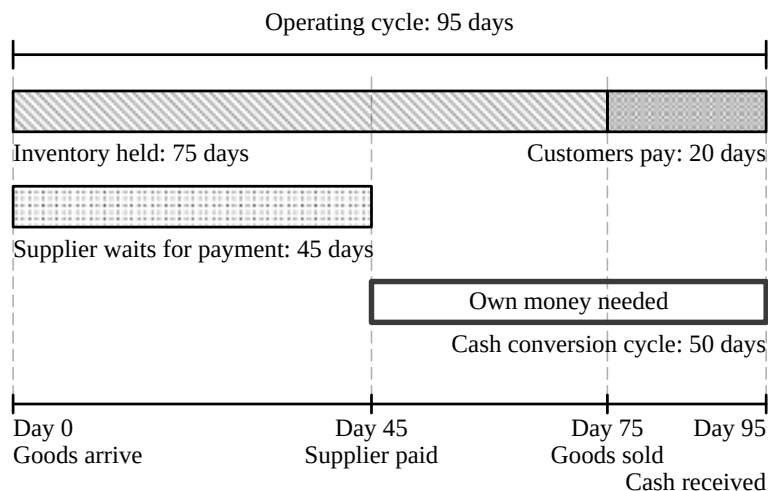


FIG. 15.1. Timeline of the cash conversion cycle for Meghna and Sons Limited

PAUSE AND CHECK 15.P2 A business has an operating cycle of 70 days and pays its suppliers after 30 days. What is its cash conversion cycle?

Three limits matter. First, the days are averages. A shop sells many products, and each has its own days. Second, a business that is paid at the till has no receivable days. A grocer who is paid in cash and pays suppliers after 30 days can have a cash conversion cycle below zero. This means customers' money arrives before the suppliers are paid. Third, a long wait for suppliers is not free. A supplier who is paid late may charge more, or may stop deliveries.

The financial manager therefore has three levers. Hold less inventory. Collect from customers sooner. Pay suppliers later, but fairly. A faster collection cuts the receivable days, so the operating cycle and the cash conversion cycle both shrink by the same number of days. A customer, by contrast, prefers more days to pay. Each party's interest is different.

Before moving on. The operating cycle is inventory days plus receivable days. The cash conversion cycle subtracts the payable days. For Meghna and Sons they are 95 days and 50 days. You should now

The cash conversion cycle counts only the days the business must supply its own money.

Cash conversion cycle. the operating cycle minus the days a business takes to pay its suppliers

The figure shows the three delays on one timeline and the gap that is the cash conversion cycle.

The cycle is an average, and it can be short, long, or even below zero.

The financial manager can shorten the cycle by holding less inventory or collecting sooner.

Recap: the cycle is inventory days plus receivable days, less payable days.

be able to say why the supplier's 45 days reduce the shop's need for its own money.

* * *

15.3 FLOAT

A cheque takes days to become usable money, and the delay has a cost.

The case file records the three delays one cheque met before the shop could use its money.

Float is money paid by one person that the receiver cannot yet use.

Float. money that has been paid by one person but cannot yet be used by the receiver

The figure places the three parts of float on one timeline, from posting to usable money.

Start with one cheque. A customer of Meghna and Sons pays a large bill by cheque. The customer has paid, but can the shop spend the money that same day? It cannot. The case file records what happened.

Case file, illustrative. Meghna and Sons Limited. On 3 February 2025 a customer posted a cheque for Tk 200,000. Mail took 2 days, processing took 1 day and clearing took 3 days. Total float was 6 days. Money unused at 10% a year costs $200,000 \times 0.10 \times 6/365 = \text{Tk } 329$.

While the cheque travelled, the customer had paid but the shop could not use the money. That delay has a name. **Float** is money that has been paid by one person but cannot yet be used by the receiver. It has three parts. Mail float is the days the cheque spends travelling to the receiver. Processing float is the days the receiver takes to record the cheque and pay it into a bank. Clearing float is the days the banks take to move the money from the payer's account to the receiver's. For the shop, $2 + 1 + 3 = 6$ days.

The next figure is a second timeline. Read it from left to right. Each unit is one day. Day 0 is the day of posting, and day 6 is the day the money can be used. Each bar is one part of float, and the line below it shows the total.

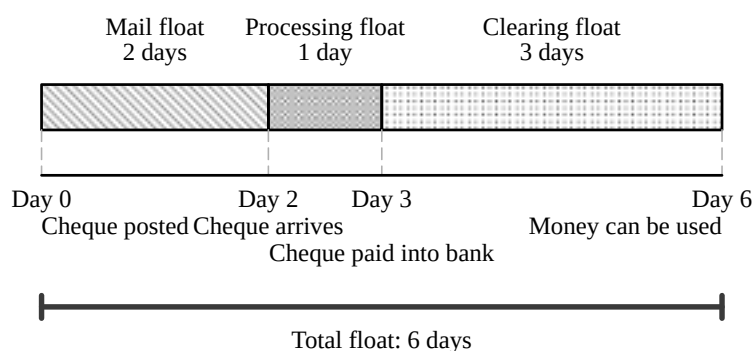


FIG. 15.2. Timeline of one cheque, from posting to usable money

The cost of float is the amount, times the interest rate, times the days lost, divided by 365.

Float shrinks with faster payment methods, and its days differ by bank and country.

Money that cannot be used has a cost, because it could have reduced a loan or earned a return. In words: cost of float = amount $\times$ yearly interest rate $\times$ days of float $\div$ 365. Here, $\text{Tk } 200,000 \times 0.10 \times 6 \div 365 = \text{Tk } 329$. The sum is small for one cheque. A business that receives hundreds of cheques loses much more. Float can also help the payer, who keeps the use of the money until the cheque clears. What the payer gains, the receiver loses.

These days are the case file's illustration. The time a cheque needs depends on the banks and the country, and payment by electronic transfer can remove most of it.

Before moving on. Float is money paid by one person that the receiver cannot yet use. Its parts are mail, processing and clearing. Each day of float costs the receiver the use of the money.

* * *

15.4 ESTIMATING WORKING CAPITAL NEEDS

How much money does the cycle hold? Knowing that it lasts 50 days is not enough, because a manager must know how many taka to find. We turn each of the three items into taka. The idea is simple to state. One day of trading carries a certain amount, and the item holds that many days of it.

There are three rules, each using a 365-day year.¹ Inventory is held at cost, so $\text{inventory} = \text{cost of sales} \times \text{inventory days} \div 365$. Accounts receivable is what customers owe at the selling price, so $\text{receivables} = \text{sales} \times \text{receivable days} \div 365$. Accounts payable is owed at cost, so $\text{payables} = \text{cost of sales} \times \text{payable days} \div 365$. The net investment in the cycle is $\text{inventory} + \text{receivables} - \text{payables}$.

Case file, illustrative. Meghna and Sons Limited, 1 September 2025. Using a 365-day year, inventory was Tk 1,294,521, receivables (on sales) were Tk 493,151 and payables were Tk 776,712. The net investment in the cycle was Tk 1,010,959.

So the shop must find about Tk 1,010,959 of its own money to keep the cycle turning. Its suppliers fund Tk 776,712 of the goods and the debts, and only for 45 days at a time. The full working is shown in section 15.5. How a business pays for this money is the work of Chapter 16, 'Short-Term Financing and Matching', section 16.4 'The Matching Principle'.

The need grows with trading. If sales and cost of sales rise by one fifth and the days stay the same, each of the three items rises by one fifth, and so does the net investment.

Three cautions apply. The estimate uses yearly averages, so a busy season can need more than the average. It leaves out cash and the other current assets. And the days can change: a customer who pays later raises the need at once.

Before moving on. Inventory and payables use cost of sales, and receivables use sales. Multiply by the days and divide by 365. Add inventory and receivables, then subtract payables. For Meghna and Sons the answer is Tk 1,010,959.

* * *

WORKED EXAMPLES: FROM DAYS TO TAKA

We work four examples. The first uses small, round numbers, so the method is the only new thing. The second uses the case file. The third changes the situation, and the fourth combines the ideas of this chapter. Each follows one order: given, find, formula, steps, answer, check.

Example 1: clean numbers. Given. A small trader holds inventory for 30 days. Customers pay in 10 days, and suppliers are paid in 20 days. Yearly sales are Tk 365,000, and cost of sales is Tk 219,000. *Find.* The cash conversion cycle and the net investment in the cycle. *Formula.* Cash conversion cycle = $\text{inventory days} + \text{receivable days} - \text{payable days}$. Net investment = $(\text{cost of sales} \times$

¹ Some businesses use a 360-day year to make the arithmetic easier. This book uses 365 days throughout.

Recap: float is paid money that cannot yet be used, in three parts.

The days can be turned into taka to show how much money the cycle needs.

Each item equals one day's amount multiplied by the days it is held.

The case file gives the three amounts and the net investment for Meghna and Sons.

The shop must find about Tk 1,010,959 of its own money, because its suppliers fund only part of the cycle.

The need grows in step with trading: more sales means more money in the cycle.

The estimate uses averages and three items only, so it is a guide and not an exact budget.

Recap: turn each item into taka using days, then add inventory and receivables and subtract payables.

Four examples apply the method, from small numbers to a combined problem.

Example 1 gives the data, the question and the formulas.

The steps take each item in turn and then combine them.

The answer is 20 days and Tk 16,000, and the check explains the Tk 4,000 above cost.

Example 2 repeats the method with the case file figures.

The steps carry two decimal places and round only at the end.

The cycle is 50 days and the net investment is Tk 1,010,959, which passes a reasonableness check.

Four slips cause most wrong answers in this kind of problem.

Example 3 changes the situation: the suppliers ask to be paid sooner.

The steps change only the payables and compare the new need with the old.

The cycle grows by 15 days and the need by Tk 258,904, as the check confirms.

Example 4 combines the cash needed with the cost of float.

The steps find the float, price it, and add it to the net investment.

About Tk 147,945 is always in float, costing about Tk 14,795 a year.

inventory days $\div$ 365) + (sales $\times$ receivable days $\div$ 365) – (cost of sales $\times$ payable days $\div$ 365).

Steps. Cycle: $30 + 10 - 20 = 20$ days. Inventory: $219,000 \times 30 \div 365 = \text{Tk } 18,000$. Receivables: $365,000 \times 10 \div 365 = \text{Tk } 10,000$. Payables: $219,000 \times 20 \div 365 = \text{Tk } 12,000$. Net investment: $18,000 + 10,000 - 12,000 = \text{Tk } 16,000$.

Answer. The cycle is 20 days, and the trader needs Tk 16,000 of its own money in it. *Check.* Cost of sales is Tk 600 a day, and 20 days at that rate is Tk 12,000. The answer is Tk 4,000 higher because receivables are valued at the selling price: $10 \text{ days} \times (\text{Tk } 1,000 - \text{Tk } 600) = \text{Tk } 4,000$.

Example 2: realistic numbers. *Given.* Sales Tk 9,000,000 and cost of sales Tk 6,300,000. Inventory is held 75 days, customers pay in 20 days and suppliers are paid in 45 days. *Find.* The same two answers. *Formula.* The same as in Example 1.

Steps. Cycle: $75 + 20 - 45 = 50$ days. Inventory: $6,300,000 \times 75 \div 365 = 472,500,000 \div 365 = \text{Tk } 1,294,520.55$. Receivables: $9,000,000 \times 20 \div 365 = 180,000,000 \div 365 = \text{Tk } 493,150.68$. Payables: $6,300,000 \times 45 \div 365 = 283,500,000 \div 365 = \text{Tk } 776,712.33$. Net investment: $1,294,520.55 + 493,150.68 - 776,712.33 = \text{Tk } 1,010,958.90$, which rounds to Tk 1,010,959.²

Answer. The cycle is 50 days, and the shop needs Tk 1,010,959 of its own money in it. *Check.* Fifty days of cost of sales is about Tk 863,014. The answer is higher because receivables are counted at the selling price, which includes the shop's profit.

Common slips are these. Using sales where cost of sales belongs, for inventory and payables. Forgetting to divide by 365. Adding payables instead of subtracting them. Mixing periods, such as months for one item and days for another. Write the units beside every number, and check the sign of each line.

Example 3: a reshaped case. This is an exercise built on the case file, not part of it. *Given.* Suppose the suppliers of Meghna and Sons shorten the payment time from 45 days to 30 days. All other figures stay as in Example 2. *Find.* The new cycle and the extra money needed. *Formula.* The same as before.

Steps. New cycle: $75 + 20 - 30 = 65$ days. New payables: $6,300,000 \times 30 \div 365 = 189,000,000 \div 365 = \text{Tk } 517,808.22$. New net investment: $1,294,520.55 + 493,150.68 - 517,808.22 = \text{Tk } 1,269,863.01$, which rounds to Tk 1,269,863. Extra money: $1,269,863 - 1,010,959 = \text{Tk } 258,904$.

Answer. The cycle lengthens by 15 days to 65 days, and the shop must find Tk 258,904 more. *Check.* One day of cost of sales is about Tk 17,260. Fifteen days $\times$ Tk 17,260 is about Tk 258,900, which agrees.

Example 4: a combined problem. *Given.* The case file figures. Suppose every sale of the year is paid by cheque, and each cheque has the 6 days of float found in section 15.3. The interest rate is 10% a year. *Find.* The money held in float on average, its yearly cost, and the total need. *Formula.* Money in float = sales $\times$ float days $\div$ 365. Yearly cost = money in float $\times$ interest rate. Total need = net investment + money in float.

Steps. Money in float: $9,000,000 \times 6 \div 365 = 54,000,000 \div 365 = \text{Tk } 147,945.21$. Yearly cost: $147,945.21 \times 0.10 = \text{Tk } 14,794.52$. Total need: $1,010,958.90 + 147,945.21 = \text{Tk } 1,158,904.11$.

Answer. About Tk 147,945 is always in float, which costs about Tk 14,795 a year. The total need is about Tk 1,158,904. *Check.* One cheque of Tk 200,000 costs Tk 329. Sales are 45 times larger,

² Each amount is rounded to the nearest taka in the case file. Adding the rounded parts gives Tk 1,010,960. The unrounded total rounds to Tk 1,010,959, and that is the figure we use.

since $9,000,000 \div 200,000 = 45$. Then $45 \times 329 = \text{Tk } 14,805$, close to the answer.

Before moving on. The method never changed: days, then taka, then add and subtract. A change in one day count moves the need by that many days of the right amount. Float adds more days of money that cannot be used.

Recap: the same method handles clean numbers, the case file, a changed case and a combined case.

* * *

SUMMARY

Working capital is money held in current assets such as inventory and accounts receivable. Net working capital is current assets minus current liabilities. The operating cycle is inventory days plus receivable days, and the cash conversion cycle subtracts the payable days. For Meghna and Sons the cycle is 50 days and the net investment is Tk 1,010,959. Float is money paid but not yet usable. For one cheque it lasted 6 days and cost Tk 329.

Working capital is money held in short-term assets, and the cash conversion cycle measures how long the business must fund it.

THINK IT THROUGH

15.T1 Meghna and Sons expects sales and cost of sales to rise by one fifth, with the days unchanged. Estimate the net investment in the cycle, and the extra money needed. Explain in plain words why the need rises.

WHERE PEOPLE GO WRONG

Wrong: a business that makes a profit must have money. *Right:* profit is counted when goods are sold, but cash arrives only when customers pay. Money can sit in inventory and accounts receivable while bills fall due.

A profit in the reports is not cash in the drawer.

Wrong: a long cash conversion cycle shows a badly run business. *Right:* some trades, such as selling books to schools, have long cycles. The danger is a long cycle with no plan to fund it.

A long cycle is not a failure; it is a need that must be funded.

Wrong: the later a supplier is paid, the better. *Right:* more payable days shorten the cycle, but a supplier who is paid late may raise prices or stop deliveries.

Paying suppliers later helps only while the supplier agrees to wait.

Wrong: float is the bank's problem. *Right:* the payer keeps the use of the money while the cheque travels, and the receiver loses it. The loss is measured in days and in interest.

Float helps the payer and costs the receiver.

IN HISTORY

W. T. Grant was a chain of variety stores in the United States. William Thomas Grant opened the first of them in Lynn, Massachusetts, in 1906. By the 1970s the chain was large and national, and its reports still showed profit. In October 1975 it filed for reorganisation under Chapter 11 of the United States bankruptcy law, which lets a business keep trading while it tries to settle its debts. It was liquidated soon afterwards.

A large American retailer reported profits yet failed, because its cash was held in credit sales and inventory.

In 1980 two authors, James Largay and Clyde Stickney, published a study of the failure in the *Financial Analysts Journal*. They found that for about ten years before the collapse the chain's operations had not produced cash. The money was tied up in credit sales and inventory, the same items you met in this chapter, until none was left to pay the bills. They argued that watching cash flows would have revealed the danger

Two authors later showed that the chain's operations had not produced cash for years.

years before the usual ratios did. A profit in the reports does not pay a bill. Cash does.

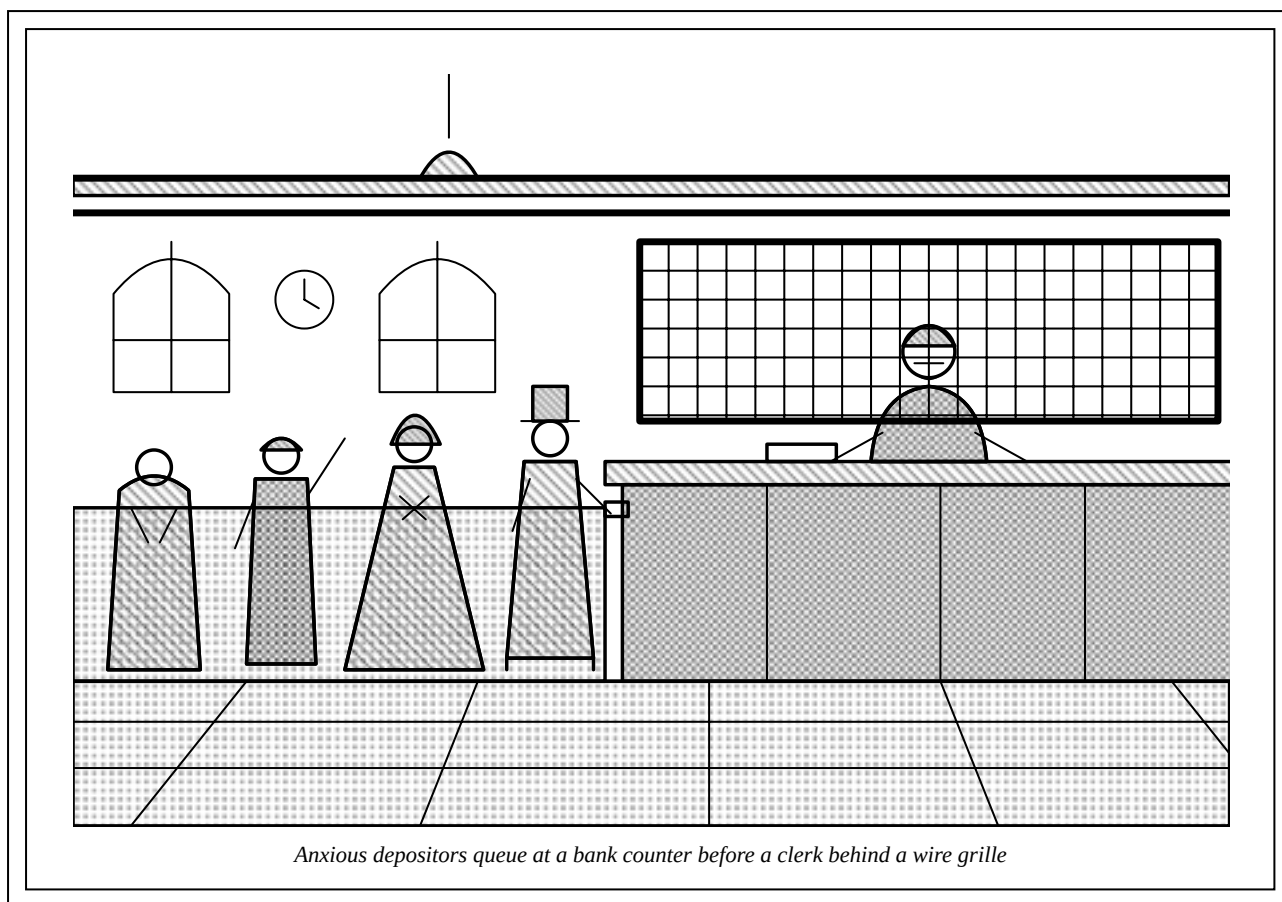
REVIEW QUESTIONS

15.1 A business holds inventory for 60 days, its customers pay in 25 days and its suppliers are paid in 35 days. Find the operating cycle and the cash conversion cycle.

15.2 The business in question 15.1 has yearly sales of Tk 10,950,000 and cost of sales of Tk 7,300,000. Using a 365-day year, find the money it invests in the cycle.

15.3 A customer posts a cheque. Mail takes 3 days, processing takes 2 days and clearing takes 2 days. Name the three parts of float and find the days lost.

15.4 Explain in your own words why a faster collection from customers shortens the cash conversion cycle.



CHAPTER XVI

Short-Term Financing and Matching

Why can a sound bank fail on the day its customers ask for their money?



This chapter sorts financing by the time for which a business holds the money. You will price trade credit, the most common short-term source, and learn when a supplier's discount is worth taking. You will then apply the matching principle to decide which kind of money suits which need.

YOU WILL NEED

Chapter 15, section 15.1 'What Working Capital Is' and section 15.2 'The Cash Conversion Cycle'.

A loan that does not fit the time of the need can ruin a sound plan.

The opposite mistake is to borrow for many years to meet a need that lasts weeks.

Every business must choose how long to hold borrowed money, and each choice has a cost and a danger.

Financing is sorted first by how long the money is held.

Term. Word in use. In daily life, a period of time, such as a school term. In finance, the length of time of a loan, or a condition of a contract, as in credit terms.

There are three spans: up to one year, one to five years, and more than five years.

Short-term financing. money borrowed or raised for up to one year

Intermediate-term financing. money borrowed or raised for between one and five years

Long-term financing. money borrowed or raised for more than five years

Each span has its usual sources, shown in a table.

Suppose you need a motorcycle to start a delivery round. A friend lends you the money but says she must have it back in one month. The motorcycle will earn its cost only slowly, over five years. At the end of the month you have not earned enough to repay her. What happens now? You must find a new lender in a hurry, or sell the motorcycle at a poor price. Your delivery round was sound, but the loan did not fit it.

Now reverse the story. You borrow money for ten years to buy a stock of toys that you will sell in six weeks. After the sale you hold idle cash, yet you still pay interest every month for nine more years. Both mistakes come from one error: the time for which you hold the money does not match the time for which you need it.

Businesses face this choice every day. Money can be raised for a few days or for many years, and each choice brings its own cost and its own danger. This chapter shows how to sort financing by time, how to price the most common short-term source, which is credit from suppliers, and how to decide which kind of money suits which need. It ends with a bank that made the first mistake on a very large scale.

* * *

16.1 FINANCING BY TIME SPAN

Every business needs funds, and Chapter 15 showed that working capital is the money tied up in day-to-day operations.¹ The source of funds matters as much as the amount. The first way to sort sources is by time: for how long will the business hold the money before it must repay it? Three spans are in common use, and the dividing lines are one year and five years.

Each span has a name. **Short-term financing** means money borrowed or raised for up to one year. For example, a bank overdraft that is cleared in two months is short-term financing. **Intermediate-term financing** means money borrowed or raised for between one and five years, such as a loan for a delivery van repaid over three years. **Long-term financing** means money borrowed or raised for more than five years, such as the money that the owners put into the business as equity.

The usual sources follow the spans. The table below names them. Read each row from left to right: the first cell gives the span, the second gives its length, and the third lists the sources that businesses commonly use for it.

¹ Chapter 15, 'Working Capital and the Cash Cycle', section 15.1 'What Working Capital Is': working capital is defined there.

FINANCING SPANS AND THEIR USUAL SOURCES

SPAN	LENGTH	USUAL SOURCES
Short-term	Up to 1 year	Trade credit, bank overdraft, short bank loans
Intermediate-term	1 to 5 years	Loans for equipment or vehicles, repaid in instalments
Long-term	More than 5 years	Bonds, common stock, profit kept in the business, long loans

The lines between the spans are not walls. A bank may renew an overdraft each year, so short-term money can stay with a business for many years. But the bank is free to refuse, and that freedom is the danger. The label describes the agreement, not the hope of the borrower. The long-term instruments, such as loans and bonds, are taught in Chapter 17.²

Viewpoint changes the picture. A lender who gives short-term money keeps the right to ask for it back soon, so the lender takes a smaller risk and usually asks for a lower promise of reward. The owner of the business sees the reverse: short money is easy to obtain but must be found again and again. Long-term money gives the owner peace of mind, but the owner must pay for it every month and cannot return it early without agreement. Each span is therefore a trade between ease, safety and cost.

Before moving on. Financing is sorted by time: up to one year is short, one to five years is intermediate, and more than five years is long. Each span has its usual sources. A short-term label is only as safe as the lender's promise to renew.

* * *

16.2 TRADE CREDIT

Imagine that a supplier sends you books worth Tk 500,000 and offers two choices. You may pay Tk 490,000 within 10 days, or Tk 500,000 within 40 days. The saving is only 2%. Is it worth hurrying to pay? Many people say no, because 2% seems small. We shall find that the answer depends on a short calculation, and that the answer is often surprising.

Start with the idea of delay. **Trade credit** means a supplier allowing a business to pay for goods some days after delivery. A bookshop that receives books on the 1st and pays on the 30th has used the supplier's money for 29 days without paying interest. In the shop's records the unpaid invoice is an account payable. Trade credit is the most common source of short-term financing, because almost every business buys on credit. To the supplier it is a way to win customers. To the buyer it is a loan that appears with each purchase.

Suppliers state their conditions in a standard code. **Credit terms** means the discount offered and the days allowed for payment, written like 2/10, net 40. Read the code in three parts. The first number, 2, is the discount in per cent. The second number, 10, is the number of days in which the discount is available. The word net with the last number, 40, means that the full invoice is due by day 40. So 2/10, net 40 says: take 2% off if you pay within 10 days; otherwise pay the full amount by day 40.

The label describes the agreement, not how long the business hopes to use the money.

Owners and lenders see the three spans from different sides.

Recap of the section on spans.

A supplier's early-payment discount looks small, yet refusing it can be very costly.

Trade credit lets a business pay a supplier after the goods arrive.

Trade credit. a supplier allowing a business to pay for goods some days after delivery

Credit terms are a short code that states the discount and the days allowed.

Credit terms. the discount offered and the days allowed for payment, written like 2/10, net 40

² Chapter 17, 'Long-Term Financing and the Cost of Capital', section 17.1 'Term Loans and Debt Instruments': long-term debt instruments are taught there.

A cash discount rewards early payment with a lower price.

Cash. Word in use. In daily life, coins and notes. In finance, money in hand or in a bank account, ready to be spent.

Cash discount. a reduction in price for paying a supplier early

If you refuse the discount, you borrow from the supplier; we first find the price of that loan.

The cost of giving up a discount has a standard formula.

A cost above the interest rate of other sources means the discount should be taken.

Two sets of terms can be compared by their yearly costs.

Recap of the section on trade credit.

The reduction itself has a name. A **cash discount** means a reduction in price for paying a supplier early. On a Tk 500,000 invoice at 2%, the cash discount is Tk 10,000. A supplier offers it because early money is useful: the supplier can pay its own bills and does not wait for slow customers.

Now the key question. If you do not take the discount, you in effect borrow money from the supplier for the extra days. What is the interest rate on that loan? Use a small example first. The invoice is Tk 10,000 on terms 2/10, net 30. If you pay on day 10, you pay Tk 9,800. If you wait until day 30, you pay Tk 10,000. By waiting 20 more days you keep Tk 9,800 in your hands, and the price of doing so is Tk 200. The cost for the period is $200 \div 9,800 = 2.04\%$. A year holds $365 \div 20 = 18.25$ such periods, so the cost for the year is about $2.04\% \times 18.25 = 37.24\%$.

The same steps give a formula. In words: the cost of giving up the discount equals the discount divided by what you pay if you take it, multiplied by the number of periods in a year. In symbols, $\text{cost} = d \div (1 - d) \times 365 \div (N - D)$. Here d is the discount as a decimal, so 2% is 0.02, D is the last day for the discount, and N is the day on which the full invoice is due. Check with the example: $0.02 \div 0.98 \times 365 \div 20 = 0.020408 \times 18.25 = 0.3724$, or 37.24%. The result is an approximation, because it does not allow for interest earning further interest.

PAUSE AND CHECK 16.P1 A supplier offers terms 1/10, net 30. What is the cost of giving up the discount, as a yearly rate?

What does the cost tell us? Compare it with the interest rate of the cheapest other source. If a bank charges 14% and giving up the discount costs 37.24%, the business should take the discount, borrowing from the bank if it lacks the cash. If giving up the discount costs less than the bank charges, the business should wait. Three cautions apply. Paying after the net date is not free: the supplier may add charges, refuse future credit or raise prices. Where no discount is offered, trade credit appears free, but the price may already include the cost of the delay. And the formula assumes that the full invoice is paid on the net date.

The formula also lets us compare suppliers. Take terms 2/10, net 30 and terms 1/10, net 40, both on an invoice of Tk 10,000. For the first, the cost of giving up the discount is $0.02 \div 0.98 \times 365 \div 20 = 37.24\%$. For the second, it is $0.01 \div 0.99 \times 365 \div 30 = 0.010101 \times 12.1667 = 12.29\%$. The first offer punishes late payment far more, because its discount is larger and its extra days are fewer. A business that must choose which invoice to pay early should pay the one with the higher cost of giving up the discount.

Before moving on. Trade credit is a supplier's permission to pay later. Credit terms such as 2/10, net 40 state the discount and the days. Giving up a cash discount is a costly loan, and its yearly cost is $d \div (1 - d) \times 365 \div (N - D)$.

* * *

16.3 OTHER SHORT-TERM SOURCES

When suppliers cannot supply enough, the business turns to a bank. A **bank overdraft** means a bank allowing an account to go below zero up to an agreed limit, with interest charged on the amount used. Suppose the limit is Tk 100,000 and the business uses Tk 50,000 for 30 days at 12%. The interest is $50,000 \times 0.12 \times 30 \div 365 = \text{Tk } 493.15$. Interest runs only on the amount used and only for the days used, which suits needs that rise and fall. A short loan is different: the bank lends a fixed amount for a fixed period, often a few months, and interest runs on the whole amount until repayment. Banks may refuse to renew either one, and rates and rules differ by bank and by country.

Before moving on. An overdraft charges interest only on the amount used. A short loan charges interest on the whole amount for the whole period. Both are short-term, and both depend on the bank's willingness to continue.

* * *

16.4 THE MATCHING PRINCIPLE

Return to the opening puzzle. The friend's one-month loan was wrong for a five-year motorcycle, and the ten-year loan was wrong for a six-week stock of toys. A rule can prevent both errors. First, though, we must sort a business's needs by time, because some needs never end and others come and go.

A bookshop shows the difference. It always needs some stock, some money in the till and some money owed by customers, whatever the month. **Permanent working capital** means the minimum working capital a business needs all year. Before the school term opens, the shop must also buy extra books. **Seasonal working capital** means extra working capital needed only at the busiest times of the year. The size of both comes from the cash conversion cycle, which sets how much money is tied up in operations.³

The figure below shows both kinds of need. To read it, look at the horizontal axis, which shows the twelve months of the year, and at the height of the shaded area, which shows the money the business needs at that time, in taka. The lower band is the permanent need, which stays level all year. The upper band is the seasonal need, which appears for three months and then disappears.

Banks supply short-term money as overdrafts and as short loans.

Bank overdraft. a bank allowing an account to go below zero up to an agreed limit, with interest charged on the amount used

Recap of the section on banks.

Return to the motorcycle puzzle: the money must fit the need.

Permanent needs stay all year, while seasonal needs appear at the busiest times.

Permanent working capital. the minimum working capital a business needs all year

Seasonal working capital. extra working capital needed only at the busiest times of the year

The figure shows a level permanent need and a seasonal peak above it.

³ Chapter 15, 'Working Capital and the Cash Cycle', section 15.2 'The Cash Conversion Cycle': the cycle sets the size of the need.

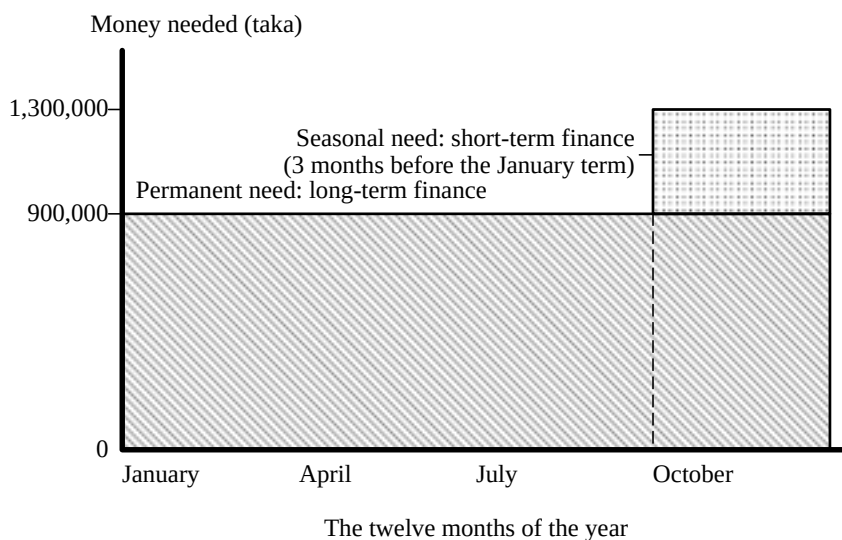


FIG. 16.1. *Permanent and seasonal needs over twelve months, with the finance that matches each*

The matching principle ties each kind of need to a kind of money.

Matching principle. the rule that short-term needs are financed with short-term funds and long-term needs with long-term funds

The rule follows from the figure. The **matching principle** is the rule that short-term needs are financed with short-term funds and long-term needs with long-term funds. In the figure the lower band calls for long-term finance, and the upper band calls for short-term finance. Two reasons support the rule. The first is safety: money borrowed for three months to buy a machine that lasts ten years must be replaced three months later, and the lender may refuse. The second is cost: money borrowed for ten years to cover a three-month peak must be paid for through nine idle months.

IN SHORT

Match the time. Permanent needs are paid for with long-term funds, and seasonal needs with short-term funds.

A small example with round numbers shows why matching saves interest.

A small example with round numbers shows the saving. A shop needs Tk 100,000 all year and a further Tk 50,000 for two months. Long-term money costs 10% a year and short-term money costs 12% a year. If the shop borrows Tk 50,000 short-term for two months, the interest is $50,000 \times 0.12 \times 2 \div 12 = \text{Tk } 1,000$. If it carries the same Tk 50,000 on a long-term loan for the whole year, the interest is $50,000 \times 0.10 = \text{Tk } 5,000$. The short-term rate is higher, yet the matched choice costs Tk 4,000 less, because the money is held for only two months.

The principle is a guide, and a business may choose to lean towards safety or towards low cost.

The principle is a guide, not a law. A cautious business may use more long-term money than the rule needs, so that it is never forced to refinance in a hurry. The price is interest on idle cash. A bolder business may rely more on short-term money, because short-term interest rates are often lower. The price is the danger that lenders refuse to renew. Neither choice is wrong in itself, but each should be made knowingly. A lender prefers to know when it will be repaid, and an owner prefers a low cost and no sudden shortage. Matching serves both.

Before moving on. Needs are permanent or seasonal. The matching principle pays for permanent needs with long-term funds and for seasonal needs with short-term funds. The reasons are safety and cost, and the rule is a guide.

* * *

Recap of the section on matching.

WORKED CASE: A DISCOUNT AND A SEASONAL PEAK

Case file, illustrative. Meghna and Sons Limited, 1 October 2025. A supplier invoice of Tk 500,000 on terms 2/10, net 40. Taking the discount saves Tk 10,000 and costs Tk 490,000 on day 10. Cost of giving up the discount = $2/98 \times 365/30 = 24.83\%$ a year (approximation). A bank overdraft costs 14%: borrowing Tk 490,000 for 30 days costs Tk 5,638, which is less than the Tk 10,000 discount, so take the discount and borrow. Seasonal need: permanent current assets Tk 900,000 financed long term; a seasonal peak of Tk 400,000 for 3 months before the January term. Short-term borrowing at 14% costs Tk 14,000; carrying the Tk 400,000 on a long-term loan at 11% for the whole year costs Tk 44,000.

Given. Invoice Tk 500,000; terms 2/10, net 40; overdraft interest rate 14% a year. *Find.* The yearly cost of giving up the discount, and whether to take the discount. *Formula.* $\text{Cost} = d \div (1 - d) \times 365 \div (N - D)$. *Steps.* First, the discount is $0.02 \times 500,000 = \text{Tk } 10,000$, so the payment on day 10 is Tk 490,000. Second, the extra days are $N - D = 40 - 10 = 30$. Third, $\text{cost} = 0.02 \div 0.98 \times 365 \div 30 = 0.020408 \times 12.1667 = 0.2483$. *Answer.* The cost of giving up the discount is 24.83% a year, which is above the overdraft rate of 14%, so the discount should be taken. *Check.* $\text{Tk } 10,000 \div \text{Tk } 490,000 = 2.04\%$ for 30 days. Twelve such periods give about 24.5%, close to 24.83%, because a year holds slightly more than twelve periods of 30 days.

Given. Borrowing Tk 490,000 for 30 days at 14%. *Find.* The interest cost, to compare with the Tk 10,000 discount. *Formula.* $\text{Interest} = \text{amount} \times \text{yearly rate} \times \text{days} \div 365$. *Steps.* $490,000 \times 0.14 = 68,600$. Then $68,600 \times 30 \div 365 = \text{Tk } 5,638$. *Answer.* The overdraft costs Tk 5,638, which is less than the Tk 10,000 discount, so the business gains $\text{Tk } 10,000 - \text{Tk } 5,638 = \text{Tk } 4,362$ by borrowing to take the discount. *Check.* $5,638 \div 490,000 = 1.15\%$, below the 2.04% that the supplier's terms charge for the same 30 days.

Given. Permanent current assets Tk 900,000 financed long term. Seasonal peak Tk 400,000 for 3 months. Short-term rate 14%; long-term rate 11%. *Find.* The interest cost of each way of financing the peak. *Formula.* $\text{Interest} = \text{amount} \times \text{yearly rate} \times \text{months} \div 12$. *Steps.* Short-term: $400,000 \times 0.14 \times 3 \div 12 = \text{Tk } 14,000$. Long-term for the whole year: $400,000 \times 0.11 \times 12 \div 12 = \text{Tk } 44,000$. *Answer.* Short-term borrowing costs Tk 14,000 and the long loan costs Tk 44,000, so matching saves Tk 30,000 for the year. *Check.* The short rate is higher, 14% against 11%, but it is paid for only 3 months instead of 12, so the total is smaller.

Many students make the same slips here, so check for them. First, using 40 days instead of 30 in the cost formula; the days that matter are the extra days, $N - D$. Second, forgetting that the bottom of the first fraction is $100\% - 2\% = 98\%$, not 100%. Third, mixing months and years: 3 months is $3 \div 12$ of a year. Fourth, comparing the cost of giving up the discount with the interest rate without checking that both are yearly rates.

* * *

SUMMARY

Financing is short-term (up to one year), intermediate-term (one to five years) or long-term (more than five years). Trade credit is a supplier's permission to pay later, and credit terms such as 2/10, net 40 state the discount and the days. Giving up a cash discount is a costly loan, with a yearly cost of $d \div (1 - d) \times 365$

The case file records a supplier discount and a seasonal need on two dates.

First question: is the discount worth taking?

Second question: what does the overdraft cost for the 30 days?

Third question: which source is cheapest for the seasonal peak?

Four slips are common in these calculations.

The chapter's main ideas in a few lines.

$\div (N - D)$. A bank overdraft charges interest on the amount used. The matching principle pays for permanent working capital with long-term funds and for seasonal working capital with short-term funds.

THINK IT THROUGH

16.T1 A toy shop has a permanent need of Tk 300,000 all year and a seasonal need of Tk 150,000 for 4 months before December. A long-term loan costs 10% a year and short-term borrowing costs 13% a year. Choose how to fund each need, compute the interest for the year, and find the interest saved compared with funding everything with the long-term loan.

WHERE PEOPLE GO WRONG

Mistake: believing short-term money is always cheaper.

Mistake: "Short-term money is always cheaper." Correction: the yearly rate of short-term money may be higher or lower than the long-term rate. What matters is the total interest, which depends on the rate and on how long the money is held.

Mistake: believing trade credit costs nothing.

Mistake: "Trade credit is free." Correction: it is free only when the buyer pays on the last day without a discount lost. If the buyer gives up a cash discount, the credit can cost more than a bank loan, as the worked case showed with 24.83% against 14%.

Mistake: taking the discount only when cash is in hand.

Mistake: "I can take the discount only if I have the cash." Correction: if giving up the discount costs more than the bank charges, the business should borrow to take it. The worked case gained Tk 4,362 by doing so.

Mistake: thinking the matching principle must be exact.

Mistake: "Matching means every need is paid for by money of exactly the same length." Correction: matching is a guide. A business may hold some extra long-term money for safety, and it pays interest on idle cash as the price.

Mistake: treating the formula result as an exact rate.

Mistake: "The cost of giving up the discount is exact." Correction: the formula is an approximation, because it does not allow for interest earning further interest. It is accurate enough to compare with a bank's interest rate, but it is not a precise measure.

IN HISTORY

In 2007 a British bank that borrowed short to lend long met a queue of depositors.

Northern Rock was a bank in the United Kingdom that lent money to people buying homes. A loan for a home, called a mortgage, is repaid over many years. Yet the bank had funded much of this long lending with money borrowed for short periods from lenders in the money markets, and with sales of mortgages to investors, rather than with savings from ordinary customers.

Only a small part of the bank's lending came from its savers.

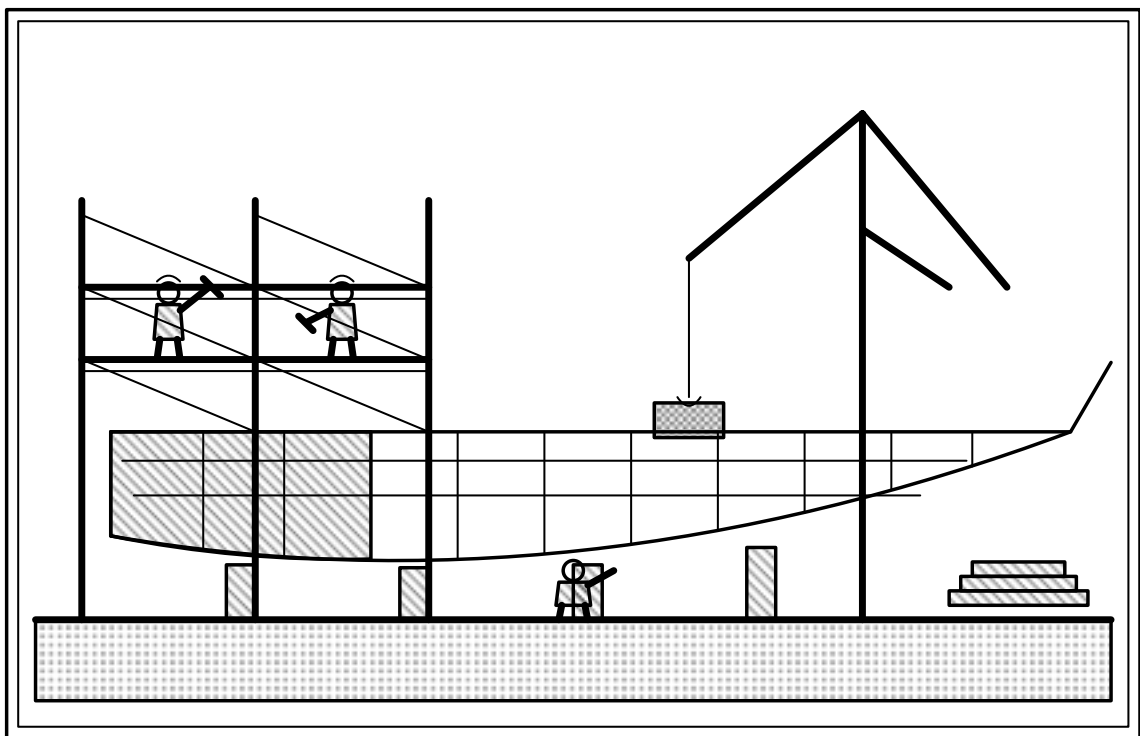
Reports at the time said that only a small part of the money the bank lent came from its savers (Channel 4 News, 2007). For years this plan let the bank borrow cheaply and offer cheap mortgages. It worked while lenders were willing to renew, and it failed when they were not.

When short lenders stopped renewing, the bank had to ask for help, and savers queued.

On 9 August 2007 the markets for short-term borrowing all but froze, and Northern Rock found that it could not renew its short-term loans (Shin, 2009). The bank's difficulty became public on 13 September 2007, when the BBC reported that it was seeking support from the Bank of England, the country's central bank. On the morning of 14 September 2007, depositors queued outside its branches to take out their savings. The lines continued until 17 September. The lesson matches this chapter: a bank that holds long assets but depends on short money is exposed when lenders stop renewing.

REVIEW QUESTIONS

- 16.1 An invoice of Tk 10,000 carries terms 2/10, net 30. Find the cost of giving up the discount as a yearly rate.
- 16.2 Find the cost of giving up the discount on terms 3/10, net 45.
- 16.3 An invoice of Tk 200,000 carries terms 2/10, net 40. The bank charges 13% a year. Should the business borrow from the bank to take the discount? Compare the interest with the discount.
- 16.4 Compare the terms 3/10, net 30 and 1/10, net 60. Which is more costly to give up, and by how much?
- 16.5 Define short, intermediate and long spans of financing, and give one source for each.
- 16.6 A toy shop has a December peak in stock. Apply the matching principle: say how it should fund its permanent need and its December need.
- 16.7 Explain the difference between permanent and seasonal working capital, using a seller of school books as your example.
- 16.8 Explain one difference between a bank overdraft and a short loan, with a reason why a business might prefer the overdraft.



A shipyard with an iron steamship half built on the slips

CHAPTER XVII

Long-Term Financing and the Cost of Capital

Who lends money for five years, and what do they ask in return?



This chapter surveys the long-term funds a business can raise: loans, owners' money and shares sold in the capital market. It names the institutions that supply them in Bangladesh and shows how to compute what the funds cost.

A half-built ship needs money that stays for years, not weeks.

You will need: Chapter 16, 'Short-Term Financing and Matching', section 16.1 'Financing by Time Span'; Chapter 13, 'Valuing Bonds', section 13.1 'What a Bond Is'; Chapter 14, 'Valuing Common and Preferred Stock', section 14.1 'Rights of Shareholders'.

The answer is money that stays as long as the asset takes to pay for itself, and it has a price.

The chapter surveys the sources of long-term funds and then computes their combined cost.

A lender will wait five years if it is paid along the way and holds rights that protect it.

A term loan is repaid in instalments over a fixed period longer than one year.

Recall: term loans and debt, assumed from earlier chapters; time spans of financing, Chapter 16, section 16.1 'Financing by Time Span'.

Term. Word in use. In daily life, a period of time such as a school term. In lending, the length of time a loan runs before it is fully repaid.

Term loan. a loan repaid in instalments over a fixed period of more than one year

The case file records the new term loan and the two protections the bank asks for.

Collateral is an asset pledged to the lender, who may take it if the loan is not repaid.

Collateral. an asset a borrower pledges so the lender can take it if the loan is not repaid

In a shipyard, an iron steamship rests on the slips with half its hull still open to the sky. The builder has paid for iron, rivets and wages for many months, and the ship will earn nothing until it is launched, fitted and sailing. Suppose the builder had borrowed the money for ninety days. On day ninety the lender would ask for it back, and the half-built ship could be neither sold nor repaid. What kind of money does such a builder need?

The answer is money that stays for as long as the ship takes to pay for itself: years, not weeks. This is long-term financing, and it is harder to find. The lender must trust the borrower for longer, and must be told what happens if things go wrong. The builder must also pay for it. Every lender and every owner who puts money in expects something back, and the sum of those expectations is a cost the business must earn.

This chapter has two parts. The first surveys where long-term money comes from: loans, owners' funds and shares sold to the public, with the institutions in Bangladesh that supply them. The second is one calculation, the cost of the whole mix of funds, which a business uses as the rate for judging its projects. Our case is Meghna and Sons Limited, which needs a machine for Project Press, the printing project you met in Chapter 8.

* * *

17.1 TERM LOANS AND DEBT INSTRUMENTS

Why would a bank wait five years for its money? Because the borrower pays interest and returns part of the money along the way, and because the bank holds rights that protect it if payments stop. Those rights are the heart of long-term lending.

The commonest long-term debt is the **term loan**. A term loan is a loan repaid in instalments over a fixed period of more than one year. An *instalment* is one of the regular payments that together repay the loan. The agreed length of time is the term. Each instalment usually holds some interest and some repayment of the amount borrowed. Meghna and Sons plans to buy a press machine and asks a bank for a term loan.

Case file, illustrative. Meghna and Sons Limited, 15 November 2025. A new term loan of Tk 600,000 over 5 years at 10% needs yearly instalments of Tk 158,278.49. Collateral: the machine. Covenant: owners' equity must stay above Tk 2,000,000.

A lender who waits five years wants a way out if the borrower fails. **Collateral** is an asset a borrower pledges so the lender can take it if the loan is not repaid. In the case file, the machine is the collateral. If the shop stopped paying, the bank could take the machine and sell it. Collateral lowers the lender's risk, and so it often lowers the interest rate.

The lender may also limit what the borrower does while the loan lasts. A **covenant** is a promise in a loan agreement that limits what the borrower may do. In the case file, owners' equity must stay above Tk 2,000,000. Other covenants may limit new borrowing or the profit paid out to owners. The reason is the same each time: money paid out, or lost, reduces the cushion that protects the lender. If a covenant is broken, the lender may ask for early repayment.

The instalment of Tk 158,278.49 comes from the payment method you learned in Chapter 5, 'Annuities'. Section 17.6 works it through in full.

A business that wants money from many lenders issues bonds. Chapter 13, section 13.1 'What a Bond Is', describes them. Bonds differ in what backs them. A **debenture** is a bond backed only by the general standing of the borrower, with no particular asset pledged. A holder of a debenture trusts the business's record and its profit. A mortgage is different. It is a loan secured by a pledge of land or buildings, which is a particular kind of collateral. A debenture holder therefore carries more risk than a lender with collateral, and usually asks a higher interest rate.

PAUSE AND CHECK 17.P1 A bank lends Tk 300,000 and takes a pledge over the borrower's shop building. Is this a debenture? Give a reason.

These protections have limits. A machine may sell for less than the debt. A covenant may stop a business from acting quickly. And debt must be repaid whether or not the business makes a profit, which is why too much of it is dangerous. The lender wants safety. The borrower wants freedom and a low cost. A loan agreement is the balance between the two.

Before moving on. A term loan runs for more than one year and is repaid in instalments. Collateral is an asset the lender may take. A covenant is a promise that limits the borrower. A debenture is a bond with no particular asset pledged.

* * *

17.2 OWNERS' FUNDS AS LONG-TERM FINANCE

Debt is not the only long-term money. Owners supply it too, in two ways. The first needs no new cash from anyone.

Retained earnings are the profits a business keeps instead of paying them out. Each year, Meghna and Sons may keep part of its profit to buy inventory, fittings or machines. The money stays as long as the business needs it. It is convenient, but it is not free. The owners could have taken the money and used it elsewhere, so they expect the same return on it as on any other money they leave in. For that reason the case file counts retained earnings inside common equity.

The second way is to issue new stock. Common stock gives its owners a share in the profit that rises and falls with the business, and the right to vote. Preferred stock usually pays a fixed dividend and gives fewer voting rights. Neither has to be repaid on a fixed date, which is its great advantage over debt. A new issue of common stock also brings new owners, who share the profit and the votes of the old ones.

The table compares four sources of long-term money. Read each row as one source. The second column asks whether the money must be repaid on a fixed date, and the third shows what the supplier is paid.

A covenant is a promise in the loan agreement that limits what the borrower may do.

Covenant. a promise in a loan agreement that limits what the borrower may do

The instalment is found by the payment method of Chapter 5, shown in section 17.6.

A debenture is a bond with no particular asset pledged, so its holder trusts the borrower's standing.

Debenture. a bond backed only by the general standing of the borrower, with no particular asset pledged

Collateral and covenants lower the lender's risk, but they do not remove it, and debt must be repaid whatever the profit.

Recap: a term loan is repaid in instalments; collateral and covenants protect the lender; a debenture pledges nothing in particular.

Owners can supply long-term money by leaving profit in the business or by paying in new money.

Retained earnings are profit kept in the business, and the owners expect a return on it.

Recall: retained earnings, assumed from earlier chapters.

New common or preferred stock brings new money from owners who expect a return.

Recall: rights of shareholders, Chapter 14, section 14.1 'Rights of Shareholders'.

The table compares four sources by repayment and by the return each one is paid.

FOUR SOURCES OF LONG-TERM MONEY

SOURCE	REPAID ON A FIXED DATE?	RETURN PAID
Term loan	Yes, in instalments	Interest set by the agreement
Preferred stock	No	A fixed dividend (12% in the case file)
Common stock	No	Dividends that vary with profit
Retained earnings	No	None paid, but owners expect 14%

Owner money carries no repayment date, but it asks for a higher return and may share control.

Recap: retained earnings and new stock are owner money, with no repayment date and a higher return.

Owners' funds have limits. They cost more than debt, because the owners take more risk. A new issue costs fees and may reduce the control of the existing owners. Seen from the lender's side, a thick layer of owners' funds is a comfort, which is why covenants such as the one in 17.1 protect equity.

Before moving on. Retained earnings are profit kept in the business. New common and preferred stock bring new owner money. None of these has a fixed repayment date, and all of them expect a return.

* * *

17.3 RAISING FUNDS IN THE CAPITAL MARKET

A business that needs more than one bank or its owners can supply may sell securities to the public.

The capital market is where long-term loans and shares are issued and traded.

Recall: financial market, assumed from earlier chapters.

Capital. Word in use. In daily life, money or wealth, or the main city of a country. In finance, the long-term funds a business uses, as in cost of capital.

Capital market. the market where long-term loans and shares are issued and traded

An initial public offering is the first sale of a business's shares to the general public.

Initial public offering. the first sale of a business shares to the general public

A rights issue offers new shares to existing shareholders in proportion to what they already hold.

Rights issue. an offer of new shares to existing shareholders in proportion to what they already hold

A bank loan comes from one lender. What if a business needs more money than a single bank will lend, and its present owners cannot supply more? It may offer shares or bonds to many people. For that it needs a place where such offers are made and later traded.

That place is the **capital market**. The capital market is the market where long-term loans and shares are issued and traded. Securities are first sold there by the business that issues them. Later they may be traded between investors, who do not give any money to the business itself. A buyer who wishes to leave can sell to another buyer, and this makes people more willing to buy at the start.

Suppose a printing business owned by one family decides to sell 30% of its shares to people it does not know. This is an **initial public offering**: the first sale of a business shares to the general public. After it, the shares are usually listed on a stock exchange, so that they can be bought and sold. A later offer to the public by a business already listed is called a public issue. Both bring in new owners and new money, and both are costly and carefully regulated.

There is a quieter way to raise owners' money. A **rights issue** is an offer of new shares to existing shareholders in proportion to what they already hold. Take a business with 1,000 shares, of which one owner holds 100, or 10%. It offers 1 new share for every 5 held, so 200 new shares in all. The owner may buy 20 of them. Then the owner holds 120 of 1,200 shares, still 10%. An owner who does not buy loses a part of that share.

The two issues compare as follows. A public issue reaches many new buyers and can raise large sums, but it brings fees, disclosure and new owners. A rights issue is cheaper and keeps control with the old owners, but it raises only as much as they are willing to pay.

The rules for issuing shares differ by country. In Bangladesh, the Bangladesh Securities and Exchange Commission regulates the capital market, as the next section shows.

Before moving on. The capital market is where long-term loans and shares are issued and traded. An initial public offering is a first sale to the public. A rights issue offers new shares to existing owners in proportion to their holdings.

* * *

17.4 LONG-TERM LENDERS AND INVESTORS IN BANGLADESH

Where could Meghna and Sons find five years of money in Dhaka? Several kinds of lender and market venue matter. The names below were checked on 4 October 2026, and names can change, so confirm them before relying on them. Other countries have different names, but the same kinds of supplier appear in most of them.

The table lists the institutions. Read each row as one name, the kind of body it is, and what it offers to a business that needs long-term funds.

SUPPLIERS OF LONG-TERM FUNDS AND MARKET VENUES IN BANGLADESH

NAME	KIND	WHAT IT SUPPLIES
Commercial banks	Banks	Term loans, usually against collateral
Bangladesh Development Bank Limited	Development bank	Long-term loans to industrial projects
Investment Corporation of Bangladesh	State-owned investment institution	Money invested in shares and other securities
Infrastructure Development Company Limited (IDCOL)	Government-owned finance company	Long-term loans for infrastructure and renewable energy
Dhaka Stock Exchange and Chittagong Stock Exchange	Stock exchanges	Places where shares are issued and traded
Bangladesh Securities and Exchange Commission	Regulator	No funds; rules for issuing and trading securities

For a small family business, the nearest source is a commercial bank. Its term loan looks like the one in the case file: instalments, collateral and covenants. A bank decides by the borrower's record and the collateral on offer. For a new business with few assets, this can be difficult.

Bangladesh Development Bank Limited was incorporated on 16 November 2009. It joined two public institutions set up in 1972, Bangladesh Shilpa Bank and Bangladesh Shilpa Rin Sangstha, which lent to industry. Its purpose is the same as theirs: long-term loans for industrial projects. IDCOL works in a similar way for infrastructure and renewable energy.

A public issue brings in new owners; a rights issue keeps control with the old ones.

Rules for issuing shares differ by country, and Bangladesh has its own regulator.

Recap: the capital market issues and trades long-term funds; an initial public offering is the first sale to the public; a rights issue goes to existing owners first.

Bangladesh has commercial banks, development institutions, an investment institution and two stock exchanges.

The table lists each institution, its kind, and what it supplies.

The commercial bank is the nearest source for a small business, and it lends against collateral.

Development institutions were set up to lend for longer terms to projects the state wishes to encourage.

The investment corporation, the exchanges and the regulator make up the capital market side.

A business matches its need to the supplier: a bank for a machine loan, the exchange for a large public issue.

Recap: banks and development institutions lend; the investment corporation invests; the exchanges are the venue; the Commission regulates.

Every source of money has a price, and the business must earn enough to pay them all.

Recall: required return and interest rate, assumed from earlier chapters.

The cost of capital is one average rate for the whole mix of funds.

Cost of capital. the average return a business must earn on its funds to satisfy those who supplied them

Debt costs its interest rate less the tax it saves; preferred stock and common equity cost the return they require.

The case file lists the three sources of funds, their costs and their weights.

A weight is each source's share of the total funds.

The figure shows the funds as one bar, split by hatching into the three weights.

The Investment Corporation of Bangladesh was created by an ordinance in 1976. It is a state-owned institution that invests in shares and other securities, and so supplies money to the capital market. The Dhaka Stock Exchange and the Chittagong Stock Exchange, which was set up in 1995, are the two places where shares are issued and traded. The Bangladesh Securities and Exchange Commission was established on 8 June 1993. It regulates the capital market and approves the rules under which a business may offer shares to the public.

A business matches its need to the supplier. Meghna and Sons Limited, with a machine to buy, would start with a bank. A much larger business that wished to raise money from the public would work through an exchange and under the Commission's rules.

Before moving on. Banks and development institutions lend long-term money. The Investment Corporation of Bangladesh invests in securities. The two stock exchanges are the venues, and the Commission regulates them.

* * *

17.5 THE COST OF CAPITAL

Meghna and Sons raised money from three sources, and each one asks for a different return. How much must the business earn, on average, to keep all three satisfied? That average has a name.

The **cost of capital** is the average return a business must earn on its funds to satisfy those who supplied them. It is a rate, such as 12.85%, and not an amount of money. We find it in three steps: the cost of each source, the weight of each source, and the sum.

Start with the cost of each source. The bank charges 10% a year. But interest is paid before tax is counted, so it lowers the profit that is taxed. At the assumed tax rate of 25%, the state in effect bears a quarter of the interest.¹ The after-tax cost of the loan is $10\% \times (1 - 0.25) = 10\% \times 0.75 = 7.5\%$. Preferred stock costs its dividend rate, 12%, because dividends are not taken off before tax. Common equity costs the return its owners require, 14%, and this includes retained earnings.

Case file, illustrative. Meghna and Sons Limited, 15 November 2025. Long-term funds: bank loans outstanding Tk 500,000 at 10% (after tax $10\% \times 0.75 = 7.5\%$); preferred stock Tk 100,000 at 12%; common equity including retained earnings Tk 2,400,000 with a required return of 14%. Total Tk 3,000,000; weights 16.67%, 3.33% and 80.00%.

The sources differ in size, so they cannot count equally. The weight of a source is its share of the total. Debt: $500,000 \div 3,000,000 = 0.1667$, or 16.67%. Preferred stock: $100,000 \div 3,000,000 = 0.0333$, or 3.33%. Common equity: $2,400,000 \div 3,000,000 = 0.80$, or 80.00%. The weights add to 100%.

The figure below is one horizontal bar. The whole bar stands for the total funds, Tk 3,000,000. Each hatched part is one source, and its width shows its share of the total. Read the label under each part for its weight and its cost.

¹ The tax rate of 25% is assumed for illustration only. Tax rules differ by country and change over time.

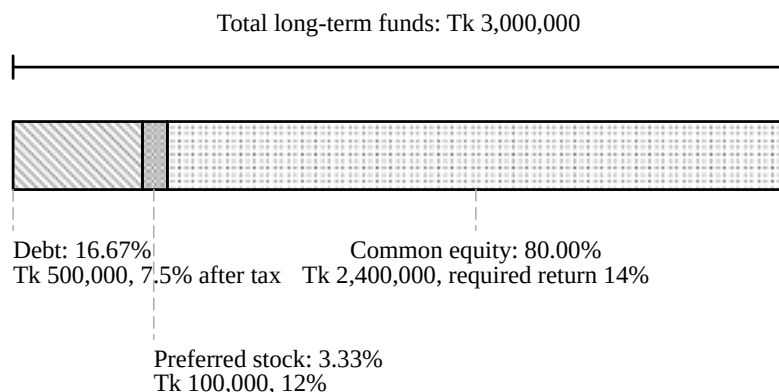


FIG. 17.1. The long-term funds of Meghna and Sons Limited, split by weight

PAUSE AND CHECK 17.P2 A business has Tk 200,000 of debt and Tk 800,000 of common equity. What is the weight of each source?

Now we combine them. The **weighted average cost of capital** is the cost of each source of funds multiplied by its share of the total and added together. In symbols: $K = (W_d \times K_d) + (W_p \times K_p) + (W_e \times K_e)$. Here K is the weighted average cost, W stands for a weight and K with a letter stands for a cost. The letters d , p and e mean debt after tax, preferred stock and common equity.

Case file, illustrative. Meghna and Sons Limited, 15 November 2025. Weighted average cost of capital = $0.1667 \times 7.5\% + 0.0333 \times 12\% + 0.80 \times 14\% = 1.25\% + 0.40\% + 11.20\% = 12.85\%$. Project Press (Chapter 8 flows) at 12.85%: net present value Tk 172,667, still positive.

Why compute it? A project must earn at least what its funds cost. So the cost of capital serves as the discount rate in net present value, as in Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value'. A positive net present value at 12.85% means Project Press earns more than all its suppliers of funds require.

Four cautions apply. The weights here use the amounts in the case file, and not current market prices. The cost of equity is an estimate, not a bill that arrives. The rate suits projects as risky as the business itself, and a riskier project needs a higher rate. And the cost changes when the mix changes: more debt lowers the average at first, but lenders and owners may ask for more as debt grows.

Before moving on. Find the after-tax cost of debt, and the costs of preferred stock and common equity. Find each weight. Multiply each cost by its weight and add. For Meghna and Sons the result is 12.85%, and Project Press still has a positive net present value.

* * *

The weighted average cost of capital multiplies each cost by its weight and adds the results.

Weighted average cost of capital. the cost of each source of funds multiplied by its share of the total and added together

The case file gives the result, 12.85%, and the test of Project Press against it.

The cost of capital is the discount rate for judging projects: a project must earn at least this much.

The result is an estimate that depends on the weights, the costs and the risk of the project.

Recap: find each cost, find each weight, multiply and add.

WORKED EXAMPLES: FROM COSTS TO A SINGLE RATE

Four examples apply the method, from clean numbers to a combined problem.

Example 1 uses two sources and round numbers so that the method is the only new thing.

Example 2 repeats the method with the three sources in the case file.

The steps find the after-tax cost, the weights and the products in turn.

Example 3 changes the situation: the new term loan is added to the funds.

Example 4 finds a loan instalment and then tests a project at the cost of capital.

The steps work the instalment first and the project second.

We work four examples. The first uses clean numbers. The second uses the case file. The third changes the situation, and the fourth combines the loan payment with the use of the cost of capital as a discount rate. Each example follows one order: given, find, formula, steps, answer, check.

Example 1: clean numbers. *Given.* Debt of Tk 400,000 costs 8% after tax. Common equity of Tk 600,000 costs 12%. *Find.* The weighted average cost of capital. *Formula.* $K = (W_d \times K_d) + (W_e \times K_e)$, where W is a weight and K is a cost. *Steps.* Total funds: $400,000 + 600,000 = \text{Tk } 1,000,000$. Weights: $400,000 \div 1,000,000 = 0.40$, and $600,000 \div 1,000,000 = 0.60$. $K = 0.40 \times 8\% + 0.60 \times 12\% = 3.2\% + 7.2\% = 10.4\%$. *Answer.* The cost of capital is 10.4%. *Check.* The answer lies between 8% and 12%, and nearer to 12%, because equity is the larger source.

Example 2: realistic numbers. *Given.* The case file: bank loans Tk 500,000 at 10%, preferred stock Tk 100,000 at 12%, common equity Tk 2,400,000 at 14%, and a tax rate of 25%. *Find.* The weighted average cost of capital. *Formula.* $K = (W_d \times K_d) + (W_p \times K_p) + (W_e \times K_e)$.

Steps. After-tax cost of debt: $10\% \times (1 - 0.25) = 7.5\%$. Total funds: $500,000 + 100,000 + 2,400,000 = \text{Tk } 3,000,000$. Weights: 0.1667, 0.0333 and 0.8000. Products: $0.1667 \times 7.5\% = 1.25\%$. $0.0333 \times 12\% = 0.40\%$. $0.80 \times 14\% = 11.20\%$. Sum: $1.25\% + 0.40\% + 11.20\% = 12.85\%$. *Answer.* The cost of capital is 12.85%. *Check.* The answer lies between 7.5% and 14%, and close to 14%, because common equity is 80% of the funds. A common slip is to use 10% for the debt and forget the tax: that would give 13.27%, which is too high.

Example 3: a reshaped case. This is an exercise built on the case file, not part of it. *Given.* Suppose the Tk 600,000 term loan is added to the existing funds, which are unchanged. The loan costs 7.5% after tax, and the owners' required return stays at 14%. *Find.* The new weighted average cost of capital. *Formula.* The same as in Example 2. *Steps.* Debt: $500,000 + 600,000 = \text{Tk } 1,100,000$. Total funds: $1,100,000 + 100,000 + 2,400,000 = \text{Tk } 3,600,000$. Weights: 0.3056, 0.0278 and 0.6667. Products: $0.3056 \times 7.5\% = 2.292\%$. $0.0278 \times 12\% = 0.334\%$. $0.6667 \times 14\% = 9.334\%$. Sum: $2.292\% + 0.334\% + 9.334\% = 11.96\%$. *Answer.* The cost of capital falls to 11.96%. *Check.* The debt share rose from 16.67% to 30.56%, and debt is the cheapest source, so the average falls. In practice owners may ask for more than 14% as debt rises, and then the fall would be smaller.

Example 4: a combined problem. *Given.* The term loan: Tk 600,000 over 5 years at 10%, repaid in equal instalments at each year end. A project costs Tk 200,000 now and returns Tk 225,000 in one year. The discount rate is 12.85%. *Find.* The yearly instalment, and whether the project passes. *Formula.* Instalment = amount $\times$ interest rate $\div (1 - 1 \div (1 + \text{interest rate})^n)$ multiplied by itself n times), where n is the number of years. Net present value = present value of the return – cost.

Steps. Instalment: 1.10 multiplied by itself 5 times = 1.61051. Then $1 \div 1.61051 = 0.620921$, and $1 - 0.620921 = 0.379079$. The amount times the interest rate is $600,000 \times 0.10 = 60,000$. Then $60,000 \div 0.379079 = \text{Tk } 158,278.49$, using more decimal places than shown. Project: present value = $225,000 \div 1.1285 = \text{Tk } 199,379.70$. Net present value = $199,379.70 - 200,000 = -\text{Tk } 620.30$.

Answer. The instalment is Tk 158,278.49 a year. The project has a net present value of about -Tk 620, so on these figures it should be rejected. *Check.* Five instalments repay $5 \times 158,278.49 = \text{Tk } 791,392.45$, which is Tk 191,392.45 more than the loan: this is the interest, and it is sensible for a five-year loan at 10%. At the loan's rate of 10% the project would have looked acceptable, since $225,000 \div 1.10 = \text{Tk } 204,545.45$. At the cost of capital it fails.

Before moving on. The method never changed: after-tax cost, weight, multiply, add. A change in the mix of funds changes the weights and so the answer. The result is the discount rate that a project must beat.

* * *

SUMMARY

A term loan is repaid in instalments over more than one year, and collateral and covenants protect the lender. A debenture is a bond with no particular asset pledged. Owners supply long-term money through retained earnings and new stock. The capital market issues and trades long-term loans and shares, and in Bangladesh banks, development institutions, the investment corporation, two exchanges and a regulator play their parts. The cost of capital is the weighted average of the after-tax cost of debt, the cost of preferred stock and the cost of common equity. For Meghna and Sons it is 12.85%.

The instalment is Tk 158,278.49, and the project fails by about Tk 620.

Recap: the same steps give the cost of capital for clean, realistic and changed funds, and the rate tests a project.

Long-term funds come from loans, owners and the capital market, and the business must earn their combined cost.

THINK IT THROUGH

17.T1 A new business asks why long-term funds suit its first machines and its shop fit-out, and why short-term loans do not. Explain in plain words.

WHERE PEOPLE GO WRONG

Wrong: retained earnings cost nothing, because nobody is paid. *Right:* the owners could have taken the profit and used it elsewhere, so they expect a return on it. That is why the case file counts them inside common equity at 14%.

Wrong: the loan costs 10%, so use 10% in the calculation. *Right:* interest lowers taxed profit, so the cost after a 25% tax rate is 7.5%. Use the after-tax cost, and use it for debt only.

Wrong: debt is the cheapest source, so a business should borrow as much as it can. *Right:* debt must be repaid whatever the profit, and lenders and owners ask for more as debt grows.

Wrong: a project that beats the loan's 10% is acceptable. *Right:* a project must beat the cost of all the funds, which is 12.85% for Meghna and Sons.

Retained earnings are not free money.

The cost of debt is the interest rate less the tax it saves.

Debt is cheaper, but a business cannot use only debt.

The cost of capital is a rate for the whole mix, not the rate of one loan.

IN HISTORY

Ford Motor Company was founded by Henry Ford in 1903. For half a century its shares were held by members of the Ford family and by the Ford Foundation, a charity in the United States. In January 1956 the Foundation sold about 10.2 million Ford shares to the public at \$64.50 each. It was the largest public share offering in American history up to that time. The sum raised was between \$640 million and \$660 million, depending on whether commissions are counted.

A famous offering of shares in 1956 moved a great business into public hands and created a market for its shares.

The sale was led by an investment bank, and the money went to the Foundation, not to the business.

A group of 205 banks, led by Goldman Sachs, an American investment bank, handled the sale. The money went to the Foundation, which wanted to hold less of its wealth in one business, and not to Ford. The offering shows what an initial public offering does. It moves shares into many hands and creates a market where they can be traded, which is the work of a capital market.

REVIEW QUESTIONS

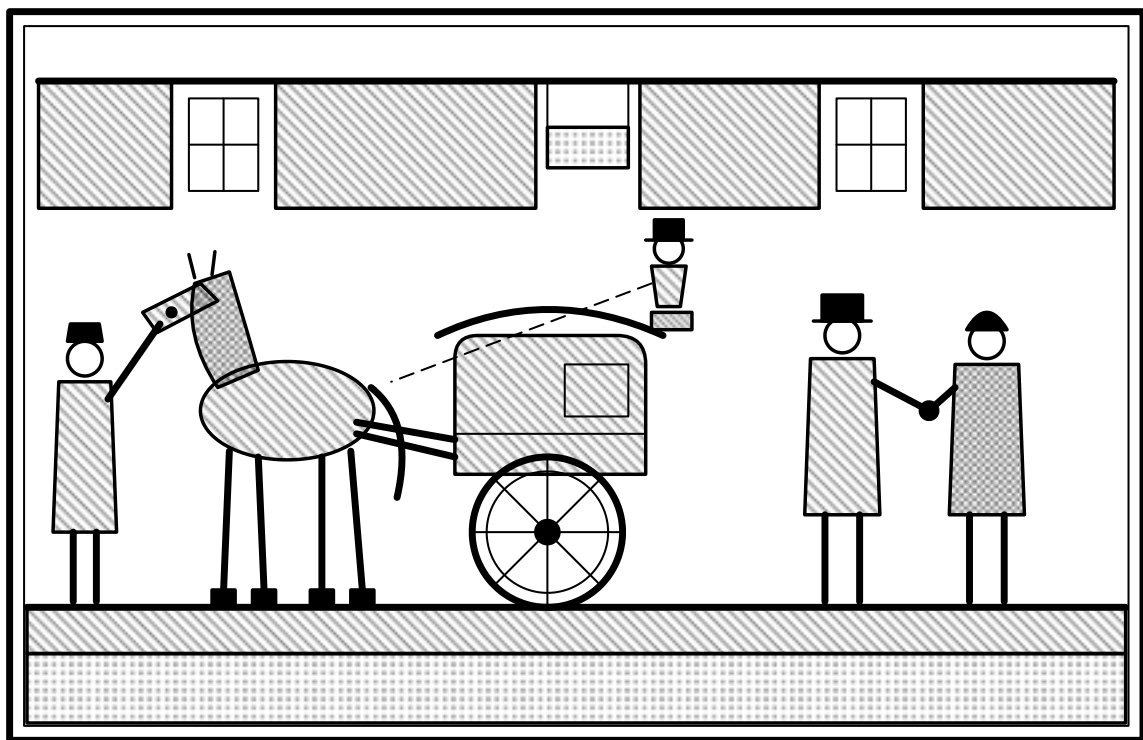
17.1 Name three conditions a lender may attach to a term loan, and say why a lender asks for each.

17.2 Compare a rights issue with a public issue. Say who may buy and what each does to the control of the existing owners.

17.3 List four types of lender or market venue for long-term funds in Bangladesh, and say what each supplies.

17.4 A loan costs 11% a year and the tax rate is 25%. Find the after-tax cost of the loan.

17.5 A business is funded 25% by debt costing 7.5% after tax, 5% by preferred stock costing 12%, and 70% by common equity costing 14%. Find the weighted average cost of capital.



A carriage-hire yard, where the owner and the hirer agree their terms

CHAPTER XVIII

Leasing

Must a business own everything it uses?



This chapter explains what a lease is, who the two parties are and how the two main kinds differ. It then shows how to compare leasing with buying by turning each year's after-tax cash outflow into a present value.

YOU WILL NEED

Chapter 17, "Long-Term Financing and the Cost of Capital", section 17.5 "The Cost of Capital", for the after-tax cost of debt; Chapter 8, "Capital Budgeting: Evaluation Techniques", section 8.2 "Net Present Value", for the method of discounting; Chapter 5, "Annuities", section 5.4 "Finding the Payment", for the payments on a loan.

A traveller hires a cab and owns nothing, yet the work is done.

Hiring and owning differ in when the money is paid.

A business meets the same puzzle with larger sums.

A visitor to London in the 1870s who wished to cross the city did not buy a carriage, a horse and a stable. He hired a cab for an hour, paid the driver and walked away owning nothing. The cab did his work as well as any carriage would have done. His neighbour, who crossed the city twice every day, took the opposite view and kept a carriage of his own.

Both men were sensible. Their choices differed because their needs differed, and because the two roads carry different costs. Hiring is paid in small amounts as time passes. Owning is paid mostly at the start, and the owner must also meet repairs, idle days and the day when the carriage is worth less than before.

A modern business meets the same puzzle with larger sums. Meghna and Sons Limited needs a second machine. The seller wants Tk 780,000 now. Another business, one that owns machines in order to let others use them, offers the same machine for a payment at the end of each year. Which road costs less? To answer, we must compare payments made in different years, with tax changing each payment. You already own the tools for that task. This last chapter brings them together.

* * *

18.1 WHAT A LEASE IS

A student who rents a room already knows the idea.

Rent. Word in use. In daily life, the money paid to live in a room or use a shop. In this chapter, the regular payment under a lease is called a lease payment.

Lease. A contract in which the owner of an asset lets another party use it for regular payments.

A lease lets one party use what another party owns.

Lessor. The owner who lets an asset out under a lease.

Lessee. The party who pays to use an asset under a lease.

Every lease has two parties: the one who owns and the one who uses.

Several reasons lead a manager to lease instead of buy.

Begin with something familiar. A student who rents a room pays the owner each month. The student may use the room but may not sell it, and must leave when the agreement ends. A lease follows the same pattern, applied to the assets of a business.

A **lease** is a contract in which the owner of an asset lets another party use it in return for regular payments. A baker, for example, may lease an oven for five years instead of buying it. The baker uses the oven every day, but the oven stays the property of its owner.

Every lease has two parties. The **lessor** is the owner who lets an asset out under a lease. The **lessee** is the party who pays to use the asset. In the cab story the cab owner is the lessor and the traveller is the lessee. Many students swap these two words, because they differ by one letter. The ending *-or* belongs to the one who does the letting; the ending *-ee* belongs to the one who receives the use.

Why would a business lease? Consider the manager's viewpoint first. A new business may need a machine on its first day but lack the cash to buy it, and a lease asks for small payments instead of one large sum. A business may need an asset for a short time only. A business may also wish to keep its borrowing power for other uses, or may fear that the asset will become out of date before it wears out.

Now consider the lessor. The lessor owns many assets of one kind, can buy them at a good price and knows how to resell them. The lessor earns a return from the payments. Tax rules matter as well, because in many countries the lease payment reduces the lessee's taxable profit, as section 18.3 shows. Rules differ from country to country, so this chapter uses the tax rate of 25% assumed for illustration in the Meghna and Sons case.¹

The lessor has reasons too, and the tax rules matter.

Leasing has limits. The lessor must earn a return, so the payments over the whole lease usually add up to more than the price of the asset. The lessee must pay even if the asset is idle. When the lease ends, any resale value belongs to the lessor, not to the lessee. Leasing is therefore a different way of paying for use, not a cheaper way by nature. Section 18.3 shows how to test whether it is cheaper in a given case.

Leasing is not free, and resale value passes to the lessor.

Before moving on. A lease lets a lessee use an asset owned by a lessor in return for regular payments. A business leases to save cash, to use an asset for a short time, or to share risk. Leasing is not free: the lessor must earn a return, and the lessee gives up any resale value.

Recap of section 18.1.

18.2 KINDS OF LEASE

Leases range from a cab hired for an hour to a machine held for most of its working life. Two main kinds help us sort them, and both depend on one question: who carries the risks of owning the asset? These risks include repairs, damage and the chance that the asset becomes out of date.

Leases range from one day to many years; risk sorts them.

An **operating lease** is a short or cancellable lease where the lessor keeps most of the risks of owning the asset. A clinic that hires a portable scanner by the month, with the right to return it, holds an operating lease. The lessor pays for repairs and bears the risk that the scanner becomes out of date.

Operating lease. A short or cancellable lease where the lessor keeps most of the risks of owning the asset.

A short lease the user may cancel, with the owner holding the risk.

A **financial lease** is a long lease that cannot be cancelled, in which the lessee in effect carries the cost and risk of owning. A bakery that signs for an oven for six years, with no right to leave early, holds a financial lease. The payments usually cover most of the oven's price plus the lessor's return, which is why the lessee in effect pays for the asset.

Financial lease. A long lease that cannot be cancelled, where the lessee in effect carries the cost and risk of owning.

A long, fixed lease where the user bears the burden of owning.

The two kinds are easier to remember side by side. The table below compares them on four features. Read each row from left to right: the feature, then how an operating lease treats it, then how a financial lease treats it.

The table sets the two kinds side by side on four features.

OPERATING LEASE AND FINANCIAL LEASE COMPARED

FEATURE	OPERATING LEASE	FINANCIAL LEASE
Length	Short, often well below the asset's working life	Long, close to the asset's working life
Cancellation	Usually allowed	Not allowed
Repairs and risk of becoming out of date	Mostly with the lessor	Mostly with the lessee
Typical use	Cabs, copiers, scanners for a short need	Machines and vehicles kept for years

The five-year lease offered to Meghna and Sons is treated here as a financial lease. The comparison with buying in section 18.3 suits financial leases best. An

The lease-or-buy test fits financial leases best.

¹ Names and legal details of leasing contracts differ between countries. The ideas in this chapter are common to all of them.

Sale and leaseback. Selling an asset to a lessor and at once leasing it back.

Selling an asset and leasing it back turns it into cash.

Accounting rules change the records, not the contract.

Recap of section 18.2.

The question: which road costs the business less?

Recall: net present value, Chapter 8, section 8.2 "Net Present Value".

After-tax cash outflow. Cash paid out in a year after taking away the tax saved by that payment.

Tax saved lowers the true cost of each payment.

For leasing, the whole payment is a deduction.

Buying with a loan has two tax savings; repayment of the loan is not one.

Recall: after-tax cost of debt, Chapter 17, section 17.5 "The Cost of Capital".

The discount rate is the after-tax cost of borrowing.

operating lease also gives the lessee a service, such as repairs, so a comparison of cost alone would leave out part of what is bought.

A third arrangement is a **sale and leaseback**, which means selling an asset to a lessor and at once leasing it back. A baker who owns a delivery van may sell it to a lessor for cash and then lease the same van back. The business receives cash today and keeps using the van, but it now makes lease payments for as long as it needs the van.

Accounting rules affect how a lease appears in a business's records, but they do not change what the contract is.² The risk question stays useful, because it still tells us who bears the burden of owning.

Before moving on. An operating lease is short, can be cancelled and leaves most risk with the lessor. A financial lease is long, cannot be cancelled and leaves most risk with the lessee. A sale and leaseback turns an owned asset into cash while the business keeps using it.

18.3 THE LEASE OR BUY DECISION

Suppose a business has decided that it needs a machine. The investment decision is made. One question remains: should the business lease the machine, or buy it with money borrowed from a bank? The answer cannot come from adding the payments, because a taka paid in year five is worth less today than a taka paid in year one. The answer must come from present values, using the method of net present value.

Tax changes each payment, so we need one more idea. The **after-tax cash outflow** is the cash paid out in a year after taking away the tax saved by that payment. Suppose a lease payment is Tk 100 and the tax rate is 25%. The payment lowers taxable profit by Tk 100, so tax falls by Tk 25, and the after-tax cash outflow is Tk 75.

PAUSE AND CHECK 18.P1 A lease payment is Tk 60,000 and the tax rate is 25%. What is the after-tax cash outflow?

For leasing, the rule is direct. The lessee deducts the whole lease payment from taxable profit. In words: the after-tax cash outflow of leasing equals the lease payment multiplied by one minus the tax rate.

For buying with a loan, the rule needs more care. Each yearly instalment repays part of the loan and pays interest. Only the interest is deducted from taxable profit. The business may also deduct depreciation each year, although depreciation is not a cash payment. Many students find this step confusing, because the instalment is cash but depreciation is not. The explanation is that depreciation sends no cash out, yet it lowers the tax bill, so it still leaves the business with more cash. The after-tax cash outflow of buying therefore equals the instalment minus the tax rate multiplied by the sum of interest and depreciation.

Which discount rate should we use? Both streams are fixed by contract, so they carry about the same risk as a loan. The fitting rate is therefore the after-tax cost of debt: the interest rate multiplied by one minus the tax rate. At an interest rate of 10% and tax of 25%, this is $10\% \times 0.75 = 7.5\%$.

² Under International Financial Reporting Standard 16, which applies to reporting periods beginning on or after 1 January 2019, a lessee records most leases as an asset and a liability. Standards differ by place.

The decision follows four steps. First, find the after-tax cash outflow of leasing in each year. Second, find the after-tax cash outflow of buying in each year, using the loan payments of Chapter 5. Third, convert each year's outflow into a present value at the after-tax cost of debt, and add. Fourth, compare the two totals: the lower total is the cheaper road. If the machine can be sold at the end, subtract the present value of the after-tax sale proceeds from the total for buying.

A resale value changes the comparison, so we note it now. Suppose, in a different case, that a machine can be sold for Tk 10,000 at the end of year 5, with no tax on the sale. That sum reaches the business only if it owns the machine. Its present value at 7.5% is $10,000 \times 0.696559 = \text{Tk } 6,966$, and this amount is subtracted from the present value of buying. The lessor, who keeps the machine, receives the resale value, which is one reason a lessor can offer a lease at all.

The test has limits. It counts cost only. It does not weigh the service that an operating lease may include, the risk of the asset becoming out of date, or the chance that the lessee earns no profit against which to deduct the payments. When the gap is small, such as two or three per cent, the manager should weigh these other facts before choosing.

Before moving on. The lease-or-buy decision compares present values of after-tax cash outflows at the after-tax cost of debt. Leasing deducts the whole payment; buying deducts interest and depreciation. The lower present value wins, if cost is all that matters.

* * *

WORKED EXAMPLE: LEASE OR BUY THE SECOND MACHINE

We first practise the method on small, clean numbers with no tax, and then repeat it on the case file.

Given. A delivery bicycle can be leased for Tk 4,000 at the end of each of 2 years, or bought now for Tk 7,000. There is no tax and no resale value. The discount rate is 10%. *Find.* The cheaper road. *Formula.* Present value is each payment divided by one plus the rate, once for every year of waiting. In symbols: $PV = P_1 \div (1 + r) + P_2 \div (1 + r)^2$, where P is a payment and r is the discount rate. *Steps.* $4,000 \div 1.10 = \text{Tk } 3,636.36$. $4,000 \div 1.21 = \text{Tk } 3,305.79$. The sum is Tk 6,942.15. Buying costs Tk 7,000.00 today. *Answer.* Leasing is cheaper by Tk 57.85. *Check.* The lease payments add to Tk 8,000, more than Tk 7,000, yet the second payment is worth less today, so a lower present value is sensible.

Case file, illustrative. 3 December 2025. Meghna and Sons Limited needs a second machine costing Tk 780,000. *Given.* The lease asks Tk 210,000 at each year end for 5 years. To buy, the business borrows Tk 780,000 at 10% a year and repays it in 5 equal year-end instalments of Tk 205,762.04. Depreciation is Tk 156,000 a year, and no resale value is assumed. Tax is 25%, assumed for illustration only. The discount rate is the after-tax cost of debt, 7.5%.

Find. The present value of leasing and of buying, and the lower of the two. *Formula.* In words: the after-tax outflow of leasing is the lease payment times one minus the tax rate; the after-tax outflow of buying is the instalment minus the tax rate times interest plus depreciation; the present value is the sum of each year's outflow divided by one plus the rate for each year of waiting. In symbols: $L = P \times (1 - T)$; $B = I - T \times (X + D)$; $PV = C_1 \div (1 + r) + C_2 \div (1 + r)^2 + \dots + C_5 \div (1 + r)^5$. Here L and B are the after-tax outflows of leasing and buying, P is the lease payment, T the tax rate, I the instalment, X the

Recall: loan payments, Chapter 5, section 5.4 "Finding the Payment".

Four steps make the decision orderly.

A resale value would lower the cost of buying.

The test counts cost only, so a close result needs more thought.

Recap of section 18.3.

We learn the method on small numbers first, with no tax.

A bicycle example in the standard order.

Recall: loan payments, Chapter 5, section 5.4 "Finding the Payment".

The case file gives every figure for the second machine.

Find: the present value of each road, and the cheaper one.

Step 1 finds the lease outflow; step 2 finds the interest each year.

interest in the year, D the depreciation, C the outflow in a year and r the discount rate.

Steps. Step 1, leasing: $210,000 \times (1 - 0.25) = \text{Tk } 157,500$ in each of the 5 years. Step 2, buying: the interest in year 1 is $10\% \times 780,000 = \text{Tk } 78,000$. The instalment of Tk 205,762.04 repays Tk 127,762.04 of the loan, leaving Tk 652,237.96 owed, so year 2 interest is $10\% \times 652,237.96 = \text{Tk } 65,223.80$. The same method gives the later years. The table that follows is read across each row: the year, the interest, the tax saved (25% of interest plus Tk 156,000 of depreciation) and the after-tax outflow, which is the instalment minus the tax saved.

AFTER-TAX OUTFLOW OF BUYING, IN TAKA TO THE NEAREST TAKA

YEAR	INTEREST	TAX SAVED	AFTER-TAX OUTFLOW
1	78,000	58,500	147,262
2	65,224	55,306	150,456
3	51,170	51,792	153,970
4	35,711	47,928	157,834
5	18,706	43,676	162,086

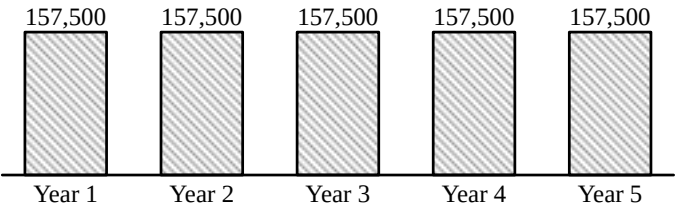
Step 3 converts each outflow to a present value at 7.5%.

Step 3, present values at 7.5%. Leasing is an ordinary annuity, so the 5-year annuity factor of 4.045885 applies: $157,500 \times 4.045885 = \text{Tk } 637,227$. For buying, each year's outflow is multiplied by its present value factor: $147,262 \times 0.930233 = 136,988$; $150,456 \times 0.865333 = 130,195$; $153,970 \times 0.804961 = 123,940$; $157,834 \times 0.748801 = 118,186$; $162,086 \times 0.696559 = 112,902$. The present values add to Tk 622,210; the yearly figures here are rounded, so their sum can differ from this total by a taka.

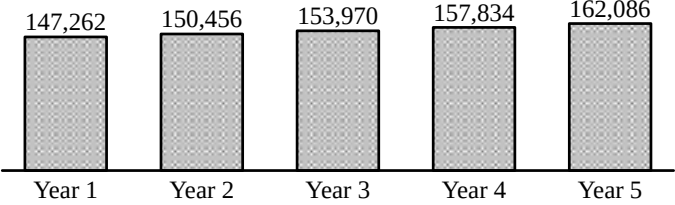
The figure shows the two outflow streams side by side.

Figure 1 draws the two streams. To read it, look at one row at a time: each bar is one year's after-tax outflow, and a taller bar means more cash paid out that year. The top row, with line hatching, is leasing; the bottom row, with cross hatching, is buying. Bars above one another belong to the same year.

Leasing: after-tax outflow each year (Tk)



Buying: after-tax outflow each year (Tk)



Same scale in both rows; lines hatching = leasing, cross hatching = buying

FIG. 18.1. Yearly after-tax outflows of leasing and of buying, aligned by year

Answer and check: buying is cheaper by about Tk 15,000.

Answer. Leasing has a present value of Tk 637,227 and buying has Tk 622,210. Buying costs Tk 15,017 less, so on cost alone the business should buy. Check. The five outflows of leasing add to Tk 787,500 and those of buying to Tk

771,608, so buying is lower before and after discounting; the sign of the answer is sensible. The gap is about 2.4% of the lease figure, which is small, so other facts could change the choice.

From the owners' viewpoint, the saving of Tk 15,017 is a gain in present value, because the same machine does the same work for less. The lender's viewpoint differs: buying with a loan adds Tk 780,000 of debt, while leasing adds no loan but does add a fixed promise to pay each year. A manager should weigh that fixed promise before signing either contract.

Common slips are these. Deducting tax on the whole instalment wrongly treats loan repayment as a tax saving. Leaving out depreciation overstates the cost of buying. Discounting at the loan rate of 10% instead of 7.5% mixes a before-tax rate with after-tax cash. Placing the year-5 payment in year 4 mixes up the periods.

The owners' viewpoint: a lower cost adds to the value of the business.

Common slips: wrong deduction, forgotten depreciation, wrong rate or period.

* * *

SUMMARY

A lease is a contract in which a lessor lets a lessee use an asset in return for regular payments. An operating lease is short and leaves risk with the lessor; a financial lease is long and leaves risk with the lessee. To choose between leasing and buying, find each year's after-tax cash outflow, convert it to a present value at the after-tax cost of debt, and choose the lower total. For the second machine of Meghna and Sons Limited, buying costs Tk 15,017 less in present value.

The chapter in a few lines.

THINK IT THROUGH

18.T1 A clinic needs a scanner for 3 years. Lease P asks Tk 60,000 at each year end for 3 years. Lease Q asks Tk 45,000 at each year end for 2 years, after which the clinic must hire a scanner for the third year at Tk 70,000. Tax is 25% and the discount rate is 7.5%. Which is cheaper in present value?

18.T2 Name the kind of arrangement in each case: a school hires a minibus for one week; a bakery signs a six-year lease of an oven that cannot be cancelled; a printer sells its press to a lessor and leases it back.

WHERE PEOPLE GO WRONG

Wrong: leasing is cheaper because no large sum is paid at the start. *Correct:* the lessor must earn a return, so total payments usually exceed the price. Only a present value comparison shows which road costs less.

Wrong idea: leasing is cheaper because no cash is paid at the start.

Wrong: the tax saved on buying is the tax rate times the whole instalment. *Correct:* only interest and depreciation reduce taxable profit; the part of the instalment that repays the loan does not.

Wrong idea: tax applies to the whole loan instalment.

Wrong: the lessee owns the asset when the lease ends. *Correct:* the asset stays with the lessor unless the contract says otherwise, which is why this chapter assumes no resale value for the lessee.

Wrong idea: the lessee owns the asset at the end of a lease.

Wrong: the difference between the two kinds of lease is only length. *Correct:* length matters, but the key question is who carries the risks of owning the asset.

Wrong idea: the two kinds of lease are defined by length alone.

Leasing is old, but the lease business is modern.

Why leasing grew: scarce credit and favourable tax rules.

IN HISTORY

Some historians trace leasing to the ancient Near East, around 2000 BC, when owners let farm tools to farmers under written agreements. In 1877 the Bell Telephone Company began to let telephones to customers instead of selling them. The United States Leasing Corporation, formed in San Francisco in 1952, is usually named as the first business to treat equipment leasing as its main work. Leasing then spread to Europe in the late 1950s and early 1960s, helped in the United States by tax rules that favoured it.

Why did leasing grow so fast? One reason, in the United States, was a shortage of medium-term credit, which left businesses wanting machines they could not pay for at once. Another was the tax treatment of lease payments and depreciation, which differs from country to country and can favour the lessor, the lessee or both. Banks and manufacturers later formed their own leasing businesses. A lease is therefore both a way of financing and a way of sharing tax savings, which is why the comparison in this chapter always counts tax.

REVIEW QUESTIONS

18.1 A rickshaw owner lets a puller use a rickshaw each day for an agreed payment under a contract. Name the lessor and the lessee, and say what a lease is.

18.2 Explain the difference between an operating lease and a financial lease, with one example of each.

18.3 A lease payment is Tk 80,000 a year and the tax rate is 25%. Find the after-tax cash outflow and say why it is lower than the payment.

18.4 State the four steps of the lease-or-buy decision in order.

18.5 A clinic can lease a scanner for Tk 100,000 at each year end for 3 years, or buy it with a loan. The after-tax outflows of buying are Tk 70,000, Tk 72,000 and Tk 74,000. The loan interest rate is 8% and the tax rate is 25%. Find the discount rate and the present value of each road, then decide.

GLOSSARY

Accounts payable. Money a business owes to suppliers for goods bought on credit. One business's accounts payable is its supplier's accounts receivable. (*Chapter XV*)

Accounts receivable. Money owed to a business by customers who bought on credit. A school that took books in March and will pay in April creates accounts receivable for the bookshop. (*Chapter XV*)

After-tax cash outflow. Cash paid out in a year after taking away the tax saved by that payment. Example: a lease payment of Tk 100 at a tax rate of 25% saves Tk 25 of tax, so the after-tax cash outflow is Tk 75. (*Chapter XVIII*)

Agency conflict. A clash between what a principal wants and what an agent chooses to do. A manager paid on sales may choose discounts that raise sales, while the owners want profit. (*Chapter III*)

Agency cost. The loss in value, plus the cost of prevention, that comes from an agency conflict. If a bad choice costs the owners Tk 105,000 and checking the records costs more, both amounts count. (*Chapter III*)

Agent. A person who acts for a principal and takes decisions in the principal's name. A hired branch manager is an agent of the owners. (*Chapter III*)

Amortisation. Repaying a loan by equal payments that cover the interest and part of the amount borrowed. The instalment equals the loan divided by the present value factor, and the loan falls to zero at the last payment. (*Chapter V*)

Annuity. A series of equal payments made at equal time intervals for a set number of periods. Three deposits of Tk 1,000, one at the end of each of three years, form an annuity, as do monthly loan instalments over two years. (*Chapter V*)

Annuity due. An annuity whose payments fall at the start of each period. Rent paid in advance is an example. Each payment comes one period earlier, so the annuity due is worth $1 + i$ times an ordinary annuity. (*Chapter V*)

Asset. Something of value that a business owns and uses to earn money, such as books for sale, shelves or a bank balance. Each asset is the result of a use of funds, so on any one date the assets equal the sources. (*Chapter I*)

Bank overdraft. A bank allowing an account to go below zero up to an agreed limit, with interest charged on the amount used. Interest runs only for the days used. For example, using Tk 50,000 for 30 days at 12% costs Tk 493.15. (*Chapter XVI*)

Beta. A number showing how strongly an investment moves with the whole market. The whole market has a beta of 1. An investment with a beta of 1.5 tends to rise or fall about 15% when the market return changes by 10%. (*Chapter X*)

Bond. A loan sold to investors, repaid at a set date with fixed interest along the way. The business that borrows is the issuer. For example, a business may divide a loan of Tk 5,000,000 into 5,000 bonds of Tk 1,000 each. (*Chapter XIII*)

Business function. One main kind of work in a business, such as selling, buying or making, managing people, and managing money. Finance is one business function. A small business may give several functions to one person; a larger one gives each its own team. (*Chapter I*)

adds interest every six months has a compounding period of six months. (*Chapter IV*)

Contribution margin. The selling price per unit minus the variable operating cost per unit. It is the amount each unit sold contributes towards the fixed operating cost and then towards profit. A pack sold at Tk 90 with a variable cost of Tk 50 has a contribution margin of Tk 40. (*Chapter XI*)

Cost of capital. The average return a business must earn on its funds to satisfy those who supplied them. It is a rate, not an amount, and it serves as the discount rate for judging projects. For Meghna and Sons Limited it is 12.85%. (*Chapter XVII*)

Coupon interest. The fixed interest a bond pays each period, set as a percentage of par value. The percentage is the coupon rate. A par value of Tk 1,000 and a coupon rate of 9% give coupon interest of Tk 90 a year. (*Chapter XIII*)

Covenant. A promise in a loan agreement that limits what the borrower may do. It may set a minimum level of owners' equity or limit new borrowing. If it is broken, the lender may ask for early repayment. (*Chapter XVII*)

Credit terms. The discount offered and the days allowed for payment, written like 2/10, net 40. This means 2% off if paid within 10 days, and otherwise the full amount due by day 40. The code lets a buyer compare the saving with the cost of waiting. (*Chapter XVI*)

Current asset. An asset expected to turn into cash within about a year. Cash, inventory and accounts receivable are examples. A delivery cart used for many years is not a current asset. (*Chapter XV*)

Current liability. An amount owed that must be paid within about a year. A bill owed to a supplier and due next week is an example. A loan repayable over five years is not. (*Chapter XV*)

Debenture. A bond backed only by the general standing of the borrower, with no particular asset pledged. Its holder trusts the borrower's record and profit, and so carries more risk than a lender with collateral. (*Chapter XVII*)

Debt. Money borrowed that must be repaid, with an extra charge for using it. The lender is promised repayment on agreed dates. The bank loan of Tk 600,000 at 10% a year is debt. (*Chapter I*)

Degree of financial leverage. The percentage change in earnings per share caused by a one percent change in EBIT. It equals EBIT divided by EBIT less interest less preferred dividend divided by one minus the tax rate. For the printing unit, $80,000 \div 44,000 = 1.82$. (*Chapter XII*)

Degree of operating leverage. The percentage change in EBIT caused by a one percent change in sales. It equals contribution divided by EBIT at the starting volume. A value of 4.00 means a 1% change in sales changes EBIT by 4%. (*Chapter XI*)

Degree of total leverage. The percentage change in earnings per share caused by a one percent change in sales. It equals the degree of operating leverage times the degree of financial leverage. For the printing unit, $4.00 \times 1.8182 = 7.27$. (*Chapter XII*)

Depreciation. The part of an asset cost charged as an expense in each year of its use. With the straight-line method, the yearly charge is cost less resale value, divided by years. No cash leaves in those years. (*Chapter VII*)

Discount bond. A bond that sells for less than its par value. This happens when the required return is above the coupon rate. The case bond at 12% sells for Tk 891.86. (*Chapter XIII*)

Discount rate. The rate used to turn a future amount into a present value. It is the same number as the interest rate when the money could earn that rate, used for working backwards. (*Chapter IV*)

Discounted cash flow. A way of valuing a series of future cash flows by turning each one into a present value and adding the results. For example, Tk 1,000 in year 1 and Tk 2,000 in year 2 at 10% are valued as Tk 909.09 plus Tk 1,652.89. (*Chapter VI*)

Discounting. Finding a present value by working backwards from a future amount at a rate, dividing by $(1 + i)$ once for each year. Tk 12,100 becomes Tk 11,000 and then Tk 10,000. (*Chapter IV*)

Diversification. Spreading money across different investments so that one loss hurts less. The investments must differ in what affects them. Ten umbrella stalls do not diversify the trader's risk from wet weather, but one umbrella stall and one sunshade stall do. (*Chapter X*)

Dividend. A share of profit paid out to the owners of a business. If a shop earns Tk 90,000 and pays Tk 30,000 to its owners, the dividend is Tk 30,000. A family may call its payout a withdrawal. (*Chapter II*)

Dividend decision. Choosing how much profit to pay out to owners and how much to keep in the business. A payout rewards owners now; keeping profit leaves more to fund growth later without asking a lender. (*Chapter II*)

Dividend growth model. A formula valuing stock as the next dividend divided by the required return minus the growth rate. It assumes the dividend grows at a steady rate for ever, and that this rate is below the required return. A next dividend of Tk 4.24, required return of 12% and growth of 6% give Tk 70.67. (*Chapter XIV*)

Earnings per share. The profit left for common stock owners divided by the number of common shares. It is found by taking interest, tax and preferred dividend from EBIT in turn. Tk 33,000 left for 5,000 shares gives earnings per share of Tk 6.60. (*Chapter XII*)

EBIT. Earnings before interest and tax: the profit from operations before paying lenders or the tax office. It equals revenue less cash costs less depreciation. The book uses EBIT in place of the term operating profit. (*Chapter VII*)

Effective annual rate. The single yearly rate that gives the same growth as compounding within the year. A rate of 12% compounded half-yearly has an effective annual rate of 12.36%. (*Chapter IV*)

Efficient market. A market in which prices already reflect the information available. New information moves prices, and since it cannot be predicted, price changes look random. Efficient does not mean that prices are always right or that investors never lose. (*Chapter X*)

Equity. Money put in by the owners, who share the business profit or loss. Owners are not promised a fixed payment. At Meghna and Sons, the family's Tk 1,800,000 is equity. Money from owners is often called capital. (*Chapter I*)

Expected value. The average outcome, found by multiplying each outcome by its probability and adding the products. It is a long-run average and not a forecast for one trial. Outcomes of 9%, 12% and 15%

with probabilities 0.25, 0.50 and 0.25 give an expected value of 12%. (*Chapter IX*)

Factor table. A printed table of ready-made multipliers for given rates and numbers of periods. A reader finds the row for the number of periods, reads the column needed, and multiplies the factor by the payment. (*Chapter V*)

Finance. The work of finding money and deciding how to use it, so that a business or a person can reach chosen aims. It has two halves: finding money, and choosing its use. A stall owner who borrows Tk 800 and spends it on onions to sell is doing finance. (*Chapter I*)

Financial control. Checking that money is spent and recorded as planned, and acting when it is not. If cloth planned at Tk 30,000 costs Tk 36,000, control finds the Tk 6,000 difference, asks why, and changes something. (*Chapter II*)

Financial lease. A long lease that cannot be cancelled, in which the lessee in effect carries the cost and risk of owning the asset. The payments usually cover most of the price plus the lessor's return. Example: a six-year oven lease with no right to leave early. (*Chapter XVIII*)

Financial leverage. The effect of fixed financing charges in making earnings per share change by more than EBIT. With no fixed charges the two change by the same percentage. With interest of Tk 2,000, a 20% change in EBIT of Tk 10,000 moves the profit for each share by 25%. (*Chapter XII*)

Financial manager. The person who plans, raises and controls the money of a business, whatever the job title. In a small shop one brother may hold the duty; in a large business a team may. The manager answers to owners, lenders and employees. (*Chapter II*)

Financial market. A place or network where buyers and sellers trade money, loans and ownership parts of businesses. A bank where one borrower meets one lender is a very small example; large markets let a business find many sources at once. (*Chapter I*)

Financial planning. Deciding in advance how much money a business will need and when. A tailor who knows she must pay Tk 38,000 for cloth and rent in the first week of March can arrange the money before the day arrives. (*Chapter II*)

Financial risk. The extra risk to owners that comes from funding part of a business by borrowing. Interest must be paid in poor years as well as good ones, so what is left for owners varies more. A bookshop with a loan of Tk 600,000 at 10% a year faces it. (*Chapter IX*)

Financing decision. Choosing how to raise the money for purchases, from owners or lenders. It asks how to pay, not what to buy. An oven may be paid for from owners' funds, a bank loan, or both. (*Chapter II*)

Firm-specific risk. Risk that affects one business only and can be reduced by diversifying. A fire at one factory or a failed new product are examples. An investor who holds many businesses feels such a loss as a small part of the whole portfolio. (*Chapter X*)

Fixed financing charge. Interest or preferred dividend that must be paid whatever the level of EBIT. It is owed to those who supplied funds, not to those who supply the trade. Interest of Tk 20,000 a year on a bank loan is a fixed financing charge. (*Chapter XII*)

Fixed operating cost. An operating cost that stays the same whatever the number of units sold. Shop rent of Tk 8,000 a year is the same for ten notebooks or a thousand, so it is a fixed operating cost. (*Chapter XI*)

Float. Money that has been paid by one person but cannot yet be used by the receiver. It has three parts: mail float, processing float and clearing float. A cheque with 2, 1 and 3 days of these parts has 6 days of float. (*Chapter XV*)

Funds. Money that a business has available to spend, save or invest at one moment. Funds may be notes in a purse or a balance held at a bank. Goods waiting to be sold are not funds until they are sold and paid for. (*Chapter I*)

Future value. What an amount today will grow to after a stated time at a stated rate, found by $FV = S \times (1 + i)^n$ to the power of n . Tk 10,000 at 10% for two years has a future value of Tk 12,100. (*Chapter IV*)

Independent projects. Projects where accepting one has no effect on whether another is accepted. A print-on-demand machine and a new set of shelves are independent: a business may accept both, either or neither. (*Chapter VII*)

Initial investment. Cash spent at the start to get a project going. It includes the price of the asset, delivery, installation and any working capital needed. It is paid at time zero, before year one begins. (*Chapter VII*)

Initial public offering. The first sale of a business shares to the general public. After it, the shares are usually listed on a stock exchange. A family printing business selling 30% of its shares to strangers for the first time is an example. (*Chapter XVII*)

Interest. The price paid for using money for a period, received by the lender. If a bank pays Tk 8,000 for the use of Tk 100,000 for a year, the Tk 8,000 is interest. (*Chapter IV*)

Interest rate. The interest for one period, shown as a percentage of the amount lent or borrowed. Tk 8,000 earned on Tk 100,000 in a year is an interest rate of 8% a year. (*Chapter IV*)

Intermediate-term financing. Money borrowed or raised for between one and five years. It suits assets that last a few years. For example, a loan for a delivery van, repaid in instalments over three years, is intermediate-term financing. (*Chapter XVI*)

Internal rate of return. The discount rate at which a project's net present value is exactly zero. It is the yearly return the project itself earns. Find it by trial between two rates and interpolation; accept when it is at least the required return. (*Chapter VIII*)

Intrinsic value. The worth of an asset worked out from the present value of its expected cash flows. An asset that pays Tk 1,100 in one year has an intrinsic value of Tk 1,000 at a required return of 10%. (*Chapter XIII*)

Inventory. Goods held for sale or for use in making goods. A bookshop's books, a tailor's cloth and a baker's flour are examples. Shopkeepers often call it stock. (*Chapter XV*)

Investment decision. Choosing which long-lasting assets or projects a business should buy. It asks what to buy, and it commits money for a long time. A bakery choosing between a new oven and a delivery van makes one. (*Chapter II*)

Lease. A contract in which the owner of an asset lets another party use it in return for regular payments. The owner keeps ownership throughout. Example: a baker pays a yearly sum to use an

oven for five years while the oven stays the property of its owner. (*Chapter XVIII*)

Lessee. The party who pays to use an asset under a lease. The lessee gains the use of the asset but does not own it. Example: a clinic that pays each month to use a scanner owned by another business. (*Chapter XVIII*)

Lessor. The owner who lets an asset out under a lease. The lessor receives the lease payments and keeps the asset, together with any resale value at the end. Example: a business that owns machines in order to let other businesses use them. (*Chapter XVIII*)

Liability. An amount a business owes to someone else. Debt is one kind of liability; an unpaid bill to a supplier is another. Assets less liabilities gives the owners' wealth in the business. (*Chapter I*)

Liquidity. How easily a business can pay its bills on time from cash it has or can quickly raise. Books on shelves are not liquid until sold, while a bank balance is. Weak liquidity can delay wages and stop deliveries. (*Chapter II*)

Long-term financing. Money borrowed or raised for more than five years. It suits needs that never end or assets that last a long time. Money that owners put into a business as equity, and a bond, are examples. (*Chapter XVI*)

Market risk. Risk that affects all investments in a market and cannot be removed by diversifying. A fall in demand across a whole country is an example. Because it cannot be removed, it is the only risk that earns a premium in the pricing model. (*Chapter X*)

Market value. The price at which something could be sold today in a market. A used bicycle that a buyer would pay Tk 6,000 for today has a market value of Tk 6,000. (*Chapter III*)

Matching principle. The rule that short-term needs are financed with short-term funds and long-term needs with long-term funds. It protects the business from having to renew money in a hurry and from paying interest on idle cash. It is a guide, not a law. (*Chapter XVI*)

Maturity. The date on which a bond is repaid. A bond with five years to maturity pays its par value, with its last coupon interest, at the end of year 5. (*Chapter XIII*)

Mixed stream. A series of payments or receipts that differ in size from one period to the next. Receipts of Tk 1,000, Tk 2,000 and Tk 3,000 at the ends of three years form a mixed stream. An annuity is the special case in which all sizes are equal. (*Chapter VI*)

Mutually exclusive projects. Projects where accepting one means rejecting the others. A machine and a branch fit-out that cannot be run together are an example, as are two scanners when only one will be bought. (*Chapter VII*)

Net present value. The present value of a project's cash inflows minus its initial investment. It shows, in taka today, how much worth the project adds. Accept a project when the value is zero or positive; among exclusive projects, choose the highest. (*Chapter VIII*)

Net working capital. Current assets minus current liabilities. If the stall holds Tk 11,200 in current assets and owes Tk 10,000, its net working capital is Tk 1,200. A negative result warns that short-term debts exceed short-term assets. (*Chapter XV*)

Operating break-even point. The sales volume at which EBIT is exactly zero. It equals the fixed operating cost divided by the contribution margin, and is stated in units. With fixed cost Tk 240,000 and contribution margin Tk 40, it is 6,000 packs. (*Chapter XI*)

Operating cash flow. Cash a project produces each year from running it, after tax. It equals EBIT, less tax on EBIT, plus depreciation, since depreciation reduced EBIT but no cash was paid. (*Chapter VII*)

Operating cycle. The days from buying stock to collecting cash from the customer. It equals the inventory days plus the receivable days. For Meghna and Sons Limited, $75 + 20 = 95$ days. (*Chapter XV*)

Operating efficiency. How little money and effort a business uses to produce each unit of sales. A tailor who spends Tk 70 for every Tk 100 of sales is more efficient than one who spends Tk 80 for the same Tk 100. (*Chapter II*)

Operating lease. A short or cancellable lease where the lessor keeps most of the risks of owning the asset, such as repairs and the chance that the asset becomes out of date. Example: a portable scanner hired by the month, with the right to return it. (*Chapter XVIII*)

Operating leverage. The effect of fixed operating costs in making EBIT change by more than sales change. The larger the fixed cost compared with the contribution, the stronger the effect. A business with no fixed operating cost has none. (*Chapter XI*)

Opportunity cost. The value of the best choice given up when you pick another. Cash kept idle in a drawer has an opportunity cost equal to the interest a bank would have paid on it. (*Chapter IV*)

Ordinary annuity. An annuity whose payments fall at the end of each period. Loan instalments and bank interest are usually of this kind. When no kind is stated, an annuity is taken to be ordinary. (*Chapter V*)

Par value. The amount printed on a bond that is repaid at maturity. It is also called face value. A bond with a par value of Tk 1,000 repays Tk 1,000 to the investor on the date of maturity. (*Chapter XIII*)

Payback period. The time a project takes to recover its initial investment from its cash flows. With equal inflows it is the initial investment divided by the yearly inflow; otherwise add the inflows into a running total. Tk 10,000 repaid by Tk 5,000 a year takes 2 years. (*Chapter VIII*)

Permanent working capital. The minimum working capital a business needs all year. A bookshop always needs some stock, some money in the till and some money owed by customers, whatever the month. This level of need is normally financed with long-term funds. (*Chapter XVI*)

Perpetuity. An annuity that never ends, such as a fund paying a scholarship of Tk 50,000 every year for ever. Its present value is the payment divided by the interest rate. (*Chapter V*)

Portfolio. A collection of investments held together by one investor. What matters is the combined result of the whole collection. A trader with an umbrella stall and a sunshade stall holds a portfolio of two stalls. (*Chapter X*)

Preferred stock. Ownership shares with a fixed dividend that is paid before any dividend on common stock. The holder cannot claim more than the fixed amount. Tk 100,000 of preferred stock at 12% promises Tk 12,000 a year ahead of common stock owners. (*Chapter XII*)

Premium bond. A bond that sells for more than its par value. This happens when the required return is below the coupon rate. The case bond at 7% sells for Tk 1,082.00. (*Chapter XIII*)

Present value. What an amount to be received later is worth today at a stated rate. At 10%, Tk 12,100 due in two years has a present value of Tk 10,000. (*Chapter IV*)

Principal. A person who hires another person to act for them. The owners of a bookshop are principals when they hire someone to run a branch for them. (*Chapter III*)

Probability. A number from 0 to 1 showing how likely an outcome is. Zero means the outcome cannot happen and one means it is certain. A fair coin landing heads has a probability of 0.5. Heavy traffic on 25 days in 100 has a probability of 0.25. (*Chapter IX*)

Probability distribution. A list of possible outcomes with the probability of each, adding up to one. Exactly one outcome must happen, so the probabilities total one. Route times of 10, 25 and 40 minutes with probabilities of 0.25, 0.50 and 0.25 form a probability distribution. (*Chapter IX*)

Profit maximisation. The aim of earning the largest profit possible in a period, usually one year. A shop that always picks the plan with the largest profit for the year follows this aim, even if the plan harms later years. (*Chapter III*)

Profitability. How much profit a business earns compared with the money invested or the sales made. A profit of Tk 20,000 on Tk 200,000 invested is 10% of the money invested; compare ratios, not amounts. (*Chapter II*)

Profitability index. The present value of a project's inflows divided by its initial investment. A value above 1 means each taka invested returns more than a taka in present value. For Press, $1,097,055 \div 900,000$ gives 1.22. (*Chapter VIII*)

Rate of return. The return shown as a percentage of the amount first invested, found by dividing the return by that amount and multiplying by 100. It lets investments of different size be compared. A return of Tk 10,000 on Tk 100,000 is 10%. (*Chapter IX*)

Relevant cash flow. Cash flow that changes only because a project is accepted. It must be cash, lie in the future and differ with the decision. A sunk cost, such as a survey already paid, is not relevant. (*Chapter VII*)

Required return. The lowest return a project must earn to be worth accepting. Also called the required rate of return or hurdle rate. It is used as the discount rate, and it reflects the risk of the project and the cost of the funds. Meghna and Sons uses 12%. (*Chapter VIII*)

Residual claim. The right to whatever is left of a business's income and assets after lenders, suppliers, staff and preferred stockholders are paid. Common stockholders hold it. When the business does well the remainder can be large, and when it does badly it can be nothing. (*Chapter XIV*)

Retained earnings. Profit kept in the business instead of being paid out to owners. If the shop earns Tk 90,000 and pays out Tk 30,000, retained earnings are Tk 60,000. They belong to the owners and may already be spent on assets. (*Chapter II*)

Return. The total gain or loss from an investment over a period, found by adding the income received to any change in value. A bond bought for Tk 100,000 that

pays Tk 6,000 and later sells for Tk 104,000 gives a return of Tk 10,000. (*Chapter IX*)

Rights issue. An offer of new shares to existing shareholders in proportion to what they already hold. An owner with 10% of the shares who buys the offered share of a 1-for-5 issue keeps 10% of the business. (*Chapter XVII*)

Risk. The chance that actual results differ from those expected, especially for the worse. The wider the range of possible results, the greater the risk. A certain loss carries no risk, because nothing is uncertain. A fixed-payment savings bond has little risk. (*Chapter IX*)

Risk premium. The extra return investors demand for taking on risk above the risk-free rate. It equals the required return less the risk-free rate. If the risk-free rate is 8% and a lender demands 12%, the risk premium is 4 percentage points. (*Chapter X*)

Risk-free rate. The return on an investment with no chance of loss, such as a short government bill. It pays for waiting alone. No investment is perfectly free of risk, but a short loan to a stable government comes close. The case file assumes 8%. (*Chapter X*)

Sale and leaseback. Selling an asset to a lessor and at once leasing it back. The business receives cash today and keeps using the asset, but makes lease payments from then on. Example: a baker sells a delivery van to a lessor and leases the same van back. (*Chapter XVIII*)

Seasonal working capital. Extra working capital needed only at the busiest times of the year. A bookshop buys extra books before the school term opens and needs the extra money for a few months. This peak is normally financed with short-term funds. (*Chapter XVI*)

Security market line. The straight line that shows required return rising with beta. It starts at the risk-free rate when beta is 0 and passes through the market return when beta is 1. An investment above the line promises more than its required return. (*Chapter X*)

Semi-strong-form efficiency. The condition in which prices reflect all public information, not only past prices. If it holds, reading a published annual report or news item after it appears brings no gain, because the price has already moved. (*Chapter X*)

Short-term financing. Money borrowed or raised for up to one year. It suits needs that last a short time. For example, a bank overdraft cleared in two months, or a supplier's trade credit of 30 days, is short-term financing. (*Chapter XVI*)

Simple interest. Interest worked out on the original amount only, so the same amount is added each period. Tk 10,000 at 10% earns Tk 1,000 every year. (*Chapter IV*)

Single sum. One payment or receipt made at one point in time. A deposit of Tk 10,000 made today is a single sum, unlike payments that repeat each year. (*Chapter IV*)

Sinking fund. Money put aside in equal deposits to reach a target sum by a set date. The deposit equals the target divided by the future value factor. A business may use one to replace a vehicle. (*Chapter V*)

Source of funds. A place a business gets money from, such as its owners or a lender. Meghna and Sons has two: the family and a bank. Each

source supplies part of the total money that the business has obtained. (*Chapter I*)

Stakeholder. Any person or group affected by a business, such as workers, customers, lenders and owners. A bookshop's stakeholders include its staff, the schools that buy from it and the bank that lent it money. (*Chapter III*)

Standard deviation. A measure of how widely outcomes spread around the expected value. Find each deviation from the expected value, square it, weight it by its probability, add the results and take the square root. A small figure means outcomes lie close to the expected value. (*Chapter IX*)

Strong-form efficiency. The condition in which prices reflect all information, public and private. If it holds, even secret knowledge held by managers is already in the price. Few people believe it holds, and trading on secret information is a crime in many countries. (*Chapter X*)

Term loan. A loan repaid in instalments over a fixed period of more than one year. Each instalment usually holds some interest and some repayment of the amount borrowed. A bank loan of Tk 600,000 repaid yearly over 5 years is an example. (*Chapter XVII*)

Terminal cash flow. Cash received at the end of a project life, such as the resale value of equipment. It also includes working capital that returns, and is reduced by tax on any gain from the sale. (*Chapter VII*)

Time value of money. The idea that money received sooner is worth more than the same amount received later, because money in hand can be put to work. Tk 100,000 today is worth more than Tk 100,000 in a year. (*Chapter IV*)

Trade credit. A supplier allowing a business to pay for goods some days after delivery. It is a loan that appears with each purchase and is the most common source of short-term financing. A bookshop that receives books on the 1st and pays on the 30th is using trade credit. (*Chapter XVI*)

Use of funds. A purpose a business spends money on, such as stock or equipment. Keeping money at the bank for later is also a use. Each use of funds usually gives the business an asset. (*Chapter I*)

Valuation. The work of finding what an asset is worth from the cash it is expected to bring. It needs the expected cash and the required return of the investor. (*Chapter XIII*)

Variable operating cost. An operating cost that rises and falls with the number of units sold. If each notebook costs Tk 12 to buy, then 100 notebooks cost Tk 1,200 and 200 notebooks cost Tk 2,400. (*Chapter XI*)

Voting right. The right of a shareholder to vote on matters put to the owners, such as choosing the directors who guide the business. Voting rights usually come with common stock, often one vote for each share, and usually not with preferred stock. (*Chapter XIV*)

Weak-form efficiency. The condition in which prices already reflect all past price information. If it holds, studying old price charts cannot show which investment will rise next, because any pattern in past prices is already in the current price. (*Chapter X*)

Wealth. The total value of what a person or business owns, after taking away what is owed. Owning goods and cash worth Tk 4,000 and owing Tk 800 gives wealth of Tk 3,200. Wealth can be large while funds are small. (*Chapter I*)

Wealth maximisation. The aim of making the market value of the owners' stake as large as possible over time. It takes account of when profits arrive and how certain they are, because buyers pay for both. (*Chapter III*)

Weighted average cost of capital. The cost of each source of funds multiplied by its share of the total and added together. Debt is counted after tax. For Meghna and Sons Limited: $1.25\% + 0.40\% + 11.20\% = 12.85\%$. (*Chapter XVII*)

Working capital. The money a business has tied up in short-term assets such as stock and money owed by customers. It is the money that moves daily from cash to goods to customers' promises and back to cash. A stall's rice, cash and customer debts are an example. (*Chapter XV*)

Yield to maturity. The yearly return an investor earns by buying a bond at its price and holding it to maturity. It is the rate that makes the present value of all payments equal the price. The case bond at Tk 920 has a yield to maturity of about 11.2%. (*Chapter XIII*)

ANSWERS TO QUESTIONS

CHAPTER I · MONEY, FINANCE AND THE FIRM

1.p1 No. The tailor has accepted an order, but no money has moved today. The cash flow into the business will come next week, when the customer pays.

1.p2 The Tk 300,000 of savings is equity, and the Tk 100,000 loan is debt. The bank must be repaid on agreed dates; the owner shares any profit or loss.

1.t1 Ticket sales of Tk 40,000 are a cash flow into the business today. The film hire of Tk 25,000 is owed next month, so it is not a cash flow today. The Tk 10,000 loan instalment is a cash flow out today.

1.t2 The net cash flow today is Tk 40,000 less Tk 10,000, which is Tk 30,000 in. The film hire of Tk 25,000 is a liability; it will become a cash flow out next month, when it is paid.

1.1 For her, finance is finding money and deciding how to use it. She might find Tk 800 by asking her cousin, and decide to spend it on onions to sell.

1.2 Recording the money received is accounting, because it records what has happened. Deciding whether to borrow for the machine is finance, because it looks forward and asks where to find money and how to use it.

1.3 The two sources are the owner's savings, which are equity, and the loan, which is debt. Two uses might be ovens and other equipment, and ingredients to bake with, or money kept at the bank.

1.4 The sale is counted in June, when the goods are taken. The money moves only on 28 July, so the cash flow into the business falls in July.

1.5 Cash flow in is Tk 3,000, and cash flow out is Tk 2,000. The net movement is Tk 3,000 less Tk 2,000, which is Tk 1,000 in.

1.6 The Tk 800 is a source of funds, and it is debt, because it must be repaid. Afterwards she owes her cousin Tk 800, which is a liability, plus any extra charge they agree.

1.7 A source might be the school fund; a use might be buying ingredients for the canteen. The canteen's daily sales bring cash flow into it.

1.8 The sources are the owners' savings, which are equity, and the bank loan, which is debt. Two assets might be medical equipment and a cash and bank balance. Finance is one business function beside the clinic's other work; it finds funds and judges the money side of the other functions' decisions.

CHAPTER II · THE FINANCIAL MANAGER'S WORK

2.p1 The decision to buy the oven for Tk 200,000 is the investment decision. Paying Tk 80,000 from owners' funds and borrowing Tk 120,000 is the financing decision.

2.p2 The dividend is Tk 25,000, the part of the profit paid out. The retained earnings are Tk 90,000 less Tk 25,000, which is Tk 65,000.

2.t1 The new oven is an investment decision. The payout to the owners is a dividend decision. Repaying the loan changes how the business is funded, so it is a financing decision.

2.t2 Repaying the loan reduces debt, so lenders supply less of the funds. A dividend takes money out to the owners, so the business keeps less profit as owners' funds. Repaying the loan leaves the Tk 150,000 of profit used inside the business; a dividend sends it out.

2.1 Planning: decide in advance what money the shop will need, such as stock for the new term. Raising: find that money from the family or a bank. Control: compare spending with the plan and act on any difference.

2.2 Buying the scanner is the investment decision; choosing a bank loan to pay for it is the financing decision. The dividend decision, how much profit to pay out and how much to keep, has not yet been made.

2.3 Planning decides in advance that Tk 30,000 will be needed for cloth and arranges it. Control compares the actual bill with Tk 30,000 afterwards, finds any difference, and acts on it.

2.4 Profit is not cash. If cash is short, suppliers may refuse deliveries, wages may be late and chances to buy cheaply may pass, even though the shop earns a profit.

2.5 Buying the fridge is an investment decision. Repaying the loan changes how the household is funded, so it is a financing decision.

2.6 The freezer is an investment decision, and paying herself is a dividend decision. The dividend is Tk 50,000, and the retained earnings are Tk 40,000. Using kept profit to pay for the freezer is also part of the financing decision.

2.7 The three decisions are investment (the machine), financing (the loan) and dividend (the payout). The more profit is paid out, the less is retained to pay for the machine, so the more the manufacturer may need to borrow.

2.8 Tailor A is more efficient, because A uses Tk 70 for each Tk 100 of sales against Tk 80. Efficiency looks at the cost per unit of sales; profitability compares profit with the money invested or the sales made.

CHAPTER III · THE GOAL OF THE FIRM AND AGENCY

3.p1 Wealth maximisation looks at the market value of the owners' stake over time. A buyer would pay less for a business that has lost its customers, so the cut lowers the owners' wealth even though this year's profit rises.

3.p2 The bonus is 5% of branch sales. It rises with sales whatever happens to profit, so Plan Y, with more sales and less profit, pays the manager more while the owners lose.

3.t1 Plan P: bonus 4% of Tk 1,000,000 = Tk 40,000; owners keep Tk 120,000 - Tk 40,000 = Tk 80,000. Plan Q: bonus 4% of Tk 1,400,000 = Tk 56,000; owners keep Tk 60,000 - Tk 56,000 = Tk 4,000. The manager prefers Q (Tk 16,000 more) and the owners prefer P. If the manager chooses Q, the owners lose Tk 80,000 - Tk 4,000 = Tk 76,000.

3.t2 The change in profit over the two years is Tk 80,000 - Tk 150,000 = minus Tk 70,000. Five per cent of that is minus Tk 3,500, a loss to him, so he would not cut restocking. The new rule lines his reward up with the owners' result.

3.1 Profit maximisation looks at one period only. It cannot see that money arriving earlier is more useful, or that one plan may be less certain than another, so a larger profit now can leave the owners worse off.

3.2 Tell the shopkeeper: the aim is to make the shop worth as much as possible if someone wished to buy it, now and in the coming years. Choices that win loyal customers and keep the shelves full raise that worth, while quick gains that lose customers lower it.

3.3 The taxi owner wants a car in good condition and a steady profit. The driver is paid per trip, so takes as many trips as possible,

perhaps driving hard and wearing out the car. Their rewards differ, so their choices differ.

3.4 Examples: pay the manager by long-term value (cost: a harder pay rule to design); give the manager a stake in the business (cost: a share of the owners' return); check records and require reports (cost: time and money for the owners).

CHAPTER IV · THE TIME VALUE OF MONEY: SINGLE SUMS

4.p1 Year 1: Tk 11,000. Year 2: Tk 12,100. Year 3: $Tk\ 12,100 \times 1.10 = Tk\ 13,310$. Check: $Tk\ 10,000 \times 1.331 = Tk\ 13,310$.

4.p2 Year by year: $Tk\ 13,310 \div 1.10 = Tk\ 12,100$, then $\div 1.10 = Tk\ 11,000$, then $\div 1.10 = Tk\ 10,000$. The present value is Tk 10,000.

4.t1 $PV = Tk\ 130,000 \div 1.07$ to the power of 5 = $Tk\ 130,000 \div 1.402552 =$ about Tk 92,688. This is above Tk 90,000, so waiting is better. Check: $Tk\ 90,000 \times 1.402552 =$ about Tk 126,230, which is below Tk 130,000.

4.t2 Factor = $21,277.79 \div 15,000 = 1.418519$. In the 6% column of the future value table this factor is in the row for 6 years, so $n = 6$ years.

4.1 Factor = 1.09 to the power of 6 = 1.677100. $FV = Tk\ 15,000 \times 1.677100 = Tk\ 25,156.50$.

4.2 Factor = $1 \div 1.11$ to the power of 4 = $1 \div 1.518070 = 0.658731$. $PV = Tk\ 80,000 \times 0.658731 = Tk\ 52,698.48$.

4.3 Simple: $Tk\ 25,000 \times (1 + 0.06 \times 8) = Tk\ 37,000$. Compound: $Tk\ 25,000 \times 1.06$ to the power of 8 = $Tk\ 25,000 \times 1.593848 = Tk\ 39,846.20$. Compound interest is Tk 2,846.20 higher.

4.4 Rate per quarter = $12\% \div 4 = 3\%$. Effective annual rate = 1.03 to the power of 4 – 1 = $1.125509 - 1 = 12.55\%$.

4.5 Opportunity cost of idle cash is the interest it could have earned. Example: Tk 50,000 kept in a drawer for a year loses the Tk 4,000 that a bank paying 8% would have given.

CHAPTER V · ANNUITIES

5.p1 Factor = $(1.10^4 - 1) \div 0.10 = (1.4641 - 1) \div 0.10 = 4.641$. Future value = $Tk\ 1,000 \times 4.641 = Tk\ 4,641$.

5.p2 $Tk\ 3,310 \times 1.10 = Tk\ 3,641$, or $Tk\ 1,331 + Tk\ 1,210 + Tk\ 1,100 = Tk\ 3,641$.

5.t1 Present value of the plan = $Tk\ 90,000 \times 3.790787 =$ about Tk 341,171. This is less than Tk 400,000, so the payment plan costs less today. The delay in paying is worth something at 10%.

5.t2 Each payment of an annuity due is made one period earlier, so it earns one more period of interest, or is discounted one period less. Every payment is therefore worth $1 + i$ times as much.

5.1 Factor = $(1.09^6 - 1) \div 0.09 = 7.523335$. Ordinary: $Tk\ 12,000 \times 7.523335 =$ about Tk 90,280. Due: $Tk\ 90,280 \times 1.09 =$ about Tk 98,405.

5.2 Factor = $(1 - 1.10^{-5}) \div 0.10 = 3.790787$. Present value = $Tk\ 45,000 \times 3.790787 =$ about Tk 170,585.

5.3 Factor = $(1.07^4 - 1) \div 0.07 = 4.439943$. Deposit = $Tk\ 300,000 \div 4.439943 =$ about Tk 67,568.

5.4 Factor = $(1 - 1.12^{-3}) \div 0.12 = 2.401831$. Payment = $Tk\ 250,000 \div 2.401831 =$ about Tk 104,087.

5.5 An ordinary annuity pays at the end of each period and an annuity due at the start. The annuity due has the higher future value, because each payment earns one more period of interest.

5.6 Present value = Tk 20,000 ÷ 0.05 = Tk 400,000. A fund of Tk 400,000 earns 5% × Tk 400,000 = Tk 20,000 a year, so paying out only the interest leaves the fund whole.

CHAPTER VI · UNEVEN CASH FLOWS

6.p1 Tk 3,000 ÷ 1.10 + Tk 2,000 ÷ 1.21 + Tk 1,000 ÷ 1.331 = Tk 2,727.27 + Tk 1,652.89 + Tk 751.31 = about Tk 5,131.48. It is worth more, because the larger receipts come sooner.

6.t1 Plan X: Tk 81,818.18 + Tk 49,586.78 + Tk 22,539.44 = Tk 153,944.40. Plan Y: Tk 27,272.73 + Tk 49,586.78 + Tk 67,618.33 = Tk 144,477.84. Plan Y costs less in present value, because its larger payments fall later.

6.t2 Money has a time value, so a taka received sooner is worth more than a taka received later. The same total, paid in a different order, gives a different present value.

6.1 Tk 18,181.82 + Tk 28,925.62 + Tk 37,565.74 = Tk 84,673.18.

6.2 Tk 20,000 × 1.21 + Tk 35,000 × 1.10 + Tk 50,000 = Tk 24,200 + Tk 38,500 + Tk 50,000 = Tk 112,700.

6.3 Annuity part: Tk 15,000 for 5 years, Tk 15,000 × 3.790787 = Tk 56,861.80. Odd part: Tk 25,000 extra in year 5, Tk 25,000 × 0.620921 = Tk 15,523.03. Present value = Tk 72,384.83.

6.4 At 9%: rising Tk 86,416.60, falling Tk 90,774.03. At 15%: rising Tk 76,732.14, falling Tk 83,093.61. The falling stream is worth more at both rates, and the gap is larger at 15%.

6.5 In an annuity all payments are equal, in a mixed stream they differ. The annuity factor assumes equal payments, so it fits only a level part of a stream; the rest must be discounted as single sums.

CHAPTER VII · CAPITAL BUDGETING: DECISIONS AND CASH FLOWS

7.p1 No. The survey was paid in December 2024 and cannot be recovered whether or not the machine is bought, so it is a sunk cost and does not change the decision.

7.t1 Asset cost 560,000 + 40,000 = Tk 600,000; initial investment 600,000 + 50,000 = Tk 650,000. Depreciation (600,000 – 100,000) ÷ 4 = 125,000. EBIT 500,000 – 260,000 – 125,000 = 115,000; tax 28,750; operating cash flow 115,000 – 28,750 + 125,000 = Tk 211,250. Terminal cash flow 100,000 + 50,000 = 150,000. Net cash flows: year 0 –650,000; years 1 to 3 211,250; year 4 361,250.

7.t2 They are mutually exclusive, because buying one means rejecting the other. The first step is to find and list the proposals, then estimate the cash flows of each scanner.

7.1 Capital budgeting is the process of planning, evaluating and choosing long-term investments. A business needs it because such decisions are long, large for the business and hard to reverse, so a mistake is costly and lasts for years.

7.2 Find proposals (which oven, which supplier); estimate cash flows (price, extra sales, running costs); evaluate and choose (judge the cash flows against the cost of funds);

implement (buy, install, train the staff); review (compare real cash flows with the forecast).

7.3 Independent projects do not affect each other, such as a new oven and a new delivery bicycle. Mutually exclusive projects allow only one, such as two different ovens for the same space, or a machine against a branch fit-out that cannot be run together.

7.4 Relevant: (a) the price, (c) the extra electricity and (d) the extra stock, because each changes only if the freezer is bought. Not relevant: (b) the report, a sunk cost, and (e) the rent, which is the same either way.

7.5 $\text{EBIT} = 240,000 - 120,000 - 40,000 = \text{Tk } 80,000$. $\text{Tax} = 0.25 \times 80,000 = \text{Tk } 20,000$. $\text{Operating cash flow} = 80,000 - 20,000 + 40,000 = \text{Tk } 100,000$. Check: $240,000 - 120,000 - 20,000 = 100,000$.

7.6 $\text{Gain} = 70,000 - 50,000 = 20,000$; $\text{tax} = 0.25 \times 20,000 = 5,000$; $\text{sale after tax} = 65,000$. Add working capital of 20,000: terminal cash flow = Tk 85,000.

CHAPTER VIII · CAPITAL BUDGETING: EVALUATION TECHNIQUES

8.p1 Project A: $60,000 \div 30,000 = 2$ years. Project B: 3 years ($3 \times 20,000 = 60,000$). Payback misses the cash after the payback point: A returns only its cost, while B returns Tk 120,000 in all.

8.p2 Present values: $12,000 \div 1.10 = 10,909$ and $12,000 \div 1.21 = 9,917$, total 20,826. $\text{NPV} = 20,826 - 20,000 = \text{Tk } 826$. It is positive, so accept.

8.t1 Payback: $X = 2.00$ years; $Y = 3 + 50,000 \div 400,000 = 3.13$ years. NPV at 10%: $X = 272,727 + 165,289 + 75,131 + 68,301 - 500,000 = \text{Tk } 81,449$; $Y = 90,909 + 123,967 + 150,263 + 273,205 - 500,000 = \text{Tk } 138,344$. Payback prefers X, net present value prefers Y. Choose Y, because it adds Tk 56,895 more value and net present value uses all the cash and its timing.

8.t2 Net present value is the present value of the inflows minus the initial investment. At the internal rate of return the discounted inflows are exactly equal to the initial investment, so the difference is zero.

8.1 After 2 years the inflows total 270,000; unrecovered 130,000. Year 3 inflow is 180,000: $130,000 \div 180,000 = 0.72$. Payback period = 2.72 years.

8.2 Present values: $109,091 + 123,967 + 135,237 + 136,603 = \text{about } 504,897$. $\text{NPV} = \text{about Tk } 104,897$. It is positive, so accept.

8.3 Payback: accept if no longer than the limit set by managers. Net present value: accept if zero or more. Internal rate of return: accept if at least the required return. Profitability index: accept if at least 1.

8.4 NPV at 20% = +4,784; at 25% = -33,920. $\text{IRR} = 20 + 5 \times 4,784 \div (4,784 + 33,920) = 20 + 0.62 = \text{about } 20.6\%$.

8.5 M: index $330,000 \div 300,000 = 1.10$; NPV = Tk 30,000. N: index $540,000 \div 500,000 = 1.08$; NPV = Tk 40,000. M ranks first by index; N ranks first by net present value.

8.6 Payback period: the time to recover the initial investment. Net present value: the worth today of the inflows minus the initial investment, in taka. Internal rate of return: the discount rate at which net present value is zero. Profitability index: present value of inflows per taka of initial investment.

8.7 Net present value should decide, because it uses all the cash flows, respects their timing and states the gain in taka. Payback ignores later cash and timing, so it serves only as a second look at how long the money is tied up.

CHAPTER IX · RETURN AND RISK

9.p1 No. The probabilities add up to $0.2 + 0.5 + 0.4 = 1.1$, and a probability distribution must add up to exactly one.

9.p2 Deviations below the expected value are negative and those above are positive, so they would cancel to zero. Squaring makes every deviation positive, so the spread shows.

9.t1 Coefficient of variation: Project X = $10,000 \div 50,000 = 0.20$; Project Y = $24,000 \div 80,000 = 0.30$. Project Y carries more risk for each unit of return.

9.1 Return = $5,600 + (82,000 - 80,000) = \text{Tk } 7,600$. Rate of return = $7,600 \div 80,000 = 0.095$, or 9.5%.

9.2 Business risk comes from the trade: sales, costs and trade conditions, for example a rise in the price of paper. Financial risk comes from borrowing: interest of Tk 60,000 a year on a loan of Tk 600,000 at 10% must be paid in poor years too, so what is left for the owners varies more.

9.3 Expected value = $0.25 \times 6 + 0.50 \times 10 + 0.25 \times 14 = 1.5 + 5.0 + 3.5 = 10\%$. Deviations $-4, 0, +4$; squares 16, 0, 16; weighted $4 + 0 + 4 = 8$; standard deviation = $\sqrt{8} = 2.83\%$.

9.4 Asset M: $3 \div 10 = 0.30$. Asset N: $9 \div 18 = 0.50$. Asset N carries more risk for each unit of return, although its expected return is higher.

CHAPTER X · RISK, REQUIRED RETURN AND MARKET EFFICIENCY

10.p1 About 10%. A beta of 2 means the investment moves about twice as far as the market, so $2 \times 5\% = 10\%$.

10.p2 Market risk premium = $10\% - 6\% = 4\%$. Required return = $6\% + 0.5 \times 4\% = 8\%$.

10.t1 Premium = $14\% - 8\% = 6\%$. Required return = $8\% + 1.0 \times 6\% = 14\%$. The expected return of 13% is below 14%, so the line lies below the security market line. Reject it.

10.1 Market risk premium = $13\% - 7\% = 6\%$. Required return = $7\% + 1.2 \times 6\% = 7\% + 7.2\% = 14.2\%$.

10.2 Firm-specific risk hits one business only, so losses at some businesses are offset by gains at others when many are held. Market risk hits all businesses at once, so holding more of them cannot remove it.

10.3 The line starts at 6% at beta 0 and rises 4 points for each whole step of beta, reaching 10% at beta 1 and 14% at beta 2. At beta 0.5 the required return is 8%; at beta 1.5 it is 12%.

10.4 Weak form: studying past price charts does not help. Semi-strong form: reading a public report after it appears does not help. Strong form: using secret information known only to managers does not help.

10.5 No. An expected return below the line is less than the required return for that much market risk, so the investment does not pay enough for the risk. The investor should reject it, or accept only if its expected return rises to the line.

CHAPTER XI · OPERATING LEVERAGE AND BREAK-EVEN ANALYSIS

11.p1 $5,000 \div 25 = 200$ units.

11.p2 DOL = $60,000 \div 20,000 = 3.00$. A fall of 10% in sales lowers EBIT by $3.00 \times 10\% = 30\%$.

11.t1 Before: $CM = 90 - 50 = \text{Tk } 40$, so break-even is $240,000 \div 40 = 6,000$ packs. After: $CM = 90 - 30 = \text{Tk } 60$, so break-even is $480,000 \div 60 = 8,000$ packs.

11.t2 Before: contribution $9,000 \times 40 = \text{Tk } 360,000$, EBIT Tk 120,000, DOL 3.00. After: contribution $9,000 \times 60 = \text{Tk } 540,000$, EBIT $540,000 - 480,000 = \text{Tk } 60,000$, DOL $540,000 \div 60,000 = 9.00$. The new machine carries more business risk, because its break-even point and its leverage are both higher.

11.1 A fixed operating cost stays the same whatever the units sold, such as shop rent. A variable operating cost grows with each unit, such as the price paid for each book bought for resale.

11.2 $CM = 150 - 95 = \text{Tk } 55$. Break-even = $330,000 \div 55 = 6,000$ units.

11.3 At 5,000 units: $5,000 \times 55 - 330,000 = -\text{Tk } 55,000$, a loss. At 6,000 units: EBIT is 0. At 7,000 units: $385,000 - 330,000 = \text{Tk } 55,000$.

11.4 Contribution = $8,000 \times 55 = \text{Tk } 440,000$. EBIT = $440,000 - 330,000 = \text{Tk } 110,000$. DOL = $440,000 \div 110,000 = 4.00$.

11.5 Near break-even EBIT is small compared with contribution, so the quotient is large. For the printing unit it is 4.00 at 8,000 packs and 3.00 at 9,000 packs. At break-even EBIT is zero and the value cannot be found.

CHAPTER XII · FINANCIAL AND TOTAL LEVERAGE

12.p1 Degree of financial leverage = $50,000 \div (50,000 - 10,000) = 50,000 \div 40,000 = 1.25$.

12.t1 A high degree of operating leverage already makes EBIT swing widely when sales change. Borrowing adds a second lever, and the two multiply, so earnings per share would swing even more. Owners who dislike a wide swing may borrow little to keep total risk lower.

12.1 Profit before tax = $150,000 - 30,000 = 120,000$. Tax = 30,000. Profit after tax = 90,000. Less preferred dividend 18,000 = 72,000. Earnings per share = $72,000 \div 10,000 = \text{Tk } 7.20$.

12.2 Preferred dividend before tax = $18,000 \div 0.75 = 24,000$. Degree of financial leverage = $150,000 \div (150,000 - 30,000 - 24,000) = 150,000 \div 96,000 = 1.56$.

12.3 Degree of total leverage = $2.50 \times 1.60 = 4.00$.

12.4 Earnings per share rises by about $4.00 \times 10\% = 40\%$. From Tk 5.00 it becomes about $\text{Tk } 5.00 \times 1.40 = \text{Tk } 7.00$.

12.5 Plan X: at EBIT 100,000, $(100,000 - 40,000) \div 4,000 = \text{Tk } 15.00$; at 60,000, $20,000 \div 4,000 = \text{Tk } 5.00$, a fall of 66.7%. Plan Y: at 100,000, $100,000 \div 8,000 = \text{Tk } 12.50$; at 60,000, $60,000 \div 8,000 = \text{Tk } 7.50$, a fall of 40%. Plan X is riskier, because its earnings per share swings more for the same fall in EBIT.

CHAPTER XIII · VALUING BONDS

13.p1 $8\% \times 500 = \text{Tk } 40$ each year.

13.p2 The investor receives $50 + 1,000 = \text{Tk } 1,050$ in one year. Value = $1,050 \div 1.10 = \text{Tk } 954.55$.

13.t1 More than the coupon rate. At 10% the value is Tk 962.09, above Tk 940. At 11% it is Tk 926.08, below Tk 940. The required return lies between 10% and 11% (about 10.6%), above 9%.

13.t2 A discount bond, because Tk 940 is below par. If the required return fell to 9%, the price would become Tk 1,000.00, equal to par.

13.1 $C = \text{Tk } 80$, $a = 3.169865$, $f = 0.683013$. Coupons: $80 \times 3.169865 = 253.589$. Par value: $1,000 \times 0.683013 = 683.013$. Value = Tk 936.60.

13.2 Discount bond: the required return is above the coupon rate, so the fixed coupon is less attractive and the price falls below par. Par bond: the two rates are equal, so the price equals par. Premium bond: the required return is below the coupon rate, so the coupon is more attractive and the price rises above par.

13.3 Coupon Tk 30, 4 periods, 4% per half-year: $a = 3.629895$, $f = 0.854804$. Coupons: $30 \times 3.629895 = 108.897$. Par value: $1,000 \times 0.854804 = 854.804$. Value = Tk 963.70.

13.4 At 10% the value is Tk 962.09; at 11% it is Tk 926.08. The fraction is $(962.09 - 950) \div (962.09 - 926.08) = 12.09 \div 36.01 = 0.336$. The yield to maturity is about 10.3%.

13.5 The coupons and the par value are fixed. A higher required return means each future payment is discounted more heavily, so its present value is smaller and the total falls.

CHAPTER XIV · VALUING COMMON AND PREFERRED STOCK

14.p1 Value = $10 \div (0.15 - 0.05) = 10 \div 0.10 = \text{Tk } 100.00$.

14.t1 The bond suits her better. It pays fixed interest by contract, and a lender is paid before owners if the business ends. Common stock pays dividends the directors may cut, and its holder has only a residual claim, paid last, so the risk of loss is higher.

14.1 $P = 9 \div 0.12 = \text{Tk } 75.00$.

14.2 $D_1 = 3.50 \times 1.05 = 3.675$. Then $r - g = 0.11 - 0.05 = 0.06$, so $P_0 = 3.675 \div 0.06 = \text{Tk } 61.25$.

14.3 If g equals r , the bottom of the fraction is zero and no value exists. If g is above r , the value is negative, which has no meaning. A business also cannot raise dividends faster than the required return for ever.

14.4 Any three: a bond holder lends and a stockholder owns; bond interest is fixed while dividends are decided by the directors; a bond gives no vote while common stock does; a lender is paid before owners; the stockholder carries more risk.

14.5 At 12%: $6 \div 0.12 = \text{Tk } 50.00$. At 15%: $6 \div 0.15 = \text{Tk } 40.00$. The value falls by Tk 10.00, because a higher required return lowers the present value.

14.6 A: $D_1 = 4.16$, so $P_0 = 4.16 \div 0.08 = \text{Tk } 52.00$. B: $D_1 = 4.28$, so $P_0 = 4.28 \div 0.05 = \text{Tk } 85.60$. The gap of Tk 33.60 arises because higher growth shrinks $r - g$ from 0.08 to 0.05.

14.7 Bond: $80 \times 2.48685 + 1,000 \times 0.75131 = 198.95 + 751.31 = \text{Tk } 950.26$. Stock: $D_1 = 2.00 \times 1.05 = 2.10$, so $P_0 = 2.10 \div 0.05 = \text{Tk } 42.00$; ten shares are Tk 420.00. Total = Tk 1,370.26.

CHAPTER XV · WORKING CAPITAL AND THE CASH CYCLE

15.p1 $40 + 15 = 55$ days.

15.p2 $70 - 30 = 40$ days. For 40 days the business must use its own money.

15.t1 Each item rises by one fifth, so the net investment is $\text{Tk } 1,010,959 \times 1.2 =$ about Tk 1,213,151. The extra money needed is about Tk 202,192. More sales means more inventory and more money owed by customers, and the amount owed to suppliers rises less than the total.

15.1 Operating cycle: $60 + 25 = 85$ days. Cash conversion cycle: $85 - 35 = 50$ days.

15.2 Inventory: $7,300,000 \times 60 \div 365 = \text{Tk } 1,200,000$. Receivables: $10,950,000 \times 25 \div 365 = \text{Tk } 750,000$. Payables: $7,300,000 \times 35 \div 365 = \text{Tk } 700,000$. Net investment: $1,200,000 + 750,000 - 700,000 = \text{Tk } 1,250,000$.

15.3 The parts are mail float, processing float and clearing float. The days lost are $3 + 2 + 2 = 7$ days.

15.4 Customers then pay in fewer days, so the receivable days fall. The operating cycle and the cash conversion cycle fall by the same number of days, and less of the business's own money is tied up.

CHAPTER XVI · SHORT-TERM FINANCING AND MATCHING

16.p1 Cost = $0.01 \div 0.99 \times 365 \div 20 = 0.010101 \times 18.25 = 0.1843$, or 18.43% a year.

16.t1 Fund the permanent Tk 300,000 long term: $300,000 \times 0.10 = \text{Tk } 30,000$. Fund the seasonal Tk 150,000 short term: $150,000 \times 0.13 \times 4 \div 12 = \text{Tk } 6,500$. Total Tk 36,500. All long term: $450,000 \times 0.10 = \text{Tk } 45,000$. Interest saved: Tk 8,500.

16.1 Cost = $0.02 \div 0.98 \times 365 \div 20 = 0.020408 \times 18.25 = 37.24\%$ a year.

16.2 Cost = $0.03 \div 0.97 \times 365 \div 35 = 0.030928 \times 10.4286 = 32.25\%$ a year.

16.3 Discount = Tk 4,000, so the payment on day 10 is Tk 196,000. Interest for 30 days = $196,000 \times 0.13 \times 30 \div 365 = \text{Tk } 2,094$, which is less than Tk 4,000. Giving up the discount costs 24.83%, above 13%. Borrow and take the discount, gaining about Tk 1,906.

16.4 3/10, net 30: $0.03 \div 0.97 \times 365 \div 20 = 56.44\%$. 1/10, net 60: $0.01 \div 0.99 \times 365 \div 50 = 7.37\%$. The first is more costly to give up, by about 49.07 percentage points.

16.5 Short: up to one year, for example trade credit or a bank overdraft. Intermediate: one to five years, for example a loan for a van repaid in instalments. Long: more than five years, for example equity from the owners or a bond.

16.6 Fund the permanent need, the stock held all year, with long-term money. Fund the extra December stock with short-term money, such as an overdraft, and repay it in January when the stock is sold.

16.7 Permanent working capital is the minimum the shop needs all year, such as basic stock and money in the till. Seasonal working capital is the extra stock and money needed only before the school term opens.

16.8 An overdraft charges interest only on the amount used and only for the days used, while a short loan charges interest on the whole amount for the whole period. A business whose need rises and falls may prefer the overdraft.

CHAPTER XVII · LONG-TERM FINANCING AND THE COST OF CAPITAL

17.p1 No. A debenture has no particular asset pledged. Here the building is pledged, so this is a loan secured by collateral, a mortgage.

17.p2 Debt: $200,000 \div 1,000,000 = 20\%$. Common equity: $800,000 \div 1,000,000 = 80\%$.

17.t1 A machine and a shop fit-out last for years and pay for themselves slowly. A short-term loan would fall due long before they have earned the money, so the

business would have to sell them or borrow again. Long-term funds stay for as long as the assets take to pay for themselves.

17.1 Any three. Collateral: the lender can take an asset if the loan is not repaid, which lowers its risk. A covenant, such as a minimum level of owners' equity: it protects the cushion behind the loan. A limit on new borrowing or on profit paid out: it stops money leaving the business. A schedule of instalments: it returns the money along the way.

17.2 A public issue offers shares to the general public. It reaches many buyers and can raise large sums, but it brings fees, disclosure and new owners. A rights issue offers new shares only to existing shareholders in proportion to their holdings. It is cheaper and keeps control with the old owners, but it raises only what they are willing to pay.

17.3 Any four. Commercial banks: term loans. Bangladesh Development Bank Limited: long-term loans to industrial projects. Investment Corporation of Bangladesh: money invested in shares and securities. IDCOL: long-term loans for infrastructure and renewable energy. Dhaka Stock Exchange and Chittagong Stock Exchange: places where shares are issued and traded.

17.4 $11\% \times (1 - 0.25) = 11\% \times 0.75 = 8.25\%$.

17.5 $0.25 \times 7.5\% = 1.875\%$. $0.05 \times 12\% = 0.60\%$. $0.70 \times 14\% = 9.80\%$. Sum: 12.275%, or about 12.28%.

CHAPTER XVIII · LEASING

18.p1 $60,000 \times (1 - 0.25) = \text{Tk } 45,000$. The payment saves Tk 15,000 of tax.

18.t1 Lease P: $60,000 \times 0.75 = \text{Tk } 45,000$ a year; $45,000 \times 2.600527 = \text{Tk } 117,024$. Lease Q: $45,000 \times 0.75 = \text{Tk } 33,750$ in years 1 and 2, worth $33,750 \times 1.795566 = \text{Tk } 60,600$; year 3 hire $70,000 \times 0.75 = \text{Tk } 52,500$, worth $52,500 \times 0.804961 = \text{Tk } 42,260$; total about Tk 102,861. Lease Q is cheaper by about Tk 14,163.

18.t2 The minibus hired for one week is an operating lease. The six-year oven lease that cannot be cancelled is a financial lease. The press sold and leased back is a sale and leaseback.

18.1 The owner is the lessor and the puller is the lessee. A lease is a contract in which the owner of an asset lets another party use it in return for regular payments.

18.2 An operating lease is short or cancellable and the lessor keeps most risks, for example a scanner hired by the month. A financial lease is long and cannot be cancelled, and the lessee in effect carries the cost and risk of owning, for example a six-year oven lease.

18.3 $80,000 \times (1 - 0.25) = \text{Tk } 60,000$. The payment lowers taxable profit by Tk 80,000, which saves Tk 20,000 of tax.

18.4 First, find the after-tax cash outflow of leasing each year. Second, find the after-tax cash outflow of buying each year. Third, convert each outflow to a present value at the after-tax cost of debt and add. Fourth, compare the totals and choose the lower.

18.5 Discount rate = $8\% \times 0.75 = 6\%$. Lease: $100,000 \times 0.75 = 75,000$ a year; $75,000 \times 2.673017 = \text{about Tk } 200,476$. Buy: $70,000 \div 1.06 + 72,000 \div 1.1236 + 74,000 \div 1.191016 = \text{about Tk } 192,249$. Buying costs about Tk 8,227 less, so buy.

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What is finance, and how are funds raised, risked and put to work? This treatise answers from the beginning. It sets out the nature and aims of finance, then follows it through the time value of money, the sources of capital, the weighing of risk against return, and the choice of investments. Each chapter opens with a puzzle, closes with review questions, and needs no earlier knowledge.